Trio Unified API User Documentation for C++ Language

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Namespace Index

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Hierarchical Index

2.1 Class Hierarchy

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Data Structure Index

3.1 Data Structures

Here are the data structures with brief descriptions:

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Chapter 5

Namespace Documentation

5.1 Trio Namespace Reference

Namespaces

- namespace UnifiedApi

5.2 Trio::UnifiedApi Namespace Reference

Data Structures

- struct DataStreamPacket

  Data used with a callback for streaming back the response of commands that return string data.
- struct FileInfo

  Data used with a callback for processing the response of Dir command.
- class ITrioConnection

  Communication object intarface.
- struct ProcessInfo

  Data used with a callback for processing the response of Process command.
- struct ProgressInfo

  Progress info helper.
- struct Range

  A structure representing a range of consecutive values.
- class TrioConnectionException

  Exception raised when error occurs in communication library.
- class TrioConnectionFactory

  Factory for creating communication instances.
- struct VersionInfo

  Version info.

Typedefs

- typedef std::function< void TRIO_CALLCONV(EventType event_id, uint64_t integer_data, const std::string↩ _view &string_data) EventCallback)

  Prototype for the callback function that will be called when events happen on the underlying connection.
- typedef std::function< void TRIO_CALLCONV(const FileInfo &file, const std::string_view &file_name) FileCallback)

  Prototype of a callback for processing the response of Dir command.
- typedef std::function< void TRIO_CALLCONV(const DataStreamPacket &data_info, const char *data) StreamHandlerCallback)

  Prototype of a callback for streaming back the response of commands that return string data.
- typedef std::function< void TRIO_CALLCONV(const ProgressInfo &progress) ProgressHandlerCallback)

  Prototype of a callback for progress report.
- typedef std::function< void TRIO_CALLCONV(const ProcessInfo &process, const std::string_view &program_name) ProcessInfoCallback)
- typedef std::function< bool TRIO_CALLCONV(const std::string_view &string) StringCallback)

Enumerations

- enum class ErrorCode : uint64_t {
  NoError = 0 , _Unknown = 0xFFFFFFFF , _Fatal = 0x8000000000 , _ApiError = 0x100000000 ,
  Api_Unknown = 0 | _ApiError | _Fatal , Fail = 1 | _ApiError | _Fatal , NotSupported = 2 | _ApiError ,
  ConnectionModule = 3 | _ApiError | _Fatal ,
  ConnectionContext = 4 | _ApiError | _Fatal , TokenTable = 5 | _ApiError | _Fatal , Timeout = 6 | _ApiError |
  _Fatal , UnknownFloatingPointFormat = 7 | _ApiError | _Fatal ,
  Remote_TokenBuffer = 8 | _ApiError | _Fatal , UnsupportedController = 10 | _ApiError | _Fatal , BadResponse
  = 11 | _ApiError | _Fatal , BadArgument = 12 | _ApiError ,
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  LowLevelError = 17 | _ApiError | _Fatal , VersionMismatch = 18 | _ApiError | _Fatal , Compression = 19 |
  _ApiError | _Fatal , Abort = 20 | _ApiError | _Fatal ,
  Disconnected = 21 | _ApiError , InvalidParamBlock = 22 | _ApiError , OutOfRange = 23 | _ApiError ,
  CapacityExceeded = 24 | _ApiError ,
  Project = 25 | _ApiError , _ApiLast = 25 | _ApiError | _Fatal , _SystemError = 0x200000000 , Sys_Unknown
  = (_SystemError | _Unknown) ,
  _ControllerError = 0x400000000 , ControllerError_COMMAND_NOT_RECOGNISED = 1 | _ControllerError ,
  ControllerError_INVALID_TRANSFER_TYPE = 2 | _ControllerError , ControllerError_FLASH_PROGRAMMING_ERROR
  = 3 | _ControllerError ,
  ControllerError_OPERAND_EXPECTED = 4 | _ControllerError , ControllerError_ASSIGNMENT_EXPECTED
  = 5 | _ControllerError , ControllerError_QUOTES_EXPECTED = 6 | _ControllerError , ControllerError_STACK_OVERFLOW
  = 7 | _ControllerError ,
  ControllerError_TOO_MANY_VARIABLES = 8 | _ControllerError , ControllerError_DIVISION_BY_ZERO = 9 |
  _ControllerError , ControllerError_EXTRA_CHARS_AT_EOLN = 10 | _ControllerError , ControllerError_CLOSE_SQUARE_BRA
  = 11 | _ControllerError ,
  ControllerError_CANNOT_MODIOFY_SPECIAL_PROGRAM = 12 | _ControllerError , ControllerError_THEN_EXPECTED_IN_
  = 13 | _ControllerError , ControllerError_FLASH_ERASE_ERROR = 14 | _ControllerError , ControllerError_START_OF_EXPRE
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  ControllerError_COMMAND_IS_COMMAND_LINE_ONLY = 24 | _ControllerError , ControllerError_COMMAND_IS_RUNTIME_
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  = 27 | _ControllerError ,

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= 31 | _ControllerError ,
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= 33 | _ControllerError , ControllerError_NO_MORE_PROGRAMS_AVAILABLE = 34 | _ControllerError ,
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ControllerError_NO_CODE_AVAILABLE_FOR_PROGRAM = 36 | _ControllerError , ControllerError_COMMAND_OUT_OF_CO
= 37 | _ControllerError , ControllerError_TOO_MANY_NESTED_STRUCTURES = 38 | _ControllerError ,
ControllerError_STRUCTURE_NESTING_ERROR = 39 | _ControllerError ,
ControllerError_ELSE_ELSEIF_ENDIF_WITHOUT_PREVIOUS_IF = 40 | _ControllerError , ControllerError_WEND_WITHOUT
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ControllerError_IDENTIFIER_EXPECTED = 43 | _ControllerError ,
ControllerError_TO_EXPECTED_AFTER_FOR = 44 | _ControllerError , ControllerError_TOO_MANY_NESTED_FOR_NEXT
= 45 | _ControllerError , ControllerError_NEXT_WITHOUT_FOR = 46 | _ControllerError , ControllerError_UNTIL_IDLE_LOADE
= 47 | _ControllerError ,
ControllerError_GOTO_GOSUB_EXEC_EXPECTED = 48 | _ControllerError , ControllerError_TOO_MANY_NESTED_GOSUB
= 49 | _ControllerError , ControllerError_RETURN_WITHOUT_GOSUB = 50 | _ControllerError ,
ControllerError_LABEL_MUST_BE_AT_START_OF_LINE = 51 | _ControllerError ,
ControllerError_CANNOT_NEST_ONE_LINE_IF = 52 | _ControllerError , ControllerError_LABEL_NOT_FOUND
= 53 | _ControllerError , ControllerError_LINE_NUMBER_CANNOT_HAVE_DECIMAL_POINT = 54 | _↩
ControllerError , ControllerError_MAXIMUM_NUMBER_OF_PROGRAM_INSTANCES_ACTIVE = 55 | _↩
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= 57 | _ControllerError , ControllerError_PROGRAM_ALREADY_EXISTS = 58 | _ControllerError ,
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ControllerError_PROGRAM_NOT_RUNNING = 76 | _ControllerError , ControllerError_PROGRAM_NOT_PAUSED_ON_THIS_
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ControllerError_DIRECTORY_STRUCTURE_INVALID = 80 | _ControllerError , ControllerError_DIRECTORY_IS_LOCKED
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| _ControllerError ,
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ControllerError_DISK_ERROR__INVALID_BOOT_SECTOR = 126 | _ControllerError , ControllerError_DISK_ERROR__INVALI
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ControllerError_FILE_ERROR__DIRECTORY_FULL = 132 | _ControllerError , ControllerError_COMMAND_ONLY_ALLOWED_
= 133 | _ControllerError , ControllerError_HASH_EXPECTED = 134 | _ControllerError , ControllerError_FOR_EXPECTED
= 135 | _ControllerError ,
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= 138 | _ControllerError , ControllerError_FILE_ALREADY_OPEN = 139 | _ControllerError ,
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= 141 | _ControllerError , ControllerError_INVALID_SYSTEM_CODE_WRONG_CONTROLLER = 142 | _↩
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ControllerError ,
ControllerError_PARAMETER_NAME_EXPECTED = 180 | _ControllerError , ControllerError_DEVICE_ERROR_NO_SDCARD_
= 181 | _ControllerError , ControllerError_DEVICE_ERROR_SDCARD_WRITE_PROTECTED = 182 | _↩
ControllerError , ControllerError_FAILED_ATTEMPT_1 = 183 | _ControllerError ,
ControllerError_FAILED_ATTEMPT_2 = 184 | _ControllerError , ControllerError_FAILED_TOO_MANY_TIMES_PLEASE_RESE
= 185 | _ControllerError , ControllerError_LINE_MUST_BE_INSIDE_FUNCTION = 186 | _ControllerError ,
ControllerError_DUPLICATE_NAME_USED_IN_LABEL_AND_VARIABLE = 187 | _ControllerError ,
ControllerError_STRUCTURE_FIELD_NAME_NOT_RECOGNISED = 188 | _ControllerError , ControllerError_ID_NOT_DEFIN
= 189 | _ControllerError , ControllerError_ROBOT_CONFIGURATION_CANNOT_BE_CHANGED_WHILE_MOVEL
= 190 | _ControllerError , ControllerError_TOOL_OFFSET_ID_NOT_DEFINED = 191 | _ControllerError ,
ControllerError_OBJECT_FRAME_ID_NOT_DEFINED = 192 | _ControllerError , ControllerError_ROBOT_FRAME_ID_NOT_D
= 193 | _ControllerError , ControllerError_COMMAND_NOT_PERMITTED_IN_FUNCTION = 194 | _↩
ControllerError , ControllerError_TOO_MANY_FUNCTION_PARAMETERS = 195 | _ControllerError ,
ControllerError_OPERATION_CANNOT_BE_PERFORMED_UNTIL_IDLE = 196 | _ControllerError ,
ControllerError_SINGULARITY = 197 | _ControllerError , ControllerError_TOOL_CANNOT_BE_MODIFIED_OR_DELETED_W
= 198 | _ControllerError , ControllerError_OBJECT_FRAME_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED
= 199 | _ControllerError ,
ControllerError_ROBOT_FRAME_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED = 200 |
_ControllerError , ControllerError_ROBOT_MOVES_CANNOT_BE_PERFORMED_UNTIL_JOG_OPERATION_IS_DISABLED
= 201 | _ControllerError , ControllerError_CYCLIC_INCLUDE_DEPENDENCY = 202 | _ControllerError ,
ControllerError_OBJECT_FRAME_ALREADY_ATTACHED = 203 | _ControllerError ,
ControllerError_OBJECT_FRAME_HAS_NOT_BEEN_ATTACHED = 204 | _ControllerError , ControllerError_OBJECT_FRAME_
= 205 | _ControllerError , ControllerError_BASE_AXIS_DOES_NOT_BELONG_TO_ANY_KINEMATIC_GROUP
= 206 | _ControllerError , ControllerError_SYNC_IN_PROCESS = 207 | _ControllerError ,
ControllerError_ROBOT_PROGRAM_STOPPED = 208 | _ControllerError , ControllerError_TARGET_POINT_NOT_ACTIVATED
= 209 | _ControllerError , ControllerError_PROGRAM_FAILED_TO_RUN = 210 | _ControllerError ,
ControllerError_RETURN_STATEMENT_EXPECTED_BEFORE_ENDFUNC = 211 | _ControllerError ,
ControllerError_CONTINUE_WITHOUT_PREVIOUS_FOR_WHILE_OR_REPEAT = 212 | _Controller↩
Error , ControllerError_EXIT_WITHOUT_PREVIOUS_FOR_WHILE_OR_REPEAT = 213 | _ControllerError
, ControllerError_TOO_MANY_EXIT_STATEMENTS_WITHIN_SINGLE_BLOCK = 214 | _ControllerError ,
ControllerError_CASE_OR_END_CASE_WITHOUT_PREVIOUS_SELECT_CASE = 215 | _ControllerError ,
ControllerError_FIXED_INTEGER_OR_ELSE_EXPECTED = 216 | _ControllerError , ControllerError_TOO_MANY_CASE_OPT
= 217 | _ControllerError , ControllerError_ROBOTICS_RUNTIME_LIMITED_TO_1_HOUR = 218 | _↩
ControllerError , ControllerError_COLLISION_OBJECTS_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED
= 219 | _ControllerError ,
ControllerError_TOO_MANY_COLLISION_OBJECTS_ACTIVE = 220 | _ControllerError , ControllerError_ATYPE_MUST_BE_C
= 221 | _ControllerError , ControllerError_COMMAND_NOT_ALLOWED_IN_FRAME = 222 | _ControllerError
, ControllerError_COMMAND_LINE_TOO_LONG = 223 | _ControllerError ,
ControllerError_TABLE_LOCATION_WRITE_PROTECTED = 224 | _ControllerError , ControllerError_ENCRYPTION_AUTHOR
= 225 | _ControllerError , ControllerError_NO_FREE_ENCRYPTION_AUTHORITY = 226 | _ControllerError ,
ControllerError_LOCAL_TARGET_VALUE_HAS_NOT_BEEN_DEFINED = 227 | _ControllerError ,
ControllerError_COMMAND_NOT_PERMITTED_IN_BASIC_LIBRARY = 228 | _ControllerError , ControllerError_INVALID_MO
= 229 | _ControllerError , ControllerError_FUNCTION_EXPECTED = 230 | _ControllerError , ControllerError_CAM_TABLE_ID_
= 231 | _ControllerError ,
ControllerError_RUNNING_OF_RPS_PROGRAMS_INHIBITED_USE_DISABLE_ROBOT = 232 | _↩
ControllerError , ControllerError_U_AND_V_ORIENTATION_INHIBITED_WHILE_WRIST_SINGULARITY
= 233 | _ControllerError , ControllerError_TARGET_POINT_OUT_OF_ROBOT_WORKING_AREA = 234 |
_ControllerError , ControllerError_AXIS_LOCKED_TO_A_ROBOT = 235 | _ControllerError ,
ControllerError_ROBOT_CONTROLLED_BY_ANOTHER_PROCESS = 236 | _ControllerError , ControllerError_ROBOT_NOT_
= 237 | _ControllerError , ControllerError_COMMAND_INHIBITED_ROBOT_MUST_BE_USED_TO_ACTIVATE_A_KINEMATIC
= 238 | _ControllerError , ControllerError_UNDEFINED_VARIABLE_IN_DIM_INITIALISER = 239 | _↩
ControllerError ,
ControllerError_THE_SPECIFIED_ROBOT_CONFIGURATION_IS_NOT_ALLOWED = 240 | _Controller↩
Error , ControllerError_TOO_MANY_AXES_ACTIVATED = 241 | _ControllerError , ControllerError_MULTIPLE_MOVE_TRIGGE

= 242 | _ControllerError , ControllerError_MOVE_TRIGGER_RESPONSE_NOT_SPECIFIED = 243 | _↩
ControllerError ,
ControllerError_MOVE_TRIGGER_NOT_SPECIFIED = 244 | _ControllerError , ControllerError_INVALID_ARRAY_DIMENSION
= 245 | _ControllerError , ControllerError_DUPLICATE_VR_INDEX = 246 | _ControllerError , ControllerError_CONSTANT_VALU
= 247 | _ControllerError ,
ControllerError_RPS_KERNEL_V2_NOT_ENABLED = 248 | _ControllerError , ControllerError_INCLUDE_FILE_MUST_CONTA
= 249 | _ControllerError , ControllerError_TARGET_TAUGHT_WITH_ANOTHER_ROBOT = 250 | _↩
ControllerError , ControllerError_TARGET_ROBOT_CONFIG_MISMATCH = 251 | _ControllerError ,
ControllerError_TARGET_TOOL_OFFSET_MISMATCH = 252 | _ControllerError , ControllerError_TARGET_OBJECT_FRAME_
= 253 | _ControllerError , ControllerError_TARGET_ROBOT_FRAME_MISMATCH = 254 | _ControllerError ,
ControllerError_INVALID_FRAME_NUMBER = 255 | _ControllerError ,
ControllerError_FRAME_NOT_SUPPORTED_BY_RPS = 256 | _ControllerError , ControllerError_FORWARD_TRANSFORM_F
= 257 | _ControllerError , ControllerError_INVERSE_TRANSFORM_FUNCTION_NOT_DEFINED = 258 |
_ControllerError , ControllerError_INVALID_MOVEOPTION = 259 | _ControllerError ,
ControllerError_MAX_OPEN_KEYWORDS_REACHED = 260 | _ControllerError , ControllerError_KEYWORD_INACTIVE
= 261 | _ControllerError , ControllerError_KEYWORD_FUNCTION_NOT_DEFINED = 262 | _ControllerError
, ControllerError_KEYWORD_SAME_AS_SYSTEM_WORD = 263 | _ControllerError ,
ControllerError_UNEXPECTED_CHARACTERS_AFTER_LINE_CONTINUATION = 264 | _Controller↩
Error , ControllerError_THIS_COMMAND_CAN_BE_EXECUTED_ONLY_ONCE = 265 | _ControllerError ,
ControllerError_TABLE_MEMORY_CONFLICT = 266 | _ControllerError , ControllerError_MAX_OPEN_MOVES_REACHED
= 267 | _ControllerError ,
ControllerError_USER_DEFINED_ERROR = 268 | _ControllerError , ControllerError_INVALID_FORMAT_SPECIFIER
= 269 | _ControllerError , ControllerError_OPEN_MOVE_FUNCTIONS_NOT_DEFINED = 270 | _Controller↩
Error , ControllerError_TOO_MANY_CHARS_IN_STR = 271 | _ControllerError ,
ControllerError_INVALID_OPEN_ARCH_FUNC_DEFINITION = 272 | _ControllerError , ControllerError_PROCESS_IN_USE
= 273 | _ControllerError , ControllerError_VR_LOCATION_WRITE_PROTECTED = 274 | _ControllerError }

Enumeration of error codes used in Unified API.

- enum class EventType { Error , Warning , Message }

  Events that may occur on a connection.

- enum class ConnectionType : uint32_t { Unknown = 0x00 , TCP = 0x01 , PCMCAT = 0x02 , FLEX7 = 0x04 }

  Connection types.

- enum class ProgressEvent { Start , Progress , Finish }

  Progress event.

- enum class Co_ObjectType {
  Boolean = 1 , Integer8 = 2 , Integer16 = 3 , Integer32 = 4 ,
  Unsigned8 = 5 , Unsigned16 = 6 , Unsigned32 = 7 , Real32 = 8 ,
  Real64 = 17 , Integer64 = 21 , Unsigned64 = 27 }

  Co Read/Write operation object types.

- enum class EthercatState { Initial = 0 , PreOperational = 1 , SafeOperational = 2 , Operational = 3 }

  EtherCAT state.

- enum class EthercatStartOption : uint32_t { None = 0x0000 , PrintOutProgress = 0x0001 , ForceRescan = 0x0002 , ExpressStartup = 0x0004 }

  EtherCAT start option.

- enum class FileTransferOption : uint32_t { None = 0x0000 , CompressOnly = 0x0001 , CompressDecompress = 0x0002 }

  File transfer option.

- enum class FileType : int32_t {
  Undefined = -1 , BASIC = 0 , EncryptedBASIC = 1 , GCode = 2 ,
  TextFile = 3 , IecProgram = 4 , FifoFile = 5 , TempFile = 6 ,
  Project = 7 , Binary = 8 , MCConfig = 9 , IecLibrary = 10 ,
  HmiDesign = 11 , BASICLibrary = 12 , EncryptedBASICibrary = 13 , HmiLibrary = 14 ,
  TempVolatileFile = 15 , Robot = 16 , RobotBasic = 17 , TempTextFile = 18 ,
  RobotLibrary = 19 , RobotBasicLibrary = 20 , Recipe = 21 , Alarms = 22 ,
  RealtimeBASICLibrary = 23 , EncryptedRealtimeBasicLibrary = 24 , EncryptedIecProgram = 25 ,
  EncryptedIecLibrary = 26 ,
  __Size }

File types on the Motion Coordinator.

- enum class FileFlags : uint32_t { None = 0x00 , Editable = 0x01 , Autorun = 0x02 }

  Flags related to files and programs in internal storage on the Motion Coordinator.

- enum class ConnectMode : uint32_t { MPOS = 0 , DPOS = 1 }

  Mode for CONNECT command.

- enum class PSwitchMode : uint32_t { None = 0x00 , Invert = 0x02 , DPOS = 0x04 , Mask = Invert | DPOS }

  Flags PSWITCH command in single/cyclic mode.

- enum class PSwitchMultiMode : uint32_t {
  None = 0x00 , DPOS = 0x04 , TimerMode = 0x10 , NegativeDirection = 0x20 ,
  ZOut = 0x40 , Mask = DPOS | TimerMode | NegativeDirection | ZOut }

  Flags PSWITCH command in multi/sequence mode.

- enum class CamBoxLinkOptions : uint32_t {
  None = 0 , LinkMark = 1 , LinkAbsolute = 2 , LinkRepeat = 4 ,
  LinkPattern = 8 , LinkPositiveMove = 32 , LinkForceStart = 128 , LinkMarkB = 256 ,
  LinkR_Mark = 512 , LinkThreshold = 16384 }

  Link options in CamBox command.

- enum class AxisType : uint16_t {
  None = 0 , Stepper = 1 , Servo = 2 , Encoder = 3 ,
  StepperEncoderInt = 4 , Resolver = 5 , VOut = 6 , SSI = 7 ,
  CAN = 8 , CAN_Remote = 9 , PSwitch = 10 , DLink_Remote = 11 ,
  Servo2 = 12 , Yaskawa = 13 , EncoderOut = 14 , HighestMotionType = EncoderOut ,
  Dummy15 = 15 , Dummy16 = 16 , Dummy17 = 17 , DS402PosType1 = 18 ,
  DS402VelType1 = 19 , First_Comms = 20 , CAN_Comms = First_Comms , USB_Comms = 21 ,
  DLink_Comms = 22 , Profibus_Comms = 23 , Dummy24 = 24 , Dummy25 = 25 ,
  DS402PosType2 = 26 , DS402VelType2 = 27 , Dummy28 = 28 , Dummy29 = 29 ,
  DACout_ANALin = 30 , ANALout_ANALin = 31 , Dummy32 = 32 , Dummy33 = 33 ,
  Dummy34 = 34 , SDout_PWMout = 35 , SDout_PWMout2 = 36 , PWMout_SDFREQin = 37 ,
  PWMout_SDin = 37 , Dummy38 = 38 , Dummy39 = 39 , MechatroPos = 40 ,
  MechatroSpeed = 41 , MechatroTorque = 42 , SDout_Zout = 43 , DACout_QUADin = 44 ,
  QUADout_Zout = 45 , DACout_TAMAin = 46 , DACout_ENDATin = 47 , DACout_SSIin = 48 ,
  MechatroInv = 49 , PanasonicPos = 50 , PanasonicSpeed = 51 , PanasonicTorque = 52 ,
  SercosSpeed = 53 , SercosPos = 54 , SercosTorque = 55 , SercosOpen = 56 ,
  SercosSpeedReg = 57 , SercosPosReg = 58 , SercosEncoder = 59 , DACout_SDin = 60 ,
  SLM32 = 61 , PLM32 = 62 , SDout_Zin = 63 , QUADout_Zin = 64 ,
  EthercatPos = 65 , EthercatSpeed = 66 , EthercatTorque = 67 , EthercatOpenSpeed = 68 ,
  EthercatEncoderIn = 69 , EthercatInAnalOut = 70 , EthercatPosTrio = 71 , EthercatSpeedTrio = 72 ,
  EthercatDummy5 = 73 , EthercatDummy6 = 74 , DACout_SSISLAVEin = 75 , QUADin = 76 ,
  DACout_QUADin_Zout = 77 , SDout_VFFDACout = 78 , SDin = 79 , G5Pos = 80 ,
  G5Speed = 81 , G5Torque = 82 , Jacquard = 83 , QUADout_VFFDACout = 84 ,
  AxesDiff = 85 , TAMAin = 86 , ENDATin = 87 , SSIin = 88 ,
  SDFREQout_ANALin = 89 , DACout_SDFREQin = 90 , AFFDACout = 91 , DACout_BISSin = 92 ,
  BISSin = 93 , DACPWMout_QUADin = 94 , SDout_TAMAin = 95 , Dummy96 = 96 ,
  QUADin_Zout = 97 , Dummy98 = 98 , Dummy99 = 99 , SDout_HD = 100 ,
  QUADout_HD = 101 , DACout_QUADin_HD = 102 , DACout_SDin_HD = 103 , QUADin_HD = 104 ,
  SSIin_HD = 105 , TAMAin_HD = 106 , Dummy107 = 107 , Dummy108 = 108 ,
  Dummy109 = 109 , SDout_SDin_HD = 110 , SDout_QUADin_HD = 111 , QUADout_SDin_HD = 112 ,
  QUADout_QUADin_HD = 113 , SDout_TAMAin_HD = 114 , SDout_ENDATin_HD = 115 , SDout_SSIin_HD
  = 116 ,
  SDout_BISSin_HD = 117 , QUADout_TAMAin_HD = 118 , QUADout_ENDATin_HD = 119 , QUADout_SSIin_HD
  = 120 ,
  QUADout_BISSin_HD = 121 , Dummy122 = 122 , Dummy123 = 123 , Dummy124 = 124 ,
  Dummy125 = 125 , Dummy126 = 126 , Dummy127 = 127 , Dummy128 = 128 ,
  Dummy129 = 129 , SLMvirtual = 200 , PLMvirtual = 201 }

  Enumeration of known axis types used with ATYPE axis parameter.

- enum class AxisMode : uint32_t {
  NONE = 0 , NOT_USED_1 = ((uint32_t)1 << 0) , ALLOW_CONNECT_MOVES_IN_LIMIT = ((uint32_t)1
  << 1) , DIRECTION_MODE_3D = ((uint32_t)1 << 2) ,

CANCEL_ADDAX_ON_LIMIT = ((uint32_t)1 << 3) , DISABLE_LOOKAHEAD = ((uint32_t)1 << 4) , NOT_USED_2 = ((uint32_t)1 << 5) , USE_NON_SIGNEXTENDED_ANALOGUE_FEEDBACK = ((uint32↩_t)1 << 6) ,
NOT_USED_3 = ((uint32_t)1 << 7) , SOFTWARE_LIMIT_USES_DPOS = ((uint32_t)1 << 8) , DISABLE_FE_ACCUMULATION_DURING_MOVE = ((uint32_t)1 << 9) , USE_FASTDEC_WHEN_CANCELLING_RPS_JOG = ((uint32_t)1 << 10) ,
DELAY_POSITION_EVALUATION_TO_LOAD_TIME = ((uint32_t)1 << 11) , DIRECTION_SHORTEST = ((uint32_t)1 << 12) , MONITOR_FE = ((uint32_t)1 << 13) , OVERRIDE_FE_LIMITS = ((uint32_t)1 << 14) ,
DUAL_AXIS_AVERAGE_POS = ((uint32_t)1 << 15) , INVERT_AXIS_DIRECTION = ((uint32_t)1 << 16) , CANCEL_MTYPE_ON_MASTER_IF_ADDAX_HITS_LIM = ((uint32_t)1 << 17) }

Enumeration of axis mode flags used with AXIS_MODE axis parameter.

- enum class CornerMode : uint32_t {
NONE = 0 , RESERVED = ((uint32_t)1 << 0) , AUTO_SPEED_CONTROL = ((uint32_t)1 << 1) , CORNER_STATE_MACHINE = ((uint32_t)1 << 2) ,
AUTO_RADIUS_SPEED_CONTROL = ((uint32_t)1 << 3) , AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE = ((uint32_t)1 << 4) , AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE_USE_ACCEL = ((uint32_t)1 << 5) , AUTO_BLEND_MODE_0_USING_CORNER_RADIUS = ((uint32_t)1 << 6) ,
AUTO_BLEND_MODE_1_USING_CORNER_RADIUS = ((uint32_t)1 << 7) }

  Enumeration of corner mode flags used with CORNER_MODE axis parameter.

- enum class CornerState : uint32_t { NONE = 0 , HELD = ((uint32_t)1 << 0) , LOAD = ((uint32_t)1 << 1) , LOAD_ZERO_SPEED = ((uint32_t)1 << 2) }

  Enumeration of corner state flags used with CORNER_STATE axis parameter.

- enum class MotionType : uint32_t {
IDLE = 0 , MOVE = 1 , MOVEABS = 2 , MHELICAL = 3 ,
MOVECIRC = 4 , MOVEMODIFY = 5 , MOVESP = 6 , MOVEABSSP = 7 ,
MOVECIRCSP = 8 , MHELICALSP = 9 , FORWARD = 10 , REVERSE = 11 ,
DATUM = 12 , CAM = 13 , FORWARD_JOG = 14 , REVERSE_JOG = 15 ,
WETCAM = 16 , CAMBOX = 20 , CONNECT = 21 , MOVELINK = 22 ,
CONNPATH = 23 , FLEXLINK = 24 , CAMLINK = 25 , COMPENSATE_XY = 26 ,
PICKLINK = 27 , CAMIN = 28 , MOVETANG = 30 , MSPHERICAL = 31 ,
MTRANSCIRC = 32 , MOVESEQ = 33 , MOVEABSSEQ = 34 , MOVETURN = 35 ,
MOVEPICK = 36 , MOVECONTOUR = 37 , MOVEDISPENSE = 38 , MSPHERICALSP = 40 ,
MTRANSCIRCSP = 41 , MOVESEQSP = 42 , MOVEABSSEQSP = 43 , MOVEJ = 44 ,
MOVEL = 45 , MOVEC = 46 , MOVEBLEND = 47 , MOVEJREL = 48 ,
MOVELREL = 49 , MOVETEST = 50 , MOVEP = 51 , MOVEU = 52 ,
MOVEB = 53 , MOVESPCONT = 60 , MOVEABSSPCONT = 61 , GEARIN = 62 ,
ACCPROFILE = 70 , VELPROFILE = 71 , POSPROFILE = 72 , COMBINEAXES = 73 ,
OPENMOVE_BASE = 100 , OPENMOVE_1 = OPENMOVE_BASE , OPENMOVE_2 = OPENMOVE_BASE + 1 , OPENMOVE_3 = OPENMOVE_BASE + 2 ,
OPENMOVE_4 = OPENMOVE_BASE + 3 , OPENMOVE_5 = OPENMOVE_BASE + 4 }

  Motion types returned by MTYPE and NTYPE axis parameters.

- enum class AxisStatus : uint64_t {
NONE = 0 , LOOKAHEAD_OVERRIDE = ((uint64_t)1 << 0) , FE_WARNING = ((uint64_t)1 << 1) , COMMS_ERROR = ((uint64_t)1 << 2) ,
REMOTE_DRIVE_ERROR = ((uint64_t)1 << 3) , IN_FORWARD_LIMIT = ((uint64_t)1 << 4) , IN_REVERSE_LIMIT = ((uint64_t)1 << 5) , DATUMING = ((uint64_t)1 << 6) ,
FEEDHOLD = ((uint64_t)1 << 7) , FE_EXCEEDS_LIMIT = ((uint64_t)1 << 8) , FORWARD_SOFTWARE_LIMIT = ((uint64_t)1 << 9) , REVERSE_SOFTWARE_LIMIT = ((uint64_t)1 << 10) ,
CANCEL_CURRENT = ((uint64_t)1 << 11) , PULSE_OVER_SPEED = ((uint64_t)1 << 12) , OVERRIDE_TO_ZERO = ((uint64_t)1 << 13) , MOVE_CANCEL_DONE = ((uint64_t)1 << 14) ,
VOLUME_LIMIT = ((uint64_t)1 << 15) , AXIS_FS_LIMIT = ((uint64_t)1 << 16) , AXIS_RS_LIMIT = ((uint64_t)1 << 17) , ENCODER_OVER_I = ((uint64_t)1 << 18) ,
PSWITCH_FIFO_NOT_EMPTY = ((uint64_t)1 << 19) , PSWITCH_FIFO_FULL = ((uint64_t)1 << 20) , KINEMATICS_1HOUR = ((uint64_t)1 << 21) , BISS_CLEAN_WARNING = ((uint64_t)1 << 22) ,
WORLD_FS_LIMIT = ((uint64_t)1 << 23) , WORLD_RS_LIMIT = ((uint64_t)1 << 24) , CANCEL_CURRENT_AT_SPEED = ((uint64_t)1 << 25) , REMOTE_NODE_STATUS_ERROR = ((uint64_t)1 << 26) ,

PSWITCH_FIFO_HALF_FULL = ((uint64_t)1 << 27) , MOVEADD = ((uint64_t)1 << 28) , CANCEL_MOVEADD = ((uint64_t)1 << 29) , REMOTE_NODE_ERROR2 = ((uint64_t)1 << 30) , SINGULARITY = ((uint64_t)1 << 31) , OPEN_FRAME_ERROR = ((uint64_t)1 << 32) , VP_ERROR = ((uint64_t)1 << 33) , CANCEL_BUFFERED = ((uint64_t)1 << 34) , CANCEL_BUFFERED_DECEL_INTO_BUFFERED_MOVES = ((uint64_t)1 << 35) , CANCEL_BUFFERED_DECEL_WITHIN_ = ((uint64_t)1 << 36) , CANCEL_LAST_BUFFERED = ((uint64_t)1 << 37) }

- enum class RobotStatus : uint32_t {
  NONE = 0 , WORLD_FS_LIMIT = ((uint32_t)1 << 0) , WORLD_RS_LIMIT = ((uint32_t)1 << 1) , ROBOT_FS_LIMIT = ((uint32_t)1 << 2) , ROBOT_RS_LIMIT = ((uint32_t)1 << 3) , TCP_FS_LIMIT = ((uint32_t)1 << 4) , TCP_RS_LIMIT = ((uint32↩_t)1 << 5) , FE_WARNING = ((uint32_t)1 << 6) , FE_EXCEEDS_LIMIT = ((uint32_t)1 << 7) , WRIST_SINGULARITY = ((uint32_t)1 << 8) , ALIGNMENT_SINGULARITY = ((uint32_t)1 << 9) , ELBOW_SINGULARITY = ((uint32_t)1 << 10) , MAX_SPEED_LIMIT = ((uint32_t)1 << 11) , COLLIDED = ((uint32_t)1 << 12) , NOT_IN_POS = ((uint32_t)1 << 13) , ELBOW_CLOSED_SINGULARITY = ((uint32_t)1 << 14) , DISALLOW_JOG = ((uint32_t)1 << 15) }
- enum class SyncControl : int32_t {
  WAITING_TO_START = -1 , NO_SYNC = 0 , STARTING_SYNC = 1 , PERFORMING_SYNC = 2 , SYNCHRONISED = 3 , STOPPING_SYNC = 4 , STARTING_INTERP_MOVE = 5 , PERFORMING_INTERP_MOVE = 6 , STARTING_RESYNC = 10 , PERFORMING_RESYNC = 11 , STARTING_RESYNC_TO_USERFRAME = 20 , PERFORMING_RESYNC_TO_USERFRAME = 21 }
- enum class RegistMode : int32_t {
  Cancel = 0 , ZAMarkRising = 1 , ZAMarkFalling = 2 , RAInputRising = 3 , RAInputFalling = 4 , RAInputRising_ZBMarkRising = 6 , RAInputRising_ZBMarkFalling = 7 , RAInputFalling_ZBMarkRising = 8 , RAInputFalling_ZBMarkFalling = 9 , RAInputRising_RBInputRising = 10 , RAInputRising_RBInputFalling = 11 , RAInputFalling_RBInputRising = 12 , RAInputFalling_RBInputFalling = 13 , ZBMarkRising = 14 , ZBMarkFalling = 15 , RBInputRising = 16 , RBInputFalling = 17 , HardwareRegistration = 20 , TimeInterpolatedRegistration = 21 , MultiChannelRegistration = 22 , EnableFilter = 23 , DisableFilter = 24 , InsideOpenCloseWindow = 256 , OutsideOpenCloseWindow = 768 }
- enum class RegistEdge : int32_t { Rising = 0 , Falling = 1 }
- enum class RegistWindow : int32_t {
  No = 0 , InsideOpenCloseWin = 1 , OutsideOpenCloseWin = 2 , VirtualInputHigh = 3 , VirtualInputLow = 4 }
- enum class ProcessStatus : uint32_t {
  Stopped = 0 , Ready , Stepping , Paused , Pausing , Aborted , Suspend , Suspended , Sleeping }

  The execution state of the process.
- enum class FileExistsFlags : int32_t {
  NoFile = 0 , HasFileNotOpen = -1 , HasFileOpenForReading = -2 , HasFileOpenForWriting = -3 , HasFileOpenForReadingAndWriting = -4 }

  Enumeration of results of IS_FILE/IS_PROG command.
- enum class SecurityCodeType : int32_t { Controller = 0 , OEM = 1 , User = 2 }

  The security code type used with project decryption key functions.

5.2.1 Typedef Documentation

5.2.1.1 EventCallback

```
typedef std::function<void TRIO_CALLCONV(EventType event_id, uint64_t integer_data, const std::string_view& string_data) Trio::UnifiedApi::EventCallback)
```

Prototype for the callback function that will be called when events happen on the underlying connection.

Definition at line 2385 of file Trio_UnifiedApi_CPP.h.

5.2.1.2 FileCallback

```
typedef std::function<void TRIO_CALLCONV(const FileInfo& file, const std::string_view& file_↩
name) Trio::UnifiedApi::FileCallback)
```

Prototype of a callback for processing the response of Dir command.

Definition at line 2429 of file Trio_UnifiedApi_CPP.h.

5.2.1.3 ProcessInfoCallback

```
typedef std::function<void TRIO_CALLCONV(const ProcessInfo& process, const std::string_view&
program_name) Trio::UnifiedApi::ProcessInfoCallback)
```

Definition at line 2555 of file Trio_UnifiedApi_CPP.h.

5.2.1.4 ProgressHandlerCallback

```
typedef std::function<void TRIO_CALLCONV(const ProgressInfo& progress) Trio::UnifiedApi::↩
ProgressHandlerCallback)
```

Prototype of a callback for progress report.

Definition at line 2488 of file Trio_UnifiedApi_CPP.h.

5.2.1.5 StreamHandlerCallback

```
typedef std::function<void TRIO_CALLCONV(const DataStreamPacket& data_info, const char* data)
Trio::UnifiedApi::StreamHandlerCallback)
```

Prototype of a callback for streaming back the response of commands that return string data.

Definition at line 2457 of file Trio_UnifiedApi_CPP.h.

5.2.1.6 StringCallback

```
typedef std::function<bool TRIO_CALLCONV(const std::string_view& string) Trio::UnifiedApi::↩
StringCallback)
```

Definition at line 2557 of file Trio_UnifiedApi_CPP.h.

5.2.2 Enumeration Type Documentation

5.2.2.1 AxisMode

```
enum class Trio::UnifiedApi::AxisMode : uint32_t [strong]
```

Enumeration of axis mode flags used with AXIS_MODE axis parameter.

Enumerator

NONE	
NOT_USED_1	
ALLOW_CONNECT_MOVES_IN_LIMIT	
DIRECTION_MODE_3D	
CANCEL_ADDAX_ON_LIMIT	
DISABLE_LOOKAHEAD	
NOT_USED_2	
USE_NON_SIGNEXTENDED_ANALOGUE_FEEDBACK	
NOT_USED_3	
SOFTWARE_LIMIT_USES_DPOS	
DISABLE_FE_ACCUMULATION_DURING_MOVE	
USE_FASTDEC_WHEN_CANCELLING_RPS_JOG	
DELAY_POSITION_EVALUATION_TO_LOAD_TIME	
DIRECTION_SHORTEST	
MONITOR_FE	
OVERRIDE_FE_LIMITS	
DUAL_AXIS_AVERAGE_POS	
INVERT_AXIS_DIRECTION	
CANCEL_MTYPE_ON_MASTER_IF_ADDAX_HITS_LIM	

Definition at line 1568 of file Trio_UnifiedApi_CPP.h.

5.2.2.2 AxisStatus

```
enum class Trio::UnifiedApi::AxisStatus : uint64_t [strong]
```

Enumerator

NONE	
LOOKAHEAD_OVERRIDE	'L' - Speed limit active
FE_WARNING	'W' - Following error warning range
COMMS_ERROR	'A' - Communications error to remote drive
REMOTE_DRIVE_ERROR	'M' - Remote drive error (EtherCAT Use: Set if status word Fault Flag raised)
IN_FORWARD_LIMIT	'F' - In forward hardware limit
IN_REVERSE_LIMIT	'R' - In reverse hardware limit
DATUMING	'D' - Datuming in progress
FEEDHOLD	'H' - Feedhold active
FE_EXCEEDS_LIMIT	'E' - Following error exceeds limit
FORWARD_SOFTWARE_LIMIT	'X' - Forward software limit active
REVERSE_SOFTWARE_LIMIT	'Y' - Reverse software limit active
CANCEL_CURRENT	'C' - Cancelling the move (from CANCEL(0), CANCEL(2) and CANCEL(5), the latter two also setting AS_CANCEL_BUFFERED and AS_CANCEL_MOVEADD respectively)
PULSE_OVER_SPEED	'O' - Pulse output axis overspeed
OVERRIDE_TO_ZERO	'T' - Tangential move decelerating
MOVE_CANCEL_DONE	' ' - TODO: Doesn't seem to be used

Enumerator

VOLUME_LIMIT	'V' - Volume limit active (TODO: I can't see where this is used - Remove?)
AXIS_FS_LIMIT	'I' - Axis forward software limit active
AXIS_RS_LIMIT	'J' - Axis reverse software limit active
ENCODER_OVER_I	'P' - Encoder poer supply overload
PSWITCH_FIFO_NOT_EMPTY	'N' - Hardware P-switch FIFO not empty
PSWITCH_FIFO_FULL	'B' - Hardware P-switch FIFO full
KINEMATICS_1HOUR	'K' - Kinematic frame stopped (due to 1 hour timeout)
BISS_CLEAN_WARNING	' ' - BISS encoder warning flag
WORLD_FS_LIMIT	' ' - World forward software limit active (TODO: This is a duplicate of RS_WORLD_FS_LIMIT)
WORLD_RS_LIMIT	' ' - World reverse software limit active (TODO: This is a duplicate of RS_WORLD_RS_LIMIT)
CANCEL_CURRENT_AT_SPEED	' ' - Mode 3 cancel status (from CANCEL(3))
REMOTE_NODE_STATUS_ERROR	' ' - Remote node status error (EtherCAT Use: Set based on masked flags in status word)
PSWITCH_FIFO_HALF_FULL	' ' - Hardware P-switch FIFO half full
MOVEADD	' ' - MOVEADD additional offset active (axis has MOVEADD in progress)
CANCEL_MOVEADD	' ' - MOVEADD is being cancelled (from CANCEL(5))
REMOTE_NODE_ERROR2	' ' - EtherCAT emergency message received from remote drive (EtherCAT Use: Set if emergency message received)
SINGULARITY	' ' - TODO: Doesn't seem to be used (all the code is inside a #if 0)
OPEN_FRAME_ERROR	' ' - A user frame has thrown an error
VP_ERROR	' ' - Indicates that the VPU is in an error state
CANCEL_BUFFERED	' ' - Cancel all buffered moves (from CANCEL(1) or CANCEL(2), the latter also setting AS_CANCEL_CURRENT)
CANCEL_BUFFERED_DECEL_INTO_↩ BUFFERED_MOVES	' ' - Signifies that deceleration will stray into the buffered moves (stopping distance bigger than distance to go)
CANCEL_BUFFERED_DECEL_WITHIN_↩ CURRENT_MOVE	' ' - Signifies that deceleration will finish at the end of the current move (stopping distance less than distance to go)
CANCEL_LAST_BUFFERED	' ' - Cancel last buffered move (from CANCEL(4))

Definition at line 1864 of file Trio_UnifiedApi_CPP.h.

5.2.2.3 AxisType

```
enum class Trio::UnifiedApi::AxisType : uint16_t [strong]
```

Enumeration of known axis types used with ATYPE axis parameter.

Enumerator

None	

Enumerator

Stepper	
Servo	
Encoder	
StepperEncoderInt	
Resolver	
VOut	
SSI	
CAN	
CAN_Remote	
PSwitch	
DLink_Remote	
Servo2	
Yaskawa	
EncoderOut	
HighestMotionType	
Dummy15	
Dummy16	
Dummy17	
DS402PosType1	
DS402VelType1	
First_Comms	
CAN_Comms	
USB_Comms	
DLink_Comms	
Profibus_Comms	
Dummy24	
Dummy25	
DS402PosType2	
DS402VelType2	
Dummy28	
Dummy29	
DACout_ANALin	
ANALout_ANALin	
Dummy32	
Dummy33	
Dummy34	
SDout_PWMout	
SDout_PWMout2	PWM variant that uses path distance to generate output.
PWMout_SDFREQin	
PWMout_SDin	
Dummy38	
Dummy39	
MechatroPos	This is the first MC464 usable type.
MechatroSpeed	
MechatroTorque	
SDout_Zout	Standard stepper axis.
DACout_QUADin	Standard analogue servo axis with encoder feedback.
QUADout_Zout	

Enumerator

DACout_TAMAin	
DACout_ENDATin	
DACout_SSIin	
MechatroInv	
PanasonicPos	
PanasonicSpeed	
PanasonicTorque	
SercosSpeed	
SercosPos	
SercosTorque	
SercosOpen	
SercosSpeedReg	
SercosPosReg	
SercosEncoder	
DACout_SDin	
SLM32	
PLM32	
SDout_Zin	
QUADout_Zin	
EthercatPos	
EthercatSpeed	
EthercatTorque	
EthercatOpenSpeed	
EthercatEncoderIn	
EthercatInAnalOut	
EthercatPosTrio	
EthercatSpeedTrio	
EthercatDummy5	
EthercatDummy6	
DACout_SSISLAVEin	
QUADin	Standard encoder only axis.
DACout_QUADin_Zout	
SDout_VFFDACout	
SDin	
G5Pos	
G5Speed	
G5Torque	
Jacquard	
QUADout_VFFDACout	
AxesDiff	
TAMAin	
ENDATin	
SSIin	
SDFREQout_ANALin	
DACout_SDFREQin	
AFFDACout	
DACout_BISSin	
BISSin	
DACPWMout_QUADin	

Enumerator

SDout_TAMAin	
Dummy96	
QUADin_Zout	
Dummy98	
Dummy99	
SDout_HD	
QUADout_HD	
DACout_QUADin_HD	
DACout_SDin_HD	
QUADin_HD	
SSIin_HD	
TAMAin_HD	
Dummy107	
Dummy108	
Dummy109	
SDout_SDin_HD	
SDout_QUADin_HD	
QUADout_SDin_HD	
QUADout_QUADin_HD	
SDout_TAMAin_HD	
SDout_ENDATin_HD	
SDout_SSIin_HD	
SDout_BISSin_HD	
QUADout_TAMAin_HD	
QUADout_ENDATin_HD	
QUADout_SSIin_HD	
QUADout_BISSin_HD	
Dummy122	
Dummy123	
Dummy124	
Dummy125	
Dummy126	
Dummy127	
Dummy128	
Dummy129	
SLMvirtual	
PLMvirtual	

Definition at line 1277 of file Trio_UnifiedApi_CPP.h.

5.2.2.4 CamBoxLinkOptions

```
enum class Trio::UnifiedApi::CamBoxLinkOptions : uint32_t [strong]
```

Link options in CamBox command.

Enumerator

None	No flags are set.
LinkMark	link commences exactly when registration event MARK occurs on link axis
LinkAbsolute	link commences at an absolute position on link axis
LinkRepeat	CAMBOX repeats automatically and bi-directionally when this bit is set.
LinkPattern	PATTERN mode. Advanced use of CAMBOX: allows multiple scale values to be used.
LinkPositiveMove	Link is only active during a positive move on the link axis.
LinkForceStart	Forces the profile to start at a defined point in the link_dist.
LinkMarkB	link commences exactly when registration event MARKB occurs on link axis
LinkR_Mark	link commences exactly when registration event R_MARK occurs on link axis.
LinkThreshold	Link is only active during a negative move on the link axisLink is active during positive movement from a threshold.

Definition at line 1221 of file Trio_UnifiedApi_CPP.h.

5.2.2.5 Co_ObjectType

```
enum class Trio::UnifiedApi::Co_ObjectType [strong]
```

Co Read/Write operation object types.

Enumerator

Boolean	Boolean type.
Integer8	8 bit integer
Integer16	16 bit integer
Integer32	32 bit integer
Unsigned8	8 bit unsigned integer
Unsigned16	16 bit unsigned integer
Unsigned32	32 bit unsigned integer
Real32	32 bit floating point
Real64	64 bit floating point
Integer64	64 bit integer
Unsigned64	64 bit unsigned integer

Definition at line 825 of file Trio_UnifiedApi_CPP.h.

5.2.2.6 ConnectionType

```
enum class Trio::UnifiedApi::ConnectionType : uint32_t [strong]
```

Connection types.

Enumerator

Unknown	Unknown connection type. Returned when there is no connection context.
TCP	TCP connection via Remote protocol.
PCMCAT	Shared memory connection for PC-MCAT devices.
FLEX7	Shared memory connection for Flex7 devices.

Definition at line 778 of file Trio_UnifiedApi_CPP.h.

5.2.2.7 ConnectMode

```
enum class Trio::UnifiedApi::ConnectMode : uint32_t [strong]
```

Mode for CONNECT command.

Enumerator

MPOS	Measured position (MPOS) is followed by connected axis.
DPOS	Demand position (DPOS) is followed by connected axis.

Definition at line 1137 of file Trio_UnifiedApi_CPP.h.

5.2.2.8 CornerMode

```
enum class Trio::UnifiedApi::CornerMode : uint32_t [strong]
```

Enumeration of corner mode flags used with CORNER_MODE axis parameter.

Enumerator

NONE	
RESERVED	
AUTO_SPEED_CONTROL	
CORNER_STATE_MACHINE	
AUTO_RADIUS_SPEED_CONTROL	
AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE	
AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE_USE_ACCEL	
AUTO_BLEND_MODE_0_USING_CORNER_RADIUS	
AUTO_BLEND_MODE_1_USING_CORNER_RADIUS	

Definition at line 1612 of file Trio_UnifiedApi_CPP.h.

5.2.2.9 CornerState

```
enum class Trio::UnifiedApi::CornerState : uint32_t [strong]
```

Enumeration of corner state flags used with CORNER_STATE axis parameter.

Enumerator

NONE	
HELD	
LOAD	
LOAD_ZERO_SPEED	

Definition at line 1636 of file Trio_UnifiedApi_CPP.h.

5.2.2.10 ErrorCode

```
enum class Trio::UnifiedApi::ErrorCode : uint64_t [strong]
```

Enumeration of error codes used in Unified API.

Enumerator

NoError	No error.
_Unknown	An error code that cannot be determined.
_Fatal	A fatal error flag.
_ApiError	UnifiedAPI error flag. The actual error code is in the lowest 32 bits.
Api_Unknown	An UnifiedAPI error that cannot be determined.
Fail	General error.
NotSupported	Error indicating that operation is not supported.
ConnectionModule	Requested connection type (module) cannot be load/initialized or is missing.
ConnectionContext	A bad connection context is used.
TokenTable	TOKENTABLE reading failed.
Timeout	The operation timed out.
UnknownFloatingPointFormat	The data is of unknown floating point format.
Remote_TokenBuffer	Error in the token buffer for the Remote protocol.
UnsupportedController	This Motion Coordinator model/version is not supported.
BadResponse	Bad/unrecognized response received.
BadArgument	A bad argument has been passed.
Reopen	The connection or object is already opened.
BufferTooSmall	The passed buffer is too small.
SocketRead	The read to the socket failed.
SocketWrite	The write to the socket failed.
LowLevelError	Low level API call failed.
VersionMismatch	Versions of modules do not match.
Compression	Compress/Decompress data error.
Abort	The operation got aborted.
Disconnected	No connection.
InvalidParamBlock	The parameter block is invalid.
OutOfRange	Value is out of range.
CapacityExceeded	Capacity exceeded.
Project	Project error.
_ApiLast	The last API error.
_SystemError	A system error flag. The actual system error code is in the lowest 32 bits.
Sys_Unknown	Unknown system error.
_ControllerError	A controller error flag. The actual controller error code is in the lowest 32 bits.
ControllerError_COMMAND_NOT_RECOGNISED	Controller Error codes.
ControllerError_INVALID_TRANSFER_TYPE	
ControllerError_FLASH_PROGRAMMING_ERROR	

Enumerator

ControllerError_OPERAND_EXPECTED	
ControllerError_ASSIGNMENT_EXPECTED	
ControllerError_QUOTES_EXPECTED	
ControllerError_STACK_OVERFLOW	
ControllerError_TOO_MANY_VARIABLES	
ControllerError_DIVISION_BY_ZERO	
ControllerError_EXTRA_CHARS_AT_EOLN	
ControllerError_CLOSE_SQUARE_BRACKET_↩ EXPECTED_IN_PRINT	
ControllerError_CANNOT_MODIOFY_SPECIAL_↩ PROGRAM	
ControllerError_THEN_EXPECTED_IN_IF_ELSEIF	
ControllerError_FLASH_ERASE_ERROR	
ControllerError_START_OF_EXPRESSION_↩ EXPECTED	
ControllerError_CLOSE_ROUND_BRACKET_↩ EXPECTED	
ControllerError_COMMA_EXPECTED	
ControllerError_COMMAND_LINE_BROKEN_BY_↩ ESC	
ControllerError_PARAMETER_OUT_OF_RANGE	
ControllerError_NO_PROCESS_AVAILABLE	
ControllerError_VALUE_IS_READ_ONLY	
ControllerError_MODIFIER_NOT_ALLOWED	
ControllerError_REMOTE_AXIS_IS_IN_USE	
ControllerError_COMMAND_IS_COMMAND_LINE↩ _ONLY	
ControllerError_COMMAND_IS_RUNTIME_ONLY	
ControllerError_LABEL_EXPECTED	
ControllerError_PROGRAM_NOT_FOUND	
ControllerError_DUPLICATE_IDENTIFIER	
ControllerError_PROGRAM_IS_LOCKED	
ControllerError_PROGRAMS_RUNNING	
ControllerError_PROGRAM_IS_STOPPED	
ControllerError_CANNOT_SELECT_PROGRAM	
ControllerError_NO_PROGRAM_SELECTED	
ControllerError_NO_MORE_PROGRAMS_↩ AVAILABLE	
ControllerError_OUT_OF_MEMORY	
ControllerError_NO_CODE_AVAILABLE_FOR_↩ PROGRAM	
ControllerError_COMMAND_OUT_OF_CONTEXT	
ControllerError_TOO_MANY_NESTED_↩ STRUCTURES	
ControllerError_STRUCTURE_NESTING_ERROR	
ControllerError_ELSE_ELSEIF_ENDIF_WITHOUT↩ _PREVIOUS_IF	
ControllerError_WEND_WITHOUT_PREVIOUS_↩ WHILE	
ControllerError_UNTIL_WITHOUT_PREVIOUS_↩ REPEAT	

Enumerator

ControllerError_IDENTIFIER_EXPECTED	
ControllerError_TO_EXPECTED_AFTER_FOR	
ControllerError_TOO_MANY_NESTED_FOR_NEXT	
ControllerError_NEXT_WITHOUT_FOR	
ControllerError_UNTIL_IDLE_LOADED_↩ EXPECTED_AFTER_WAIT	
ControllerError_GOTO_GOSUB_EXEC_EXPECTED	
ControllerError_TOO_MANY_NESTED_GOSUB	
ControllerError_RETURN_WITHOUT_GOSUB	
ControllerError_LABEL_MUST_BE_AT_START_↩ OF_LINE	
ControllerError_CANNOT_NEST_ONE_LINE_IF	
ControllerError_LABEL_NOT_FOUND	
ControllerError_LINE_NUMBER_CANNOT_HAVE↩ _DECIMAL_POINT	
ControllerError_MAXIMUM_NUMBER_OF_↩ PROGRAM_INSTANCES_ACTIVE	
ControllerError_INVALID_USE_OF_DOLAR_SIGN	
ControllerError_VR_X_OR_STRING_VARIABLE_↩ EXPECTED	
ControllerError_PROGRAM_ALREADY_EXISTS	
ControllerError_PROCESS_ALREADY_SELECTED	
ControllerError_DUPLICATE_AXES_NOT_↩ PERMITTED	
ControllerError_PLC_TYPE_IS_INVALID	
ControllerError_EVALUATION_ERROR	
ControllerError_RESERVED_KEYWORD_NOT_↩ AVAILABLE_ON_THIS_CONTROLLER	
ControllerError_VARIABLE_NOT_FOUND	
ControllerError_TABLE_INDEX_RANGE_ERROR	
ControllerError_FEATURES_ENABLED_DO_NOT↩ _ALLOW_ATYPE_CHANGE	
ControllerError_INVALID_LINE_NUMBER	
ControllerError_STRING_EXCEEDS_PERMITTED↩ _LENGTH	
ControllerError_SCOPE_PERIOD_SHOULD_↩ EXCEED_NUMBER_OF_AIN_PARAMS	
ControllerError_VALUE_IS_INCORRECT	
ControllerError_INVALID_IO_CHANNEL	
ControllerError_FLASH_VALUE_CANNOT_BE_SET	
ControllerError_DIRECTORY_NOT_LOCKED	
ControllerError_DIRECTORY_ALREADY_LOCKED	
ControllerError_PROGRAM_NOT_RUNNING_ON↩ _THIS_PROCESS	
ControllerError_PROGRAM_NOT_RUNNING	
ControllerError_PROGRAM_NOT_PAUSED_ON_↩ THIS_PROCESS	
ControllerError_PROGRAM_NOT_PAUSED	
ControllerError_COMMAND_NOT_ALLOWED_↩ WHEN_RUNNING_MOTION_PERFECT	

Enumerator

ControllerError_DIRECTORY_STRUCTURE_↩ INVALID	
ControllerError_DIRECTORY_IS_LOCKED	
ControllerError_CANNOT_EDIT_PROGRAM	
ControllerError_TOO_MANY_NESTED_OPERANDS	
ControllerError_CANNOT_RESET_WHEN_DRIVE↩ _SERVO_ON	
ControllerError_FLASH_STICK_BLANK	
ControllerError_FLASH_STICK_NOT_AVAILABLE↩ _ON_THIS_CONTROLLER	
ControllerError_SLAVE_ERROR	
ControllerError_MASTER_ERROR	
ControllerError_NETWORK_TIMEOUT	
ControllerError_NETWORK_PROTOCOL_ERROR	
ControllerError_GLOBAL_DEFINITION_IS_↩ DIFFERENT	
ControllerError_INVALID_PROGRAM_NAME	
ControllerError_PROGRAM_CORRUPT	
ControllerError_MORE_THAN_ONE_PROGRAM_↩ RUNNING_WHEN_TRYING_TO_SET_GLOBAL_↩ OR_CONSTANT	
ControllerError_PROGRAM_ENCRYPTED	
ControllerError_BASIC_TOKEN_DEFINITION_↩ INCORRECT	
ControllerError_OPEN_BRACKET_EXPECTED	
ControllerError_NUMBER_EXPECTED	
ControllerError_AS_EXPECTED	
ControllerError_DATA_TYPE_EXPECTED	
ControllerError_STRING_EXPECTED	
ControllerError_DOWNLOAD_ABORT_OR_↩ TIMEOUT	
ControllerError_CANNOT_SPECIFY_PROGRAM_↩ TYPE_FOR_AN_EXISTING_PROGRAM	
ControllerError_INVALID_COFF_IMAGE_FILE	
ControllerError_VARIABLE_DEFINED_OUTSIDE_↩ INCLUDE_FILE	
ControllerError_COMMAND_NOT_ALLOWED_↩ WITHIN_INCLUDE_FILE	
ControllerError_SERIAL_NUMBER_MUST_BE_↩ MINUS_1	
ControllerError_APPEND_BLOCK_INCONSISTENT	
ControllerError_INVALID_RANGE_SPECIFIED	
ControllerError_TOO_MANY_ITEMS_DEFINED_↩ FOR_BLOCK	
ControllerError_INVALID_MSPHERICAL_INPUT	
ControllerError_TOO_MANY_LABELS	
ControllerError_SYMBOL_TABLE_LOCKED	
ControllerError_INCORRECT_SYMBOL_TYPE	
ControllerError_VARIABLES_NOT_PERMITTED_↩ ON_COMMAND_LINE	
ControllerError_INVALID_PROGRAM_TYPE	

Enumerator

ControllerError_PARAMETER_EXPECTED	
ControllerError_FIRMWARE_ERROR__DEVICE_↩ IN_USE	
ControllerError_DEVICE_ERROR__TIMEOUT_↩ WAITING_FOR_DEVICE	
ControllerError_DEVICE_ERROR__COMMAND_↩ NOT_SUPPORTED_BY_DEVICE	
ControllerError_DEVICE_ERROR__CRC_ERROR	
ControllerError_DEVICE_ERROR__ERROR_↩ WRITING_TO_DEVICE	
ControllerError_DEVICE_ERROR__INVALID_↩ RESPONSE_FROM_DEVICE	
ControllerError_FIRMWARE_ERROR__CANNOT_↩ REFERENCE_DATA_OUTSIDE_CURRENT_BLOCK	
ControllerError_DISK_ERROR__INVALID_MBR	
ControllerError_DISK_ERROR__INVALID_BOOT_↩ SECTOR	
ControllerError_DISK_ERROR__INVALID_↩ SECTOR_CLUSTER_REFERENCE	
ControllerError_FILE_ERROR__DISK_FULL	
ControllerError_FILE_ERROR__FILE_NOT_FOUND	
ControllerError_FILE_ERROR__FILENAME_↩ ALREADY_EXISTS	
ControllerError_FILE_ERROR__INVALID_FILENAME	
ControllerError_FILE_ERROR__DIRECTORY_FULL	
ControllerError_COMMAND_ONLY_ALLOWED_↩ WHEN_RUNNING_MOTION_PERFECT	
ControllerError_HASH_EXPECTED	
ControllerError_FOR_EXPECTED	
ControllerError_INPUT__OUTPUT__APPEND__↩ FIFO_READ__FIFO_WRITE_EXPECTED	
ControllerError_FILE_NOT_OPEN	
ControllerError_END_OF_FILE	
ControllerError_FILE_ALREADY_OPEN	
ControllerError_INVALID_STORAGE_AREA	
ControllerError_INVALID_FLOATING_POINT_↩ OPERATION	
ControllerError_INVALID_SYSTEM_CODE_↩ WRONG_CONTROLLER	
ControllerError_IEC_ERROR__INVALID_↩ VARIABLE_ACCESS	
ControllerError_NUMERICAL_ERROR__NAN_USED	
ControllerError_NUMERICAL_ERROR__INF_USED	
ControllerError_NUMERICAL_ERROR__↩ SUBNORMAL_VALUE_USED	
ControllerError_MAC_EEPROM_IS_LOCKED	
ControllerError_INVALID_MIX_OF_DATA_TYPES	
ControllerError_INVALID_STARTUP_↩ CONFIGURATION_COMMAND	
ControllerError_SYMBOL_IS_NOT_A_VARIABLE	

Enumerator

ControllerError_KINEMATICS_RUNTIME_↩ LIMITED_TO_1_HOUR_FEC_22	
ControllerError_IEC_RUNTIME_LIMITED_TO_1_↩ HOUR_FEC_21	
ControllerError_COMMAND_NOT_ALLOWED_↩ WITH_CURRENT_ATYPE	
ControllerError_WILDCARD_LENGTH_MUST_BE_1	
ControllerError_INCOMPATIBLE_ARRAY_↩ DIMENSIONS	
ControllerError_MATRIX_IS_SINGULAR	
ControllerError_PROGRAM_IS_NOT_↩ EXECUTABLE_TYPE	
ControllerError_DISK_ERROR__FORMAT_MUST↩ _BE_FAT32	
ControllerError_PROGRAM_IS_STOPPED_HALT_↩ FORCED	
ControllerError_VARIABLE_NOT_DEFINED_↩ USING_DIM_OR_GLOBAL_CONSTANT	
ControllerError_VARIABLE_NOT_DEFINED_↩ USING_DIM_WITHIN_INCLUDE_FILE	
ControllerError_PARAMETER_CAN_ONLY_BE_↩ CHANGED_USING_MC_CONFIG	
ControllerError_IEC_TASK_MEMORY_↩ ALLOCATION_ERROR	
ControllerError_IEC_IO_BINDING_INDEX_OUT_↩ OF_RANGE	
ControllerError_IEC_VR_BINDING_INDEX_OUT_↩ OF_RANGE	
ControllerError_IEC_TABLE_BINDING_INDEX_↩ OUT_OF_RANGE	
ControllerError_FAILED_OPEN_CONNECTION	
ControllerError_MCCONFIG_CHANGED_EX_↩ REQUIRED	
ControllerError_NO_RESPONSE_FROM_CAN_↩ MODULE	
ControllerError_INVALID_FPGA_VERSION	
ControllerError_VARIABLE_NOT_CREATED_↩ USING_GLOBAL_CONSTANT	
ControllerError_PARAMETER_CAN_ONLY_BE_↩ CHANGED_USING_MC_CONFIG_OR_↩ COMMAND_LINE	
ControllerError_ETHERCAT_AXES_CANNOT_BE_↩ SET_USING_ATYPE_PLEASE_USE_DRIVE_MODE	
ControllerError_BIT_NUMBER_OUT_OF_RANGE	
ControllerError_FUNCTION_NAME_EXPECTED	
ControllerError_FUNCTION_MUST_BE_DEFINED↩ _WITHIN_A_BASIC_LIBRARY	
ControllerError_TOO_MANY_FUNCTIONS_↩ DEFINED	
ControllerError_MISSING_ENDFUNC	

Enumerator

ControllerError_ENDFUNC_WITHOUT_↩ PREVIOUS_FUNCTION	
ControllerError_PARAMETER_NAME_EXPECTED	
ControllerError_DEVICE_ERROR_NO_SDCARD_↩ INSERTED	
ControllerError_DEVICE_ERROR_SDCARD_↩ WRITE_PROTECTED	
ControllerError_FAILED_ATTEMPT_1	
ControllerError_FAILED_ATTEMPT_2	
ControllerError_FAILED_TOO_MANY_TIMES_↩ PLEASE_RESET	
ControllerError_LINE_MUST_BE_INSIDE_↩ FUNCTION	
ControllerError_DUPLICATE_NAME_USED_IN_↩ LABEL_AND_VARIABLE	
ControllerError_STRUCTURE_FIELD_NAME_↩ NOT_RECOGNISED	
ControllerError_ID_NOT_DEFINED	
ControllerError_ROBOT_CONFIGURATION_↩ CANNOT_BE_CHANGED_WHILE_MOVEL	
ControllerError_TOOL_OFFSET_ID_NOT_DEFINED	
ControllerError_OBJECT_FRAME_ID_NOT_↩ DEFINED	
ControllerError_ROBOT_FRAME_ID_NOT_DEFINED	
ControllerError_COMMAND_NOT_PERMITTED_↩ IN_FUNCTION	
ControllerError_TOO_MANY_FUNCTION_↩ PARAMETERS	
ControllerError_OPERATION_CANNOT_BE_↩ PERFORMED_UNTIL_IDLE	
ControllerError_SINGULARITY	
ControllerError_TOOL_CANNOT_BE_MODIFIED_↩ OR_DELETED_WHILE_SELECTED	
ControllerError_OBJECT_FRAME_CANNOT_BE_↩ MODIFIED_OR_DELETED_WHILE_SELECTED	
ControllerError_ROBOT_FRAME_CANNOT_BE_↩ MODIFIED_OR_DELETED_WHILE_SELECTED	
ControllerError_ROBOT_MOVES_CANNOT_BE_↩ PERFORMED_UNTIL_JOG_OPERATION_IS_↩ DISABLED	
ControllerError_CYCLIC_INCLUDE_DEPENDENCY	
ControllerError_OBJECT_FRAME_ALREADY_↩ ATTACHED	
ControllerError_OBJECT_FRAME_HAS_NOT_↩ BEEN_ATTACHED	
ControllerError_OBJECT_FRAME_ALREADY_↩ SYNCED	
ControllerError_BASE_AXIS_DOES_NOT_↩ BELONG_TO_ANY_KINEMATIC_GROUP	
ControllerError_SYNC_IN_PROCESS	
ControllerError_ROBOT_PROGRAM_STOPPED	
ControllerError_TARGET_POINT_NOT_ACTIVATED	

Enumerator

ControllerError_PROGRAM_FAILED_TO_RUN	
ControllerError_RETURN_STATEMENT_↩ EXPECTED_BEFORE_ENDFUNC	
ControllerError_CONTINUE_WITHOUT_↩ PREVIOUS_FOR_WHILE_OR_REPEAT	
ControllerError_EXIT_WITHOUT_PREVIOUS_↩ FOR_WHILE_OR_REPEAT	
ControllerError_TOO_MANY_EXIT_STATEMENTS↩ _WITHIN_SINGLE_BLOCK	
ControllerError_CASE_OR_END_CASE_↩ WITHOUT_PREVIOUS_SELECT_CASE	
ControllerError_FIXED_INTEGER_OR_ELSE_↩ EXPECTED	
ControllerError_TOO_MANY_CASE_OPTIONS	
ControllerError_ROBOTICS_RUNTIME_LIMITED_↩ TO_1_HOUR	
ControllerError_COLLISION_OBJECTS_CANNOT↩ _BE_MODIFIED_OR_DELETED_WHILE_↩ SELECTED	
ControllerError_TOO_MANY_COLLISION_↩ OBJECTS_ACTIVE	
ControllerError_ATYPE_MUST_BE_↩ CONFIGURED_USING_MC_CONFIG_↩ PARAMETER_DRIVE_MODE	
ControllerError_COMMAND_NOT_ALLOWED_IN_↩ FRAME	
ControllerError_COMMAND_LINE_TOO_LONG	
ControllerError_TABLE_LOCATION_WRITE_↩ PROTECTED	
ControllerError_ENCRYPTION_AUTHORITY_NOT↩ _FOUND	
ControllerError_NO_FREE_ENCRYPTION_↩ AUTHORITY	
ControllerError_LOCAL_TARGET_VALUE_HAS_↩ NOT_BEEN_DEFINED	
ControllerError_COMMAND_NOT_PERMITTED_↩ IN_BASIC_LIBRARY	
ControllerError_INVALID_MOVEC_INPUT	
ControllerError_FUNCTION_EXPECTED	
ControllerError_CAM_TABLE_ID_NOT_DEFINED	
ControllerError_RUNNING_OF_RPS_↩ PROGRAMS_INHIBITED_USE_DISABLE_ROBOT	
ControllerError_U_AND_V_ORIENTATION_↩ INHIBITED_WHILE_WRIST_SINGULARITY	
ControllerError_TARGET_POINT_OUT_OF_↩ ROBOT_WORKING_AREA	
ControllerError_AXIS_LOCKED_TO_A_ROBOT	
ControllerError_ROBOT_CONTROLLED_BY_↩ ANOTHER_PROCESS	
ControllerError_ROBOT_NOT_DEFINED_USING_↩ KINEMATIC_GROUP	

Enumerator

ControllerError_COMMAND_INHIBITED_ROBOT_↩ MUST_BE_USED_TO_ACTIVATE_A_KINEMATIC_↩ GROUP	
ControllerError_UNDEFINED_VARIABLE_IN_DIM↩ _INITIALISER	
ControllerError_THE_SPECIFIED_ROBOT_↩ CONFIGURATION_IS_NOT_ALLOWED	
ControllerError_TOO_MANY_AXES_ACTIVATED	
ControllerError_MULTIPLE_MOVE_TRIGGERS_↩ SPECIFIED	
ControllerError_MOVE_TRIGGER_RESPONSE_↩ NOT_SPECIFIED	
ControllerError_MOVE_TRIGGER_NOT_SPECIFIED	
ControllerError_INVALID_ARRAY_DIMENSION	
ControllerError_DUPLICATE_VR_INDEX	
ControllerError_CONSTANT_VALUE_EXPECTED	
ControllerError_RPS_KERNEL_V2_NOT_ENABLED	
ControllerError_INCLUDE_FILE_MUST_CONTAIN↩ _DEFCONST_DIM_STATEMENTS_ONLY	
ControllerError_TARGET_TAUGHT_WITH_↩ ANOTHER_ROBOT	
ControllerError_TARGET_ROBOT_CONFIG_↩ MISMATCH	
ControllerError_TARGET_TOOL_OFFSET_↩ MISMATCH	
ControllerError_TARGET_OBJECT_FRAME_↩ MISMATCH	
ControllerError_TARGET_ROBOT_FRAME_↩ MISMATCH	
ControllerError_INVALID_FRAME_NUMBER	
ControllerError_FRAME_NOT_SUPPORTED_BY_↩ RPS	
ControllerError_FORWARD_TRANSFORM_↩ FUNCTION_NOT_DEFINED	
ControllerError_INVERSE_TRANSFORM_↩ FUNCTION_NOT_DEFINED	
ControllerError_INVALID_MOVEOPTION	
ControllerError_MAX_OPEN_KEYWORDS_↩ REACHED	
ControllerError_KEYWORD_INACTIVE	
ControllerError_KEYWORD_FUNCTION_NOT_↩ DEFINED	
ControllerError_KEYWORD_SAME_AS_SYSTEM↩ _WORD	
ControllerError_UNEXPECTED_CHARACTERS_↩ AFTER_LINE_CONTINUATION	
ControllerError_THIS_COMMAND_CAN_BE_↩ EXECUTED_ONLY_ONCE	
ControllerError_TABLE_MEMORY_CONFLICT	
ControllerError_MAX_OPEN_MOVES_REACHED	
ControllerError_USER_DEFINED_ERROR	
ControllerError_INVALID_FORMAT_SPECIFIER	

Enumerator

ControllerError_OPEN_MOVE_FUNCTIONS_NOT↩ _DEFINED	
ControllerError_TOO_MANY_CHARS_IN_STR	
ControllerError_INVALID_OPEN_ARCH_FUNC_↩ DEFINITION	
ControllerError_PROCESS_IN_USE	
ControllerError_VR_LOCATION_WRITE_↩ PROTECTED	

Definition at line 35 of file Trio_UnifiedApi_CPP.h.

5.2.2.11 EthercatStartOption

```
enum class Trio::UnifiedApi::EthercatStartOption : uint32_t [strong]
```

EtherCAT start option.

Enumerator

None	No option is selected.
PrintOutProgress	Print out progress information on - Print out progress information (default) off - Do not print progress information.
ForceRescan	Network re-scan off - Normal state change (default) on - Force a re-scan of the network.
ExpressStartup	Start-up type on - Express start-up; do not wait for PLL to lock between state changes off - Standard start-up.

Definition at line 912 of file Trio_UnifiedApi_CPP.h.

5.2.2.12 EthercatState

```
enum class Trio::UnifiedApi::EthercatState [strong]
```

EtherCAT state.

Enumerator

Initial	Initial.
PreOperational	Pre operational.
SafeOperational	Safe operational.
Operational	Operational.

Definition at line 886 of file Trio_UnifiedApi_CPP.h.

5.2.2.13 EventType

```
enum class Trio::UnifiedApi::EventType [strong]
```

Events that may occur on a connection.

Enumerator

Error	Error processing a command.
Warning	Warning for non critical problem.
Message	Information about the command being processed.

Definition at line 757 of file Trio_UnifiedApi_CPP.h.

5.2.2.14 FileExistsFlags

```
enum class Trio::UnifiedApi::FileExistsFlags : int32_t [strong]
```

Enumeration of results of IS_FILE/IS_PROG command.

Enumerator

NoFile	File or program does not exist on Motion Coordinator.
HasFileNotOpen	File or program exists and is not open.
HasFileOpenForReading	File or program exists and is open for reading.
HasFileOpenForWriting	File or program exists and is open for writing.
HasFileOpenForReadingAndWriting	File or program exists and is open for reading and writing.

Definition at line 2320 of file Trio_UnifiedApi_CPP.h.

5.2.2.15 FileFlags

```
enum class Trio::UnifiedApi::FileFlags : uint32_t [strong]
```

Flags related to files and programs in internal storage on the Motion Coordinator.

Enumerator

None	No flags are set.
Editable	If set file is editable on the controller.
Autorun	If set program can be auto-run.

Definition at line 1116 of file Trio_UnifiedApi_CPP.h.

5.2.2.16 FileTransferOption

```
enum class Trio::UnifiedApi::FileTransferOption : uint32_t [strong]
```

File transfer option.

Enumerator

None	No option is selected.
CompressOnly	The data gets compressed and transferred. The other side receives and stores the compressed data.
CompressDecompress	The data gets compressed, transferred and decompressed. This saves bandwidth.

Definition at line 944 of file Trio_UnifiedApi_CPP.h.

5.2.2.17 FileType

```
enum class Trio::UnifiedApi::FileType : int32_t [strong]
```

File types on the Motion Coordinator.

Enumerator

Undefined	The file type is not recognized.
BASIC	TrioBASIC program.
EncryptedBASIC	Encrypted TrioBASIC program.
GCode	GCode program.
TextFile	Unknown contents.
IecProgram	IEC program.
FifoFile	FIFO (not valid in project)
TempFile	Temporary File (not valid in project)
Project	Project file.
Binary	Container for anything that the controller does not understand.
MCConfig	Special for the MC_CONFIG file.
IecLibrary	IEC library.
HmiDesign	HMI.
BASICLibrary	Trio BASIC library.
EncryptedBASICibrary	Encrypted Trio BASIC library.
HmiLibrary	HMI Library.
TempVolatileFile	Volatile temporary controller file stored in RAM (not valid in project)
Robot	Robot program.
RobotBasic	Robot basic program.
TempTextFile	Temporary Text File (not valid in project)
RobotLibrary	Robot library.
RobotBasicLibrary	Robot basic library.
Recipe	Recipe data.
Alarms	Alarm data.
RealtimeBASICLibrary	Real-time basic library.
EncryptedRealtimeBasicLibrary	Encrypted real-time basic library.
EncryptedIecProgram	Encrypted IEC program.
EncryptedIecLibrary	Encrypted IEC library.
__Size	Not a file type. Used to size the FileType enumeration.

Definition at line 965 of file Trio_UnifiedApi_CPP.h.

5.2.2.18 MotionType

```
enum class Trio::UnifiedApi::MotionType : uint32_t [strong]
```

Motion types returned by MTYPE and NTYPE axis parameters.

Enumerator

IDLE	
MOVE	
MOVEABS	
MHELICAL	
MOVECIRC	
MOVEMODIFY	
MOVESP	Lookahead scannable.
MOVEABSSP	Lookahead scannable.
MOVECIRCSP	Lookahead scannable.
MHELICALSP	Lookahead scannable.
FORWARD	
REVERSE	
DATUM	
CAM	
FORWARD_JOG	
REVERSE_JOG	
WETCAM	
CAMBOX	Linked move (no VPU required)
CONNECT	Linked move (no VPU required)
MOVELINK	Linked move (no VPU required)
CONNPATH	Linked move (no VPU required)
FLEXLINK	Linked move (no VPU required)
CAMLINK	Linked move (no VPU required)
COMPENSATE_XY	Linked move (no VPU required)
PICKLINK	Linked move (no VPU required)
CAMIN	Linked move (no VPU required)
MOVETANG	??? From FORWARD to less than this number, no lookahead
MSPHERICAL	
MTRANSCIRC	
MOVESEQ	
MOVEABSSEQ	
MOVETURN	
MOVEPICK	
MOVECONTOUR	
MOVEDISPENSE	
MSPHERICALSP	Lookahead scannable.
MTRANSCIRCSP	Lookahead scannable.
MOVESEQSP	Lookahead scannable.
MOVEABSSEQSP	Lookahead scannable.
MOVEJ	MoveJoint for RPS.
MOVEL	MoveLinear for RPS.
MOVEC	MoveCircular for RPS.
MOVEBLEND	MoveBlend for RPS.
MOVEJREL	MoveJointRelative for RPS.

Enumerator

MOVELREL	MoveLinearRelative for RPS.
MOVETEST	
MOVEP	MovePick for RPS. Note : if you don't see an implementation, I have just reserved these MTYPES to implement them (within 3 months) - SV.
MOVEU	MoveUndulate for RPS.
MOVEB	MoveBezier for RPS. Note : if you don't see an implementation, I have just reserved these MTYPES to implement them (within 3 months) - SV.
MOVESPCONT	Lookahead scannable.
MOVEABSSPCONT	Lookahead scannable.
GEARIN	
ACCPROFILE	
VELPROFILE	
POSPROFILE	
COMBINEAXES	
OPENMOVE_BASE	
OPENMOVE_1	
OPENMOVE_2	
OPENMOVE_3	
OPENMOVE_4	
OPENMOVE_5	

Definition at line 1650 of file Trio_UnifiedApi_CPP.h.

5.2.2.19 ProcessStatus

```
enum class Trio::UnifiedApi::ProcessStatus : uint32_t [strong]
```

The execution state of the process.

Enumerator

Stopped	No program assigned to the process. not called by scheduler.
Ready	Program assigned and being processed by the scheduler.
Stepping	Step through the program (one command) and return to Paused.
Paused	A program executing halted.
Pausing	Program about to be paused.
Aborted	A request has been made for the program to stop.
Suspend	Request to suspend program.
Suspended	Program is suspended.
Sleeping	Program is sleeping.

Definition at line 2269 of file Trio_UnifiedApi_CPP.h.

5.2.2.20 ProgressEvent

```
enum class Trio::UnifiedApi::ProgressEvent [strong]
```

Progress event.

Enumerator

Start	Progress started.
Progress	Progress.
Finish	Progress finished.

Definition at line 804 of file Trio_UnifiedApi_CPP.h.

5.2.2.21 PSwitchMode

```
enum class Trio::UnifiedApi::PSwitchMode : uint32_t [strong]
```

Flags PSWITCH command in single/cyclic mode.

Enumerator

None	No flags are set.
Invert	If set - set position should be bigger than reset position If not set - set position should be less than reset position.
DPOS	If set DPOS is used If not set MPOS is used.
Mask	Mask of allowed bits.

Definition at line 1153 of file Trio_UnifiedApi_CPP.h.

5.2.2.22 PSwitchMultiMode

```
enum class Trio::UnifiedApi::PSwitchMultiMode : uint32_t [strong]
```

Flags PSWITCH command in multi/sequence mode.

Enumerator

None	No flags are set.
DPOS	If set DPOS is used If not set MPOS is used.
TimerMode	If set timer mode If not set set and reset positions mode.
NegativeDirection	If set negative direction If not set positive direction.
ZOut	If set Z output mode If not set digital output mode.
Mask	Mask of allowed bits.

Definition at line 1181 of file Trio_UnifiedApi_CPP.h.

5.2.2.23 RegistEdge

```
enum class Trio::UnifiedApi::RegistEdge : int32_t [strong]
```

Enumerator

Rising	
Falling	

Definition at line 2246 of file Trio_UnifiedApi_CPP.h.

5.2.2.24 RegistMode

```
enum class Trio::UnifiedApi::RegistMode : int32_t [strong]
```

Enumerator

Cancel	Cancel any previous registration commands.
ZAMarkRising	Z Mark rising into REG_POS.
ZAMarkFalling	Z Mark falling into REG_POS.
RAInputRising	RA input rising into REG_POS.
RAInputFalling	RA input falling into REG_POS.
RAInputRising_ZBMarkRising	RA input rising into REG_POS and Z Mark rising into REG_POSB.
RAInputRising_ZBMarkFalling	RA input rising into REG_POS and Z Mark falling into REG_POSB.
RAInputFalling_ZBMarkRising	RA input falling into REG_POS and Z Mark rising into REG_POSB.
RAInputFalling_ZBMarkFalling	RA input falling into REG_POS and Z Mark falling into REG_POSB.
RAInputRising_RBInputRising	RA input rising into REG_POS and RB input rising into REG_POSB.
RAInputRising_RBInputFalling	RA input rising into REG_POS and RB input rising into REG_POSB.
RAInputFalling_RBInputRising	RA input falling into REG_POS and RB input falling into REG_POSB.
RAInputFalling_RBInputFalling	RA input falling into REG_POS and RB input falling into REG_POSB.
ZBMarkRising	Z Mark rising into REG_POSB.
ZBMarkFalling	Z Mark falling into REG_POSB.
RBInputRising	RB input rising into REG_POSB.
RBInputFalling	RB input falling into REG_POSB.
HardwareRegistration	Replacement mode for 1..16 with additional configuration options.
TimeInterpolatedRegistration	Time interpolated registration.
MultiChannelRegistration	Replacement for modes 32..39 and 64..71.
EnableFilter	This mode assigns a 2.4 microseconds minimum pulse width to the axis. This affects any REGIST mode that is used.
DisableFilter	This mode assigns a 0.15 microseconds minimum pulse width to the axis (default). This affects any REGIST mode that is used.
InsideOpenCloseWindow	Position must be inside OPEN_WIN .. CLOSE_WIN. To be combined with modes 1..17.
OutsideOpenCloseWindow	Position must be outside OPEN_WIN .. CLOSE_WIN. To be combined with modes 1..17.

Definition at line 2123 of file Trio_UnifiedApi_CPP.h.

5.2.2.25 RegistWindow

```
enum class Trio::UnifiedApi::RegistWindow : int32_t [strong]
```

Enumerator

No	
InsideOpenCloseWin	
OutsideOpenCloseWin	
VirtualInputHigh	
VirtualInputLow	

Definition at line 2253 of file Trio_UnifiedApi_CPP.h.

5.2.2.26 RobotStatus

```
enum class Trio::UnifiedApi::RobotStatus : uint32_t [strong]
```

Enumerator

NONE	
WORLD_FS_LIMIT	
WORLD_RS_LIMIT	
ROBOT_FS_LIMIT	
ROBOT_RS_LIMIT	
TCP_FS_LIMIT	
TCP_RS_LIMIT	
FE_WARNING	
FE_EXCEEDS_LIMIT	
WRIST_SINGULARITY	
ALIGNMENT_SINGULARITY	
ELBOW_SINGULARITY	
MAX_SPEED_LIMIT	
COLLIDED	
NOT_IN_POS	
ELBOW_CLOSED_SINGULARITY	
DISALLOW_JOG	

Definition at line 2059 of file Trio_UnifiedApi_CPP.h.

5.2.2.27 SecurityCodeType

```
enum class Trio::UnifiedApi::SecurityCodeType : int32_t [strong]
```

The security code type used with project decryption key functions.

Enumerator

Controller	Controller security code.
OEM	OEM security code.
User	User security code.

Definition at line 2351 of file Trio_UnifiedApi_CPP.h.

5.2.2.28 SyncControl

```
enum class Trio::UnifiedApi::SyncControl : int32_t [strong]
```

Enumerator

WAITING_TO_START	
NO_SYNC	
STARTING_SYNC	
PERFORMING_SYNC	
SYNCHRONISED	
STOPPING_SYNC	
STARTING_INTERP_MOVE	
PERFORMING_INTERP_MOVE	
STARTING_RESYNC	
PERFORMING_RESYNC	
STARTING_RESYNC_TO_USERFRAME	
PERFORMING_RESYNC_TO_USERFRAME	

Definition at line 2096 of file Trio_UnifiedApi_CPP.h.

Chapter 6

Data Structure Documentation

6.1 Trio::UnifiedApi::DataStreamPacket Struct Reference

Data used with a callback for streaming back the response of commands that return string data.

```
#include <Trio_UnifiedApi_CPP.h>
```

Data Fields

- uint32_t total_length

  The length of the stream, if available. 0 otherwise.
- uint32_t packet_number

  The packet number.
- uint32_t packet_count

  The total count of the packets of the stream, if available. 0 otherwise.
- uint32_t data_length

  The number of bytes.

6.1.1 Detailed Description

Data used with a callback for streaming back the response of commands that return string data.

Definition at line 2434 of file Trio_UnifiedApi_CPP.h.

6.1.2 Field Documentation

6.1.2.1 data_length

```
uint32_t Trio::UnifiedApi::DataStreamPacket::data_length
```

The number of bytes.

Definition at line 2451 of file Trio_UnifiedApi_CPP.h.

6.1.2.2 packet_count

```
uint32_t Trio::UnifiedApi::DataStreamPacket::packet_count
```

The total count of the packets of the stream, if available. 0 otherwise.

Definition at line 2447 of file Trio_UnifiedApi_CPP.h.

6.1.2.3 packet_number

```
uint32_t Trio::UnifiedApi::DataStreamPacket::packet_number
```

The packet number.

Definition at line 2443 of file Trio_UnifiedApi_CPP.h.

6.1.2.4 total_length

```
uint32_t Trio::UnifiedApi::DataStreamPacket::total_length
```

The length of the stream, if available. 0 otherwise.

Definition at line 2439 of file Trio_UnifiedApi_CPP.h.

The documentation for this struct was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.2 Trio::UnifiedApi::FileInfo Struct Reference

Data used with a callback for processing the response of Dir command.

```
#include <Trio_UnifiedApi_CPP.h>
```

Public Member Functions

- FileInfo ()

Data Fields

- FileType type

  Type of the file on the controller.
- uint32_t size

  Text code size.
- uint32_t compiled_size

  Compiled binary size.
- int32_t autorun_process

  Autorun process number.
- FileFlags flags

  Flags related to files and programs on the controller.

6.2.1 Detailed Description

Data used with a callback for processing the response of Dir command.

NOTE: (BD) There are mirror structurs in Trio_UnifiedApi_C.h, Trio_UnifiedApi_CPP.h and ITrioConnection.cs make sure they are in sync when doing changes here

Definition at line 2394 of file Trio_UnifiedApi_CPP.h.

6.2.2 Constructor & Destructor Documentation

6.2.2.1 FileInfo()

```
Trio::UnifiedApi::FileInfo::FileInfo ( ) [inline]
```

Definition at line 2417 of file Trio_UnifiedApi_CPP.h.

6.2.3 Field Documentation

6.2.3.1 autorun_process

```
int32_t Trio::UnifiedApi::FileInfo::autorun_process
```

Autorun process number.

Definition at line 2411 of file Trio_UnifiedApi_CPP.h.

6.2.3.2 compiled_size

```
uint32_t Trio::UnifiedApi::FileInfo::compiled_size
```

Compiled binary size.

Definition at line 2407 of file Trio_UnifiedApi_CPP.h.

6.2.3.3 flags

```
FileFlags Trio::UnifiedApi::FileInfo::flags
```

Flags related to files and programs on the controller.

Definition at line 2415 of file Trio_UnifiedApi_CPP.h.

6.2.3.4 size

```
uint32_t Trio::UnifiedApi::FileInfo::size
```

Text code size.

Definition at line 2403 of file Trio_UnifiedApi_CPP.h.

6.2.3.5 type

```
FileType Trio::UnifiedApi::FileInfo::type
```

Type of the file on the controller.

Definition at line 2399 of file Trio_UnifiedApi_CPP.h.

The documentation for this struct was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.3 Trio::UnifiedApi::ITrioConnection Class Reference

Communication object intarface.

```
#include <Trio_UnifiedApi_CPP.h>
```

Public Member Functions

- virtual void DeleteMe ()=0

  Publicly exposed method for communication object destruction Calling it is the proper way of destoring ITrioConnection instance After calling this method other methods should not be called.
- virtual void SetTcpHostName (const std::string_view &host)=0

  Set host name or IP to connection object. Valid only for TCP connection.
- virtual std::string_view GetTcpHostName ()=0

  Get current host name or IP of connection object. Valid only for TCP connection.
- virtual void SetTcpPort (uint16_t port)=0

  Set TCP port to connection object. Valid only for TCP connection.
- virtual uint16_t GetTcpPort ()=0

  Get current TCP port from connection object. Valid only for TCP connection.
- virtual void SetTcpCommandTimeout (uint32_t timeout_ms)=0

  Set new timeout for command execution. Valid only for TCP connection.
- virtual uint32_t GetTcpCommandTimeout ()=0

  Get current timeout for command execution. Valid only for TCP connection.
- virtual void SetTcpConnectTimeout (uint32_t timeout_ms)=0

  Set new timeout for opening TCP connection. Valid only for TCP connection.
- virtual uint32_t GetTcpConnectTimeout ()=0

  Get current timeout for opening TCP connection. Valid only for TCP connection.
- virtual bool IsConnected () const =0

  Check if a connection to the Motion Coordinator was opened.
- virtual ConnectionType GetConnectionType () const =0

  Get connection type.
- virtual VersionInfo GetApiVersion ()=0

  Get Unified API version info.
- virtual std::string_view GetApiVersionStr ()=0

  Get Unified API version string.
- virtual double GetFirmwareVersionNumber ()=0

  Get the firmware version of the Motion Coordinator.

- virtual void OpenConnection ()=0

  Open a connection to the Motion Coordinator.
- virtual void CloseConnection ()=0

  Closes previously open connection to the Motion Coordinator.
- virtual void Ex (int32_t mode=0)=0

  Perform software reset on Motion Coordinator.
- virtual void Cancel (int32_t mode, int32_t axis=-1)=0

  Used to cancel current or buffered axis commands on an axis or an interpolating axis group.
- virtual void Connect (double ratio, int32_t linkAxis, ConnectMode mode=ConnectMode::MPOS, int32_t axis=-1)=0

  Links the demand position of the base axis to the measured movements of the driving axes to produce an electronic gearbox.
- virtual void ConnPath (double ratio, int32_t linkAxis, int32_t axis=-1)=0

  Enables you to link to the path of an interpolated movement by linking the demand position of the base axis, to the interpolated path distance of the driving axis.
- virtual void PSwitch (int32_t switchNumber, PSwitchMode mode, int32_t axis, uint32_t outputNumber, bool outputState, double setPosition, double resetPosition)=0

  The PSWITCH command allows an output to be set when a predefined position is reached, and to be reset when a second position is reached.
- virtual void PSwitchOff (int32_t switchNumber, bool hold=false)=0

  The PSWITCHOFF command is used to disable.
- virtual void PSwitchMulti (int32_t switchNumber, PSwitchMultiMode mode, int32_t axis, uint32_t output↩Number, bool outputState, uint32_t tableStart, uint32_t tableEnd, uint32_t pulseTime=0)=0

  The PSWITCHMULTI command allows an output to be set that turns the output on and off at selected positions loaded from the TABLE memory.
- virtual void Execute (const std::string_view &command)=0

  Execute arbitrary command.
- virtual void ExecuteWithResponse (const std::string_view &command, char *buff, uint32_t &buff_size)=0

  Execute arbitrary command with response.
- virtual void ExecuteWithResponse (const std::string_view &command, StreamHandlerCallback &&handler)=0

  Execute arbitrary command with response.
- virtual std::string ExecuteWithResponse (const std::string_view &command)=0

  Execute arbitrary command with response.
- virtual void SetAxisBase (int32_t axis)=0

  The BASE command is used to direct all subsequent motion commands and axis parameter read/writes to a particular axis, or group of axes. The default setting is a sequence: 0, 1, 2, 3...
- virtual void SetMultiAxisBase (const int32_t axes[ ], uint32_t axisCount)=0

  Set multiple base axes.
- virtual void Datum (int32_t sequence, int32_t axis=-1)=0

  Performs one of 6 homing sequences to locate an axis to an absolute position.
- virtual void DefPos (double pos, int32_t axis=-1)=0

  Defines the current axis position as a new absolute value.
- virtual void DefPosMulti (const double pos[ ], uint32_t posCount, int32_t axis=-1)=0

  Defines the current axis position as a new absolute value to multiple axes.
- virtual void Forward (int32_t axis=-1)=0

  Sets continuous forward movement.
- virtual void Reverse (int32_t axis=-1)=0

  Sets continuous reverse movement.
- virtual void MoveRel (double pos, int32_t axis=-1)=0

  Incremental move of single axis.
- virtual void MoveRelMulti (const double dist[ ], uint32_t distCount, int32_t axis=-1)=0

  Incremental move of multiple axes.

- virtual void MoveRelSp (double pos, int32_t axis=-1)=0

  Incremental move of single axis with speed changed between moves.
- virtual void MoveRelSpMulti (const double dist[ ], uint32_t distCount, int32_t axis=-1)=0

  Incremental move of multiple axes with speed changed between moves.
- virtual void MoveAbs (double pos, int32_t axis=-1)=0

  Absolute positional move of single axis.
- virtual void MoveAbsMulti (const double pos[ ], uint32_t posCount, int32_t axis=-1)=0

  Absolute positional move of multiple axes.
- virtual void MoveAbsSp (double pos, int32_t axis=-1)=0

  Absolute positional move of single axis with speed changed between moves.
- virtual void MoveAbsSpMulti (const double pos[ ], uint32_t posCount, int32_t axis=-1)=0

  Absolute positional move of multiple axes with speed changed between moves.
- virtual void MoveCirc (int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double transitionAngle=0, int32_t tableOutput=-1)=0

  Moves 2 orthogonal axes in such a way as to produce a circular arc at the tool point. The length and radius of the arc are defined by the five parameters in the command line. The move parameters are always relative to the end of the last specified move. This is the start position on the circle circumference. Axis 1 is the current BASE axis. Axis 2 is the next axis in the BASE array. The first 4 distance parameters are scaled according to the current unit conversion factor for the BASE axis.
- virtual void MoveCircSp (int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double transitionAngle=0, int32_t tableOutput=0)=0

  Works as MoveCirc and additionally allows vector speed to be changed when using multiple moves in the look ahead buffer when MERGE = ON , using additional parameters FORCE_SPEED and ENDMOVE_SPEED. Optional parameters ta and output allow the use of transition curves.
- virtual void MoveHelical (int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double linearDistance, int32_t mode, double transitionAngle=0, int32_t tableOutput=0)=0

  Performs a helical move. Moves 2 orthogonal axes in such a way as to produce a circular arc at the tool point with a simultaneous linear move on a third axis. The first 5 parameters are similar to those of an MOVECIRC command.The sixth parameter defines the simultaneous linear move.
- virtual void MoveHelicalSp (int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double linearDistance, int32_t mode, double transitionAngle=0, int32_t tableOutput=0)=0

  Performs a helical move the same as MHELICAL and additionally allows vector speed to be changed when using multiple moves in the buffer. Uses additional axis parameters FORCE_SPEED, ENDMOVE_SPEED and STARTMOVE_SPEED.
- virtual void MoveSpherical (int32_t axis, double endMidX, double endMidY, double endMidZ, double centreX, double centreY, double centreZ, int32_t mode=0, int32_t gtpi=0, double rotau=0, double rotav=0, double rotaw=0, double rotap=0)=0

  Moves the three axis group defined in BASE along a spherical path with a vector speed determined by the SPEED set in the first axis of the BASE array. There are 4 modes of operation with the option of finishing the move at an endpoint different to the start, or returning to the start point to complete a circle. The path of the movement in 3D space can be defined either by specifying a point somewhere along the path, or by specifying the centre of the sphere.
- virtual void MoveSphericalSp (int32_t axis, double endMidX, double endMidY, double endMidZ, double centreX, double centreY, double centreZ, int32_t mode=0, int32_t gtpi=0, double rotau=0, double rotav=0, double rotaw=0, double rotap=0)=0

  Performs a spherical move the same as MSPHERICAL and additionally allows vector speed to be changed when using multiple moves in the look ahead buffer when MERGE = ON, using additional parameters FORCE_SPEED, ENDMOVE_SPEED and STARTMOVE_SPEED.
- virtual void MoveRelSeq (int32_t axis, uint32_t tablePointer, uint32_t nAxes, uint32_t nPoints, int32_t options, double radius, int32_t tableOutput=0, double transitionAngle=0)=0

  Function allows a sequence of relative movements to be loaded via TABLE values. The moves can be automatically merged together using a circular or spherical arc. Movement on 2 to 6 axes may be controlled in this way.
- virtual void MoveAbsSeq (int32_t axis, uint32_t tablePointer, uint32_t nAxes, uint32_t nPoints, int32_t options, double radius, int32_t tableOutput=0, double transitionAngle=0)=0

Function allows a sequence of absolute movements to be loaded via TABLE values. The moves can be automatically merged together using a circular or spherical arc. Movement on 2 to 6 axes may be controlled in this way.

- virtual void MoveModify (double position, int32_t axis=-1)=0

  MOVEMODIFY will change the absolute end position of a single axis MOVE , MOVEABS , MOVESP , MOVEABSSP or MOVEMODIFY that is in the last position in the movement buffer. If there is no motion command in the movement buffers or the last movement is not a single axis linear move, then MOVEMODIFY is loaded.

- virtual void MovePick (int32_t axis, int32_t mode, double x, double y, double z, double withdraw, double overlapControlA=0.5, double overlapControlB=0.5, double u=0, double v=0, double w=0, double withdraw↩Control=1, double xyControl=1, double approachControl=1)=0

  MOVEPICK is a new move type aimed at making it much simpler to program fast pick and place operations.

- virtual void MoveLink (int32_t axis, double distance, double link_dist, double link_acc, double link_dec, int32↩_t link_axis=-1, int32_t link_options=0, double link_pos=0, double base_dist=0, double y_dist=0, double z↩_dist=0, double u_dist=0, double v_dist=0, double w_dist=0)=0

  The linked move command is designed for controlling movements such as:

- virtual void SetMotion (int32_t axis, double speed, double accel, double decel, double jerk)=0

  Function used to pre-set the parameter list for a subsequent move using just one command.

- virtual void SetMotion (int32_t axis, double speed, double accel, double decel)=0

  Function used to pre-set the parameter list for a subsequent move using just one command.

- virtual void SetMotion (int32_t axis, double speed, double accel)=0

  Function used to pre-set the parameter list for a subsequent move using just one command.

- virtual void SetMotion (int32_t axis, double speed)=0

  Function used to pre-set the parameter list for a subsequent move using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel, double forceJerk, double forceRamp, double forceDwell)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel, double forceJerk, double forceRamp)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel, double forceJerk)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed, double endMove↩Speed)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed, double startMoveSpeed)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetMotionSp (int32_t axis, double forceSpeed)=0

  SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

- virtual void SetTableValue (uint32_t index, double value)=0

  Set single table value.

- virtual double GetTableValue (uint32_t index)=0

  Get single table value.
- virtual void SetMultiTableValues (uint32_t start_index, uint32_t count, const double values[ ])=0

  Set multiple table entries.
- virtual void GetMultiTableValues (uint32_t start_index, uint32_t count, double values[ ])=0

  Get multiple table entries.
- virtual void SetVrValue (uint32_t index, double value)=0

  Set single VR value.
- virtual double GetVrValue (uint32_t index)=0

  Get single VR value.
- virtual void SetMultiVrValues (uint32_t start_index, uint32_t count, const double values[ ])=0

  Set multiple VR entries.
- virtual void GetMultiVrValues (uint32_t start_index, uint32_t count, double values[ ])=0

  Get multiple VR entries.
- virtual void SetInvertIN (uint32_t input, bool invert)=0

  The INVERT_IN command allows the input channels to be individually inverted in software.
- virtual bool GetInvertIN (uint32_t input)=0

  Gets the invert status set by INVERT_IN command.
- virtual void RapidStop (int32_t mode=0)=0

  The RAPIDSTOP command operates like the CANCEL command but does so for ALL axes using a subset of cancel modes.
- virtual void Dir (FileCallback &&handler)=0

  Function returns the list of programs and files in internal memory of the Motion Coordinator.
- virtual void SetVrString (uint32_t vr, const std::string_view &str)=0

  Set a null-terminated string into consecutive VRs.
- virtual uint32_t GetVrStringLen (uint32_t vr)=0

  Get the length of a null-terminated string from consecutive VRs.
- virtual void GetVrString (uint32_t vr, char *buff, uint32_t &buff_size)=0

  Get a null-terminated string from consecutive VRs.
- virtual void GetVrString (uint32_t vr, StreamHandlerCallback &&handler)=0

  Get a null-terminated string from consecutive VRs using callback function to re-construct the string from received chunks.
- virtual std::string GetVrString (uint32_t vr)=0

  Get a null-terminated string from consecutive VRs.
- virtual int32_t GetAIn (uint32_t channel)=0

  Get a value from an Analogue Input.
- virtual int32_t GetAOut (uint32_t channel)=0

  Get the state of the Analogue Output.
- virtual void SetAOut (uint32_t channel, int32_t value)=0

  Set the state of the Analogue Output. Note that the actual change will happen on the next servo cycle. Therefore in immediate read may give a wrong result on some (fast) systems.
- virtual uint32_t GetDIn (uint32_t firstChannel, uint32_t lastChannel)=0

  Get the state of the Digital Inputs.
- virtual bool GetDIn (uint32_t channel)=0

  Get the state of the Digital Input.
- virtual uint32_t GetDIn32 ()=0

  Get the state of the first 32 Digital Inputs.
- virtual uint32_t GetDOut (uint32_t firstChannel, uint32_t lastChannel)=0

  Get the state of the Digital Outputs.
- virtual bool GetDOut (uint32_t channel)=0

  Get the state of the Digital Output.

- virtual uint32_t GetDOut32 ()=0

  Get the state of the first 32 Digital Outputs.
- virtual void SetDOut (uint32_t firstChannel, uint32_t lastChannel, uint32_t states)=0

  Set the state of the Digital Outputs.
- virtual void SetDOut (uint32_t channel, bool state)=0

  Set the state of the Digital Output. Note that the actual change will happen on the next servo cycle. Therefore in immediate read may give a wrong result on some (fast) systems.
- virtual void SetDOut32 (uint32_t states32)=0

  Set the state of the first 32 Digital Outputs. Note that the actual change will happen on the next servo cycle. Therefore in immediate read may give a wrong result on some (fast) systems.
- virtual void StopProgram (const std::string_view &name)=0

  Stops a program at its current line.
- virtual void StopProgram (const std::string_view &name, int32_t process_number)=0

  Stops a program at its current line.
- virtual void RunProgram (const std::string_view &name)=0

  Runs a program.
- virtual void RunProgram (const std::string_view &name, int32_t process_number)=0

  Runs a program.
- virtual uint16_t GetRemoteFileCRC (const std::string_view &name)=0

  Get CRC16 of a file or program on Motion Coordinator.
- virtual uint16_t GetLocalFileCRC (const std::string_view &path, bool isBinary)=0

  Get CRC16 of a local file.
- virtual FileExistsFlags FileExists (const std::string_view &name)=0

  Check if file or program exists on Motion Coordinator.
- virtual void AddDac (int32_t axis, int32_t secondAxis)=0

  Adds the output from the servo control block of a secondary axis to the output of the base axis. The resulting DAC↩_OUT of the base axis is then the sum of the two control loop outputs.
- virtual void AddDac (int32_t secondAxis)=0

  Adds the output from the servo control block of a secondary axis to the output of the base axis. The resulting DAC↩_OUT of the base axis is then the sum of the two control loop outputs. Axis previously set by SetAxisBase() is subject to superimposition of DAC_OUT value.
- virtual void AddAx (int32_t axis, int32_t linkAxis)=0

  Function is used to superimpose 2 or more movements to build up a more complex movement profile.
- virtual void AddAx (int32_t linkAxis)=0

  Function is used to superimpose 2 or more movements to build up a more complex movement profile. Axis previously set by SetAxisBase() is subject to superimposition.
- virtual void ClearVariables ()=0

  Sets all global (numbered) variables and VR values to 0 and sets local variables on the process on which command is run to 0.
- virtual void DeleteFrameGroup (uint32_t group)=0

  Delete frame group. When the group is deleted FRAME is set to 0. See the TrioBASIC FRAME_GROUP documentation for more information.
- virtual void SetFrameGroup (uint32_t group, uint32_t tableOffset, const int32_t axes[], uint32_t axisCount)=0

  Define new frame group.
- virtual void SetFrameGroup (uint32_t group, uint32_t tableOffset, const std::vector< int32_t > &axes)=0

  Define new frame group.
- virtual void GetFrameGroup (uint32_t group, uint32_t &tableOffset, int32_t axes[], uint32_t &axisCount)=0

  Get frame group info.
- virtual void GetFrameGroup (uint32_t group, uint32_t &tableOffset, std::vector< int32_t > &axes)=0

  Get frame group info.
- virtual void Ethercat_StartNetwork (int32_t slot, EthercatStartOption options, uint32_t MAC_retries)=0

  Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.

- virtual void Ethercat_StartNetwork (int32_t slot, EthercatStartOption options)=0

  Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.
- virtual void Ethercat_StartNetwork (int32_t slot)=0

  Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.
- virtual void Ethercat_StopNetwork (int32_t slot)=0

  Closedown the EtherCAT network.
- virtual void Ethercat_CheckSlaveOnline_VR (int32_t slot, int32_t slave, uint32_t out_vr_index)=0

  Check whether given slave is online. See Trio BASIC help for more details.
- virtual bool Ethercat_CheckSlaveOnline (int32_t slot, int32_t slave)=0

  Check whether given slave is online. See Trio BASIC help for more details.
- virtual void Ethercat_CheckNumberOfSlaves_VR (int32_t slot, uint32_t out_vr_index)=0

  Determine the number of slaves on the bus. See Trio BASIC help for more details.
- virtual uint32_t Ethercat_CheckNumberOfSlaves (int32_t slot)=0

  Determine the number of slaves on the bus. See Trio BASIC help for more details.
- virtual void Ethercat_GetSlaveAddress_VR (int32_t slot, int32_t slave, uint32_t out_vr_index)=0

  Determine the configured address of the slave identified by its position on the network. See Trio BASIC help for more details.
- virtual int32_t Ethercat_GetSlaveAddress (int32_t slot, int32_t slave)=0

  Determine the configured address of the slave identified by its position on the network. See Trio BASIC help for more details.
- virtual void Ethercat_GetSlaveAxis_VR (int32_t slot, int32_t slave, uint32_t out_vr_index)=0

  Determine the axis of the slave identified by its position on the network. See Trio BASIC help for more details.
- virtual int32_t Ethercat_GetSlaveAxis (int32_t slot, int32_t slave)=0

  Determine the axis of the slave identified by its position on the network. See Trio BASIC help for more details.
- virtual void Ethercat_SetState (int32_t slot, EthercatState state, uint32_t display)=0

  This function controls the EtherCAT State Machine (ESM). It requests the master change to given EtherCAT 'state', and hence changes all slaves to the same state. When a change to a higher state is made, the EtherCAT network will progress to the new state through the in-between states to allow correct starting of the network. See Trio BASIC help for more details.
- virtual void Ethercat_GetState_VR (int32_t slot, uint32_t out_vr_index)=0

  Gets the present state of the EtherCAT running on the defined slot. See Trio BASIC help for more details.
- virtual EthercatState Ethercat_GetState (int32_t slot)=0

  Gets the present state of the EtherCAT running on the defined slot. See Trio BASIC help for more details.
- virtual void Ethercat_SetSlave_ESC_Register (int32_t slot, int32_t slave, uint32_t esc_offset, uint32_t data↩_length, const char *data)=0

  Low-level write to the slave's ESC registers, addressing method used is auto-increment addressing. See Trio BASIC help for more details.
- virtual void Ethercat_GetSlave_ESC_Register (int32_t slot, int32_t slave, uint32_t esc_offset, uint32_t data↩_length, uint32_t out_vr_index)=0

  Low-level read from the slave's ESC registers, addressing method used is auto-increment addressing. See Trio BASIC help for more details.
- virtual void Ethercat_SendResetSequence (int32_t axis, uint32_t mode, uint32_t timeout)=0

  Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed. See Trio BASIC help for more details.
- virtual void Ethercat_SendResetSequence (int32_t axis, uint32_t mode)=0

  Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed. See Trio BASIC help for more details.
- virtual void Ethercat_SendResetSequence (int32_t axis)=0

  Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed. See Trio BASIC help for more details.
- virtual void Ethercat_SetStatusMaskAndTriggerPattern_VR (int32_t slot, int32_t addr, uint32_t vr_index)=0

This command enables the user to define a CoE Statusword (0x6041:00) mask and trigger pattern for raising the controller AXISSTATUS 'Remote Drive Status word Error' flag (bit 26.) The mask defines which bits within the CoE status word are relevant, and the trigger pattern defines the state of these bits(0 or 1) which will trigger the controller AXISSTATUS flag. If it is necessary to reset the watchdog(master enable) when this AXISISTATUS flag is set then the axis ERRORMASK must be set accordingly. The slave mask and pattern can be set and modified from preop to operational state of the network(after the configured addresses have been assigned.) It must be reset if the network is restarted. The slave must support the CoE statusword(0x6041:00) within the PDO cyclic data. If the Mask is zero the controller will not execute any runtime check of the CoE statusword. If the watchdog must be cleared to OFF when the 'remote drive statusword error' flag is raised then bit 26 of the axis ERRORMASK parameter must be set. See Trio BASIC help for more details.

- virtual void Ethercat_SetStatusMaskAndTriggerPattern (int32_t slot, int32_t addr, uint32_t mask, uint32_↩ t trigger_pattern)=0

  This command enables the user to define a CoE Statusword (0x6041:00) mask and trigger pattern for raising the controller AXISSTATUS 'Remote Drive Status word Error' flag (bit 26.) The mask defines which bits within the CoE status word are relevant, and the trigger pattern defines the state of these bits(0 or 1) which will trigger the controller AXISSTATUS flag. If it is necessary to reset the watchdog(master enable) when this AXISISTATUS flag is set then the axis ERRORMASK must be set accordingly. The slave mask and pattern can be set and modified from preop to operational state of the network(after the configured addresses have been assigned.) It must be reset if the network is restarted. The slave must support the CoE statusword(0x6041:00) within the PDO cyclic data. If the Mask is zero the controller will not execute any runtime check of the CoE statusword. If the watchdog must be cleared to OFF when the 'remote drive statusword error' flag is raised then bit 26 of the axis ERRORMASK parameter must be set. See Trio BASIC help for more details.

- virtual void Ethercat_GetStatusMaskAndTriggerPattern_VR (int32_t slot, int32_t addr, uint32_t out_vr_↩ index)=0

  This command enables the user to display and check the mask and trigger pattern values set (if any) for a given slave. See Trio BASIC help for more details.

- virtual void Ethercat_GetStatusMaskAndTriggerPattern (int32_t slot, int32_t addr, uint32_t &mask, uint32_t &trigger_pattern)=0

  This command enables the user to display and check the mask and trigger pattern values set (if any) for a given slave. See Trio BASIC help for more details.

- virtual void Ethercat_DisplayNetworkConfig (int32_t slot)=0

  Displays the network configuration to the command line terminal in Motion Perfect. See Trio BASIC help for more details.

- virtual void Ethercat_SetSync0StartTime_VR (uint32_t vr_index)=0

  Set the network default Sync0 start time offset. The offset value is from the given VR. The default start time is 1 second(1000000 microseconds).Add or subtract the required offset in microseconds.For example 1000150 is an offset of 150 microseconds and 999930 is an offset of -70 microseconds.The offset must not exceed ±50 % of the servo period. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

- virtual void Ethercat_SetSync0StartTime (int32_t offset)=0

  Set the network default Sync0 start time offset. If offset value is directly provided as a parameter. The default start time is 1 second(1000000 microseconds).Add or subtract the required offset in microseconds.For example 1000150 is an offset of 150 microseconds and 999930 is an offset of -70 microseconds.The offset must not exceed ±50 % of the servo period. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

- virtual void Ethercat_SetDistrClockMaster_VR (int32_t slot, uint32_t function, uint32_t vr_index)=0

  Set the EtherCAT master to enable the first DC enabled node to be the Distributed Clocks master. As this requires the Trio Motion Coordinator to synchronise to the external DC master, it does not function on P876, MC4N-ECAT or MC664(X) as these have FPGA based fixed timing hardware for the EtherCAT port. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

- virtual void Ethercat_SetDistrClockMaster (int32_t slot, uint32_t function, uint32_t value)=0

  Set the EtherCAT master to enable the first DC enabled node to be the Distributed Clocks master. As this requires the Trio Motion Coordinator to synchronise to the external DC master, it does not function on P876, MC4N-ECAT or MC664(X) as these have FPGA based fixed timing hardware for the EtherCAT port. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

- virtual void Ethercat_SetEmergencyMsgControl (int32_t slot, uint32_t mode)=0

Set the response of the master to an emergency telegram transmitted by a slave node. This function enables the user to determine whether emergency messages should trigger a system error alarm and cause the WDOG to be dropped. See Trio BASIC help for more details.

- virtual void Ethercat_GetEmergencyMsgControl_VR (int32_t slot, uint32_t out_vr_index)=0

  Get the current value; indicating the response of the master to an emergency telegram transmitted by a slave node. See Trio BASIC help for more details.

- virtual uint32_t Ethercat_GetEmergencyMsgControl (int32_t slot)=0

  Get the current value; indicating the response of the master to an emergency telegram transmitted by a slave node. See Trio BASIC help for more details.

- virtual void Ethercat_CoRead (int32_t slot, int32_t address, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, int32_t out_vr_index)=0

  This function gets a CANopen-over-EtherCAT (CoE) object from the remote drive or I/O device. The object's index and sub-index are used to request a value and that value is either placed in the VR or is displayed in the Motion Perfect terminal if the VR number is set to -1. See Trio BASIC help for more details.

- virtual void Ethercat_CoWrite_VR (int32_t slot, int32_t address, uint16_t object_index, uint8_t object_↩subindex, Co_ObjectType type, uint32_t vr_index)=0

  This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. The value comes from a VR. Use this function for integer values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void Ethercat_CoWrite_Value (int32_t slot, int32_t address, uint16_t object_index, uint8_t object_↩subindex, Co_ObjectType type, int64_t value)=0

  This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for integer values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void Ethercat_CoWrite_Value (int32_t slot, int32_t address, uint16_t object_index, uint8_t object_↩subindex, Co_ObjectType type, double value)=0

  This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for floating-point values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void Ethercat_CoReadAxis (int32_t axis_number, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, int32_t out_vr_index)=0

  This function gets a CANopen-over-EtherCAT (CoE) object from the remote drive or I/O device. The object's index and sub-index are used to request a value and that value is either placed in the VR or is displayed in the Motion Perfect terminal if the VR number is set to -1. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void Ethercat_CoWriteAxis_VR (int32_t axis_number, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, uint32_t vr_index)=0

  This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. The value comes from a VR. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void Ethercat_CoWriteAxis_Value (int32_t axis_number, uint16_t object_index, uint8_t object_↩subindex, Co_ObjectType type, int64_t value)=0

  This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for integer values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void Ethercat_CoWriteAxis_Value (int32_t axis_number, uint16_t object_index, uint8_t object_↩subindex, Co_ObjectType type, double value)=0

  This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for floating-point values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

- virtual void UploadFile (const std::string_view &local_file, const std::string_view &remote_file, FileType remote_file_type, FileTransferOption opt, ProgressHandlerCallback &&handler)=0

  Uploads a file to the Motion Coordinator.

- virtual void DownloadFile (const std::string_view &local_file, const std::string_view &remote_file, ProgressHandlerCallback &&handler)=0

  Downloads a file from the Motion Coordinator.

- virtual void Delete (const std::string_view &remote_file, bool compiled_code_only=false)=0

  Deletes a file on the Motion Coordinator.
- virtual void New (const std::string_view &target)=0

  Deletes a file, all files or the TABLE data the Motion Coordinator.
- virtual void Select (const std::string_view &remote_file, FileType file_type)=0

  Makes the named program the currently selected program on the Motion Coordinator, if the named program does not exist then it makes a program of that name. The selected program cannot be changed when programs are running. When a program is selected any previously selected program is compiled. When program_type is specified is to make a program of that name and type, it cannot be possible to change the type of an existing program.
- virtual void BacklashDisable (int32_t axis=-1)=0

  This axis function disables backlash compensation.
- virtual void Backlash (double distance, double speed, double acceleration, int32_t axis=-1)=0

  This axis function allows backlash compensation to be loaded. This is achieved by applying an offset move when the motor demand is in one direction, then reversing the offset move when the motor demand is in the opposite direction. These moves are superimposed on the commanded axis movements. The backlash compensation is applied after a reversal of the direction of change of the DPOS parameter. The backlash compensation can be seen in the AXIS_↩ DPOS axis parameter.This is effectively DPOS + backlash compensation.
- virtual void EncoderRatio (int64_t mpos_count, int64_t input_count, int32_t axis=-1)=0

  This command allows the incoming encoder count to be scaled by a non integer ratio : MPOS = (mpos_count / input↩ _count) x encoder_edges_input When using the servo loop you will need to adjust the gains to maintain performance and stability. Unlike the UNITS parameter, which only affects the scaling seen by the user programs, ENCODER_↩ RATIO affects all motion commands. ENCODER_RATIO does not replace UNITS.Only use ENCODER_RATIO where absolutely necessary.PP_STEP and ENCODER_RATIO cannot be used at the same time on the same axis.
- virtual void StepRatio (int64_t output_count, int64_t dpos_count, int32_t axis=-1)=0

  This command sets up an integer ratio for the axis' demanded position. Usually this is used on a stepper (pulse+direction) output. It can also be applied to a networked axis in positon mode, such as EtherCAT and RTEX position servo. Every servo-period the change in position is passed through the STEP_RATIO function before it goes to the step pulse or network drive output. The STEP_RATIO function operates before the divide by 16 factor(divide by 256 in Flexslice P375) in the stepper axis.This maintains the good timing resolution of the stepper output circuit. STEP_RATIO does not replace UNITS.
- virtual void Cam (uint32_t tableStart, uint32_t tableStop, double tableMultiplier, double linkDistance, int32_t axis=-1)=0

  This command is used to generate movement of an axis according to a table of positions which define a movement profile.
- virtual void CamBox (uint32_t tableStart, uint32_t tableStop, double tableMultiplier, double linkDistance, int32_t linkAxis, CamBoxLinkOptions linkOptions, double linkPos=0, double offsetStart=0, int32_t axis=-1)=0

  This command is used to generate movement of an axis according to a table of POSITIONS which define the movement profile.
- virtual void CamBoxMulti (uint32_t tableStart, uint32_t axisCount, int32_t axis=-1)=0

  This command is used to generate movement of multiple axas according to a table of POSITIONS which define the movement profile.
- virtual void Regist (RegistMode mode, int32_t axis=-1)=0

  The REGIST command initiates a capture of an axis position when it sees a registration input or the Z mark on the encoder. Once a registration event is captured MARK is set and the position and speed at the event can be read back.
- virtual void Regist (RegistMode mode, uint32_t channel, uint32_t source, RegistEdge edge, RegistWindow window, uint32_t repeatCount=0, uint32_t tableStart=0, int32_t axis=-1)=0

  The REGIST command initiates a capture of an axis position when it sees a registration input or the Z mark on the encoder. Once a registration event is captured MARK is set and the position and speed at the event can be read back.
- virtual int32_t ParamBlockDefineVr (const uint32_t indices[ ], uint32_t count)=0

  Define or re-define a parameter block of VR values.
- virtual int32_t ParamBlockDefineVr (const std::vector< uint32_t > &indices)=0

  Define or re-define a parameter block of VR values.
- virtual void ParamBlockAppendVr (const uint32_t indices[ ], uint32_t count, int32_t blk)=0

  Append to a parameter block of VR values.

- virtual void ParamBlockAppendVr (const std::vector< uint32_t > &indices, int32_t blk)=0

  Append to a parameter block of VR values.
- virtual int32_t ParamBlockDefineTable (const uint32_t indices[ ], uint32_t count)=0

  Define or re-define a parameter block of TABLE values.
- virtual int32_t ParamBlockDefineTable (const std::vector< uint32_t > &indices)=0

  Define or re-define a parameter block of TABLE values.
- virtual void ParamBlockAppendTable (const uint32_t indices[ ], uint32_t count, int32_t blk)=0

  Append a parameter block of TABLE values.
- virtual void ParamBlockAppendTable (const std::vector< uint32_t > &indices, int32_t blk)=0

  Append a parameter block of TABLE values.
- virtual int32_t ParamBlockDefineAxis (const std::string_view axisParams[ ], uint32_t count)=0

  Define or re-define a parameter block of Axis parameter values.
- virtual int32_t ParamBlockDefineAxis (const std::vector< std::string_view > &axisParams)=0

  Define or re-define a parameter block of Axis parameter values.
- virtual void ParamBlockAppendAxis (const std::string_view axisParams[ ], uint32_t count, int32_t blk)=0

  Append a parameter block of Axis parameter values.
- virtual void ParamBlockAppendAxis (const std::vector< std::string_view > &axisParams, int32_t blk)=0

  Append a parameter block of Axis parameter values.
- virtual int32_t ParamBlockDefineSystem (const std::string_view sysParams[ ], uint32_t count)=0

  Define or re-define a parameter block of System parameter values.
- virtual int32_t ParamBlockDefineSystem (const std::vector< std::string_view > &sysParams)=0

  Define or re-define a parameter block of System parameter values.
- virtual void ParamBlockAppendSystem (const std::string_view sysParams[ ], uint32_t count, int32_t blk)=0

  Append to a parameter block of System parameter values.
- virtual void ParamBlockAppendSystem (const std::vector< std::string_view > &sysParams, int32_t blk)=0

  Append to a parameter block of System parameter values.
- virtual int32_t ParamBlockDefineProgVar (const std::string_view &program, int32_t process, const std↩ ::string_view variables[ ], uint32_t count)=0

  Define or re-define a parameter block of Program variables.
- virtual int32_t ParamBlockDefineProgVar (const std::string_view &program, int32_t process, const std↩ ::vector< std::string_view > &variables)=0

  Define or re-define a parameter block of Program variables.
- virtual void ParamBlockAppendProgVar (const std::string_view &program, int32_t process, const std::string↩ _view variables[ ], uint32_t count, int32_t blk)=0

  Append to a parameter block of Program variables.
- virtual void ParamBlockAppendProgVar (const std::string_view &program, int32_t process, const std::vector< std::string_view > &variables, int32_t blk)=0

  Append to a parameter block of Program variables.
- virtual int32_t ParamBlockDefineIO (const Range ranges[ ], uint32_t count)=0

  Define or re-define a parameter block of Digital IO.
- virtual int32_t ParamBlockDefineIO (const std::vector< Range > &ranges)=0

  Define or re-define a parameter block of Digital IO.
- virtual void ParamBlockAppendIO (const Range ranges[ ], uint32_t count, int32_t blk)=0

  Append to a parameter block of Digital IO.
- virtual void ParamBlockAppendIO (const std::vector< Range > &ranges, int32_t blk)=0

  Append to a parameter block of Digital IO.
- virtual void ParamBlockDelete (int32_t blk)=0

  Deletes a defined parameter block.
- virtual void ParamBlockDeleteAll ()=0

  Deletes all defined parameter blocks.
- virtual uint64_t ParamBlockFetchChangedVr (int32_t blk, double values[ ], uint32_t count)=0

Fetches only changed values of a VR parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedVr (int32_t blk, std::vector< double > &values)=0

  Fetches only changed values of a VR parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedTable (int32_t blk, double values[ ], uint32_t count)=0

  Fetches only changed values of a TABLE parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedTable (int32_t blk, std::vector< double > &values)=0

  Fetches only changed values of a TABLE parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedAxisParam (int32_t blk, int32_t axis, double values[ ], uint32_↩t count)=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedAxisParam (int32_t blk, int32_t axis, std::vector< double > &values)=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchChangedAxisParam (int32_t blk, const Range &axes, double values[ ], uint32_t count, uint64_t changed_bitmasks[ ])=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchChangedAxisParam (int32_t blk, const Range &axes, std::vector< double > &values, std::vector< uint64_t > &changed_bitmasks)=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchChangedAxisParam (int32_t blk, const int32_t axes[ ], uint32_t axis_count, double values[ ], uint32_t value_count, uint64_t changed_bitmasks[ ])=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchChangedAxisParam (int32_t blk, const std::vector< int32_t > &axes, std↩::vector< double > &values, std::vector< uint64_t > &changed_bitmasks)=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchChangedAxisParam (int32_t blk, const Range ranges[ ], uint32_t range_count, double values[ ], uint32_t value_count, uint64_t changed_bitmasks[ ])=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchChangedAxisParam (int32_t blk, const std::vector< Range > &ranges, std↩::vector< double > &values, std::vector< uint64_t > &changed_bitmasks)=0

  Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedSystem (int32_t blk, double values[ ], uint32_t count)=0

  Fetches only changed values of a System parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedSystem (int32_t blk, std::vector< double > &values)=0

  Fetches only changed values of a System parameter block. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedProgVar (int32_t blk, const char *values[ ], uint32_t count)=0

  Fetches only changed values of a block of program variables. A value is considered changed if it differs from the previous fetch.

- virtual uint64_t ParamBlockFetchChangedProgVar (int32_t blk, std::vector< std::string_view > &values)=0

  Fetches only changed values of a block of program variables. A value is considered changed if it differs from the previous fetch.

- virtual void ParamBlockFetchAllVr (int32_t blk, double values[ ], uint32_t count)=0

  Fetches all values in a VR parameter block.
- virtual void ParamBlockFetchAllVr (int32_t blk, std::vector< double > &values)=0

  Fetches all values in a VR parameter block.
- virtual void ParamBlockFetchAllTable (int32_t blk, double values[ ], uint32_t count)=0

  Fetches all values in a TABLE parameter block.
- virtual void ParamBlockFetchAllTable (int32_t blk, std::vector< double > &values)=0

  Fetches all values in a TABLE parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, int32_t axis, double values[ ], uint32_t count)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, int32_t axis, std::vector< double > &values)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, const Range &axes, double values[ ], uint32_↩t count)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, const Range &axes, std::vector< double > &values)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, const int32_t axes[ ], uint32_t axis_count, double values[ ], uint32_t value_count)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, const std::vector< int32_t > &axes, std::vector< double > &values)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, const Range ranges[ ], uint32_t range_count, double values[ ], uint32_t value_count)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllAxisParam (int32_t blk, const std::vector< Range > &ranges, std::vector< double > &values)=0

  Fetches all values in an Axis parameter block.
- virtual void ParamBlockFetchAllSystem (int32_t blk, double values[ ], uint32_t count)=0

  Fetches all values in a System parameter block.
- virtual void ParamBlockFetchAllSystem (int32_t blk, std::vector< double > &values)=0

  Fetches all values in a System parameter block.
- virtual void ParamBlockFetchAllProgVar (int32_t blk, const char *values[ ], uint32_t count)=0

  Fetches all values in a block of program variables.
- virtual void ParamBlockFetchAllProgVar (int32_t blk, std::vector< std::string_view > &values)=0

  Fetches all values in a block of program variables.
- virtual double ParamBlockGetProgDoubleValue (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual int64_t ParamBlockGetProgInt64Value (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual uint64_t ParamBlockGetProgUInt64Value (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual int32_t ParamBlockGetProgInt32Value (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual uint32_t ParamBlockGetProgUInt32Value (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual bool ParamBlockGetProgBoolValue (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual void ParamBlockGetProgStrValue (int32_t blk, uint32_t index, char *buff, uint32_t &buff_size)=0

Retrieves a fetched value from a parameter block.

- virtual std::string_view ParamBlockGetProgStrValue (int32_t blk, uint32_t index)=0

  Retrieves a fetched value from a parameter block.
- virtual uint64_t ParamBlockFetchChangedIO (int32_t blk, bool outputs, std::vector< uint64_t > &values)=0

  Fetches only changed values in a Digital IO parameter block.
- virtual uint64_t ParamBlockFetchChangedIO (int32_t blk, bool outputs, uint64_t values[ ], uint32_t count)=0

  Fetches only changed values in a Digital IO parameter block.
- virtual void ParamBlockFetchAllIO (int32_t blk, bool outputs, std::vector< uint64_t > &values)=0

  Fetches all values in a Digital IO parameter block.
- virtual void ParamBlockFetchAllIO (int32_t blk, bool outputs, uint64_t values[ ], uint32_t count)=0

  Fetches all values in a Digital IO parameter block.
- virtual void SetVrBit (uint16_t bit, uint32_t vr_variable)=0

  The function can be used to set the value of a single bit within a VR variable. All other bits are unchanged.
- virtual void ClearVrBit (uint16_t bit, uint32_t vr_variable)=0

  The function can be used to clear the value of a single bit within a VR variable.
- virtual bool GetVrBit (uint16_t bit, uint32_t vr_variable)=0

  The function can be used to test the value of a single bit within a VR variable.
- virtual void Halt ()=0

  Halts execution of all running programs.
- virtual void GetProcessList (ProcessInfoCallback &&handler)=0

  Function returns the list of processes on the Motion Coordinator.
- virtual void ScopeOff ()=0

  Disable SCOPE.
- virtual void ScopeOn ()=0

  Enable SCOPE with previous settings.
- virtual void ScopeOn (uint32_t period_cycles)=0

  Enable SCOPE with previous settings and new capture period.
- virtual void ScopeOn (uint32_t period_cycles, uint32_t table_start, uint32_t table_end, const std::string_view prms[ ], uint32_t prm_count)=0

  Enable SCOPE and configure for capture to TABLE range.
- virtual void ScopeOn (uint32_t period_cycles, const std::string_view &fifo_name, uint32_t fifo_size, uint32_t sample_count, const std::string_view prms[ ], uint32_t prm_count)=0

  Enable SCOPE and configure for capture to FIFO.
- virtual void Trigger (bool rearm=false)=0

  Starts a previously set up SCOPE command. This allows you to manually start the scope capture.
- virtual double GetSystemParameterDoubleValue (const std::string_view &param_name)=0

  Get double value of system parameter by name.
- virtual void SetSystemParameterDoubleValue (const std::string_view &param_name, double value)=0

  Set double value to system parameter by name.
- virtual int64_t GetSystemParameterInt64Value (const std::string_view &param_name)=0

  Get 64-bit signed integer value of system parameter by name.
- virtual void SetSystemParameterInt64Value (const std::string_view &param_name, int64_t value)=0

  Set 64 bit signed integer value to system parameter by name.
- virtual uint64_t GetSystemParameterUInt64Value (const std::string_view &param_name)=0

  Get 64 bit unsigned integer value of system parameter by name.
- virtual void SetSystemParameterUInt64Value (const std::string_view &param_name, uint64_t value)=0

  Set 64 bit unsigned integer value to system parameter by name.
- virtual int32_t GetSystemParameterInt32Value (const std::string_view &param_name)=0

  Get 32-bit signed integer value of system parameter by name.
- virtual void SetSystemParameterInt32Value (const std::string_view &param_name, int32_t value)=0

  Set 32-bit signed integer value to system parameter by name.

- virtual uint32_t GetSystemParameterUInt32Value (const std::string_view &param_name)=0

  Get 32-bit unsigned integer value of system parameter by name.
- virtual void SetSystemParameterUInt32Value (const std::string_view &param_name, uint32_t value)=0

  Set 32 bit unsigned integer value to system parameter by name.
- virtual bool GetSystemParameterBoolValue (const std::string_view &param_name)=0

  Get boolean value of system parameter by name.
- virtual void SetSystemParameterBoolValue (const std::string_view &param_name, bool value)=0

  Set boolean value to system parameter by name.
- virtual void GetSystemParameterStringValue (const std::string_view &param_name, char *buff, uint32_t &buff_size)=0

  Get string value of system parameter by name, into provided buffer with size.
- virtual std::string GetSystemParameterStringValue (const std::string_view &param_name)=0

  Get string value of system parameter by name.
- virtual void SetSystemParameterStringValue (const std::string_view &param_name, const std::string_view &value)=0

  Set string value to system parameter by name.
- virtual double GetAxisParameterDoubleValue (const std::string_view &param_name, int32_t axis)=0

  Get double value of axis parameter by name.
- virtual void SetAxisParameterDoubleValue (const std::string_view &param_name, int32_t axis, double value)=0

  Set double value to axis parameter by name.
- virtual int64_t GetAxisParameterInt64Value (const std::string_view &param_name, int32_t axis)=0

  Get 64-bit signed integer value of axis parameter by name.
- virtual void SetAxisParameterInt64Value (const std::string_view &param_name, int32_t axis, int64_t value)=0

  Set 64 bit signed integer value to axis parameter by name.
- virtual uint64_t GetAxisParameterUInt64Value (const std::string_view &param_name, int32_t axis)=0

  Get 64 bit unsigned integer value of axis parameter by name.
- virtual void SetAxisParameterUInt64Value (const std::string_view &param_name, int32_t axis, uint64_t value)=0

  Set 64 bit unsigned integer value to axis parameter by name.
- virtual int32_t GetAxisParameterInt32Value (const std::string_view &param_name, int32_t axis)=0

  Get 32-bit signed integer value of axis parameter by name.
- virtual void SetAxisParameterInt32Value (const std::string_view &param_name, int32_t axis, int32_t value)=0

  Set 32-bit signed integer value to axis parameter by name.
- virtual uint32_t GetAxisParameterUInt32Value (const std::string_view &param_name, int32_t axis)=0

  Get 32-bit unsigned integer value of axis parameter by name.
- virtual void SetAxisParameterUInt32Value (const std::string_view &param_name, int32_t axis, uint32_t value)=0

  Set 32 bit unsigned integer value to axis parameter by name.
- virtual bool GetAxisParameterBoolValue (const std::string_view &param_name, int32_t axis)=0

  Get boolean value of axis parameter by name.
- virtual void SetAxisParameterBoolValue (const std::string_view &param_name, int32_t axis, bool value)=0

  Set boolean value to axis parameter by name.
- virtual double GetProcessParameterDoubleValue (const std::string_view &param_name, int32_t process)=0

  Get double value of process parameter by name.
- virtual void SetProcessParameterDoubleValue (const std::string_view &param_name, int32_t process, double value)=0

  Set double value to process parameter by name.
- virtual int64_t GetProcessParameterInt64Value (const std::string_view &param_name, int32_t process)=0

  Get 64-bit signed integer value of process parameter by name.
- virtual void SetProcessParameterInt64Value (const std::string_view &param_name, int32_t process, int64_t value)=0

Set 64 bit signed integer value to process parameter by name.

- virtual uint64_t GetProcessParameterUInt64Value (const std::string_view &param_name, int32_t process)=0

  Get 64 bit unsigned integer value of process parameter by name.
- virtual void SetProcessParameterUInt64Value (const std::string_view &param_name, int32_t process, uint64_t value)=0

  Set 64 bit unsigned integer value to process parameter by name.
- virtual int32_t GetProcessParameterInt32Value (const std::string_view &param_name, int32_t process)=0

  Get 32-bit signed integer value of process parameter by name.
- virtual void SetProcessParameterInt32Value (const std::string_view &param_name, int32_t process, int32_t value)=0

  Set 32-bit signed integer value to process parameter by name.
- virtual uint32_t GetProcessParameterUInt32Value (const std::string_view &param_name, int32_t process)=0

  Get 32-bit unsigned integer value of process parameter by name.
- virtual void SetProcessParameterUInt32Value (const std::string_view &param_name, int32_t process, uint32_t value)=0

  Set 32 bit unsigned integer value to process parameter by name.
- virtual bool GetProcessParameterBoolValue (const std::string_view &param_name, int32_t process)=0

  Get boolean value of process parameter by name.
- virtual void SetProcessParameterBoolValue (const std::string_view &param_name, int32_t process, bool value)=0

  Set boolean value to process parameter by name.
- virtual double GetSlotParameterDoubleValue (const std::string_view &param_name, int32_t slot)=0

  Get double value of slot parameter by name.
- virtual void SetSlotParameterDoubleValue (const std::string_view &param_name, int32_t slot, double value)=0

  Set double value to slot parameter by name.
- virtual int64_t GetSlotParameterInt64Value (const std::string_view &param_name, int32_t slot)=0

  Get 64-bit signed integer value of slot parameter by name.
- virtual void SetSlotParameterInt64Value (const std::string_view &param_name, int32_t slot, int64_t value)=0

  Set 64 bit signed integer value to slot parameter by name.
- virtual uint64_t GetSlotParameterUInt64Value (const std::string_view &param_name, int32_t slot)=0

  Get 64 bit unsigned integer value of slot parameter by name.
- virtual void SetSlotParameterUInt64Value (const std::string_view &param_name, int32_t slot, uint64_t value)=0

  Set 64 bit unsigned integer value to slot parameter by name.
- virtual int32_t GetSlotParameterInt32Value (const std::string_view &param_name, int32_t slot)=0

  Get 32-bit signed integer value of slot parameter by name.
- virtual void SetSlotParameterInt32Value (const std::string_view &param_name, int32_t slot, int32_t value)=0

  Set 32-bit signed integer value to slot parameter by name.
- virtual uint32_t GetSlotParameterUInt32Value (const std::string_view &param_name, int32_t slot)=0

  Get 32-bit unsigned integer value of slot parameter by name.
- virtual void SetSlotParameterUInt32Value (const std::string_view &param_name, int32_t slot, uint32_t value)=0

  Set 32 bit unsigned integer value to slot parameter by name.
- virtual bool GetSlotParameterBoolValue (const std::string_view &param_name, int32_t slot)=0

  Get boolean value of slot parameter by name.
- virtual void SetSlotParameterBoolValue (const std::string_view &param_name, int32_t slot, bool value)=0

  Set boolean value to slot parameter by name.
- virtual double GetPortParameterDoubleValue (const std::string_view &param_name, int32_t port)=0

  Get double value of port parameter by name.
- virtual void SetPortParameterDoubleValue (const std::string_view &param_name, int32_t port, double value)=0

Set double value to port parameter by name.

- virtual int64_t GetPortParameterInt64Value (const std::string_view &param_name, int32_t port)=0

  Get 64-bit signed integer value of port parameter by name.
- virtual void SetPortParameterInt64Value (const std::string_view &param_name, int32_t port, int64_t value)=0

  Set 64 bit signed integer value to port parameter by name.
- virtual uint64_t GetPortParameterUInt64Value (const std::string_view &param_name, int32_t port)=0

  Get 64 bit unsigned integer value of port parameter by name.
- virtual void SetPortParameterUInt64Value (const std::string_view &param_name, int32_t port, uint64_↩ t value)=0

  Set 64 bit unsigned integer value to port parameter by name.
- virtual int32_t GetPortParameterInt32Value (const std::string_view &param_name, int32_t port)=0

  Get 32-bit signed integer value of port parameter by name.
- virtual void SetPortParameterInt32Value (const std::string_view &param_name, int32_t port, int32_t value)=0

  Set 32-bit signed integer value to port parameter by name.
- virtual uint32_t GetPortParameterUInt32Value (const std::string_view &param_name, int32_t port)=0

  Get 32-bit unsigned integer value of port parameter by name.
- virtual void SetPortParameterUInt32Value (const std::string_view &param_name, int32_t port, uint32_↩ t value)=0

  Set 32 bit unsigned integer value to port parameter by name.
- virtual bool GetPortParameterBoolValue (const std::string_view &param_name, int32_t port)=0

  Get boolean value of port parameter by name.
- virtual void SetPortParameterBoolValue (const std::string_view &param_name, int32_t port, bool value)=0

  Set boolean value to port parameter by name.
- virtual void LoadProject (const std::string_view &project_file, StringCallback &&uknown_project_key_↩ handler)=0

  Loads a Motion Perfect project on the Motion Coordinator. All the running programs and motion are stopped. All the existing content on the Motion Coordinator is deleted.
- virtual void ProjectKey (const std::string_view &key, SecurityCodeType security_code=SecurityCodeType::Controller)=0

  Sets a project key on the controller.
- virtual void ProjectKeyRegistryList (StringCallback &&handler)=0

  List the authorities that have project keys stored on the Motion Coordinator.
- virtual void ProjectKeyRegistryStore (const std::string_view &authority, const std::string_view &key, SecurityCodeType security_code=SecurityCodeType::Controller)=0

  Stores a project key on the Motion Coordinator under the specified authority name.
- virtual void ProjectKeyRegistryClear ()=0

  Remove all non-default encryption authorities from database. The default (pre-installed) authorities are not removed.
- virtual void ProjectKeyRegistryUse (const std::string_view &authority)=0

  Loads a project key for use (project/program load) on the Motion Coordinator.
- virtual uint32_t GetSystemParameter_AIN0 ()=0

  Getter for AIN0 system parameter.
- virtual void SetSystemParameter_AIN0 (uint32_t value)=0

  Setter for AIN0 system parameter.
- virtual uint32_t GetSystemParameter_AIN1 ()=0

  Getter for AIN1 system parameter.
- virtual void SetSystemParameter_AIN1 (uint32_t value)=0

  Setter for AIN1 system parameter.
- virtual uint32_t GetSystemParameter_AIN2 ()=0

  Getter for AIN2 system parameter.
- virtual void SetSystemParameter_AIN2 (uint32_t value)=0

  Setter for AIN2 system parameter.
- virtual uint32_t GetSystemParameter_AIN3 ()=0

Getter for AIN3 system parameter.

- virtual void SetSystemParameter_AIN3 (uint32_t value)=0

  Setter for AIN3 system parameter.
- virtual uint32_t GetSystemParameter_AOUT0 ()=0

  Getter for AOUT0 system parameter.
- virtual void SetSystemParameter_AOUT0 (uint32_t value)=0

  Setter for AOUT0 system parameter.
- virtual uint32_t GetSystemParameter_AOUT1 ()=0

  Getter for AOUT1 system parameter.
- virtual void SetSystemParameter_AOUT1 (uint32_t value)=0

  Setter for AOUT1 system parameter.
- virtual uint32_t GetSystemParameter_AOUT2 ()=0

  Getter for AOUT2 system parameter.
- virtual void SetSystemParameter_AOUT2 (uint32_t value)=0

  Setter for AOUT2 system parameter.
- virtual uint32_t GetSystemParameter_AOUT3 ()=0

  Getter for AOUT3 system parameter.
- virtual void SetSystemParameter_AOUT3 (uint32_t value)=0

  Setter for AOUT3 system parameter.
- virtual uint32_t GetSystemParameter_AINBI0 ()=0

  Getter for AINBI0 system parameter.
- virtual void SetSystemParameter_AINBI0 (uint32_t value)=0

  Setter for AINBI0 system parameter.
- virtual uint32_t GetSystemParameter_AINBI1 ()=0

  Getter for AINBI1 system parameter.
- virtual void SetSystemParameter_AINBI1 (uint32_t value)=0

  Setter for AINBI1 system parameter.
- virtual uint32_t GetSystemParameter_AINBI2 ()=0

  Getter for AINBI2 system parameter.
- virtual void SetSystemParameter_AINBI2 (uint32_t value)=0

  Setter for AINBI2 system parameter.
- virtual uint32_t GetSystemParameter_AINBI3 ()=0

  Getter for AINBI3 system parameter.
- virtual void SetSystemParameter_AINBI3 (uint32_t value)=0

  Setter for AINBI3 system parameter.
- virtual uint32_t GetSystemParameter_INPUTS0 ()=0

  Getter for INPUTS0 system parameter.
- virtual uint32_t GetSystemParameter_INPUTS1 ()=0

  Getter for INPUTS1 system parameter.
- virtual uint32_t GetSystemParameter_SLOT0_TIME ()=0

  Getter for SLOT0_TIME system parameter.
- virtual void SetSystemParameter_SLOT0_TIME (uint32_t value)=0

  Setter for SLOT0_TIME system parameter.
- virtual uint32_t GetSystemParameter_SLOT1_TIME ()=0

  Getter for SLOT1_TIME system parameter.
- virtual void SetSystemParameter_SLOT1_TIME (uint32_t value)=0

  Setter for SLOT1_TIME system parameter.
- virtual uint32_t GetSystemParameter_SLOT2_TIME ()=0

  Getter for SLOT2_TIME system parameter.
- virtual void SetSystemParameter_SLOT2_TIME (uint32_t value)=0

  Setter for SLOT2_TIME system parameter.

- virtual uint32_t GetSystemParameter_SLOT3_TIME ()=0

  Getter for SLOT3_TIME system parameter.
- virtual void SetSystemParameter_SLOT3_TIME (uint32_t value)=0

  Setter for SLOT3_TIME system parameter.
- virtual bool GetSystemParameter_WDOG ()=0

  Getter for WDOG system parameter.
- virtual void SetSystemParameter_WDOG (bool value)=0

  Setter for WDOG system parameter.
- virtual uint32_t GetSystemParameter_SYSTEM_ERROR ()=0

  Getter for SYSTEM_ERROR system parameter.
- virtual int64_t GetSystemParameter_CURRENT_JITTER ()=0

  Getter for CURRENT_JITTER system parameter.
- virtual void SetSystemParameter_CURRENT_JITTER (int64_t value)=0

  Setter for CURRENT_JITTER system parameter.
- virtual int64_t GetSystemParameter_AVERAGE_JITTER ()=0

  Getter for AVERAGE_JITTER system parameter.
- virtual void SetSystemParameter_AVERAGE_JITTER (int64_t value)=0

  Setter for AVERAGE_JITTER system parameter.
- virtual int64_t GetSystemParameter_MAX_JITTER ()=0

  Getter for MAX_JITTER system parameter.
- virtual void SetSystemParameter_MAX_JITTER (int64_t value)=0

  Setter for MAX_JITTER system parameter.
- virtual int64_t GetSystemParameter_MISSED_INTERRUPTS ()=0

  Getter for MISSED_INTERRUPTS system parameter.
- virtual void SetSystemParameter_MISSED_INTERRUPTS (int64_t value)=0

  Setter for MISSED_INTERRUPTS system parameter.
- virtual int64_t GetSystemParameter_MISSED_SEND_FRAMES ()=0

  Getter for MISSED_SEND_FRAMES system parameter.
- virtual void SetSystemParameter_MISSED_SEND_FRAMES (int64_t value)=0

  Setter for MISSED_SEND_FRAMES system parameter.
- virtual int64_t GetSystemParameter_MISSED_RECV_FRAMES ()=0

  Getter for MISSED_RECV_FRAMES system parameter.
- virtual void SetSystemParameter_MISSED_RECV_FRAMES (int64_t value)=0

  Setter for MISSED_RECV_FRAMES system parameter.
- virtual int16_t GetSystemParameter_LIMIT_BUFFERED ()=0

  Getter for LIMIT_BUFFERED system parameter.
- virtual void SetSystemParameter_LIMIT_BUFFERED (int16_t value)=0

  Setter for LIMIT_BUFFERED system parameter.
- virtual uint32_t GetSystemParameter_ADDRESS ()=0

  Getter for ADDRESS system parameter.
- virtual void SetSystemParameter_ADDRESS (uint32_t value)=0

  Setter for ADDRESS system parameter.
- virtual int32_t GetSystemParameter_AUTO_ETHERCAT ()=0

  Getter for AUTO_ETHERCAT system parameter.
- virtual void SetSystemParameter_AUTO_ETHERCAT (int32_t value)=0

  Setter for AUTO_ETHERCAT system parameter.
- virtual uint32_t GetSystemParameter_CONTROL ()=0

  Getter for CONTROL system parameter.
- virtual int64_t GetSystemParameter_CPU_EXCEPTIONS ()=0

  Getter for CPU_EXCEPTIONS system parameter.
- virtual void SetSystemParameter_CPU_EXCEPTIONS (int64_t value)=0

Setter for CPU_EXCEPTIONS system parameter.

- virtual int32_t GetSystemParameter_DISPLAY ()=0

  Getter for DISPLAY system parameter.
- virtual void SetSystemParameter_DISPLAY (int32_t value)=0

  Setter for DISPLAY system parameter.
- virtual uint32_t GetSystemParameter_FLASH_STATUS ()=0

  Getter for FLASH_STATUS system parameter.
- virtual int32_t GetSystemParameter_ERROR_AXIS ()=0

  Getter for ERROR_AXIS system parameter.
- virtual int64_t GetSystemParameter_FPU_EXCEPTIONS ()=0

  Getter for FPU_EXCEPTIONS system parameter.
- virtual void SetSystemParameter_FPU_EXCEPTIONS (int64_t value)=0

  Setter for FPU_EXCEPTIONS system parameter.
- virtual int32_t GetSystemParameter_HMI_PROC ()=0

  Getter for HMI_PROC system parameter.
- virtual void SetSystemParameter_HMI_PROC (int32_t value)=0

  Setter for HMI_PROC system parameter.
- virtual uint32_t GetSystemParameter_IP_ADDRESS ()=0

  Getter for IP_ADDRESS system parameter.
- virtual void SetSystemParameter_IP_ADDRESS (uint32_t value)=0

  Setter for IP_ADDRESS system parameter.
- virtual uint32_t GetSystemParameter_IP_GATEWAY ()=0

  Getter for IP_GATEWAY system parameter.
- virtual void SetSystemParameter_IP_GATEWAY (uint32_t value)=0

  Setter for IP_GATEWAY system parameter.
- virtual uint64_t GetSystemParameter_IP_MAC ()=0

  Getter for IP_MAC system parameter.
- virtual void SetSystemParameter_IP_MAC (uint64_t value)=0

  Setter for IP_MAC system parameter.
- virtual uint32_t GetSystemParameter_IP_MEMORY_CONFIG ()=0

  Getter for IP_MEMORY_CONFIG system parameter.
- virtual void SetSystemParameter_IP_MEMORY_CONFIG (uint32_t value)=0

  Setter for IP_MEMORY_CONFIG system parameter.
- virtual uint32_t GetSystemParameter_IP_NETMASK ()=0

  Getter for IP_NETMASK system parameter.
- virtual void SetSystemParameter_IP_NETMASK (uint32_t value)=0

  Setter for IP_NETMASK system parameter.
- virtual uint32_t GetSystemParameter_IP_PROTOCOL_CONFIG ()=0

  Getter for IP_PROTOCOL_CONFIG system parameter.
- virtual void SetSystemParameter_IP_PROTOCOL_CONFIG (uint32_t value)=0

  Setter for IP_PROTOCOL_CONFIG system parameter.
- virtual uint32_t GetSystemParameter_IP_PROTOCOL_CTRL ()=0

  Getter for IP_PROTOCOL_CTRL system parameter.
- virtual void SetSystemParameter_IP_PROTOCOL_CTRL (uint32_t value)=0

  Setter for IP_PROTOCOL_CTRL system parameter.
- virtual uint64_t GetSystemParameter_IP_TCP_TIMEOUT ()=0

  Getter for IP_TCP_TIMEOUT system parameter.
- virtual void SetSystemParameter_IP_TCP_TIMEOUT (uint64_t value)=0

  Setter for IP_TCP_TIMEOUT system parameter.
- virtual uint32_t GetSystemParameter_IP_TCP_TX_THRESHOLD ()=0

  Getter for IP_TCP_TX_THRESHOLD system parameter.

- virtual void SetSystemParameter_IP_TCP_TX_THRESHOLD (uint32_t value)=0

  Setter for IP_TCP_TX_THRESHOLD system parameter.
- virtual uint32_t GetSystemParameter_IP_TCP_TX_TIMEOUT ()=0

  Getter for IP_TCP_TX_TIMEOUT system parameter.
- virtual void SetSystemParameter_IP_TCP_TX_TIMEOUT (uint32_t value)=0

  Setter for IP_TCP_TX_TIMEOUT system parameter.
- virtual int32_t GetSystemParameter_LAST_AXIS ()=0

  Getter for LAST_AXIS system parameter.
- virtual void SetSystemParameter_LAST_AXIS (int32_t value)=0

  Setter for LAST_AXIS system parameter.
- virtual uint16_t GetSystemParameter_MODBUS_TX_DELAY ()=0

  Getter for MODBUS_TX_DELAY system parameter.
- virtual void SetSystemParameter_MODBUS_TX_DELAY (uint16_t value)=0

  Setter for MODBUS_TX_DELAY system parameter.
- virtual uint64_t GetSystemParameter_MOTION_ERROR ()=0

  Getter for MOTION_ERROR system parameter.
- virtual uint32_t GetSystemParameter_MOTION_ERROR_COUNT ()=0

  Getter for MOTION_ERROR_COUNT system parameter.
- virtual int16_t GetSystemParameter_MPE_MODE ()=0

  Getter for MPE_MODE system parameter.
- virtual uint32_t GetSystemParameter_NOP ()=0

  Getter for NOP system parameter.
- virtual uint32_t GetSystemParameter_NIN ()=0

  Getter for NIN system parameter.
- virtual uint32_t GetSystemParameter_NIO ()=0

  Getter for NIO system parameter.
- virtual uint32_t GetSystemParameter_N_ANA_IN ()=0

  Getter for N_ANA_IN system parameter.
- virtual uint32_t GetSystemParameter_N_ANA_OUT ()=0

  Getter for N_ANA_OUT system parameter.
- virtual int32_t GetSystemParameter_PCMCAT_PROC ()=0

  Getter for PCMCAT_PROC system parameter.
- virtual void SetSystemParameter_PCMCAT_PROC (int32_t value)=0

  Setter for PCMCAT_PROC system parameter.
- virtual int32_t GetSystemParameter_PLC_CONFIG ()=0

  Getter for PLC_CONFIG system parameter.
- virtual void SetSystemParameter_PLC_CONFIG (int32_t value)=0

  Setter for PLC_CONFIG system parameter.
- virtual int64_t GetSystemParameter_PLC_ERROR ()=0

  Getter for PLC_ERROR system parameter.
- virtual int64_t GetSystemParameter_PLC_OVERFLOW ()=0

  Getter for PLC_OVERFLOW system parameter.
- virtual int64_t GetSystemParameter_PLC_RUN ()=0

  Getter for PLC_RUN system parameter.
- virtual int32_t GetSystemParameter_REMOTE_PROC ()=0

  Getter for REMOTE_PROC system parameter.
- virtual void SetSystemParameter_REMOTE_PROC (int32_t value)=0

  Setter for REMOTE_PROC system parameter.
- virtual uint32_t GetSystemParameter_SCHEDULE_TYPE ()=0

  Getter for SCHEDULE_TYPE system parameter.
- virtual void SetSystemParameter_SCHEDULE_TYPE (uint32_t value)=0

Setter for SCHEDULE_TYPE system parameter.

- virtual int32_t GetSystemParameter_SCOPE_CYCLE_COUNT ()=0

  Getter for SCOPE_CYCLE_COUNT system parameter.
- virtual void SetSystemParameter_SCOPE_CYCLE_COUNT (int32_t value)=0

  Setter for SCOPE_CYCLE_COUNT system parameter.
- virtual int64_t GetSystemParameter_SCOPE_DELAY ()=0

  Getter for SCOPE_DELAY system parameter.
- virtual void SetSystemParameter_SCOPE_DELAY (int64_t value)=0

  Setter for SCOPE_DELAY system parameter.
- virtual int32_t GetSystemParameter_SCOPE_POS ()=0

  Getter for SCOPE_POS system parameter.
- virtual void SetSystemParameter_SCOPE_POS (int32_t value)=0

  Setter for SCOPE_POS system parameter.
- virtual int32_t GetSystemParameter_SCOPE_TRIGGER_POS ()=0

  Getter for SCOPE_TRIGGER_POS system parameter.
- virtual int32_t GetSystemParameter_SERIAL_NUMBER ()=0

  Getter for SERIAL_NUMBER system parameter.
- virtual void SetSystemParameter_SERIAL_NUMBER (int32_t value)=0

  Setter for SERIAL_NUMBER system parameter.
- virtual int32_t GetSystemParameter_SERVO_OFFSET ()=0

  Getter for SERVO_OFFSET system parameter.
- virtual void SetSystemParameter_SERVO_OFFSET (int32_t value)=0

  Setter for SERVO_OFFSET system parameter.
- virtual int32_t GetSystemParameter_SERVO_PERIOD ()=0

  Getter for SERVO_PERIOD system parameter.
- virtual void SetSystemParameter_SERVO_PERIOD (int32_t value)=0

  Setter for SERVO_PERIOD system parameter.
- virtual uint32_t GetSystemParameter_SYSTEM_ERROR_MASK ()=0

  Getter for SYSTEM_ERROR_MASK system parameter.
- virtual void SetSystemParameter_SYSTEM_ERROR_MASK (uint32_t value)=0

  Setter for SYSTEM_ERROR_MASK system parameter.
- virtual uint32_t GetSystemParameter_SYSTEM_LOAD ()=0

  Getter for SYSTEM_LOAD system parameter.
- virtual uint32_t GetSystemParameter_SYSTEM_LOAD_MAX ()=0

  Getter for SYSTEM_LOAD_MAX system parameter.
- virtual void SetSystemParameter_SYSTEM_LOAD_MAX (uint32_t value)=0

  Setter for SYSTEM_LOAD_MAX system parameter.
- virtual int32_t GetSystemParameter_TEXT_FILE_LOADER_PROC ()=0

  Getter for TEXT_FILE_LOADER_PROC system parameter.
- virtual void SetSystemParameter_TEXT_FILE_LOADER_PROC (int32_t value)=0

  Setter for TEXT_FILE_LOADER_PROC system parameter.
- virtual int32_t GetSystemParameter_TSIZE ()=0

  Getter for TSIZE system parameter.
- virtual uint32_t GetSystemParameter_UNIT_ERROR ()=0

  Getter for UNIT_ERROR system parameter.
- virtual double GetSystemParameter_VERSION ()=0

  Getter for VERSION system parameter.
- virtual int32_t GetSystemParameter_VRSIZE ()=0

  Getter for VRSIZE system parameter.
- virtual int16_t GetSystemParameter_PROCNUMBER ()=0

  Getter for PROCNUMBER system parameter.

- virtual double GetAxisParameter_ACCEL (int32_t axis)=0

  Getter for ACCEL axis parameter.
- virtual void SetAxisParameter_ACCEL (int32_t axis, double value)=0

  Setter for ACCEL axis parameter.
- virtual int32_t GetAxisParameter_ADDAX_AXIS (int32_t axis)=0

  Getter for ADDAX_AXIS axis parameter.
- virtual double GetAxisParameter_AFF_GAIN (int32_t axis)=0

  Getter for AFF_GAIN axis parameter.
- virtual void SetAxisParameter_AFF_GAIN (int32_t axis, double value)=0

  Setter for AFF_GAIN axis parameter.
- virtual AxisType GetAxisParameter_ATYPE (int32_t axis)=0

  Getter for ATYPE axis parameter.
- virtual void SetAxisParameter_ATYPE (int32_t axis, AxisType value)=0

  Setter for ATYPE axis parameter.
- virtual double GetAxisParameter_AXIS_ACCEL (int32_t axis)=0

  Getter for AXIS_ACCEL axis parameter.
- virtual void SetAxisParameter_AXIS_ACCEL (int32_t axis, double value)=0

  Setter for AXIS_ACCEL axis parameter.
- virtual int32_t GetAxisParameter_AXIS_ADDRESS (int32_t axis)=0

  Getter for AXIS_ADDRESS axis parameter.
- virtual void SetAxisParameter_AXIS_ADDRESS (int32_t axis, int32_t value)=0

  Setter for AXIS_ADDRESS axis parameter.
- virtual double GetAxisParameter_AXIS_DECEL (int32_t axis)=0

  Getter for AXIS_DECEL axis parameter.
- virtual void SetAxisParameter_AXIS_DECEL (int32_t axis, double value)=0

  Setter for AXIS_DECEL axis parameter.
- virtual double GetAxisParameter_AXIS_DPOS (int32_t axis)=0

  Getter for AXIS_DPOS axis parameter.
- virtual bool GetAxisParameter_AXIS_ENABLE (int32_t axis)=0

  Getter for AXIS_ENABLE axis parameter.
- virtual void SetAxisParameter_AXIS_ENABLE (int32_t axis, bool value)=0

  Setter for AXIS_ENABLE axis parameter.
- virtual int32_t GetAxisParameter_AXIS_ERROR_COUNT (int32_t axis)=0

  Getter for AXIS_ERROR_COUNT axis parameter.
- virtual void SetAxisParameter_AXIS_ERROR_COUNT (int32_t axis, int32_t value)=0

  Setter for AXIS_ERROR_COUNT axis parameter.
- virtual double GetAxisParameter_AXIS_FS_LIMIT (int32_t axis)=0

  Getter for AXIS_FS_LIMIT axis parameter.
- virtual void SetAxisParameter_AXIS_FS_LIMIT (int32_t axis, double value)=0

  Setter for AXIS_FS_LIMIT axis parameter.
- virtual double GetAxisParameter_AXIS_MAX_JOG_SPEED (int32_t axis)=0

  Getter for AXIS_MAX_JOG_SPEED axis parameter.
- virtual void SetAxisParameter_AXIS_MAX_JOG_SPEED (int32_t axis, double value)=0

  Setter for AXIS_MAX_JOG_SPEED axis parameter.
- virtual AxisMode GetAxisParameter_AXIS_MODE (int32_t axis)=0

  Getter for AXIS_MODE axis parameter.
- virtual void SetAxisParameter_AXIS_MODE (int32_t axis, AxisMode value)=0

  Setter for AXIS_MODE axis parameter.
- virtual double GetAxisParameter_AXIS_RS_LIMIT (int32_t axis)=0

  Getter for AXIS_RS_LIMIT axis parameter.
- virtual void SetAxisParameter_AXIS_RS_LIMIT (int32_t axis, double value)=0

Setter for AXIS_RS_LIMIT axis parameter.

- virtual double GetAxisParameter_AXIS_SPEED (int32_t axis)=0

  Getter for AXIS_SPEED axis parameter.
- virtual void SetAxisParameter_AXIS_SPEED (int32_t axis, double value)=0

  Setter for AXIS_SPEED axis parameter.
- virtual double GetAxisParameter_AXIS_UNITS (int32_t axis)=0

  Getter for AXIS_UNITS axis parameter.
- virtual void SetAxisParameter_AXIS_UNITS (int32_t axis, double value)=0

  Setter for AXIS_UNITS axis parameter.
- virtual AxisStatus GetAxisParameter_AXISSTATUS (int32_t axis)=0

  Getter for AXISSTATUS axis parameter.
- virtual int64_t GetAxisParameter_BACKLASH_DIST (int32_t axis)=0

  Getter for BACKLASH_DIST axis parameter.
- virtual void SetAxisParameter_BACKLASH_DIST (int32_t axis, int64_t value)=0

  Setter for BACKLASH_DIST axis parameter.
- virtual double GetAxisParameter_CHANGE_DIR_LAST (int32_t axis)=0

  Getter for CHANGE_DIR_LAST axis parameter.
- virtual void SetAxisParameter_CHANGE_DIR_LAST (int32_t axis, double value)=0

  Setter for CHANGE_DIR_LAST axis parameter.
- virtual double GetAxisParameter_CLOSE_WIN (int32_t axis)=0

  Getter for CLOSE_WIN axis parameter.
- virtual void SetAxisParameter_CLOSE_WIN (int32_t axis, double value)=0

  Setter for CLOSE_WIN axis parameter.
- virtual double GetAxisParameter_CLUTCH_RATE (int32_t axis)=0

  Getter for CLUTCH_RATE axis parameter.
- virtual void SetAxisParameter_CLUTCH_RATE (int32_t axis, double value)=0

  Setter for CLUTCH_RATE axis parameter.
- virtual CornerMode GetAxisParameter_CORNER_MODE (int32_t axis)=0

  Getter for CORNER_MODE axis parameter.
- virtual void SetAxisParameter_CORNER_MODE (int32_t axis, CornerMode value)=0

  Setter for CORNER_MODE axis parameter.
- virtual CornerState GetAxisParameter_CORNER_STATE (int32_t axis)=0

  Getter for CORNER_STATE axis parameter.
- virtual void SetAxisParameter_CORNER_STATE (int32_t axis, CornerState value)=0

  Setter for CORNER_STATE axis parameter.
- virtual double GetAxisParameter_CREEP (int32_t axis)=0

  Getter for CREEP axis parameter.
- virtual void SetAxisParameter_CREEP (int32_t axis, double value)=0

  Setter for CREEP axis parameter.
- virtual double GetAxisParameter_D_GAIN (int32_t axis)=0

  Getter for D_GAIN axis parameter.
- virtual void SetAxisParameter_D_GAIN (int32_t axis, double value)=0

  Setter for D_GAIN axis parameter.
- virtual int32_t GetAxisParameter_DAC (int32_t axis)=0

  Getter for DAC axis parameter.
- virtual void SetAxisParameter_DAC (int32_t axis, int32_t value)=0

  Setter for DAC axis parameter.
- virtual int64_t GetAxisParameter_DAC_SCALE (int32_t axis)=0

  Getter for DAC_SCALE axis parameter.
- virtual void SetAxisParameter_DAC_SCALE (int32_t axis, int64_t value)=0

  Setter for DAC_SCALE axis parameter.

- virtual int32_t GetAxisParameter_DAC_OUT (int32_t axis)=0

  Getter for DAC_OUT axis parameter.
- virtual int16_t GetAxisParameter_DATUM_IN (int32_t axis)=0

  Getter for DATUM_IN axis parameter.
- virtual void SetAxisParameter_DATUM_IN (int32_t axis, int16_t value)=0

  Setter for DATUM_IN axis parameter.
- virtual double GetAxisParameter_DECEL (int32_t axis)=0

  Getter for DECEL axis parameter.
- virtual void SetAxisParameter_DECEL (int32_t axis, double value)=0

  Setter for DECEL axis parameter.
- virtual double GetAxisParameter_DECEL_ANGLE (int32_t axis)=0

  Getter for DECEL_ANGLE axis parameter.
- virtual void SetAxisParameter_DECEL_ANGLE (int32_t axis, double value)=0

  Setter for DECEL_ANGLE axis parameter.
- virtual double GetAxisParameter_DEMAND_SPEED (int32_t axis)=0

  Getter for DEMAND_SPEED axis parameter.
- virtual double GetAxisParameter_DPOS (int32_t axis)=0

  Getter for DPOS axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_CONTROL (int32_t axis)=0

  Getter for DRIVE_CONTROL axis parameter.
- virtual void SetAxisParameter_DRIVE_CONTROL (int32_t axis, int32_t value)=0

  Setter for DRIVE_CONTROL axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_CONTROLWORD (int32_t axis)=0

  Getter for DRIVE_CONTROLWORD axis parameter.
- virtual void SetAxisParameter_DRIVE_CONTROLWORD (int32_t axis, int32_t value)=0

  Setter for DRIVE_CONTROLWORD axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_CW_MODE (int32_t axis)=0

  Getter for DRIVE_CW_MODE axis parameter.
- virtual void SetAxisParameter_DRIVE_CW_MODE (int32_t axis, int32_t value)=0

  Setter for DRIVE_CW_MODE axis parameter.
- virtual int16_t GetAxisParameter_DRIVE_CURRENT (int32_t axis)=0

  Getter for DRIVE_CURRENT axis parameter.
- virtual int64_t GetAxisParameter_DRIVE_ENABLE (int32_t axis)=0

  Getter for DRIVE_ENABLE axis parameter.
- virtual void SetAxisParameter_DRIVE_ENABLE (int32_t axis, int64_t value)=0

  Setter for DRIVE_ENABLE axis parameter.
- virtual int64_t GetAxisParameter_DRIVE_FE (int32_t axis)=0

  Getter for DRIVE_FE axis parameter.
- virtual int64_t GetAxisParameter_DRIVE_FE_LIMIT (int32_t axis)=0

  Getter for DRIVE_FE_LIMIT axis parameter.
- virtual void SetAxisParameter_DRIVE_FE_LIMIT (int32_t axis, int64_t value)=0

  Setter for DRIVE_FE_LIMIT axis parameter.
- virtual uint32_t GetAxisParameter_DRIVE_INPUTS (int32_t axis)=0

  Getter for DRIVE_INPUTS axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_MODE (int32_t axis)=0

  Getter for DRIVE_MODE axis parameter.
- virtual void SetAxisParameter_DRIVE_MODE (int32_t axis, int32_t value)=0

  Setter for DRIVE_MODE axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_MONITOR (int32_t axis)=0

  Getter for DRIVE_MONITOR axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_PROFILE (int32_t axis)=0

Getter for DRIVE_PROFILE axis parameter.

- virtual void SetAxisParameter_DRIVE_PROFILE (int32_t axis, int32_t value)=0

  Setter for DRIVE_PROFILE axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_STATUS (int32_t axis)=0

  Getter for DRIVE_STATUS axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_TORQUE (int32_t axis)=0

  Getter for DRIVE_TORQUE axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_TYPE (int32_t axis)=0

  Getter for DRIVE_TYPE axis parameter.
- virtual void SetAxisParameter_DRIVE_TYPE (int32_t axis, int32_t value)=0

  Setter for DRIVE_TYPE axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_VELOCITY (int32_t axis)=0

  Getter for DRIVE_VELOCITY axis parameter.
- virtual int32_t GetAxisParameter_ENCODER (int32_t axis)=0

  Getter for ENCODER axis parameter.
- virtual uint16_t GetAxisParameter_ENCODER_BITS (int32_t axis)=0

  Getter for ENCODER_BITS axis parameter.
- virtual void SetAxisParameter_ENCODER_BITS (int32_t axis, uint16_t value)=0

  Setter for ENCODER_BITS axis parameter.
- virtual int32_t GetAxisParameter_ENCODER_CONTROL (int32_t axis)=0

  Getter for ENCODER_CONTROL axis parameter.
- virtual void SetAxisParameter_ENCODER_CONTROL (int32_t axis, int32_t value)=0

  Setter for ENCODER_CONTROL axis parameter.
- virtual double GetAxisParameter_ENCODER_FILTER (int32_t axis)=0

  Getter for ENCODER_FILTER axis parameter.
- virtual void SetAxisParameter_ENCODER_FILTER (int32_t axis, double value)=0

  Setter for ENCODER_FILTER axis parameter.
- virtual int32_t GetAxisParameter_ENCODER_ID (int32_t axis)=0

  Getter for ENCODER_ID axis parameter.
- virtual void SetAxisParameter_ENCODER_ID (int32_t axis, int32_t value)=0

  Setter for ENCODER_ID axis parameter.
- virtual int32_t GetAxisParameter_ENCODER_STATUS (int32_t axis)=0

  Getter for ENCODER_STATUS axis parameter.
- virtual void SetAxisParameter_ENCODER_STATUS (int32_t axis, int32_t value)=0

  Setter for ENCODER_STATUS axis parameter.
- virtual int32_t GetAxisParameter_ENCODER_TURNS (int32_t axis)=0

  Getter for ENCODER_TURNS axis parameter.
- virtual void SetAxisParameter_ENCODER_TURNS (int32_t axis, int32_t value)=0

  Setter for ENCODER_TURNS axis parameter.
- virtual double GetAxisParameter_END_DIR_LAST (int32_t axis)=0

  Getter for END_DIR_LAST axis parameter.
- virtual void SetAxisParameter_END_DIR_LAST (int32_t axis, double value)=0

  Setter for END_DIR_LAST axis parameter.
- virtual int64_t GetAxisParameter_ENDMOVE (int32_t axis)=0

  Getter for ENDMOVE axis parameter.
- virtual void SetAxisParameter_ENDMOVE (int32_t axis, int64_t value)=0

  Setter for ENDMOVE axis parameter.
- virtual double GetAxisParameter_ENDMOVE_BUFFER (int32_t axis)=0

  Getter for ENDMOVE_BUFFER axis parameter.
- virtual void SetAxisParameter_ENDMOVE_BUFFER (int32_t axis, double value)=0

  Setter for ENDMOVE_BUFFER axis parameter.

- virtual double GetAxisParameter_ENDMOVE_SPEED (int32_t axis)=0

  Getter for ENDMOVE_SPEED axis parameter.
- virtual void SetAxisParameter_ENDMOVE_SPEED (int32_t axis, double value)=0

  Setter for ENDMOVE_SPEED axis parameter.
- virtual AxisStatus GetAxisParameter_ERRORMASK (int32_t axis)=0

  Getter for ERRORMASK axis parameter.
- virtual void SetAxisParameter_ERRORMASK (int32_t axis, AxisStatus value)=0

  Setter for ERRORMASK axis parameter.
- virtual int16_t GetAxisParameter_FAST_JOG (int32_t axis)=0

  Getter for FAST_JOG axis parameter.
- virtual void SetAxisParameter_FAST_JOG (int32_t axis, int16_t value)=0

  Setter for FAST_JOG axis parameter.
- virtual double GetAxisParameter_FASTDEC (int32_t axis)=0

  Getter for FASTDEC axis parameter.
- virtual void SetAxisParameter_FASTDEC (int32_t axis, double value)=0

  Setter for FASTDEC axis parameter.
- virtual double GetAxisParameter_FE (int32_t axis)=0

  Getter for FE axis parameter.
- virtual double GetAxisParameter_FE_LIMIT (int32_t axis)=0

  Getter for FE_LIMIT axis parameter.
- virtual void SetAxisParameter_FE_LIMIT (int32_t axis, double value)=0

  Setter for FE_LIMIT axis parameter.
- virtual int16_t GetAxisParameter_FE_LIMIT_MODE (int32_t axis)=0

  Getter for FE_LIMIT_MODE axis parameter.
- virtual void SetAxisParameter_FE_LIMIT_MODE (int32_t axis, int16_t value)=0

  Setter for FE_LIMIT_MODE axis parameter.
- virtual double GetAxisParameter_FE_RANGE (int32_t axis)=0

  Getter for FE_RANGE axis parameter.
- virtual void SetAxisParameter_FE_RANGE (int32_t axis, double value)=0

  Setter for FE_RANGE axis parameter.
- virtual double GetAxisParameter_FE_LATCH (int32_t axis)=0

  Getter for FE_LATCH axis parameter.
- virtual int16_t GetAxisParameter_FHOLD_IN (int32_t axis)=0

  Getter for FHOLD_IN axis parameter.
- virtual void SetAxisParameter_FHOLD_IN (int32_t axis, int16_t value)=0

  Setter for FHOLD_IN axis parameter.
- virtual double GetAxisParameter_FHSPEED (int32_t axis)=0

  Getter for FHSPEED axis parameter.
- virtual void SetAxisParameter_FHSPEED (int32_t axis, double value)=0

  Setter for FHSPEED axis parameter.
- virtual int32_t GetAxisParameter_FORCE_DWELL (int32_t axis)=0

  Getter for FORCE_DWELL axis parameter.
- virtual void SetAxisParameter_FORCE_DWELL (int32_t axis, int32_t value)=0

  Setter for FORCE_DWELL axis parameter.
- virtual double GetAxisParameter_FORCE_RAMP (int32_t axis)=0

  Getter for FORCE_RAMP axis parameter.
- virtual void SetAxisParameter_FORCE_RAMP (int32_t axis, double value)=0

  Setter for FORCE_RAMP axis parameter.
- virtual double GetAxisParameter_FORCE_SPEED (int32_t axis)=0

  Getter for FORCE_SPEED axis parameter.
- virtual void SetAxisParameter_FORCE_SPEED (int32_t axis, double value)=0

Setter for FORCE_SPEED axis parameter.

- virtual double GetAxisParameter_FRAME_REP_DIST (int32_t axis)=0

  Getter for FRAME_REP_DIST axis parameter.
- virtual void SetAxisParameter_FRAME_REP_DIST (int32_t axis, double value)=0

  Setter for FRAME_REP_DIST axis parameter.
- virtual double GetAxisParameter_FRAME_SCALE (int32_t axis)=0

  Getter for FRAME_SCALE axis parameter.
- virtual void SetAxisParameter_FRAME_SCALE (int32_t axis, double value)=0

  Setter for FRAME_SCALE axis parameter.
- virtual double GetAxisParameter_FS_LIMIT (int32_t axis)=0

  Getter for FS_LIMIT axis parameter.
- virtual void SetAxisParameter_FS_LIMIT (int32_t axis, double value)=0

  Setter for FS_LIMIT axis parameter.
- virtual double GetAxisParameter_FULL_SP_RADIUS (int32_t axis)=0

  Getter for FULL_SP_RADIUS axis parameter.
- virtual void SetAxisParameter_FULL_SP_RADIUS (int32_t axis, double value)=0

  Setter for FULL_SP_RADIUS axis parameter.
- virtual int16_t GetAxisParameter_FWD_IN (int32_t axis)=0

  Getter for FWD_IN axis parameter.
- virtual void SetAxisParameter_FWD_IN (int32_t axis, int16_t value)=0

  Setter for FWD_IN axis parameter.
- virtual int16_t GetAxisParameter_FWD_JOG (int32_t axis)=0

  Getter for FWD_JOG axis parameter.
- virtual void SetAxisParameter_FWD_JOG (int32_t axis, int16_t value)=0

  Setter for FWD_JOG axis parameter.
- virtual double GetAxisParameter_I_GAIN (int32_t axis)=0

  Getter for I_GAIN axis parameter.
- virtual void SetAxisParameter_I_GAIN (int32_t axis, double value)=0

  Setter for I_GAIN axis parameter.
- virtual bool GetAxisParameter_IDLE (int32_t axis)=0

  Getter for IDLE axis parameter.
- virtual uint32_t GetAxisParameter_IN_POS (int32_t axis)=0

  Getter for IN_POS axis parameter.
- virtual void SetAxisParameter_IN_POS (int32_t axis, uint32_t value)=0

  Setter for IN_POS axis parameter.
- virtual int64_t GetAxisParameter_IN_POS_DIST (int32_t axis)=0

  Getter for IN_POS_DIST axis parameter.
- virtual void SetAxisParameter_IN_POS_DIST (int32_t axis, int64_t value)=0

  Setter for IN_POS_DIST axis parameter.
- virtual double GetAxisParameter_IN_POS_SPEED (int32_t axis)=0

  Getter for IN_POS_SPEED axis parameter.
- virtual void SetAxisParameter_IN_POS_SPEED (int32_t axis, double value)=0

  Setter for IN_POS_SPEED axis parameter.
- virtual double GetAxisParameter_INTERP_FACTOR (int32_t axis)=0

  Getter for INTERP_FACTOR axis parameter.
- virtual void SetAxisParameter_INTERP_FACTOR (int32_t axis, double value)=0

  Setter for INTERP_FACTOR axis parameter.
- virtual int32_t GetAxisParameter_INVERT_STEP (int32_t axis)=0

  Getter for INVERT_STEP axis parameter.
- virtual void SetAxisParameter_INVERT_STEP (int32_t axis, int32_t value)=0

  Setter for INVERT_STEP axis parameter.

- virtual double GetAxisParameter_JERK (int32_t axis)=0

  Getter for JERK axis parameter.
- virtual void SetAxisParameter_JERK (int32_t axis, double value)=0

  Setter for JERK axis parameter.
- virtual double GetAxisParameter_JOGSPEED (int32_t axis)=0

  Getter for JOGSPEED axis parameter.
- virtual void SetAxisParameter_JOGSPEED (int32_t axis, double value)=0

  Setter for JOGSPEED axis parameter.
- virtual int16_t GetAxisParameter_LINK_AXIS (int32_t axis)=0

  Getter for LINK_AXIS axis parameter.
- virtual uint32_t GetAxisParameter_LOADED (int32_t axis)=0

  Getter for LOADED axis parameter.
- virtual double GetAxisParameter_LOOKAHEAD_FACTOR (int32_t axis)=0

  Getter for LOOKAHEAD_FACTOR axis parameter.
- virtual void SetAxisParameter_LOOKAHEAD_FACTOR (int32_t axis, double value)=0

  Setter for LOOKAHEAD_FACTOR axis parameter.
- virtual int32_t GetAxisParameter_MARK (int32_t axis)=0

  Getter for MARK axis parameter.
- virtual int32_t GetAxisParameter_MARKB (int32_t axis)=0

  Getter for MARKB axis parameter.
- virtual bool GetAxisParameter_MERGE (int32_t axis)=0

  Getter for MERGE axis parameter.
- virtual void SetAxisParameter_MERGE (int32_t axis, bool value)=0

  Setter for MERGE axis parameter.
- virtual int32_t GetAxisParameter_MOVE_COUNT (int32_t axis)=0

  Getter for MOVE_COUNT axis parameter.
- virtual void SetAxisParameter_MOVE_COUNT (int32_t axis, int32_t value)=0

  Setter for MOVE_COUNT axis parameter.
- virtual double GetAxisParameter_MOVE_PA (int32_t axis)=0

  Getter for MOVE_PA axis parameter.
- virtual void SetAxisParameter_MOVE_PA (int32_t axis, double value)=0

  Setter for MOVE_PA axis parameter.
- virtual int32_t GetAxisParameter_MOVE_PA_CONT (int32_t axis)=0

  Getter for MOVE_PA_CONT axis parameter.
- virtual void SetAxisParameter_MOVE_PA_CONT (int32_t axis, int32_t value)=0

  Setter for MOVE_PA_CONT axis parameter.
- virtual double GetAxisParameter_MOVE_PA_IDLE (int32_t axis)=0

  Getter for MOVE_PA_IDLE axis parameter.
- virtual void SetAxisParameter_MOVE_PA_IDLE (int32_t axis, double value)=0

  Setter for MOVE_PA_IDLE axis parameter.
- virtual double GetAxisParameter_MOVE_PB (int32_t axis)=0

  Getter for MOVE_PB axis parameter.
- virtual void SetAxisParameter_MOVE_PB (int32_t axis, double value)=0

  Setter for MOVE_PB axis parameter.
- virtual int32_t GetAxisParameter_MOVE_PB_CONT (int32_t axis)=0

  Getter for MOVE_PB_CONT axis parameter.
- virtual void SetAxisParameter_MOVE_PB_CONT (int32_t axis, int32_t value)=0

  Setter for MOVE_PB_CONT axis parameter.
- virtual double GetAxisParameter_MOVE_PB_IDLE (int32_t axis)=0

  Getter for MOVE_PB_IDLE axis parameter.
- virtual void SetAxisParameter_MOVE_PB_IDLE (int32_t axis, double value)=0

Setter for MOVE_PB_IDLE axis parameter.

- virtual double GetAxisParameter_MOVELINK_MODIFY (int32_t axis)=0

  Getter for MOVELINK_MODIFY axis parameter.
- virtual void SetAxisParameter_MOVELINK_MODIFY (int32_t axis, double value)=0

  Setter for MOVELINK_MODIFY axis parameter.
- virtual int16_t GetAxisParameter_MOVES_BUFFERED (int32_t axis)=0

  Getter for MOVES_BUFFERED axis parameter.
- virtual double GetAxisParameter_MPOS (int32_t axis)=0

  Getter for MPOS axis parameter.
- virtual double GetAxisParameter_MSPEED (int32_t axis)=0

  Getter for MSPEED axis parameter.
- virtual double GetAxisParameter_MSPEED_FILTER (int32_t axis)=0

  Getter for MSPEED_FILTER axis parameter.
- virtual void SetAxisParameter_MSPEED_FILTER (int32_t axis, double value)=0

  Setter for MSPEED_FILTER axis parameter.
- virtual double GetAxisParameter_MSPEEDF (int32_t axis)=0

  Getter for MSPEEDF axis parameter.
- virtual MotionType GetAxisParameter_MTYPE (int32_t axis)=0

  Getter for MTYPE axis parameter.
- virtual MotionType GetAxisParameter_NTYPE (int32_t axis)=0

  Getter for NTYPE axis parameter.
- virtual double GetAxisParameter_OFFPOS (int32_t axis)=0

  Getter for OFFPOS axis parameter.
- virtual void SetAxisParameter_OFFPOS (int32_t axis, double value)=0

  Setter for OFFPOS axis parameter.
- virtual double GetAxisParameter_OPEN_WIN (int32_t axis)=0

  Getter for OPEN_WIN axis parameter.
- virtual void SetAxisParameter_OPEN_WIN (int32_t axis, double value)=0

  Setter for OPEN_WIN axis parameter.
- virtual int32_t GetAxisParameter_OUTLIMIT (int32_t axis)=0

  Getter for OUTLIMIT axis parameter.
- virtual void SetAxisParameter_OUTLIMIT (int32_t axis, int32_t value)=0

  Setter for OUTLIMIT axis parameter.
- virtual double GetAxisParameter_OV_GAIN (int32_t axis)=0

  Getter for OV_GAIN axis parameter.
- virtual void SetAxisParameter_OV_GAIN (int32_t axis, double value)=0

  Setter for OV_GAIN axis parameter.
- virtual double GetAxisParameter_P_GAIN (int32_t axis)=0

  Getter for P_GAIN axis parameter.
- virtual void SetAxisParameter_P_GAIN (int32_t axis, double value)=0

  Setter for P_GAIN axis parameter.
- virtual double GetAxisParameter_POS_DELAY (int32_t axis)=0

  Getter for POS_DELAY axis parameter.
- virtual void SetAxisParameter_POS_DELAY (int32_t axis, double value)=0

  Setter for POS_DELAY axis parameter.
- virtual int32_t GetAxisParameter_PP_STEP (int32_t axis)=0

  Getter for PP_STEP axis parameter.
- virtual void SetAxisParameter_PP_STEP (int32_t axis, int32_t value)=0

  Setter for PP_STEP axis parameter.
- virtual int32_t GetAxisParameter_PS_ENCODER (int32_t axis)=0

  Getter for PS_ENCODER axis parameter.

- virtual double GetAxisParameter_RAISE_ANGLE (int32_t axis)=0

  Getter for RAISE_ANGLE axis parameter.
- virtual void SetAxisParameter_RAISE_ANGLE (int32_t axis, double value)=0

  Setter for RAISE_ANGLE axis parameter.
- virtual double GetAxisParameter_REG_POS (int32_t axis)=0

  Getter for REG_POS axis parameter.
- virtual double GetAxisParameter_REG_POSB (int32_t axis)=0

  Getter for REG_POSB axis parameter.
- virtual double GetAxisParameter_REMAIN (int32_t axis)=0

  Getter for REMAIN axis parameter.
- virtual double GetAxisParameter_REP_DIST (int32_t axis)=0

  Getter for REP_DIST axis parameter.
- virtual void SetAxisParameter_REP_DIST (int32_t axis, double value)=0

  Setter for REP_DIST axis parameter.
- virtual int16_t GetAxisParameter_REP_OPTION (int32_t axis)=0

  Getter for REP_OPTION axis parameter.
- virtual void SetAxisParameter_REP_OPTION (int32_t axis, int16_t value)=0

  Setter for REP_OPTION axis parameter.
- virtual int16_t GetAxisParameter_REV_IN (int32_t axis)=0

  Getter for REV_IN axis parameter.
- virtual void SetAxisParameter_REV_IN (int32_t axis, int16_t value)=0

  Setter for REV_IN axis parameter.
- virtual int16_t GetAxisParameter_REV_JOG (int32_t axis)=0

  Getter for REV_JOG axis parameter.
- virtual void SetAxisParameter_REV_JOG (int32_t axis, int16_t value)=0

  Setter for REV_JOG axis parameter.
- virtual double GetAxisParameter_ROBOT_DPOS (int32_t axis)=0

  Getter for ROBOT_DPOS axis parameter.
- virtual double GetAxisParameter_ROBOT_FS_LIMIT (int32_t axis)=0

  Getter for ROBOT_FS_LIMIT axis parameter.
- virtual void SetAxisParameter_ROBOT_FS_LIMIT (int32_t axis, double value)=0

  Setter for ROBOT_FS_LIMIT axis parameter.
- virtual double GetAxisParameter_ROBOT_RS_LIMIT (int32_t axis)=0

  Getter for ROBOT_RS_LIMIT axis parameter.
- virtual void SetAxisParameter_ROBOT_RS_LIMIT (int32_t axis, double value)=0

  Setter for ROBOT_RS_LIMIT axis parameter.
- virtual int16_t GetAxisParameter_ROBOT_SP_MODE (int32_t axis)=0

  Getter for ROBOT_SP_MODE axis parameter.
- virtual void SetAxisParameter_ROBOT_SP_MODE (int32_t axis, int16_t value)=0

  Setter for ROBOT_SP_MODE axis parameter.
- virtual RobotStatus GetAxisParameter_ROBOTSTATUS (int32_t axis)=0

  Getter for ROBOTSTATUS axis parameter.
- virtual double GetAxisParameter_ROBOT_UNITS (int32_t axis)=0

  Getter for ROBOT_UNITS axis parameter.
- virtual void SetAxisParameter_ROBOT_UNITS (int32_t axis, double value)=0

  Setter for ROBOT_UNITS axis parameter.
- virtual double GetAxisParameter_RS_LIMIT (int32_t axis)=0

  Getter for RS_LIMIT axis parameter.
- virtual void SetAxisParameter_RS_LIMIT (int32_t axis, double value)=0

  Setter for RS_LIMIT axis parameter.
- virtual bool GetAxisParameter_SERVO (int32_t axis)=0

Getter for SERVO axis parameter.

- virtual void SetAxisParameter_SERVO (int32_t axis, bool value)=0

  Setter for SERVO axis parameter.
- virtual int16_t GetAxisParameter_SLOT_NUMBER (int32_t axis)=0

  Getter for SLOT_NUMBER axis parameter.
- virtual void SetAxisParameter_SLOT_NUMBER (int32_t axis, int16_t value)=0

  Setter for SLOT_NUMBER axis parameter.
- virtual double GetAxisParameter_SPEED (int32_t axis)=0

  Getter for SPEED axis parameter.
- virtual void SetAxisParameter_SPEED (int32_t axis, double value)=0

  Setter for SPEED axis parameter.
- virtual int32_t GetAxisParameter_SRAMP (int32_t axis)=0

  Getter for SRAMP axis parameter.
- virtual void SetAxisParameter_SRAMP (int32_t axis, int32_t value)=0

  Setter for SRAMP axis parameter.
- virtual int32_t GetAxisParameter_S_REF (int32_t axis)=0

  Getter for S_REF axis parameter.
- virtual void SetAxisParameter_S_REF (int32_t axis, int32_t value)=0

  Setter for S_REF axis parameter.
- virtual int32_t GetAxisParameter_S_REF_OUT (int32_t axis)=0

  Getter for S_REF_OUT axis parameter.
- virtual void SetAxisParameter_S_REF_OUT (int32_t axis, int32_t value)=0

  Setter for S_REF_OUT axis parameter.
- virtual double GetAxisParameter_START_DIR_LAST (int32_t axis)=0

  Getter for START_DIR_LAST axis parameter.
- virtual void SetAxisParameter_START_DIR_LAST (int32_t axis, double value)=0

  Setter for START_DIR_LAST axis parameter.
- virtual double GetAxisParameter_STARTMOVE_SPEED (int32_t axis)=0

  Getter for STARTMOVE_SPEED axis parameter.
- virtual void SetAxisParameter_STARTMOVE_SPEED (int32_t axis, double value)=0

  Setter for STARTMOVE_SPEED axis parameter.
- virtual double GetAxisParameter_STOP_ANGLE (int32_t axis)=0

  Getter for STOP_ANGLE axis parameter.
- virtual void SetAxisParameter_STOP_ANGLE (int32_t axis, double value)=0

  Setter for STOP_ANGLE axis parameter.
- virtual int32_t GetAxisParameter_SYNC_AXIS (int32_t axis)=0

  Getter for SYNC_AXIS axis parameter.
- virtual void SetAxisParameter_SYNC_AXIS (int32_t axis, int32_t value)=0

  Setter for SYNC_AXIS axis parameter.
- virtual SyncControl GetAxisParameter_SYNC_CONTROL (int32_t axis)=0

  Getter for SYNC_CONTROL axis parameter.
- virtual void SetAxisParameter_SYNC_CONTROL (int32_t axis, SyncControl value)=0

  Setter for SYNC_CONTROL axis parameter.
- virtual double GetAxisParameter_SYNC_DPOS (int32_t axis)=0

  Getter for SYNC_DPOS axis parameter.
- virtual void SetAxisParameter_SYNC_DPOS (int32_t axis, double value)=0

  Setter for SYNC_DPOS axis parameter.
- virtual int32_t GetAxisParameter_SYNC_DWELL (int32_t axis)=0

  Getter for SYNC_DWELL axis parameter.
- virtual void SetAxisParameter_SYNC_DWELL (int32_t axis, int32_t value)=0

  Setter for SYNC_DWELL axis parameter.

- virtual int32_t GetAxisParameter_SYNC_FLIGHT (int32_t axis)=0

  Getter for SYNC_FLIGHT axis parameter.
- virtual void SetAxisParameter_SYNC_FLIGHT (int32_t axis, int32_t value)=0

  Setter for SYNC_FLIGHT axis parameter.
- virtual int32_t GetAxisParameter_SYNC_PAUSE (int32_t axis)=0

  Getter for SYNC_PAUSE axis parameter.
- virtual void SetAxisParameter_SYNC_PAUSE (int32_t axis, int32_t value)=0

  Setter for SYNC_PAUSE axis parameter.
- virtual int32_t GetAxisParameter_SYNC_TIME (int32_t axis)=0

  Getter for SYNC_TIME axis parameter.
- virtual void SetAxisParameter_SYNC_TIME (int32_t axis, int32_t value)=0

  Setter for SYNC_TIME axis parameter.
- virtual int32_t GetAxisParameter_SYNC_TIMER (int32_t axis)=0

  Getter for SYNC_TIMER axis parameter.
- virtual double GetAxisParameter_SYNC_WITHDRAW (int32_t axis)=0

  Getter for SYNC_WITHDRAW axis parameter.
- virtual void SetAxisParameter_SYNC_WITHDRAW (int32_t axis, double value)=0

  Setter for SYNC_WITHDRAW axis parameter.
- virtual double GetAxisParameter_TABLE_POINTER (int32_t axis)=0

  Getter for TABLE_POINTER axis parameter.
- virtual void SetAxisParameter_TABLE_POINTER (int32_t axis, double value)=0

  Setter for TABLE_POINTER axis parameter.
- virtual double GetAxisParameter_TANG_DIRECTION (int32_t axis)=0

  Getter for TANG_DIRECTION axis parameter.
- virtual void SetAxisParameter_TANG_DIRECTION (int32_t axis, double value)=0

  Setter for TANG_DIRECTION axis parameter.
- virtual int32_t GetAxisParameter_T_REF (int32_t axis)=0

  Getter for T_REF axis parameter.
- virtual void SetAxisParameter_T_REF (int32_t axis, int32_t value)=0

  Setter for T_REF axis parameter.
- virtual int32_t GetAxisParameter_T_REF_OUT (int32_t axis)=0

  Getter for T_REF_OUT axis parameter.
- virtual void SetAxisParameter_T_REF_OUT (int32_t axis, int32_t value)=0

  Setter for T_REF_OUT axis parameter.
- virtual double GetAxisParameter_UNITS (int32_t axis)=0

  Getter for UNITS axis parameter.
- virtual void SetAxisParameter_UNITS (int32_t axis, double value)=0

  Setter for UNITS axis parameter.
- virtual double GetAxisParameter_VECTOR_BUFFERED (int32_t axis)=0

  Getter for VECTOR_BUFFERED axis parameter.
- virtual void SetAxisParameter_VECTOR_BUFFERED (int32_t axis, double value)=0

  Setter for VECTOR_BUFFERED axis parameter.
- virtual int16_t GetAxisParameter_VERIFY (int32_t axis)=0

  Getter for VERIFY axis parameter.
- virtual void SetAxisParameter_VERIFY (int32_t axis, int16_t value)=0

  Setter for VERIFY axis parameter.
- virtual double GetAxisParameter_VFF_GAIN (int32_t axis)=0

  Getter for VFF_GAIN axis parameter.
- virtual void SetAxisParameter_VFF_GAIN (int32_t axis, double value)=0

  Setter for VFF_GAIN axis parameter.
- virtual double GetAxisParameter_VP_ACCEL (int32_t axis)=0

Getter for VP_ACCEL axis parameter.

- virtual double GetAxisParameter_VP_JERK (int32_t axis)=0

  Getter for VP_JERK axis parameter.
- virtual int32_t GetAxisParameter_VP_MODE (int32_t axis)=0

  Getter for VP_MODE axis parameter.
- virtual void SetAxisParameter_VP_MODE (int32_t axis, int32_t value)=0

  Setter for VP_MODE axis parameter.
- virtual double GetAxisParameter_VP_POSITION (int32_t axis)=0

  Getter for VP_POSITION axis parameter.
- virtual double GetAxisParameter_VP_SPEED (int32_t axis)=0

  Getter for VP_SPEED axis parameter.
- virtual double GetAxisParameter_WORLD_ACCEL (int32_t axis)=0

  Getter for WORLD_ACCEL axis parameter.
- virtual void SetAxisParameter_WORLD_ACCEL (int32_t axis, double value)=0

  Setter for WORLD_ACCEL axis parameter.
- virtual double GetAxisParameter_WORLD_DECEL (int32_t axis)=0

  Getter for WORLD_DECEL axis parameter.
- virtual void SetAxisParameter_WORLD_DECEL (int32_t axis, double value)=0

  Setter for WORLD_DECEL axis parameter.
- virtual double GetAxisParameter_WORLD_DPOS (int32_t axis)=0

  Getter for WORLD_DPOS axis parameter.
- virtual double GetAxisParameter_WORLD_FS_LIMIT (int32_t axis)=0

  Getter for WORLD_FS_LIMIT axis parameter.
- virtual void SetAxisParameter_WORLD_FS_LIMIT (int32_t axis, double value)=0

  Setter for WORLD_FS_LIMIT axis parameter.
- virtual double GetAxisParameter_WORLD_JOGSPEED (int32_t axis)=0

  Getter for WORLD_JOGSPEED axis parameter.
- virtual void SetAxisParameter_WORLD_JOGSPEED (int32_t axis, double value)=0

  Setter for WORLD_JOGSPEED axis parameter.
- virtual double GetAxisParameter_WORLD_RS_LIMIT (int32_t axis)=0

  Getter for WORLD_RS_LIMIT axis parameter.
- virtual void SetAxisParameter_WORLD_RS_LIMIT (int32_t axis, double value)=0

  Setter for WORLD_RS_LIMIT axis parameter.
- virtual double GetAxisParameter_WORLD_SPEED (int32_t axis)=0

  Getter for WORLD_SPEED axis parameter.
- virtual void SetAxisParameter_WORLD_SPEED (int32_t axis, double value)=0

  Setter for WORLD_SPEED axis parameter.
- virtual double GetAxisParameter_WORLD_UNITS (int32_t axis)=0

  Getter for WORLD_UNITS axis parameter.
- virtual void SetAxisParameter_WORLD_UNITS (int32_t axis, double value)=0

  Setter for WORLD_UNITS axis parameter.
- virtual bool GetAxisParameter_AXIS_ENABLE_OVERRIDE (int32_t axis)=0

  Getter for AXIS_ENABLE_OVERRIDE axis parameter.
- virtual void SetAxisParameter_AXIS_ENABLE_OVERRIDE (int32_t axis, bool value)=0

  Setter for AXIS_ENABLE_OVERRIDE axis parameter.
- virtual int16_t GetAxisParameter_AXIS_DISPLAY (int32_t axis)=0

  Getter for AXIS_DISPLAY axis parameter.
- virtual void SetAxisParameter_AXIS_DISPLAY (int32_t axis, int16_t value)=0

  Setter for AXIS_DISPLAY axis parameter.
- virtual int32_t GetAxisParameter_AXIS_Z_OUTPUT (int32_t axis)=0

  Getter for AXIS_Z_OUTPUT axis parameter.

- virtual void SetAxisParameter_AXIS_Z_OUTPUT (int32_t axis, int32_t value)=0

  Setter for AXIS_Z_OUTPUT axis parameter.
- virtual int64_t GetAxisParameter_DEMAND_EDGES (int32_t axis)=0

  Getter for DEMAND_EDGES axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_INDEX (int32_t axis)=0

  Getter for DRIVE_INDEX axis parameter.
- virtual void SetAxisParameter_DRIVE_INDEX (int32_t axis, int32_t value)=0

  Setter for DRIVE_INDEX axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_NEG_TORQUE (int32_t axis)=0

  Getter for DRIVE_NEG_TORQUE axis parameter.
- virtual void SetAxisParameter_DRIVE_NEG_TORQUE (int32_t axis, int32_t value)=0

  Setter for DRIVE_NEG_TORQUE axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_PARAMETER (int32_t axis)=0

  Getter for DRIVE_PARAMETER axis parameter.
- virtual void SetAxisParameter_DRIVE_PARAMETER (int32_t axis, int32_t value)=0

  Setter for DRIVE_PARAMETER axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_POS_TORQUE (int32_t axis)=0

  Getter for DRIVE_POS_TORQUE axis parameter.
- virtual void SetAxisParameter_DRIVE_POS_TORQUE (int32_t axis, int32_t value)=0

  Setter for DRIVE_POS_TORQUE axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_SET_VAL (int32_t axis)=0

  Getter for DRIVE_SET_VAL axis parameter.
- virtual void SetAxisParameter_DRIVE_SET_VAL (int32_t axis, int32_t value)=0

  Setter for DRIVE_SET_VAL axis parameter.
- virtual int32_t GetAxisParameter_DRIVE_VALUE (int32_t axis)=0

  Getter for DRIVE_VALUE axis parameter.
- virtual void SetAxisParameter_DRIVE_VALUE (int32_t axis, int32_t value)=0

  Setter for DRIVE_VALUE axis parameter.
- virtual double GetAxisParameter_FEED_OVERRIDE (int32_t axis)=0

  Getter for FEED_OVERRIDE axis parameter.
- virtual void SetAxisParameter_FEED_OVERRIDE (int32_t axis, double value)=0

  Setter for FEED_OVERRIDE axis parameter.
- virtual double GetAxisParameter_FORCE_ACCEL (int32_t axis)=0

  Getter for FORCE_ACCEL axis parameter.
- virtual void SetAxisParameter_FORCE_ACCEL (int32_t axis, double value)=0

  Setter for FORCE_ACCEL axis parameter.
- virtual double GetAxisParameter_FORCE_DECEL (int32_t axis)=0

  Getter for FORCE_DECEL axis parameter.
- virtual void SetAxisParameter_FORCE_DECEL (int32_t axis, double value)=0

  Setter for FORCE_DECEL axis parameter.
- virtual double GetAxisParameter_FORCE_JERK (int32_t axis)=0

  Getter for FORCE_JERK axis parameter.
- virtual void SetAxisParameter_FORCE_JERK (int32_t axis, double value)=0

  Setter for FORCE_JERK axis parameter.
- virtual int32_t GetAxisParameter_FRAME (int32_t axis)=0

  Getter for FRAME axis parameter.
- virtual void SetAxisParameter_FRAME (int32_t axis, int32_t value)=0

  Setter for FRAME axis parameter.
- virtual int16_t GetAxisParameter_FWD_START (int32_t axis)=0

  Getter for FWD_START axis parameter.
- virtual void SetAxisParameter_FWD_START (int32_t axis, int16_t value)=0

Setter for FWD_START axis parameter.

- virtual int32_t GetAxisParameter_POSI_SEQ_DELAY (int32_t axis)=0

  Getter for POSI_SEQ_DELAY axis parameter.
- virtual void SetAxisParameter_POSI_SEQ_DELAY (int32_t axis, int32_t value)=0

  Setter for POSI_SEQ_DELAY axis parameter.
- virtual int32_t GetAxisParameter_POSI_SEQ_MODE (int32_t axis)=0

  Getter for POSI_SEQ_MODE axis parameter.
- virtual void SetAxisParameter_POSI_SEQ_MODE (int32_t axis, int32_t value)=0

  Setter for POSI_SEQ_MODE axis parameter.
- virtual uint32_t GetAxisParameter_PWM_CYCLE (int32_t axis)=0

  Getter for PWM_CYCLE axis parameter.
- virtual void SetAxisParameter_PWM_CYCLE (int32_t axis, uint32_t value)=0

  Setter for PWM_CYCLE axis parameter.
- virtual uint32_t GetAxisParameter_PWM_MARK (int32_t axis)=0

  Getter for PWM_MARK axis parameter.
- virtual void SetAxisParameter_PWM_MARK (int32_t axis, uint32_t value)=0

  Setter for PWM_MARK axis parameter.
- virtual int32_t GetAxisParameter_REG_INPUTS (int32_t axis)=0

  Getter for REG_INPUTS axis parameter.
- virtual void SetAxisParameter_REG_INPUTS (int32_t axis, int32_t value)=0

  Setter for REG_INPUTS axis parameter.
- virtual int32_t GetAxisParameter_REGIST_CONTROL (int32_t axis)=0

  Getter for REGIST_CONTROL axis parameter.
- virtual void SetAxisParameter_REGIST_CONTROL (int32_t axis, int32_t value)=0

  Setter for REGIST_CONTROL axis parameter.
- virtual double GetAxisParameter_REGIST_DELAY (int32_t axis)=0

  Getter for REGIST_DELAY axis parameter.
- virtual void SetAxisParameter_REGIST_DELAY (int32_t axis, double value)=0

  Setter for REGIST_DELAY axis parameter.
- virtual double GetAxisParameter_REGIST_SPEED (int32_t axis)=0

  Getter for REGIST_SPEED axis parameter.
- virtual double GetAxisParameter_REGIST_SPEEDB (int32_t axis)=0

  Getter for REGIST_SPEEDB axis parameter.
- virtual int16_t GetAxisParameter_REV_START (int32_t axis)=0

  Getter for REV_START axis parameter.
- virtual void SetAxisParameter_REV_START (int32_t axis, int16_t value)=0

  Setter for REV_START axis parameter.
- virtual double GetAxisParameter_TCP_DPOS (int32_t axis)=0

  Getter for TCP_DPOS axis parameter.
- virtual double GetAxisParameter_TCP_FS_LIMIT (int32_t axis)=0

  Getter for TCP_FS_LIMIT axis parameter.
- virtual void SetAxisParameter_TCP_FS_LIMIT (int32_t axis, double value)=0

  Setter for TCP_FS_LIMIT axis parameter.
- virtual double GetAxisParameter_TCP_RS_LIMIT (int32_t axis)=0

  Getter for TCP_RS_LIMIT axis parameter.
- virtual void SetAxisParameter_TCP_RS_LIMIT (int32_t axis, double value)=0

  Setter for TCP_RS_LIMIT axis parameter.
- virtual double GetAxisParameter_TCP_UNITS (int32_t axis)=0

  Getter for TCP_UNITS axis parameter.
- virtual void SetAxisParameter_TCP_UNITS (int32_t axis, double value)=0

  Setter for TCP_UNITS axis parameter.

- virtual double GetAxisParameter_ROBOT_FE (int32_t axis)=0

  Getter for ROBOT_FE axis parameter.
- virtual double GetAxisParameter_ROBOT_FE_LATCH (int32_t axis)=0

  Getter for ROBOT_FE_LATCH axis parameter.
- virtual double GetAxisParameter_ROBOT_FE_RANGE (int32_t axis)=0

  Getter for ROBOT_FE_RANGE axis parameter.
- virtual void SetAxisParameter_ROBOT_FE_RANGE (int32_t axis, double value)=0

  Setter for ROBOT_FE_RANGE axis parameter.
- virtual ProcessStatus GetProcessParameter_PROC_STATUS (int32_t process)=0

  Getter for PROC_STATUS process parameter.
- virtual int32_t GetProcessParameter_ERROR_LINE (int32_t process)=0

  Getter for ERROR_LINE process parameter.
- virtual int32_t GetProcessParameter_INDEVICE (int32_t process)=0

  Getter for INDEVICE process parameter.
- virtual void SetProcessParameter_INDEVICE (int32_t process, int32_t value)=0

  Setter for INDEVICE process parameter.
- virtual int32_t GetProcessParameter_OUTDEVICE (int32_t process)=0

  Getter for OUTDEVICE process parameter.
- virtual void SetProcessParameter_OUTDEVICE (int32_t process, int32_t value)=0

  Setter for OUTDEVICE process parameter.
- virtual bool GetProcessParameter_PMOVE (int32_t process)=0

  Getter for PMOVE process parameter.
- virtual int32_t GetProcessParameter_PROC_LINE (int32_t process)=0

  Getter for PROC_LINE process parameter.
- virtual int32_t GetProcessParameter_RUN_ERROR (int32_t process)=0

  Getter for RUN_ERROR process parameter.
- virtual int64_t GetProcessParameter_TICKS (int32_t process)=0

  Getter for TICKS process parameter.
- virtual void SetProcessParameter_TICKS (int32_t process, int64_t value)=0

  Setter for TICKS process parameter.
- virtual int32_t GetSlotParameter_SERIAL_NUMBER (int32_t slot)=0

  Getter for SERIAL_NUMBER slot parameter.
- virtual void SetSlotParameter_SERIAL_NUMBER (int32_t slot, int32_t value)=0

  Setter for SERIAL_NUMBER slot parameter.

Static Public Member Functions

- static TRIO_DECLSPEC const std::string_view &TRIO_CALLCONV ErrorString (ErrorCode errorCode)

  Error description.
- static TRIO_DECLSPEC VersionInfo TRIO_CALLCONV GetModuleVersion ()

  Get C++ language module version info.
- static TRIO_DECLSPEC std::string_view TRIO_CALLCONV GetModuleVersionStr ()

  Get version string of C++ language module.

Protected Member Functions

- ITrioConnection ()=default

  Constructor - protected to disallow direct construction.
- virtual ~ITrioConnection ()=default

  Destructor - protected to disallow direct destruction as it must happen from inside creating module.

6.3.1 Detailed Description

Communication object intarface.

Definition at line 2606 of file Trio_UnifiedApi_CPP.h.

6.3.2 Constructor & Destructor Documentation

6.3.2.1 ITrioConnection()

```
Trio::UnifiedApi::ITrioConnection::ITrioConnection ( )  [protected], [default]
```

Constructor - protected to disallow direct construction.

6.3.2.2 ~ITrioConnection()

```
virtual Trio::UnifiedApi::ITrioConnection::~ITrioConnection ( )  [protected], [virtual], [default]
```

Destructor - protected to disallow direct destruction as it must happen from inside creating module.

6.3.3 Member Function Documentation

6.3.3.1 AddAx() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::AddAx (
            int32_t axis,
            int32_t linkAxis )  [pure virtual]
```

Function is used to superimpose 2 or more movements to build up a more complex movement profile.

Parameters

axis	Target axis for to be subject to superimposition
linkAxis	Axis to superimpose. -1 breaks the link with the other axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.2 AddAx() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::AddAx (
            int32_t linkAxis )  [pure virtual]
```

Function is used to superimpose 2 or more movements to build up a more complex movement profile. Axis previously set by SetAxisBase() is subject to superimposition.

Parameters

linkAxis	Axis to superimpose. -1 breaks the link with the other axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.3 AddDac() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::AddDac (
            int32_t axis,
            int32_t secondAxis )  [pure virtual]
```

Adds the output from the servo control block of a secondary axis to the output of the base axis. The resulting DAC_OUT of the base axis is then the sum of the two control loop outputs.

Parameters

axis	Base axis for to be subject to DAC_OUT superimposition
secondAxis	Axis which DAC_OUT value to superimpose. -1 breaks the link with the other axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.4 AddDac() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::AddDac (
            int32_t secondAxis )  [pure virtual]
```

Adds the output from the servo control block of a secondary axis to the output of the base axis. The resulting DAC_OUT of the base axis is then the sum of the two control loop outputs. Axis previously set by SetAxisBase() is subject to superimposition of DAC_OUT value.

Parameters

secondAxis	Axis which DAC_OUT value to superimpose. -1 breaks the link with the other axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.5 Backlash()

```
virtual void Trio::UnifiedApi::ITrioConnection::Backlash (
```

```
            double distance,
            double speed,
            double acceleration,
            int32_t axis = -1 )  [pure virtual]
```

This axis function allows backlash compensation to be loaded. This is achieved by applying an offset move when the motor demand is in one direction, then reversing the offset move when the motor demand is in the opposite direction. These moves are superimposed on the commanded axis movements. The backlash compensation is applied after a reversal of the direction of change of the DPOS parameter. The backlash compensation can be seen in the AXIS_DPOS axis parameter.This is effectively DPOS + backlash compensation.

Parameters

distance	The distance to be offset in user units
speed	The ACCEL / DECEL rate at which is compensation move is applied in user units
acceleration	Enables/disables the backlash compensation
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.6 BacklashDisable()

```
virtual void Trio::UnifiedApi::ITrioConnection::BacklashDisable (
            int32_t axis = -1 )  [pure virtual]
```

This axis function disables backlash compensation.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.7 Cam()

```
virtual void Trio::UnifiedApi::ITrioConnection::Cam (
            uint32_t tableStart,
            uint32_t tableStop,
            double tableMultiplier,
            double linkDistance,
            int32_t axis = -1 )  [pure virtual]
```

This command is used to generate movement of an axis according to a table of positions which define a movement profile.

Parameters

tableStart	The start position of the cam profile in the TABLE.
tableStop	The end position of the cam profile in the TABLE.
tableMultiplier	The table values are multiplied by this value to generate the positions.
linkDistance	The distance parameter relates the speed of the axis to the time taken to complete the cam profile.
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.8 CamBox()

```
virtual void Trio::UnifiedApi::ITrioConnection::CamBox (
            uint32_t tableStart,
            uint32_t tableStop,
            double tableMultiplier,
            double linkDistance,
            int32_t linkAxis,
            CamBoxLinkOptions linkOptions,
            double linkPos = 0,
            double offsetStart = 0,
            int32_t axis = -1 )  [pure virtual]
```

This command is used to generate movement of an axis according to a table of POSITIONS which define the movement profile.

Parameters

tableStart	The start position of the cam profile in the TABLE.
tableStop	The end position of the cam profile in the TABLE.
tableMultiplier	The table values are multiplied by this value to generate the positions.
linkDistance	The distance the link axis must move to complete CAMBOX profile.
linkAxis	The axis to link to.
linkOptions	Bit value options to customise how your CAMBOX operates.
linkPos	linkOptions bit 1 - the absolute position on the link axis in user UNITS where the CAMBOX is to be start. linkOptions bit 9 – the registration channel to start the movement on
offsetStart	The position defined on the link_dist where the profile will start.
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.9 CamBoxMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::CamBoxMulti (
            uint32_t tableStart,
            uint32_t axisCount,
            int32_t axis = -1 )  [pure virtual]
```

This command is used to generate movement of multiple axas according to a table of POSITIONS which define the movement profile.

Parameters

tableStart	The start index in TABLE.
axisCount	Number of axes to generate movement for.
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

That variant of CAMBOX command allow movement to be defined for with one call. Parameters are read from a TABLE range starting at *tableStart* offset. For each axis 8 parameters should be provided in the following order↩ : Offset 0 - TABLE start point for CAMBOX profile position data Offset 1 - TABLE end point for CAMBOX profile position data Offset 2 - TABLE data multiplier Offset 3 - Link distance Offset 4 - Link axis Offset 5 - Link options(as per CAMBOX single - axis command) Offset 6 - Link position / Registration channel for link options Offset 7 - Forces the profile to start at a defined point in the link distance(options bit 7) The same pattern is repeated for the number of axes to be controlled. The data will be validated and if an error is detected then no CAMBOX requests will be loaded and a run-time error will be raised.

6.3.3.10 Cancel()

```
virtual void Trio::UnifiedApi::ITrioConnection::Cancel (
            int32_t mode,
            int32_t axis = -1 )  [pure virtual]
```

Used to cancel current or buffered axis commands on an axis or an interpolating axis group.

Parameters

mode	Cancel mode
axis	Index of axis subject to cancel operation or -1 to cancel default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.11 ClearVariables()

```
virtual void Trio::UnifiedApi::ITrioConnection::ClearVariables ( )  [pure virtual]
```

Sets all global (numbered) variables and VR values to 0 and sets local variables on the process on which command is run to 0.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.12 ClearVrBit()

```
virtual void Trio::UnifiedApi::ITrioConnection::ClearVrBit (
            uint16_t bit,
            uint32_t vr_variable )  [pure virtual]
```

The function can be used to clear the value of a single bit within a VR variable.

Parameters

bit	The bit number to clear, valid range is 0 to 52.
vr_variable	The VR which to operate on.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.13 CloseConnection()

```
virtual void Trio::UnifiedApi::ITrioConnection::CloseConnection ( )  [pure virtual]
```

Closes previously open connection to the Motion Coordinator.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.14 Connect()

```
virtual void Trio::UnifiedApi::ITrioConnection::Connect (
            double ratio,
            int32_t linkAxis,
            ConnectMode mode = ConnectMode::MPOS,
            int32_t axis = -1 )  [pure virtual]
```

Links the demand position of the base axis to the measured movements of the driving axes to produce an electronic gearbox.

Parameters

ratio	This parameter holds the number of edges the base axis is required to move per increment of the driving axis. The ratio value can be either positive or negative. The ratio is always specified as an encoder edge ratio
linkAxis	This parameter specifies the axis to link to.
mode	0 The axis follows the measured position(MPOS) of the driving axis. (default) 1 The axis follows the demand position (DPOS) of the driving axis.
axis	Index of axis subject to cancel operation or -1 to cancel default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.15 ConnPath()

```
virtual void Trio::UnifiedApi::ITrioConnection::ConnPath (
            double ratio,
            int32_t linkAxis,
            int32_t axis = -1 )  [pure virtual]
```

Enables you to link to the path of an interpolated movement by linking the demand position of the base axis, to the interpolated path distance of the driving axis.

Parameters

ratio	This is the ratio between the interpolated distance moved on the driving axis to the distance moved on the base axis.
linkAxis	This parameter specifies the axis to link to.
axis	Index of axis subject to cancel operation or -1 to cancel default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.16 Datum()

```
virtual void Trio::UnifiedApi::ITrioConnection::Datum (
            int32_t sequence,
            int32_t axis = -1 )  [pure virtual]
```

Performs one of 6 homing sequences to locate an axis to an absolute position.

Parameters

sequence	Bit-mask for homing sequence to be executed
axis	Index of the axis to be subject of datum sequence or -1 to datum default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.17 DefPos()

```
virtual void Trio::UnifiedApi::ITrioConnection::DefPos (
            double pos,
            int32_t axis = -1 )  [pure virtual]
```

Defines the current axis position as a new absolute value.

Parameters

pos	Absolute position to set on axis in user units
axis	Index of the axis for which the default position to be set or -1 to set position to default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.18 DefPosMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::DefPosMulti (
            const double pos[],
            uint32_t posCount,
            int32_t axis = -1 )  [pure virtual]
```

Defines the current axis position as a new absolute value to multiple axes.

Parameters

pos	Array of absolute positions in user units to set on axes defined by SetMultiAxisBase
posCount	Number of positions contained in array.
axis	Index of the first axis for which the default position to be set or -1 to set position to default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.19 Delete()

```
virtual void Trio::UnifiedApi::ITrioConnection::Delete (
            const std::string_view & remote_file,
            bool compiled_code_only = false )  [pure virtual]
```

Deletes a file on the Motion Coordinator.

Parameters

remote_file	The name of the file
compiled_code_only	In case of a program: when true only the compiled code of the program will be deleted, the source code will remain. When false all data related to the program will be deleted

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.20 DeleteFrameGroup()

```
virtual void Trio::UnifiedApi::ITrioConnection::DeleteFrameGroup (
            uint32_t group )  [pure virtual]
```

Delete frame group. When the group is deleted FRAME is set to 0. See the TrioBASIC FRAME_GROUP documentation for more information.

Parameters

group	Frame group number to delete

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.21 DeleteMe()

```
virtual void Trio::UnifiedApi::ITrioConnection::DeleteMe ( )  [pure virtual]
```

Publicly exposed method for communication object destruction Calling it is the proper way of destoring ITrioConnection instance After calling this method other methods should not be called.

6.3.3.22 Dir()

```
virtual void Trio::UnifiedApi::ITrioConnection::Dir (
            FileCallback && handler )  [pure virtual]
```

Function returns the list of programs and files in internal memory of the Motion Coordinator.

Parameters

handler	Callback handler function executed for each item in internal memory

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.23 DownloadFile()

```
virtual void Trio::UnifiedApi::ITrioConnection::DownloadFile (
            const std::string_view & local_file,
            const std::string_view & remote_file,
            ProgressHandlerCallback && handler )  [pure virtual]
```

Downloads a file from the Motion Coordinator.

Parameters

local_file	Local source file
remote_file	Remote file name. Can be prefixed by: "RAM:" - the location of the file is in the controller RAM. "SD:" - the location of the file is on the SD card. "FIFO:" - The target file is a FIFO file on the controller. If no prefix "RAM:" is assumed.
handler	Callback handler with the progress advance

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.24 EncoderRatio()

```
virtual void Trio::UnifiedApi::ITrioConnection::EncoderRatio (
            int64_t mpos_count,
            int64_t input_count,
            int32_t axis = -1 )  [pure virtual]
```

This command allows the incoming encoder count to be scaled by a non integer ratio : MPOS = (mpos_count / input_count) x encoder_edges_input When using the servo loop you will need to adjust the gains to maintain performance and stability. Unlike the UNITS parameter, which only affects the scaling seen by the user programs, ENCODER_RATIO affects all motion commands. ENCODER_RATIO does not replace UNITS.Only use ENCODER_RATIO where absolutely necessary.PP_STEP and ENCODER_RATIO cannot be used at the same time on the same axis.

Parameters

mpos_count	An integer number which defines the numerator
input_count	An integer number which defines the denominator
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.25 ErrorString()

```
static TRIO_DECLSPEC const std::string_view &TRIO_CALLCONV Trio::UnifiedApi::ITrioConnection↩
::ErrorString (
            ErrorCode errorCode )  [static]
```

Error description.

Parameters

errorCode	Error code

Returns

Description from error code number

6.3.3.26 Ethercat_CheckNumberOfSlaves()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::Ethercat_CheckNumberOfSlaves (
            int32_t slot )  [pure virtual]
```

Determine the number of slaves on the bus. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.

Returns

On success receives the number of slaves

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.27 Ethercat_CheckNumberOfSlaves_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CheckNumberOfSlaves_VR (
            int32_t slot,
            uint32_t out_vr_index )  [pure virtual]
```

Determine the number of slaves on the bus. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.28 Ethercat_CheckSlaveOnline()

```
virtual bool Trio::UnifiedApi::ITrioConnection::Ethercat_CheckSlaveOnline (
            int32_t slot,
            int32_t slave )  [pure virtual]
```

Check whether given slave is online. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Position on the network of the slave

Returns

On success true if slave is online or false otherwise

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.29 Ethercat_CheckSlaveOnline_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CheckSlaveOnline_VR (
            int32_t slot,
            int32_t slave,
            uint32_t out_vr_index )  [pure virtual]
```

Check whether given slave is online. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Position on the network of the slave
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.30 Ethercat_CoRead()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoRead (
            int32_t slot,
            int32_t address,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            int32_t out_vr_index )  [pure virtual]
```

This function gets a CANopen-over-EtherCAT (CoE) object from the remote drive or I/O device. The object's index and sub-index are used to request a value and that value is either placed in the VR or is displayed in the Motion Perfect terminal if the VR number is set to -1. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
address	Node address of the remote device on the network
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.31 Ethercat_CoReadAxis()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoReadAxis (
            int32_t axis_number,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            int32_t out_vr_index )  [pure virtual]
```

This function gets a CANopen-over-EtherCAT (CoE) object from the remote drive or I/O device. The object's index and sub-index are used to request a value and that value is either placed in the VR or is displayed in the Motion Perfect terminal if the VR number is set to -1. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

axis_number	Axis number of the EtherCAT drive.
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.32 Ethercat_CoWrite_Value() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoWrite_Value (
            int32_t slot,
            int32_t address,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            double value )  [pure virtual]
```

This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for floating-point values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
address	Node address of the remote device on the network
object_index	CANopen Object index
object_subindex	CANopen Object sub-index

Parameters

type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
value	The value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.33 Ethercat_CoWrite_Value() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoWrite_Value (
            int32_t slot,
            int32_t address,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            int64_t value )  [pure virtual]
```

This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for integer values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
address	Node address of the remote device on the network
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
value	The value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.34 Ethercat_CoWrite_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoWrite_VR (
            int32_t slot,
            int32_t address,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            uint32_t vr_index )  [pure virtual]
```

This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. The value comes from a VR. Use this function for integer values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
address	Node address of the remote device on the network
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
vr_index	VR number between 0 and max VR where the result will be stored. (-1 means the value will be printed to the terminal)

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.35 Ethercat_CoWriteAxis_Value() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoWriteAxis_Value (
            int32_t axis_number,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            double value )  [pure virtual]
```

This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for floating-point values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

axis_number	Axis number of the EtherCAT drive.
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
value	The value.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.36 Ethercat_CoWriteAxis_Value() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoWriteAxis_Value (
            int32_t axis_number,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            int64_t value )  [pure virtual]
```

This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. Use this function for integer values. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

axis_number	Axis number of the EtherCAT drive.
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64

Parameters

value	The value.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.37 Ethercat_CoWriteAxis_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_CoWriteAxis_VR (
            int32_t axis_number,
            uint16_t object_index,
            uint8_t object_subindex,
            Co_ObjectType type,
            uint32_t vr_index )  [pure virtual]
```

This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. The value comes from a VR. Refer to the remote device's manual for a list of available objects. See Trio BASIC help for more details.

Parameters

axis_number	Axis number of the EtherCAT drive.
object_index	CANopen Object index
object_subindex	CANopen Object sub-index
type	1 : Boolean 2 : Integer 8 3 : Integer 16 4 : Integer 32 5 : Unsigned 8 6 : Unsigned 16 7 : Unsigned 32 8 : Real 32 17 : Real 64 21 : Integer 64 27 : Unsigned 64
vr_index	VR number between 0 and max VR where the value is stored.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.38 Ethercat_DisplayNetworkConfig()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_DisplayNetworkConfig (
            int32_t slot )  [pure virtual]
```

Displays the network configuration to the command line terminal in Motion Perfect. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.39 Ethercat_GetEmergencyMsgControl()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::Ethercat_GetEmergencyMsgControl (
            int32_t slot )  [pure virtual]
```

Get the current value; indicating the response of the master to an emergency telegram transmitted by a slave node. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.

Returns

On success the current value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.40 Ethercat_GetEmergencyMsgControl_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetEmergencyMsgControl_VR (
            int32_t slot,
            uint32_t out_vr_index )  [pure virtual]
```

Get the current value; indicating the response of the master to an emergency telegram transmitted by a slave node. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.41 Ethercat_GetSlave_ESC_Register()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetSlave_ESC_Register (
            int32_t slot,
            int32_t slave,
            uint32_t esc_offset,
            uint32_t data_length,
            uint32_t out_vr_index )  [pure virtual]
```

Low-level read from the slave's ESC registers, addressing method used is auto-increment addressing. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Auto-increment address
esc_offset	Register address
data_length	Number of bytes in the register size
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.42 Ethercat_GetSlaveAddress()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::Ethercat_GetSlaveAddress (
            int32_t slot,
            int32_t slave )  [pure virtual]
```

Determine the configured address of the slave identified by its position on the network. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Position on the network of the slave

Returns

On success receives the configured slave address

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.43 Ethercat_GetSlaveAddress_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetSlaveAddress_VR (
            int32_t slot,
            int32_t slave,
            uint32_t out_vr_index )  [pure virtual]
```

Determine the configured address of the slave identified by its position on the network. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Position on the network of the slave
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.44 Ethercat_GetSlaveAxis()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::Ethercat_GetSlaveAxis (
            int32_t slot,
            int32_t slave )  [pure virtual]
```

Determine the axis of the slave identified by its position on the network. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Position on the network of the slave

Returns

On success receives the axis assigned to the slave

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.45 Ethercat_GetSlaveAxis_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetSlaveAxis_VR (
            int32_t slot,
            int32_t slave,
            uint32_t out_vr_index )  [pure virtual]
```

Determine the axis of the slave identified by its position on the network. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Position on the network of the slave
out_vr_index	Result is placed in the selected VR variable

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.46 Ethercat_GetState()

```
virtual EthercatState Trio::UnifiedApi::ITrioConnection::Ethercat_GetState (
            int32_t slot )  [pure virtual]
```

Gets the present state of the EtherCAT running on the defined slot. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.

Returns

On success redeives the state. The returned value means:
0 : Initial (EtherCAT ESC value 0x01)
1 : Pre-Operational (0x02)
2 : Safe-Operational (0x04)
3 : Operational (0x08)

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.47 Ethercat_GetState_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetState_VR (
            int32_t slot,
            uint32_t out_vr_index )  [pure virtual]
```

Gets the present state of the EtherCAT running on the defined slot. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
out_vr_index	Result is placed in the selected VR variable. The returned value means: 0 : Initial (EtherCAT ESC value 0x01) 1 : Pre-Operational (0x02) 2 : Safe-Operational (0x04) 3 : Operational (0x08)

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.48 Ethercat_GetStatusMaskAndTriggerPattern()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetStatusMaskAndTriggerPattern (
            int32_t slot,
            int32_t addr,
            uint32_t & mask,
            uint32_t & trigger_pattern )  [pure virtual]
```

This command enables the user to display and check the mask and trigger pattern values set (if any) for a given slave. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port
addr	Configured address of the slave
mask	Current statusword mask
trigger_pattern	Current trigger pattern

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.49 Ethercat_GetStatusMaskAndTriggerPattern_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_GetStatusMaskAndTriggerPattern_VR (
            int32_t slot,
            int32_t addr,
            uint32_t out_vr_index )  [pure virtual]
```

This command enables the user to display and check the mask and trigger pattern values set (if any) for a given slave. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
addr	Configured address of the slave
out_vr_index	VR index in range [0..max_vr_index-2]. Index of the array of 2 VRs where the mask and the trigger_pattern will be stored.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.50 Ethercat_SendResetSequence() [1/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SendResetSequence (
            int32_t axis )  [pure virtual]
```

Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed. See Trio BASIC help for more details.

Parameters

axis	The axis number of the drive to be reset

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Default timeout is 100 ms

6.3.3.51 Ethercat_SendResetSequence() [2/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SendResetSequence (
            int32_t axis,
            uint32_t mode )  [pure virtual]
```

Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed. See Trio BASIC help for more details.

Parameters

axis	The axis number of the drive to be reset
mode	0 : The 'Fault Reset' (bit 7) of DS402 control word is set high and then set low again after a hard coded timeout. (default) 1 : Bit 7 is set high until the 'Fault Flag' (bit 3) of the status word goes low, or a timeout occurs.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Default timeout is 100 ms

6.3.3.52 Ethercat_SendResetSequence() [3/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SendResetSequence (
            int32_t axis,
            uint32_t mode,
            uint32_t timeout )  [pure virtual]
```

Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed. See Trio BASIC help for more details.

Parameters

axis	The axis number of the drive to be reset
mode	0 : The 'Fault Reset' (bit 7) of DS402 control word is set high and then set low again after a hard coded timeout. (default) 1 : Bit 7 is set high until the 'Fault Flag' (bit 3) of the status word goes low, or a timeout occurs.
timeout	Timeout in ms used during mode 1 operation. Default is 100 ms. Range is 1 to 10000 ms.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.53 Ethercat_SetDistrClockMaster()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetDistrClockMaster (
            int32_t slot,
            uint32_t function,
            uint32_t value )  [pure virtual]
```

Set the EtherCAT master to enable the first DC enabled node to be the Distributed Clocks master. As this requires the Trio Motion Coordinator to synchronise to the external DC master, it does not function on P876, MC4N-ECAT or MC664(X) as these have FPGA based fixed timing hardware for the EtherCAT port. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
function	Distributed Clocks function. 2 = use first DC enabled node as clocks master.
value	The value. 0 = use first DC enabled node as the clocks master.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.54 Ethercat_SetDistrClockMaster_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetDistrClockMaster_VR (
            int32_t slot,
            uint32_t function,
            uint32_t vr_index )  [pure virtual]
```

Set the EtherCAT master to enable the first DC enabled node to be the Distributed Clocks master. As this requires the Trio Motion Coordinator to synchronise to the external DC master, it does not function on P876, MC4N-ECAT or MC664(X) as these have FPGA based fixed timing hardware for the EtherCAT port. The function returns TRUE

/ FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
function	Distributed Clocks function. 2 = use first DC enabled node as clocks master.
vr_index	VR index in range [0..max_vr_index-1]. Use the value in the VR number specified.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.55 Ethercat_SetEmergencyMsgControl()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetEmergencyMsgControl (
            int32_t slot,
            uint32_t mode )  [pure virtual]
```

Set the response of the master to an emergency telegram transmitted by a slave node. This function enables the user to determine whether emergency messages should trigger a system error alarm and cause the WDOG to be dropped. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
mode	0 : Log error and raise a System Error and drop WDOG off. (Default) 1 : Log error only.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.56 Ethercat_SetSlave_ESC_Register()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetSlave_ESC_Register (
            int32_t slot,
            int32_t slave,
            uint32_t esc_offset,
            uint32_t data_length,
            const char * data )  [pure virtual]
```

Low-level write to the slave's ESC registers, addressing method used is auto-increment addressing. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
slave	Auto-increment address
esc_offset	Register address
data_length	Number of data bytes to write
data	Data values

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.57 Ethercat_SetState()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetState (
            int32_t slot,
            EthercatState state,
            uint32_t display )  [pure virtual]
```

This function controls the EtherCAT State Machine (ESM). It requests the master change to given EtherCAT 'state', and hence changes all slaves to the same state. When a change to a higher state is made, the EtherCAT network will progress to the new state through the in-between states to allow correct starting of the network. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
state	-1 : Reserved 0 : Initial (EtherCAT ESC value 0x01) 1 : Pre-Operational (0x02) 2 : Safe-Operational (0x04) 3 : Operational (0x08)
display	0 : Writes state change information to the standard output stream. 1 : Do not write out state change information.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.58 Ethercat_SetStatusMaskAndTriggerPattern()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetStatusMaskAndTriggerPattern (
            int32_t slot,
            int32_t addr,
            uint32_t mask,
            uint32_t trigger_pattern )  [pure virtual]
```

This command enables the user to define a CoE Statusword (0x6041:00) mask and trigger pattern for raising the controller AXISSTATUS 'Remote Drive Status word Error' flag (bit 26.) The mask defines which bits within the CoE status word are relevant, and the trigger pattern defines the state of these bits(0 or 1) which will trigger the controller AXISSTATUS flag. If it is necessary to reset the watchdog(master enable) when this AXISISTATUS flag is set then the axis ERRORMASK must be set accordingly. The slave mask and pattern can be set and modified from preop to operational state of the network(after the configured addresses have been assigned.) It must be reset if the network is restarted. The slave must support the CoE statusword(0x6041:00) within the PDO cyclic data. If the Mask is zero the controller will not execute any runtime check of the CoE statusword. If the watchdog must be cleared to OFF when the 'remote drive statusword error' flag is raised then bit 26 of the axis ERRORMASK parameter must be set. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
addr	Configured address of the slave
mask	CoE statusword mask.
trigger_pattern	Defines the state of the relevant bits which will trigger the controller AXISSTATUS 'remote drive statusword error' flag (bit 26.)

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.59 Ethercat_SetStatusMaskAndTriggerPattern_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetStatusMaskAndTriggerPattern_VR (
            int32_t slot,
            int32_t addr,
            uint32_t vr_index )  [pure virtual]
```

This command enables the user to define a CoE Statusword (0x6041:00) mask and trigger pattern for raising the controller AXISSTATUS 'Remote Drive Status word Error' flag (bit 26.) The mask defines which bits within the CoE status word are relevant, and the trigger pattern defines the state of these bits(0 or 1) which will trigger the controller AXISSTATUS flag. If it is necessary to reset the watchdog(master enable) when this AXISISTATUS flag is set then the axis ERRORMASK must be set accordingly. The slave mask and pattern can be set and modified from preop to operational state of the network(after the configured addresses have been assigned.) It must be reset if the network is restarted. The slave must support the CoE statusword(0x6041:00) within the PDO cyclic data. If the Mask is zero the controller will not execute any runtime check of the CoE statusword. If the watchdog must be cleared to OFF when the 'remote drive statusword error' flag is raised then bit 26 of the axis ERRORMASK parameter must be set. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
addr	Configured address of the slave
vr_index	VR index in range [0..max_vr_index-1]. The index of the VR which contains the mask and the next VR (vr_index+1) contains the trigger_pattern.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.60 Ethercat_SetSync0StartTime()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetSync0StartTime (
            int32_t offset )  [pure virtual]
```

Set the network default Sync0 start time offset. If offset value is directly provided as a parameter. The default start time is 1 second(1000000 microseconds).Add or subtract the required offset in microseconds.For example 1000150

is an offset of 150 microseconds and 999930 is an offset of -70 microseconds.The offset must not exceed ±50 % of the servo period. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

Parameters

offset	Direct value for the start time

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.61 Ethercat_SetSync0StartTime_VR()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_SetSync0StartTime_VR (
            uint32_t vr_index )  [pure virtual]
```

Set the network default Sync0 start time offset. The offset value is from the given VR. The default start time is 1 second(1000000 microseconds).Add or subtract the required offset in microseconds.For example 1000150 is an offset of 150 microseconds and 999930 is an offset of -70 microseconds.The offset must not exceed ±50 % of the servo period. The function returns TRUE / FALSE depending upon whether the operation was successful. This command must be run before the EtherCAT network is started. See Trio BASIC help for more details.

Parameters

vr_index	VR index in range [0..max_vr_index-1]. Use the value in the VR number specified.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.62 Ethercat_StartNetwork() [1/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_StartNetwork (
            int32_t slot )  [pure virtual]
```

Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.63 Ethercat_StartNetwork() [2/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_StartNetwork (
            int32_t slot,
            EthercatStartOption options )  [pure virtual]
```

Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
options	bit 0: 0-Do not print progress information; 1-Print out progress information (default) bit 1: 0-Normal state change (default); 1-Force a re-scan of the network bit 2: 0-Standard startup; 1-Express startup; do not wait for PLL to lock between state changes

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.64 Ethercat_StartNetwork() [3/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_StartNetwork (
            int32_t slot,
            EthercatStartOption options,
            uint32_t MAC_retries )  [pure virtual]
```

Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.

Parameters

slot	Slot number of EtherCAT port.
options	bit 0: 0-Do not print progress information; 1-Print out progress information (default) bit 1: 0-Normal state change (default); 1-Force a re-scan of the network bit 2: 0-Standard startup; 1-Express startup; do not wait for PLL to lock between state changes
MAC_retries	Sets the number of times the master attempts to restart the Ethernet auto-negotiation. Default = 2

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.65 Ethercat_StopNetwork()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ethercat_StopNetwork (
            int32_t slot )  [pure virtual]
```

Closedown the EtherCAT network.

Parameters

slot	Slot number of EtherCAT port.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.66 Ex()

```
virtual void Trio::UnifiedApi::ITrioConnection::Ex (
            int32_t mode = 0 )  [pure virtual]
```

Perform software reset on Motion Coordinator.

Parameters

mode	Reset mode. See the TrioBASIC EX documentation for more details.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.67 Execute()

```
virtual void Trio::UnifiedApi::ITrioConnection::Execute (
            const std::string_view & command )  [pure virtual]
```

Execute arbitrary command.

Parameters

command	Command to be executed

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.68 ExecuteWithResponse() [1/3]

```
virtual std::string Trio::UnifiedApi::ITrioConnection::ExecuteWithResponse (
            const std::string_view & command )  [pure virtual]
```

Execute arbitrary command with response.

Parameters

command	Command to be executed

Returns

Command response

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.69 ExecuteWithResponse() [2/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::ExecuteWithResponse (
            const std::string_view & command,
            char * buff,
            uint32_t & buff_size )  [pure virtual]
```

Execute arbitrary command with response.

Parameters

command	Command to be executed
buff	Buffer to receive the response string characters
buff_size	Size of the buffer on input and on exit required buffer size including null terminator

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.70 ExecuteWithResponse() [3/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::ExecuteWithResponse (
            const std::string_view & command,
            StreamHandlerCallback && handler )  [pure virtual]
```

Execute arbitrary command with response.

Parameters

command	Command to be executed
handler	Callback function to receive response string chunks

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.71 FileExists()

```
virtual FileExistsFlags Trio::UnifiedApi::ITrioConnection::FileExists (
            const std::string_view & name )  [pure virtual]
```

Check if file or program exists on Motion Coordinator.

Parameters

name	Name of the file or program to check. Can be prefixed by: "RAM:" - the location of the file is in the controller RAM. "SD:" - the location of the file is on the SD card. Must be a full path and file name with extension. If no prefix "RAM:" is assumed.

Returns

Result from operation.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.72 Forward()

```
virtual void Trio::UnifiedApi::ITrioConnection::Forward (
            int32_t axis = -1 )  [pure virtual]
```

Sets continuous forward movement.

Parameters

axis	Axis index to set forward movement to or -1 to apply to default base axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.73 GetAIn()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAIn (
            uint32_t channel )  [pure virtual]
```

Get a value from an Analogue Input.

Parameters

channel	Analogue input channel number

Returns

Current value of the analog input channel

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.74 GetAOut()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAOut (
            uint32_t channel )  [pure virtual]
```

Get the state of the Analogue Output.

Parameters

channel	Analogue output channel number

Returns

Current value of the analogue output channel

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.75 GetApiVersion()

```
virtual VersionInfo Trio::UnifiedApi::ITrioConnection::GetApiVersion ( )  [pure virtual]
```

Get Unified API version info.

Returns

On success version info of the Unified API

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.76 GetApiVersionStr()

```
virtual std::string_view Trio::UnifiedApi::ITrioConnection::GetApiVersionStr ( )  [pure virtual]
```

Get Unified API version string.

Returns

On success string representation of the Unified API version

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.77 GetAxisParameter_ACCEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ACCEL (
            int32_t axis )  [pure virtual]
```

Getter for ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.78 GetAxisParameter_ADDAX_AXIS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ADDAX_AXIS (
            int32_t axis )  [pure virtual]
```

Getter for ADDAX_AXIS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.79 GetAxisParameter_AFF_GAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AFF_GAIN (
            int32_t axis )  [pure virtual]
```

Getter for AFF_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.80 GetAxisParameter_ATYPE()

```
virtual AxisType Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ATYPE (
            int32_t axis )  [pure virtual]
```

Getter for ATYPE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.81 GetAxisParameter_AXIS_ACCEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_ACCEL (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.82 GetAxisParameter_AXIS_ADDRESS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_ADDRESS (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_ADDRESS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.83 GetAxisParameter_AXIS_DECEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_DECEL (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_DECEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.84 GetAxisParameter_AXIS_DISPLAY()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_DISPLAY (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_DISPLAY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.85 GetAxisParameter_AXIS_DPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_DPOS (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_DPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.86 GetAxisParameter_AXIS_ENABLE()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_ENABLE (
```

```
int32_t axis )  [pure virtual]
```

Getter for AXIS_ENABLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.87 GetAxisParameter_AXIS_ENABLE_OVERRIDE()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_ENABLE_OVERRIDE (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_ENABLE_OVERRIDE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.88 GetAxisParameter_AXIS_ERROR_COUNT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_ERROR_COUNT (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_ERROR_COUNT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.89 GetAxisParameter_AXIS_FS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_FS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.90 GetAxisParameter_AXIS_MAX_JOG_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_MAX_JOG_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_MAX_JOG_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.91 GetAxisParameter_AXIS_MODE()

```
virtual AxisMode Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_MODE (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.92 GetAxisParameter_AXIS_RS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_RS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.93 GetAxisParameter_AXIS_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.94 GetAxisParameter_AXIS_UNITS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_UNITS (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.95 GetAxisParameter_AXIS_Z_OUTPUT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXIS_Z_OUTPUT (
            int32_t axis )  [pure virtual]
```

Getter for AXIS_Z_OUTPUT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.96 GetAxisParameter_AXISSTATUS()

```
virtual AxisStatus Trio::UnifiedApi::ITrioConnection::GetAxisParameter_AXISSTATUS (
            int32_t axis )  [pure virtual]
```

Getter for AXISSTATUS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.97 GetAxisParameter_BACKLASH_DIST()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_BACKLASH_DIST (
            int32_t axis )  [pure virtual]
```

Getter for BACKLASH_DIST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.98 GetAxisParameter_CHANGE_DIR_LAST()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_CHANGE_DIR_LAST (
```

```
int32_t axis )  [pure virtual]
```

Getter for CHANGE_DIR_LAST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.99 GetAxisParameter_CLOSE_WIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_CLOSE_WIN (
            int32_t axis )  [pure virtual]
```

Getter for CLOSE_WIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.100 GetAxisParameter_CLUTCH_RATE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_CLUTCH_RATE (
            int32_t axis )  [pure virtual]
```

Getter for CLUTCH_RATE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.101 GetAxisParameter_CORNER_MODE()

```
virtual CornerMode Trio::UnifiedApi::ITrioConnection::GetAxisParameter_CORNER_MODE (
            int32_t axis )  [pure virtual]
```

Getter for CORNER_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.102 GetAxisParameter_CORNER_STATE()

```
virtual CornerState Trio::UnifiedApi::ITrioConnection::GetAxisParameter_CORNER_STATE (
            int32_t axis )  [pure virtual]
```

Getter for CORNER_STATE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.103 GetAxisParameter_CREEP()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_CREEP (
            int32_t axis )  [pure virtual]
```

Getter for CREEP axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.104 GetAxisParameter_D_GAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_D_GAIN (
            int32_t axis )  [pure virtual]
```

Getter for D_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.105 GetAxisParameter_DAC()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DAC (
            int32_t axis )  [pure virtual]
```

Getter for DAC axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.106 GetAxisParameter_DAC_OUT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DAC_OUT (
            int32_t axis )  [pure virtual]
```

Getter for DAC_OUT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.107 GetAxisParameter_DAC_SCALE()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DAC_SCALE (
            int32_t axis )  [pure virtual]
```

Getter for DAC_SCALE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.108 GetAxisParameter_DATUM_IN()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DATUM_IN (
            int32_t axis )  [pure virtual]
```

Getter for DATUM_IN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.109 GetAxisParameter_DECEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DECEL (
            int32_t axis )  [pure virtual]
```

Getter for DECEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.110 GetAxisParameter_DECEL_ANGLE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DECEL_ANGLE (
```

```
int32_t axis )  [pure virtual]
```

Getter for DECEL_ANGLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.111 GetAxisParameter_DEMAND_EDGES()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DEMAND_EDGES (
            int32_t axis )  [pure virtual]
```

Getter for DEMAND_EDGES axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.112 GetAxisParameter_DEMAND_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DEMAND_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for DEMAND_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.113 GetAxisParameter_DPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DPOS (
            int32_t axis )  [pure virtual]
```

Getter for DPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.114 GetAxisParameter_DRIVE_CONTROL()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_CONTROL (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.115 GetAxisParameter_DRIVE_CONTROLWORD()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_CONTROLWORD (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_CONTROLWORD axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.116 GetAxisParameter_DRIVE_CURRENT()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_CURRENT (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_CURRENT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.117 GetAxisParameter_DRIVE_CW_MODE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_CW_MODE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_CW_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.118 GetAxisParameter_DRIVE_ENABLE()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_ENABLE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_ENABLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.119 GetAxisParameter_DRIVE_FE()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_FE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_FE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.120 GetAxisParameter_DRIVE_FE_LIMIT()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_FE_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_FE_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.121 GetAxisParameter_DRIVE_INDEX()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_INDEX (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_INDEX axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.122 GetAxisParameter_DRIVE_INPUTS()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_INPUTS (
```

```
int32_t axis )  [pure virtual]
```

Getter for DRIVE_INPUTS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.123 GetAxisParameter_DRIVE_MODE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_MODE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.124 GetAxisParameter_DRIVE_MONITOR()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_MONITOR (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_MONITOR axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.125 GetAxisParameter_DRIVE_NEG_TORQUE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_NEG_TORQUE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_NEG_TORQUE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.126 GetAxisParameter_DRIVE_PARAMETER()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_PARAMETER (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_PARAMETER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.127 GetAxisParameter_DRIVE_POS_TORQUE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_POS_TORQUE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_POS_TORQUE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.128 GetAxisParameter_DRIVE_PROFILE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_PROFILE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_PROFILE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.129 GetAxisParameter_DRIVE_SET_VAL()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_SET_VAL (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_SET_VAL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.130 GetAxisParameter_DRIVE_STATUS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_STATUS (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_STATUS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.131 GetAxisParameter_DRIVE_TORQUE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_TORQUE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_TORQUE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.132 GetAxisParameter_DRIVE_TYPE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_TYPE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_TYPE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.133 GetAxisParameter_DRIVE_VALUE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_VALUE (
            int32_t axis )  [pure virtual]
```

Getter for DRIVE_VALUE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.134 GetAxisParameter_DRIVE_VELOCITY()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_DRIVE_VELOCITY (
```

```
int32_t axis )  [pure virtual]
```

Getter for DRIVE_VELOCITY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.135 GetAxisParameter_ENCODER()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.136 GetAxisParameter_ENCODER_BITS()

```
virtual uint16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER_BITS (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER_BITS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.137 GetAxisParameter_ENCODER_CONTROL()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER_CONTROL (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.138 GetAxisParameter_ENCODER_FILTER()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER_FILTER (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER_FILTER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.139 GetAxisParameter_ENCODER_ID()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER_ID (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER_ID axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.140 GetAxisParameter_ENCODER_STATUS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER_STATUS (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER_STATUS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.141 GetAxisParameter_ENCODER_TURNS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENCODER_TURNS (
            int32_t axis )  [pure virtual]
```

Getter for ENCODER_TURNS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.142 GetAxisParameter_END_DIR_LAST()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_END_DIR_LAST (
            int32_t axis )  [pure virtual]
```

Getter for END_DIR_LAST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.143 GetAxisParameter_ENDMOVE()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENDMOVE (
            int32_t axis )  [pure virtual]
```

Getter for ENDMOVE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.144 GetAxisParameter_ENDMOVE_BUFFER()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENDMOVE_BUFFER (
            int32_t axis )  [pure virtual]
```

Getter for ENDMOVE_BUFFER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.145 GetAxisParameter_ENDMOVE_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ENDMOVE_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for ENDMOVE_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.146 GetAxisParameter_ERRORMASK()

```
virtual AxisStatus Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ERRORMASK (
```

```
int32_t axis )  [pure virtual]
```

Getter for ERRORMASK axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.147 GetAxisParameter_FAST_JOG()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FAST_JOG (
            int32_t axis )  [pure virtual]
```

Getter for FAST_JOG axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.148 GetAxisParameter_FASTDEC()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FASTDEC (
            int32_t axis )  [pure virtual]
```

Getter for FASTDEC axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.149 GetAxisParameter_FE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FE (
            int32_t axis )  [pure virtual]
```

Getter for FE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.150 GetAxisParameter_FE_LATCH()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FE_LATCH (
            int32_t axis )  [pure virtual]
```

Getter for FE_LATCH axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.151 GetAxisParameter_FE_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FE_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for FE_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.152 GetAxisParameter_FE_LIMIT_MODE()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FE_LIMIT_MODE (
            int32_t axis )  [pure virtual]
```

Getter for FE_LIMIT_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.153 GetAxisParameter_FE_RANGE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FE_RANGE (
            int32_t axis )  [pure virtual]
```

Getter for FE_RANGE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.154 GetAxisParameter_FEED_OVERRIDE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FEED_OVERRIDE (
            int32_t axis )  [pure virtual]
```

Getter for FEED_OVERRIDE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.155 GetAxisParameter_FHOLD_IN()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FHOLD_IN (
            int32_t axis )  [pure virtual]
```

Getter for FHOLD_IN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.156 GetAxisParameter_FHSPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FHSPEED (
            int32_t axis )  [pure virtual]
```

Getter for FHSPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.157 GetAxisParameter_FORCE_ACCEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FORCE_ACCEL (
            int32_t axis )  [pure virtual]
```

Getter for FORCE_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.158 GetAxisParameter_FORCE_DECEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FORCE_DECEL (
```

```
int32_t axis )  [pure virtual]
```

Getter for FORCE_DECEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.159 GetAxisParameter_FORCE_DWELL()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FORCE_DWELL (
            int32_t axis )  [pure virtual]
```

Getter for FORCE_DWELL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.160 GetAxisParameter_FORCE_JERK()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FORCE_JERK (
            int32_t axis )  [pure virtual]
```

Getter for FORCE_JERK axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.161 GetAxisParameter_FORCE_RAMP()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FORCE_RAMP (
            int32_t axis )  [pure virtual]
```

Getter for FORCE_RAMP axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.162 GetAxisParameter_FORCE_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FORCE_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for FORCE_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.163 GetAxisParameter_FRAME()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FRAME (
            int32_t axis )  [pure virtual]
```

Getter for FRAME axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.164 GetAxisParameter_FRAME_REP_DIST()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FRAME_REP_DIST (
            int32_t axis )  [pure virtual]
```

Getter for FRAME_REP_DIST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.165 GetAxisParameter_FRAME_SCALE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FRAME_SCALE (
            int32_t axis )  [pure virtual]
```

Getter for FRAME_SCALE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.166 GetAxisParameter_FS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.167 GetAxisParameter_FULL_SP_RADIUS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FULL_SP_RADIUS (
            int32_t axis )  [pure virtual]
```

Getter for FULL_SP_RADIUS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.168 GetAxisParameter_FWD_IN()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FWD_IN (
            int32_t axis )  [pure virtual]
```

Getter for FWD_IN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.169 GetAxisParameter_FWD_JOG()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FWD_JOG (
            int32_t axis )  [pure virtual]
```

Getter for FWD_JOG axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.170 GetAxisParameter_FWD_START()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_FWD_START (
```

```
int32_t axis )  [pure virtual]
```

Getter for FWD_START axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.171 GetAxisParameter_I_GAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_I_GAIN (
            int32_t axis )  [pure virtual]
```

Getter for I_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.172 GetAxisParameter_IDLE()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetAxisParameter_IDLE (
            int32_t axis )  [pure virtual]
```

Getter for IDLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.173 GetAxisParameter_IN_POS()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_IN_POS (
            int32_t axis )  [pure virtual]
```

Getter for IN_POS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.174 GetAxisParameter_IN_POS_DIST()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_IN_POS_DIST (
            int32_t axis )  [pure virtual]
```

Getter for IN_POS_DIST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.175 GetAxisParameter_IN_POS_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_IN_POS_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for IN_POS_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.176 GetAxisParameter_INTERP_FACTOR()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_INTERP_FACTOR (
            int32_t axis )  [pure virtual]
```

Getter for INTERP_FACTOR axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.177 GetAxisParameter_INVERT_STEP()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_INVERT_STEP (
            int32_t axis )  [pure virtual]
```

Getter for INVERT_STEP axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.178 GetAxisParameter_JERK()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_JERK (
            int32_t axis )  [pure virtual]
```

Getter for JERK axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.179 GetAxisParameter_JOGSPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_JOGSPEED (
            int32_t axis )  [pure virtual]
```

Getter for JOGSPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.180 GetAxisParameter_LINK_AXIS()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_LINK_AXIS (
            int32_t axis )  [pure virtual]
```

Getter for LINK_AXIS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.181 GetAxisParameter_LOADED()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_LOADED (
            int32_t axis )  [pure virtual]
```

Getter for LOADED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.182 GetAxisParameter_LOOKAHEAD_FACTOR()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_LOOKAHEAD_FACTOR (
```

```
int32_t axis )  [pure virtual]
```

Getter for LOOKAHEAD_FACTOR axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.183 GetAxisParameter_MARK()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MARK (
            int32_t axis )  [pure virtual]
```

Getter for MARK axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.184 GetAxisParameter_MARKB()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MARKB (
            int32_t axis )  [pure virtual]
```

Getter for MARKB axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.185 GetAxisParameter_MERGE()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MERGE (
            int32_t axis )  [pure virtual]
```

Getter for MERGE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.186 GetAxisParameter_MOVE_COUNT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_COUNT (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_COUNT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.187 GetAxisParameter_MOVE_PA()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_PA (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_PA axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.188 GetAxisParameter_MOVE_PA_CONT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_PA_CONT (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_PA_CONT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.189 GetAxisParameter_MOVE_PA_IDLE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_PA_IDLE (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_PA_IDLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.190 GetAxisParameter_MOVE_PB()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_PB (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_PB axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.191 GetAxisParameter_MOVE_PB_CONT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_PB_CONT (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_PB_CONT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.192 GetAxisParameter_MOVE_PB_IDLE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVE_PB_IDLE (
            int32_t axis )  [pure virtual]
```

Getter for MOVE_PB_IDLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.193 GetAxisParameter_MOVELINK_MODIFY()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVELINK_MODIFY (
            int32_t axis )  [pure virtual]
```

Getter for MOVELINK_MODIFY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.194 GetAxisParameter_MOVES_BUFFERED()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MOVES_BUFFERED (
```

```
int32_t axis )  [pure virtual]
```

Getter for MOVES_BUFFERED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.195 GetAxisParameter_MPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MPOS (
            int32_t axis )  [pure virtual]
```

Getter for MPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.196 GetAxisParameter_MSPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MSPEED (
            int32_t axis )  [pure virtual]
```

Getter for MSPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.197 GetAxisParameter_MSPEED_FILTER()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MSPEED_FILTER (
            int32_t axis )  [pure virtual]
```

Getter for MSPEED_FILTER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.198 GetAxisParameter_MSPEEDF()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MSPEEDF (
            int32_t axis )  [pure virtual]
```

Getter for MSPEEDF axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.199 GetAxisParameter_MTYPE()

```
virtual MotionType Trio::UnifiedApi::ITrioConnection::GetAxisParameter_MTYPE (
            int32_t axis )  [pure virtual]
```

Getter for MTYPE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.200 GetAxisParameter_NTYPE()

```
virtual MotionType Trio::UnifiedApi::ITrioConnection::GetAxisParameter_NTYPE (
            int32_t axis )  [pure virtual]
```

Getter for NTYPE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.201 GetAxisParameter_OFFPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_OFFPOS (
            int32_t axis )  [pure virtual]
```

Getter for OFFPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.202 GetAxisParameter_OPEN_WIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_OPEN_WIN (
            int32_t axis )  [pure virtual]
```

Getter for OPEN_WIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.203 GetAxisParameter_OUTLIMIT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_OUTLIMIT (
            int32_t axis )  [pure virtual]
```

Getter for OUTLIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.204 GetAxisParameter_OV_GAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_OV_GAIN (
            int32_t axis )  [pure virtual]
```

Getter for OV_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.205 GetAxisParameter_P_GAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_P_GAIN (
            int32_t axis )  [pure virtual]
```

Getter for P_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.206 GetAxisParameter_POS_DELAY()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_POS_DELAY (
```

```
int32_t axis )  [pure virtual]
```

Getter for POS_DELAY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.207 GetAxisParameter_POSI_SEQ_DELAY()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_POSI_SEQ_DELAY (
            int32_t axis )  [pure virtual]
```

Getter for POSI_SEQ_DELAY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.208 GetAxisParameter_POSI_SEQ_MODE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_POSI_SEQ_MODE (
            int32_t axis )  [pure virtual]
```

Getter for POSI_SEQ_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.209 GetAxisParameter_PP_STEP()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_PP_STEP (
            int32_t axis )  [pure virtual]
```

Getter for PP_STEP axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.210 GetAxisParameter_PS_ENCODER()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_PS_ENCODER (
            int32_t axis )  [pure virtual]
```

Getter for PS_ENCODER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.211 GetAxisParameter_PWM_CYCLE()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_PWM_CYCLE (
            int32_t axis )  [pure virtual]
```

Getter for PWM_CYCLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.212 GetAxisParameter_PWM_MARK()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_PWM_MARK (
            int32_t axis )  [pure virtual]
```

Getter for PWM_MARK axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.213 GetAxisParameter_RAISE_ANGLE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_RAISE_ANGLE (
            int32_t axis )  [pure virtual]
```

Getter for RAISE_ANGLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.214 GetAxisParameter_REG_INPUTS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REG_INPUTS (
            int32_t axis )  [pure virtual]
```

Getter for REG_INPUTS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.215 GetAxisParameter_REG_POS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REG_POS (
            int32_t axis )  [pure virtual]
```

Getter for REG_POS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.216 GetAxisParameter_REG_POSB()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REG_POSB (
            int32_t axis )  [pure virtual]
```

Getter for REG_POSB axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.217 GetAxisParameter_REGIST_CONTROL()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REGIST_CONTROL (
            int32_t axis )  [pure virtual]
```

Getter for REGIST_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.218 GetAxisParameter_REGIST_DELAY()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REGIST_DELAY (
```

```
int32_t axis )  [pure virtual]
```

Getter for REGIST_DELAY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.219 GetAxisParameter_REGIST_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REGIST_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for REGIST_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.220 GetAxisParameter_REGIST_SPEEDB()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REGIST_SPEEDB (
            int32_t axis )  [pure virtual]
```

Getter for REGIST_SPEEDB axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.221 GetAxisParameter_REMAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REMAIN (
            int32_t axis )  [pure virtual]
```

Getter for REMAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.222 GetAxisParameter_REP_DIST()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REP_DIST (
            int32_t axis )  [pure virtual]
```

Getter for REP_DIST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.223 GetAxisParameter_REP_OPTION()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REP_OPTION (
            int32_t axis )  [pure virtual]
```

Getter for REP_OPTION axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.224 GetAxisParameter_REV_IN()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REV_IN (
            int32_t axis )  [pure virtual]
```

Getter for REV_IN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.225 GetAxisParameter_REV_JOG()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REV_JOG (
            int32_t axis )  [pure virtual]
```

Getter for REV_JOG axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.226 GetAxisParameter_REV_START()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_REV_START (
            int32_t axis )  [pure virtual]
```

Getter for REV_START axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.227 GetAxisParameter_ROBOT_DPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_DPOS (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_DPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.228 GetAxisParameter_ROBOT_FE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_FE (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_FE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.229 GetAxisParameter_ROBOT_FE_LATCH()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_FE_LATCH (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_FE_LATCH axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.230 GetAxisParameter_ROBOT_FE_RANGE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_FE_RANGE (
```

```
int32_t axis )  [pure virtual]
```

Getter for ROBOT_FE_RANGE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.231 GetAxisParameter_ROBOT_FS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_FS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.232 GetAxisParameter_ROBOT_RS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_RS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.233 GetAxisParameter_ROBOT_SP_MODE()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_SP_MODE (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_SP_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.234 GetAxisParameter_ROBOT_UNITS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOT_UNITS (
            int32_t axis )  [pure virtual]
```

Getter for ROBOT_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.235 GetAxisParameter_ROBOTSTATUS()

```
virtual RobotStatus Trio::UnifiedApi::ITrioConnection::GetAxisParameter_ROBOTSTATUS (
            int32_t axis )  [pure virtual]
```

Getter for ROBOTSTATUS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.236 GetAxisParameter_RS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_RS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.237 GetAxisParameter_S_REF()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_S_REF (
            int32_t axis )  [pure virtual]
```

Getter for S_REF axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.238 GetAxisParameter_S_REF_OUT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_S_REF_OUT (
            int32_t axis )  [pure virtual]
```

Getter for S_REF_OUT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.239 GetAxisParameter_SERVO()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SERVO (
            int32_t axis )  [pure virtual]
```

Getter for SERVO axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.240 GetAxisParameter_SLOT_NUMBER()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SLOT_NUMBER (
            int32_t axis )  [pure virtual]
```

Getter for SLOT_NUMBER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.241 GetAxisParameter_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.242 GetAxisParameter_SRAMP()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SRAMP (
```

```
int32_t axis )  [pure virtual]
```

Getter for SRAMP axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.243 GetAxisParameter_START_DIR_LAST()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_START_DIR_LAST (
            int32_t axis )  [pure virtual]
```

Getter for START_DIR_LAST axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.244 GetAxisParameter_STARTMOVE_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_STARTMOVE_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for STARTMOVE_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.245 GetAxisParameter_STOP_ANGLE()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_STOP_ANGLE (
            int32_t axis )  [pure virtual]
```

Getter for STOP_ANGLE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.246 GetAxisParameter_SYNC_AXIS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_AXIS (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_AXIS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.247 GetAxisParameter_SYNC_CONTROL()

```
virtual SyncControl Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_CONTROL (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.248 GetAxisParameter_SYNC_DPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_DPOS (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_DPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.249 GetAxisParameter_SYNC_DWELL()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_DWELL (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_DWELL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.250 GetAxisParameter_SYNC_FLIGHT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_FLIGHT (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_FLIGHT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.251 GetAxisParameter_SYNC_PAUSE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_PAUSE (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_PAUSE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.252 GetAxisParameter_SYNC_TIME()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_TIME (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_TIME axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.253 GetAxisParameter_SYNC_TIMER()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_TIMER (
            int32_t axis )  [pure virtual]
```

Getter for SYNC_TIMER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.254 GetAxisParameter_SYNC_WITHDRAW()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_SYNC_WITHDRAW (
```

```
            int32_t axis )  [pure virtual]
```

Getter for SYNC_WITHDRAW axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.255 GetAxisParameter_T_REF()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_T_REF (
            int32_t axis )  [pure virtual]
```

Getter for T_REF axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.256 GetAxisParameter_T_REF_OUT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_T_REF_OUT (
            int32_t axis )  [pure virtual]
```

Getter for T_REF_OUT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.257 GetAxisParameter_TABLE_POINTER()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_TABLE_POINTER (
            int32_t axis )  [pure virtual]
```

Getter for TABLE_POINTER axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.258 GetAxisParameter_TANG_DIRECTION()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_TANG_DIRECTION (
            int32_t axis )  [pure virtual]
```

Getter for TANG_DIRECTION axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.259 GetAxisParameter_TCP_DPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_TCP_DPOS (
            int32_t axis )  [pure virtual]
```

Getter for TCP_DPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.260 GetAxisParameter_TCP_FS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_TCP_FS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for TCP_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.261 GetAxisParameter_TCP_RS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_TCP_RS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for TCP_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.262 GetAxisParameter_TCP_UNITS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_TCP_UNITS (
            int32_t axis )  [pure virtual]
```

Getter for TCP_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.263 GetAxisParameter_UNITS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_UNITS (
            int32_t axis )  [pure virtual]
```

Getter for UNITS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.264 GetAxisParameter_VECTOR_BUFFERED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VECTOR_BUFFERED (
            int32_t axis )  [pure virtual]
```

Getter for VECTOR_BUFFERED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.265 GetAxisParameter_VERIFY()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VERIFY (
            int32_t axis )  [pure virtual]
```

Getter for VERIFY axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.266 GetAxisParameter_VFF_GAIN()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VFF_GAIN (
```

```
int32_t axis )  [pure virtual]
```

Getter for VFF_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.267 GetAxisParameter_VP_ACCEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VP_ACCEL (
            int32_t axis )  [pure virtual]
```

Getter for VP_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.268 GetAxisParameter_VP_JERK()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VP_JERK (
            int32_t axis )  [pure virtual]
```

Getter for VP_JERK axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.269 GetAxisParameter_VP_MODE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VP_MODE (
            int32_t axis )  [pure virtual]
```

Getter for VP_MODE axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.270 GetAxisParameter_VP_POSITION()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VP_POSITION (
            int32_t axis )  [pure virtual]
```

Getter for VP_POSITION axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.271 GetAxisParameter_VP_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_VP_SPEED (
            int32_t axis )  [pure virtual]
```

Getter for VP_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.272 GetAxisParameter_WORLD_ACCEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_ACCEL (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.273 GetAxisParameter_WORLD_DECEL()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_DECEL (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_DECEL axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.274 GetAxisParameter_WORLD_DPOS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_DPOS (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_DPOS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.275 GetAxisParameter_WORLD_FS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_FS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.276 GetAxisParameter_WORLD_JOGSPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_JOGSPEED (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_JOGSPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.277 GetAxisParameter_WORLD_RS_LIMIT()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_RS_LIMIT (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.278 GetAxisParameter_WORLD_SPEED()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_SPEED (
```

```
            int32_t axis )  [pure virtual]
```

Getter for WORLD_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.279 GetAxisParameter_WORLD_UNITS()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameter_WORLD_UNITS (
            int32_t axis )  [pure virtual]
```

Getter for WORLD_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.280 GetAxisParameterBoolValue()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetAxisParameterBoolValue (
            const std::string_view & param_name,
            int32_t axis )  [pure virtual]
```

Get boolean value of axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.281 GetAxisParameterDoubleValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetAxisParameterDoubleValue (
            const std::string_view & param_name,
            int32_t axis )  [pure virtual]
```

Get double value of axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.282 GetAxisParameterInt32Value()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameterInt32Value (
            const std::string_view & param_name,
            int32_t axis )  [pure virtual]
```

Get 32-bit signed integer value of axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.283 GetAxisParameterInt64Value()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameterInt64Value (
            const std::string_view & param_name,
            int32_t axis )  [pure virtual]
```

Get 64-bit signed integer value of axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.284 GetAxisParameterUInt32Value()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetAxisParameterUInt32Value (
            const std::string_view & param_name,
            int32_t axis )  [pure virtual]
```

Get 32-bit unsigned integer value of axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.285 GetAxisParameterUInt64Value()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetAxisParameterUInt64Value (
            const std::string_view & param_name,
            int32_t axis )  [pure virtual]
```

Get 64 bit unsigned integer value of axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.286 GetConnectionType()

```
virtual ConnectionType Trio::UnifiedApi::ITrioConnection::GetConnectionType ( ) const  [pure
virtual]
```

Get connection type.

Returns

Connection type or current connection instance or Unknown if connection is disposed

6.3.3.287 GetDIn() [1/2]

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetDIn (
            uint32_t channel )  [pure virtual]
```

Get the state of the Digital Input.

Parameters

channel	Digital input channel number

Returns

Current state of the digital input channel

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.288 GetDIn() [2/2]

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetDIn (
            uint32_t firstChannel,
            uint32_t lastChannel )  [pure virtual]
```

Get the state of the Digital Inputs.

Parameters

firstChannel	First digital input channel number
lastChannel	Last digital input channel number. The difference between last and first channel should be at most 32

Returns

Binary sum of digital input states

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.289 GetDIn32()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetDIn32 ( )  [pure virtual]
```

Get the state of the first 32 Digital Inputs.

Returns

Bit sum of the states of the first 32 digital input channels

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.290 GetDOut() [1/2]

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetDOut (
            uint32_t channel )  [pure virtual]
```

Get the state of the Digital Output.

Parameters

channel	Digital output channel number

Returns

Current state of the digital output channel

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.291 GetDOut() [2/2]

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetDOut (
            uint32_t firstChannel,
            uint32_t lastChannel )  [pure virtual]
```

Get the state of the Digital Outputs.

Parameters

firstChannel	First digital output channel number
lastChannel	Last digital output channel number. The difference between last and first channel should be at most 32

Returns

Binary sum of digital output states

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.292 GetDOut32()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetDOut32 ( )  [pure virtual]
```

Get the state of the first 32 Digital Outputs.

Returns

Bit sum of the states of the first 32 digital output channels

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.293 GetFirmwareVersionNumber()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetFirmwareVersionNumber ( ) [pure virtual]
```

Get the firmware version of the Motion Coordinator.

Returns

On success Motion Coordinator version as a floating point number

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.294 GetFrameGroup() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::GetFrameGroup (
            uint32_t group,
            uint32_t & tableOffset,
            int32_t axes[],
            uint32_t & axisCount ) [pure virtual]
```

Get frame group info.

Parameters

group	Frame group number to get info
tableOffset	Current start offset of TABLE range used by the frame group
axes	Array of axis indices currently belonging to the frame group
axisCount	Number of axes currently belonging in the group

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.295 GetFrameGroup() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::GetFrameGroup (
            uint32_t group,
            uint32_t & tableOffset,
            std::vector< int32_t > & axes ) [pure virtual]
```

Get frame group info.

Parameters

group	Frame group number to get info
tableOffset	Current start offset of TABLE range used by the frame group
axes	Array of axis indices currently belonging to the frame group

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.296 GetInvertIN()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetInvertIN (
            uint32_t input )  [pure virtual]
```

Gets the invert status set by INVERT_IN command.

Parameters

input	The input to invert

Returns

True if inverted, false if not inverted

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.297 GetLocalFileCRC()

```
virtual uint16_t Trio::UnifiedApi::ITrioConnection::GetLocalFileCRC (
            const std::string_view & path,
            bool isBinary )  [pure virtual]
```

Get CRC16 of a local file.

Parameters

isBinary	If true file is treated as binary and checksum is calculated directly on all bytes. If false the file is treated as text and checksum is calculated line by line as it is done inside Motion Coordinator for programs and text files

Returns

CRC-16 of the file.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.298 GetModuleVersion()

```
static TRIO_DECLSPEC VersionInfo TRIO_CALLCONV Trio::UnifiedApi::ITrioConnection::GetModule↩
Version ( ) [static]
```

Get C++ language module version info.

Returns

On success version info of C++ language module

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.299 GetModuleVersionStr()

```
static TRIO_DECLSPEC std::string_view TRIO_CALLCONV Trio::UnifiedApi::ITrioConnection::Get↩
ModuleVersionStr ( ) [static]
```

Get version string of C++ language module.

Returns

On success string representation of C++ language module version

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.300 GetMultiTableValues()

```
virtual void Trio::UnifiedApi::ITrioConnection::GetMultiTableValues (
            uint32_t start_index,
            uint32_t count,
            double values[] ) [pure virtual]
```

Get multiple table entries.

Parameters

start_index	Start index of the table range to be read
count	Number of table entries to be read
values	Placeholder array for the values to be read

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.301 GetMultiVrValues()

```
virtual void Trio::UnifiedApi::ITrioConnection::GetMultiVrValues (
            uint32_t start_index,
            uint32_t count,
            double values[] )  [pure virtual]
```

Get multiple VR entries.

Parameters

start_index	Start index of the VR range to be read
count	Number of VR entries to be read
values	Placeholder array for the values to be read

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.302 GetPortParameterBoolValue()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetPortParameterBoolValue (
            const std::string_view & param_name,
            int32_t port )  [pure virtual]
```

Get boolean value of port parameter by name.

Parameters

param_name	Parameter name
port	Port index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.303 GetPortParameterDoubleValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetPortParameterDoubleValue (
            const std::string_view & param_name,
            int32_t port )  [pure virtual]
```

Get double value of port parameter by name.

Parameters

param_name	Parameter name
port	Port index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.304 GetPortParameterInt32Value()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetPortParameterInt32Value (
            const std::string_view & param_name,
            int32_t port )  [pure virtual]
```

Get 32-bit signed integer value of port parameter by name.

Parameters

param_name	Parameter name
port	Port index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.305 GetPortParameterInt64Value()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetPortParameterInt64Value (
            const std::string_view & param_name,
            int32_t port )  [pure virtual]
```

Get 64-bit signed integer value of port parameter by name.

Parameters

param_name	Parameter name
port	Port index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.306 GetPortParameterUInt32Value()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetPortParameterUInt32Value (
            const std::string_view & param_name,
            int32_t port )  [pure virtual]
```

Get 32-bit unsigned integer value of port parameter by name.

Parameters

param_name	Parameter name
port	Port index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.307 GetPortParameterUInt64Value()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetPortParameterUInt64Value (
            const std::string_view & param_name,
            int32_t port )  [pure virtual]
```

Get 64 bit unsigned integer value of port parameter by name.

Parameters

param_name	Parameter name
port	Port index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.308 GetProcessList()

```
virtual void Trio::UnifiedApi::ITrioConnection::GetProcessList (
            ProcessInfoCallback && handler )  [pure virtual]
```

Function returns the list of processes on the Motion Coordinator.

Parameters

handler	Callback handler function executed for each process item

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.309 GetProcessParameter_ERROR_LINE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameter_ERROR_LINE (
            int32_t process )  [pure virtual]
```

Getter for ERROR_LINE process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.310 GetProcessParameter_INDEVICE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameter_INDEVICE (
            int32_t process )  [pure virtual]
```

Getter for INDEVICE process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.311 GetProcessParameter_OUTDEVICE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameter_OUTDEVICE (
            int32_t process )  [pure virtual]
```

Getter for OUTDEVICE process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.312 GetProcessParameter_PMOVE()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetProcessParameter_PMOVE (
            int32_t process )  [pure virtual]
```

Getter for PMOVE process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.313 GetProcessParameter_PROC_LINE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameter_PROC_LINE (
            int32_t process )  [pure virtual]
```

Getter for PROC_LINE process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.314 GetProcessParameter_PROC_STATUS()

```
virtual ProcessStatus Trio::UnifiedApi::ITrioConnection::GetProcessParameter_PROC_STATUS (
            int32_t process )  [pure virtual]
```

Getter for PROC_STATUS process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.315 GetProcessParameter_RUN_ERROR()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameter_RUN_ERROR (
            int32_t process )  [pure virtual]
```

Getter for RUN_ERROR process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.316 GetProcessParameter_TICKS()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetProcessParameter_TICKS (
            int32_t process )  [pure virtual]
```

Getter for TICKS process parameter.

Parameters

process	Target process whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.317 GetProcessParameterBoolValue()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetProcessParameterBoolValue (
            const std::string_view & param_name,
            int32_t process )  [pure virtual]
```

Get boolean value of process parameter by name.

Parameters

param_name	Parameter name
process	Process index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.318 GetProcessParameterDoubleValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetProcessParameterDoubleValue (
            const std::string_view & param_name,
            int32_t process )  [pure virtual]
```

Get double value of process parameter by name.

Parameters

param_name	Parameter name
process	Process index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.319 GetProcessParameterInt32Value()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameterInt32Value (
            const std::string_view & param_name,
            int32_t process )  [pure virtual]
```

Get 32-bit signed integer value of process parameter by name.

Parameters

param_name	Parameter name
process	Process index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.320 GetProcessParameterInt64Value()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetProcessParameterInt64Value (
            const std::string_view & param_name,
            int32_t process )  [pure virtual]
```

Get 64-bit signed integer value of process parameter by name.

Parameters

param_name	Parameter name
process	Process index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.321 GetProcessParameterUInt32Value()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetProcessParameterUInt32Value (
            const std::string_view & param_name,
            int32_t process )  [pure virtual]
```

Get 32-bit unsigned integer value of process parameter by name.

Parameters

param_name	Parameter name
process	Process index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.322 GetProcessParameterUInt64Value()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetProcessParameterUInt64Value (
            const std::string_view & param_name,
            int32_t process )  [pure virtual]
```

Get 64 bit unsigned integer value of process parameter by name.

Parameters

param_name	Parameter name
process	Process index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.323 GetRemoteFileCRC()

```
virtual uint16_t Trio::UnifiedApi::ITrioConnection::GetRemoteFileCRC (
            const std::string_view & name )  [pure virtual]
```

Get CRC16 of a file or program on Motion Coordinator.

Parameters

name	Name of the program to calculate the checksum for.

Returns

CRC-16 of the program.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.324 GetSlotParameter_SERIAL_NUMBER()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSlotParameter_SERIAL_NUMBER (
            int32_t slot )  [pure virtual]
```

Getter for SERIAL_NUMBER slot parameter.

Parameters

slot	Target slot whose parameter is retrieved

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.325 GetSlotParameterBoolValue()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetSlotParameterBoolValue (
            const std::string_view & param_name,
            int32_t slot )  [pure virtual]
```

Get boolean value of slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.326 GetSlotParameterDoubleValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetSlotParameterDoubleValue (
            const std::string_view & param_name,
            int32_t slot )  [pure virtual]
```

Get double value of slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.327 GetSlotParameterInt32Value()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSlotParameterInt32Value (
            const std::string_view & param_name,
            int32_t slot )  [pure virtual]
```

Get 32-bit signed integer value of slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.328 GetSlotParameterInt64Value()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSlotParameterInt64Value (
            const std::string_view & param_name,
            int32_t slot )  [pure virtual]
```

Get 64-bit signed integer value of slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.329 GetSlotParameterUInt32Value()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSlotParameterUInt32Value (
            const std::string_view & param_name,
            int32_t slot )  [pure virtual]
```

Get 32-bit unsigned integer value of slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.330 GetSlotParameterUInt64Value()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetSlotParameterUInt64Value (
            const std::string_view & param_name,
            int32_t slot )  [pure virtual]
```

Get 64 bit unsigned integer value of slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.331 GetSystemParameter_ADDRESS()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_ADDRESS ( ) [pure virtual]
```

Getter for ADDRESS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.332 GetSystemParameter_AIN0()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AIN0 ( ) [pure virtual]
```

Getter for AIN0 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.333 GetSystemParameter_AIN1()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AIN1 ( ) [pure virtual]
```

Getter for AIN1 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.334 GetSystemParameter_AIN2()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AIN2 ( ) [pure virtual]
```

Getter for AIN2 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.335 GetSystemParameter_AIN3()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AIN3 ( ) [pure virtual]
```

Getter for AIN3 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.336 GetSystemParameter_AINBI0()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AINBI0 ( ) [pure virtual]
```

Getter for AINBI0 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.337 GetSystemParameter_AINBI1()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AINBI1 ( ) [pure virtual]
```

Getter for AINBI1 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.338 GetSystemParameter_AINBI2()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AINBI2 ( ) [pure virtual]
```

Getter for AINBI2 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.339 GetSystemParameter_AINBI3()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AINBI3 ( ) [pure virtual]
```

Getter for AINBI3 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.340 GetSystemParameter_AOUT0()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AOUT0 ( ) [pure virtual]
```

Getter for AOUT0 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.341 GetSystemParameter_AOUT1()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AOUT1 ( ) [pure virtual]
```

Getter for AOUT1 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.342 GetSystemParameter_AOUT2()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AOUT2 ( ) [pure virtual]
```

Getter for AOUT2 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.343 GetSystemParameter_AOUT3()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AOUT3 ( ) [pure virtual]
```

Getter for AOUT3 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.344 GetSystemParameter_AUTO_ETHERCAT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AUTO_ETHERCAT ( ) [pure virtual]
```

Getter for AUTO_ETHERCAT system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.345 GetSystemParameter_AVERAGE_JITTER()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_AVERAGE_JITTER ( ) [pure virtual]
```

Getter for AVERAGE_JITTER system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.346 GetSystemParameter_CONTROL()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_CONTROL ( ) [pure virtual]
```

Getter for CONTROL system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.347 GetSystemParameter_CPU_EXCEPTIONS()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_CPU_EXCEPTIONS ( ) [pure virtual]
```

Getter for CPU_EXCEPTIONS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.348 GetSystemParameter_CURRENT_JITTER()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_CURRENT_JITTER ( ) [pure virtual]
```

Getter for CURRENT_JITTER system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.349 GetSystemParameter_DISPLAY()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_DISPLAY ( ) [pure virtual]
```

Getter for DISPLAY system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.350 GetSystemParameter_ERROR_AXIS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_ERROR_AXIS ( ) [pure virtual]
```

Getter for ERROR_AXIS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.351 GetSystemParameter_FLASH_STATUS()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_FLASH_STATUS ( ) [pure virtual]
```

Getter for FLASH_STATUS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.352 GetSystemParameter_FPU_EXCEPTIONS()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_FPU_EXCEPTIONS ( ) [pure virtual]
```

Getter for FPU_EXCEPTIONS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.353 GetSystemParameter_HMI_PROC()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_HMI_PROC ( ) [pure
virtual]
```

Getter for HMI_PROC system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.354 GetSystemParameter_INPUTS0()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_INPUTS0 ( ) [pure
virtual]
```

Getter for INPUTS0 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.355 GetSystemParameter_INPUTS1()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_INPUTS1 ( ) [pure
virtual]
```

Getter for INPUTS1 system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.356 GetSystemParameter_IP_ADDRESS()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_ADDRESS ( ) [pure virtual]
```

Getter for IP_ADDRESS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.357 GetSystemParameter_IP_GATEWAY()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_GATEWAY ( ) [pure virtual]
```

Getter for IP_GATEWAY system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.358 GetSystemParameter_IP_MAC()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_MAC ( ) [pure virtual]
```

Getter for IP_MAC system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.359 GetSystemParameter_IP_MEMORY_CONFIG()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_MEMORY_CONFIG ( )
[pure virtual]
```

Getter for IP_MEMORY_CONFIG system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.360 GetSystemParameter_IP_NETMASK()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_NETMASK ( ) [pure
virtual]
```

Getter for IP_NETMASK system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.361 GetSystemParameter_IP_PROTOCOL_CONFIG()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_PROTOCOL_CONFIG ( )
[pure virtual]
```

Getter for IP_PROTOCOL_CONFIG system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.362 GetSystemParameter_IP_PROTOCOL_CTRL()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_PROTOCOL_CTRL ( )
[pure virtual]
```

Getter for IP_PROTOCOL_CTRL system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.363 GetSystemParameter_IP_TCP_TIMEOUT()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_TCP_TIMEOUT ( )
[pure virtual]
```

Getter for IP_TCP_TIMEOUT system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.364 GetSystemParameter_IP_TCP_TX_THRESHOLD()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_TCP_TX_THRESHOLD ( )
[pure virtual]
```

Getter for IP_TCP_TX_THRESHOLD system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.365 GetSystemParameter_IP_TCP_TX_TIMEOUT()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_IP_TCP_TX_TIMEOUT ( )
[pure virtual]
```

Getter for IP_TCP_TX_TIMEOUT system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.366 GetSystemParameter_LAST_AXIS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_LAST_AXIS ( ) [pure
virtual]
```

Getter for LAST_AXIS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.367 GetSystemParameter_LIMIT_BUFFERED()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_LIMIT_BUFFERED ( ) [pure
virtual]
```

Getter for LIMIT_BUFFERED system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.368 GetSystemParameter_MAX_JITTER()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MAX_JITTER ( ) [pure virtual]
```

Getter for MAX_JITTER system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.369 GetSystemParameter_MISSED_INTERRUPTS()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MISSED_INTERRUPTS ( ) [pure virtual]
```

Getter for MISSED_INTERRUPTS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.370 GetSystemParameter_MISSED_RECV_FRAMES()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MISSED_RECV_FRAMES ( ) [pure virtual]
```

Getter for MISSED_RECV_FRAMES system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.371 GetSystemParameter_MISSED_SEND_FRAMES()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MISSED_SEND_FRAMES ( )
[pure virtual]
```

Getter for MISSED_SEND_FRAMES system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.372 GetSystemParameter_MODBUS_TX_DELAY()

```
virtual uint16_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MODBUS_TX_DELAY ( )
[pure virtual]
```

Getter for MODBUS_TX_DELAY system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.373 GetSystemParameter_MOTION_ERROR()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MOTION_ERROR ( ) [pure
virtual]
```

Getter for MOTION_ERROR system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.374 GetSystemParameter_MOTION_ERROR_COUNT()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MOTION_ERROR_COUNT ( )
[pure virtual]
```

Getter for MOTION_ERROR_COUNT system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.375 GetSystemParameter_MPE_MODE()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_MPE_MODE ( ) [pure
virtual]
```

Getter for MPE_MODE system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.376 GetSystemParameter_N_ANA_IN()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_N_ANA_IN ( ) [pure
virtual]
```

Getter for N_ANA_IN system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.377 GetSystemParameter_N_ANA_OUT()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_N_ANA_OUT ( ) [pure virtual]
```

Getter for N_ANA_OUT system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.378 GetSystemParameter_NIN()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_NIN ( ) [pure virtual]
```

Getter for NIN system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.379 GetSystemParameter_NIO()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_NIO ( ) [pure virtual]
```

Getter for NIO system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.380 GetSystemParameter_NOP()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_NOP ( ) [pure virtual]
```

Getter for NOP system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.381 GetSystemParameter_PCMCAT_PROC()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_PCMCAT_PROC ( ) [pure
virtual]
```

Getter for PCMCAT_PROC system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.382 GetSystemParameter_PLC_CONFIG()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_PLC_CONFIG ( ) [pure
virtual]
```

Getter for PLC_CONFIG system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.383 GetSystemParameter_PLC_ERROR()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_PLC_ERROR ( ) [pure virtual]
```

Getter for PLC_ERROR system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.384 GetSystemParameter_PLC_OVERFLOW()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_PLC_OVERFLOW ( ) [pure virtual]
```

Getter for PLC_OVERFLOW system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.385 GetSystemParameter_PLC_RUN()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_PLC_RUN ( ) [pure virtual]
```

Getter for PLC_RUN system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.386 GetSystemParameter_PROCNUMBER()

```
virtual int16_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_PROCNUMBER ( ) [pure virtual]
```

Getter for PROCNUMBER system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.387 GetSystemParameter_REMOTE_PROC()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_REMOTE_PROC ( ) [pure virtual]
```

Getter for REMOTE_PROC system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.388 GetSystemParameter_SCHEDULE_TYPE()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SCHEDULE_TYPE ( ) [pure virtual]
```

Getter for SCHEDULE_TYPE system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.389 GetSystemParameter_SCOPE_CYCLE_COUNT()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SCOPE_CYCLE_COUNT ( )
[pure virtual]
```

Getter for SCOPE_CYCLE_COUNT system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.390 GetSystemParameter_SCOPE_DELAY()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SCOPE_DELAY ( ) [pure
virtual]
```

Getter for SCOPE_DELAY system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.391 GetSystemParameter_SCOPE_POS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SCOPE_POS ( ) [pure
virtual]
```

Getter for SCOPE_POS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.392 GetSystemParameter_SCOPE_TRIGGER_POS()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SCOPE_TRIGGER_POS ( )
[pure virtual]
```

Getter for SCOPE_TRIGGER_POS system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.393 GetSystemParameter_SERIAL_NUMBER()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SERIAL_NUMBER ( ) [pure
virtual]
```

Getter for SERIAL_NUMBER system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.394 GetSystemParameter_SERVO_OFFSET()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SERVO_OFFSET ( ) [pure
virtual]
```

Getter for SERVO_OFFSET system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.395 GetSystemParameter_SERVO_PERIOD()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SERVO_PERIOD ( ) [pure virtual]
```

Getter for SERVO_PERIOD system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.396 GetSystemParameter_SLOT0_TIME()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SLOT0_TIME ( ) [pure virtual]
```

Getter for SLOT0_TIME system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.397 GetSystemParameter_SLOT1_TIME()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SLOT1_TIME ( ) [pure virtual]
```

Getter for SLOT1_TIME system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.398 GetSystemParameter_SLOT2_TIME()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SLOT2_TIME ( ) [pure virtual]
```

Getter for SLOT2_TIME system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.399 GetSystemParameter_SLOT3_TIME()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SLOT3_TIME ( ) [pure virtual]
```

Getter for SLOT3_TIME system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.400 GetSystemParameter_SYSTEM_ERROR()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SYSTEM_ERROR ( ) [pure virtual]
```

Getter for SYSTEM_ERROR system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.401 GetSystemParameter_SYSTEM_ERROR_MASK()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SYSTEM_ERROR_MASK ( )
[pure virtual]
```

Getter for SYSTEM_ERROR_MASK system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.402 GetSystemParameter_SYSTEM_LOAD()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SYSTEM_LOAD ( ) [pure
virtual]
```

Getter for SYSTEM_LOAD system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.403 GetSystemParameter_SYSTEM_LOAD_MAX()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_SYSTEM_LOAD_MAX ( )
[pure virtual]
```

Getter for SYSTEM_LOAD_MAX system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.404 GetSystemParameter_TEXT_FILE_LOADER_PROC()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_TEXT_FILE_LOADER_PROC (
) [pure virtual]
```

Getter for TEXT_FILE_LOADER_PROC system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.405 GetSystemParameter_TSIZE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_TSIZE ( ) [pure virtual]
```

Getter for TSIZE system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.406 GetSystemParameter_UNIT_ERROR()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_UNIT_ERROR ( ) [pure
virtual]
```

Getter for UNIT_ERROR system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.407 GetSystemParameter_VERSION()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetSystemParameter_VERSION ( ) [pure virtual]
```

Getter for VERSION system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.408 GetSystemParameter_VRSIZE()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameter_VRSIZE ( ) [pure virtual]
```

Getter for VRSIZE system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.409 GetSystemParameter_WDOG()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetSystemParameter_WDOG ( ) [pure virtual]
```

Getter for WDOG system parameter.

Returns

Current value of the parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.410 GetSystemParameterBoolValue()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetSystemParameterBoolValue (
            const std::string_view & param_name )  [pure virtual]
```

Get boolean value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.411 GetSystemParameterDoubleValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetSystemParameterDoubleValue (
            const std::string_view & param_name )  [pure virtual]
```

Get double value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.412 GetSystemParameterInt32Value()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameterInt32Value (
            const std::string_view & param_name )  [pure virtual]
```

Get 32-bit signed integer value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.413 GetSystemParameterInt64Value()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameterInt64Value (
            const std::string_view & param_name )  [pure virtual]
```

Get 64-bit signed integer value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.414 GetSystemParameterStringValue() [1/2]

```
virtual std::string Trio::UnifiedApi::ITrioConnection::GetSystemParameterStringValue (
            const std::string_view & param_name )  [pure virtual]
```

Get string value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.415 GetSystemParameterStringValue() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::GetSystemParameterStringValue (
            const std::string_view & param_name,
            char * buff,
            uint32_t & buff_size )  [pure virtual]
```

Get string value of system parameter by name, into provided buffer with size.

Parameters

param_name	Parameter name
buff	Buffer to receive the response string value
buff_size	Size of the buffer on input and on exit required buffer size including null terminator

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.416 GetSystemParameterUInt32Value()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetSystemParameterUInt32Value (
            const std::string_view & param_name )  [pure virtual]
```

Get 32-bit unsigned integer value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.417 GetSystemParameterUInt64Value()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::GetSystemParameterUInt64Value (
            const std::string_view & param_name )  [pure virtual]
```

Get 64 bit unsigned integer value of system parameter by name.

Parameters

param_name	Parameter name

Returns

Variable to receive the value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.418 GetTableValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetTableValue (
            uint32_t index )  [pure virtual]
```

Get single table value.

Parameters

index	Index of the table entry to be read

Returns

On success the value of table entry returned

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.419 GetTcpCommandTimeout()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetTcpCommandTimeout ( )  [pure virtual]
```

Get current timeout for command execution. Valid only for TCP connection.

Returns

Current value of the command execution timeout in milliseconds

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.420 GetTcpConnectTimeout()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetTcpConnectTimeout ( ) [pure virtual]
```

Get current timeout for opening TCP connection. Valid only for TCP connection.

Returns

Current value of the open connection timeout in milliseconds

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.421 GetTcpHostName()

```
virtual std::string_view Trio::UnifiedApi::ITrioConnection::GetTcpHostName ( ) [pure virtual]
```

Get current host name or IP of connection object. Valid only for TCP connection.

Returns

Current value of the host name

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.422 GetTcpPort()

```
virtual uint16_t Trio::UnifiedApi::ITrioConnection::GetTcpPort ( ) [pure virtual]
```

Get current TCP port from connection object. Valid only for TCP connection.

Returns

Current value of the TCP port

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.423 GetVrBit()

```
virtual bool Trio::UnifiedApi::ITrioConnection::GetVrBit (
```

```
            uint16_t bit,
            uint32_t vr_variable )  [pure virtual]
```

The function can be used to test the value of a single bit within a VR variable.

Parameters

bit	The bit number to clear, valid range is 0 to 52.
vr_variable	The VR which to operate on.

Returns

On success the value VR entry returned

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.424 GetVrString() [1/3]

```
virtual std::string Trio::UnifiedApi::ITrioConnection::GetVrString (
            uint32_t vr )  [pure virtual]
```

Get a null-terminated string from consecutive VRs.

Parameters

vr	Starting VR index

Returns

On success the string stored in consecutive VR entries

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.425 GetVrString() [2/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::GetVrString (
            uint32_t vr,
            char * buff,
            uint32_t & buff_size )  [pure virtual]
```

Get a null-terminated string from consecutive VRs.

Parameters

vr	Starting VR index
buff	Buffer to receive the string characters. If NULL function succeeds and returns expected buffers size in buff_size parameter including null terminator character
buff_size	Size of the buffer on input and on exit required buffer size including null terminator

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.426 GetVrString() [3/3]

```
virtual void Trio::UnifiedApi::ITrioConnection::GetVrString (
            uint32_t vr,
            StreamHandlerCallback && handler )  [pure virtual]
```

Get a null-terminated string from consecutive VRs using callback function to re-construct the string from received chunks.

Parameters

vr	Starting VR index
handler	Callback function to receive string chunks

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.427 GetVrStringLen()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::GetVrStringLen (
            uint32_t vr )  [pure virtual]
```

Get the length of a null-terminated string from consecutive VRs.

Parameters

vr	Starting VR index

Returns

Length of the string

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.428 GetVrValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::GetVrValue (
            uint32_t index )  [pure virtual]
```

Get single VR value.

Parameters

index	Index of the VR entry to be read

Returns

On success the value VR entry returned

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.429 Halt()

```
virtual void Trio::UnifiedApi::ITrioConnection::Halt ( )  [pure virtual]
```

Halts execution of all running programs.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.430 IsConnected()

```
virtual bool Trio::UnifiedApi::ITrioConnection::IsConnected ( ) const  [pure virtual]
```

Check if a connection to the Motion Coordinator was opened.

Returns

If true if there is active connection to the Motion Coordinator or false otherwise.

6.3.3.431 LoadProject()

```
virtual void Trio::UnifiedApi::ITrioConnection::LoadProject (
            const std::string_view & project_file,
            StringCallback && uknown_project_key_handler )  [pure virtual]
```

Loads a Motion Perfect project on the Motion Coordinator. All the running programs and motion are stopped. All the existing content on the Motion Coordinator is deleted.

Parameters

project_file	The project file name. The folder should contain all the other project content - as saved by Motion Perfect.
uknown_project_key_handler	A handler receiving the project key authority name for which a project key needs to be set. It is expected the user to register the key for the passed authority. For example by using ProjectKeyRegistryStore or ProjectKey functions. If the uknown_project_key_handler returns false the project load will be cancelled and will exit with error

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.432 MoveAbs()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveAbs (
            double pos,
            int32_t axis = -1 )  [pure virtual]
```

Absolute positional move of single axis.

Parameters

pos	Absolute position to move axis to in user units
axis	Index of the axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.433 MoveAbsMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveAbsMulti (
            const double pos[],
            uint32_t posCount,
            int32_t axis = -1 )  [pure virtual]
```

Absolute positional move of multiple axes.

Parameters

pos	Array of absolute positions in user units to move axes defined by SetMultiAxisBase
posCount	Number of positions contained in array.
axis	Base axis for this command. If value is -1 then the default base axis is used.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.434 MoveAbsSeq()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveAbsSeq (
            int32_t axis,
            uint32_t tablePointer,
            uint32_t nAxes,
            uint32_t nPoints,
            int32_t options,
            double radius,
            int32_t tableOutput = 0,
            double transitionAngle = 0 )  [pure virtual]
```

Function allows a sequence of absolute movements to be loaded via TABLE values. The moves can be automatically merged together using a circular or spherical arc. Movement on 2 to 6 axes may be controlled in this way.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used
tablePointer	Start of the move sequence in the TABLE data.
nAxes	Number of axes 2 to 6
nPoints	The number of points, each point requires 2 or 3 table values
options	Modifiers for the movements. See the TrioBASIC MOVEABSSEQ documentation for more details.
radius	The merging/filleting radius to be applied. 0 for no filleting
tableOutput	Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.
transitionAngle	Transition angle (in Radians) for any filleted arcs that are generated. Set to 0 if no transition curve is to be applied.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.435 MoveAbsSp()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveAbsSp (
            double pos,
            int32_t axis = -1 )  [pure virtual]
```

Absolute positional move of single axis with speed changed between moves.

Parameters

pos	Absolute position to move axis to in user units
axis	Index of the axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.436 MoveAbsSpMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveAbsSpMulti (
            const double pos[],
            uint32_t posCount,
            int32_t axis = -1 )  [pure virtual]
```

Absolute positional move of multiple axes with speed changed between moves.

Parameters

pos	Array of absolute positions in user units to move axes defined by SetMultiAxisBase
posCount	Number of positions contained in array.
axis	Index of the first axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.437 MoveCirc()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveCirc (
            int32_t axis,
            double finishBase,
            double finishNext,
            double centreBase,
            double centreNext,
            int32_t direction,
            double transitionAngle = 0,
            int32_t tableOutput = -1 )  [pure virtual]
```

Moves 2 orthogonal axes in such a way as to produce a circular arc at the tool point. The length and radius of the arc are defined by the five parameters in the command line. The move parameters are always relative to the end of the last specified move. This is the start position on the circle circumference. Axis 1 is the current BASE axis. Axis 2 is the next axis in the BASE array. The first 4 distance parameters are scaled according to the current unit conversion factor for the BASE axis.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used.
finishBase	Position on BASE axis to finish at.
finishNext	Position on next axis in BASE array to finish at.
centreBase	Position on BASE about which to move.
centreNext	Position on next axis in BASE array about which to move.

Parameters

direction	0 Arc is interpolated in an anti - clockwise direction 1 Arc is interpolated in a clockwise direction 2 Arc is interpolated using the shortest path to endpoint 3 Arc is interpolated using the longest path to endpoint
transitionAngle	Transition angle (in Radians) for any filleted arcs that are generated. Set to 0 if no transition curve is to be applied.
tableOutput	Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.438 MoveCircSp()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveCircSp (
            int32_t axis,
            double finishBase,
            double finishNext,
            double centreBase,
            double centreNext,
            int32_t direction,
            double transitionAngle = 0,
            int32_t tableOutput = 0 )  [pure virtual]
```

Works as MoveCirc and additionally allows vector speed to be changed when using multiple moves in the look ahead buffer when MERGE = ON , using additional parameters FORCE_SPEED and ENDMOVE_SPEED. Optional parameters ta and output allow the use of transition curves.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used.
finishBase	Position on BASE axis to finish at.
finishNext	Position on next axis in BASE array to finish at.
centreBase	Position on BASE about which to move.
centreNext	Position on next axis in BASE array about which to move.
direction	0 Arc is interpolated in an anti - clockwise direction 1 Arc is interpolated in a clockwise direction 2 Arc is interpolated using the shortest path to endpoint 3 Arc is interpolated using the longest path to endpoint
transitionAngle	Transition angle (in Radians) for any filleted arcs that are generated. Set to 0 if no transition curve is to be applied.
tableOutput	Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.439 MoveHelical()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveHelical (
            int32_t axis,
            double finishBase,
            double finishNext,
            double centreBase,
            double centreNext,
            int32_t direction,
            double linearDistance,
            int32_t mode,
            double transitionAngle = 0,
            int32_t tableOutput = 0 )  [pure virtual]
```

Performs a helical move. Moves 2 orthogonal axes in such a way as to produce a circular arc at the tool point with a simultaneous linear move on a third axis. The first 5 parameters are similar to those of an MOVECIRC command.The sixth parameter defines the simultaneous linear move.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used.
finishBase	Position on the base axis to finish at.
finishNext	Position on the next axis in the base vector to finish at.
centreBase	Position on the base axis about which to move.
centreNext	Position on the next axis in the base vector about which to move.
direction	Direction in which to move around the arc.
linearDistance	Distance to move on the third axis in the base vector.
mode	Mode of operation. See the MHELICAL command documentation for more details.
transitionAngle	Transition angle (in Radians) for any filleted arcs that are generated. Set to 0 if no transition curve is to be applied.
tableOutput	Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.440 MoveHelicalSp()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveHelicalSp (
            int32_t axis,
            double finishBase,
            double finishNext,
            double centreBase,
            double centreNext,
            int32_t direction,
            double linearDistance,
            int32_t mode,
            double transitionAngle = 0,
            int32_t tableOutput = 0 )  [pure virtual]
```

Performs a helical move the same as MHELICAL and additionally allows vector speed to be changed when using multiple moves in the buffer. Uses additional axis parameters FORCE_SPEED, ENDMOVE_SPEED and STARTMOVE_SPEED.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used.
finishBase	Position on the base axis to finish at.
finishNext	Position on the next axis in the base vector to finish at.
centreBase	Position on the base axis about which to move.
centreNext	Position on the next axis in the base vector about which to move.
direction	Direction in which to move around the arc.
linearDistance	Distance to move on the third axis in the base vector.
mode	Mode of operation. See the MHELICAL command documentation for more details.
transitionAngle	Transition angle (in Radians) for any filleted arcs that are generated. Set to 0 if no transition curve is to be applied.
tableOutput	Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.441 MoveLink()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveLink (
            int32_t axis,
            double distance,
            double link_dist,
            double link_acc,
            double link_dec,
            int32_t link_axis = -1,
            int32_t link_options = 0,
            double link_pos = 0,
            double base_dist = 0,
            double y_dist = 0,
            double z_dist = 0,
            double u_dist = 0,
            double v_dist = 0,
            double w_dist = 0 )  [pure virtual]
```

The linked move command is designed for controlling movements such as:

- Synchronisation to conveyors
- Flying shears
- Thread chasing, tapping etc.
- Coil winding The motion consists of a linear movement with separately variable acceleration and deceleration phases linked via a software gearbox to the MEASURED position(MPOS) of another axis.The command uses the BASE and AXIS, and unit conversion factors in a similar way to other move commands. The “link” axis may move in either direction to drive the output motion.The link distances specified are always positive.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used

Parameters

distance	Incremental distance in user units to be moved on the current base axis, as a result of the measured movement on the “input” axis which drives the move.
link_dist	Positive incremental distance in user units which is required to be measured on the “link” axis to result in the motion on the base axis.
link_acc	Positive incremental distance in user units on the input axis over which the base axis accelerates.
link_dec	Positive incremental distance in user units on the input axis over which the base axis decelerates.
link_axis	Specifies the axis to “link” to. It should be set to a value between 0 and the number of available axes.
link_options	Bit value options to customise how your MOVELINK operates: Bit 0 (1) - link commences exactly when registration event MARK occurs on link axis Bit 1 (2) - link commences at an absolute position on link axis(see link_pos for start position) Bit 2 (4) - MOVELINK repeats automatically and bi - directionally when this bit is set. (This mode can be cleared by setting bit 1 of the REP_OPTION axis parameter) Bit 4 (16)- If this bit is set the MOVELINK acceleration and deceleration phases are constructed using an “S” speed profile not a trapezoidal speed profile Bit 5 (32)- Link is only active during a positive move on the link axis Bit 8 (256) - link commences exactly when registration event MARKB occurs on link axis Bit 9 (512) - link commences exactly when registration event R_MARK occurs on link axis. (see link_pos for channel number) Bit 10 and 11 - S-Ramp mode selection for use when bit 4 is set.Figure in() shows peak acceleration compared to linear ramp. 0 - SIN speed Profile(1.55) 1 - Power 9 polynomial speed(2.42) 2 - Power 7 polynomial speed(2.16) 3 - Power 5 polynomial speed(1.86) Bit 13 - Slave axis follows master axis DPOS Bit 14 - Link is active during positive movement from a threshold.The threshold is updated to the highest positive value.
link_pos	Link_option bit 1 - the absolute position on the link axis in user UNITS where the MOVELINK is to be start. link_option bit 9 – the registration channel to start the movement on.
base_dist	Firmware V2.0253 onwards. The “base distance” is the distance that is to be travelled at the base speed and is part of the total first “distance” parameter. See Examples 4 and 5 below. MOVELINK_MODIFY cannot be used when the base distance is in use.
y_dist	Same as base_dist, for 2nd base axis
z_dist	Same as base_dist, for 3rd base axis
u_dist	Same as base_dist, for 4th base axis
v_dist	Same as base_dist, for 5th base axis
w_dist	Same as base_dist, for 6th base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.442 MoveModify()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveModify (
            double position,
            int32_t axis = -1 )  [pure virtual]
```

MOVEMODIFY will change the absolute end position of a single axis MOVE , MOVEABS , MOVESP , MOVEABSSP or MOVEMODIFY that is in the last position in the movement buffer. If there is no motion command in the movement buffers or the last movement is not a single axis linear move, then MOVEMODIFY is loaded.

Parameters

position	Absolute position for the current move to complete at
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.443 MovePick()

```
virtual void Trio::UnifiedApi::ITrioConnection::MovePick (
            int32_t axis,
            int32_t mode,
            double x,
            double y,
            double z,
            double withdraw,
            double overlapControlA = 0.5,
            double overlapControlB = 0.5,
            double u = 0,
            double v = 0,
            double w = 0,
            double withdrawControl = 1,
            double xyControl = 1,
            double approachControl = 1 )  [pure virtual]
```

MOVEPICK is a new move type aimed at making it much simpler to program fast pick and place operations.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used
mode	Specify the number of axes 0 = xyz, 1 = xyzu, 2 = xyzuv, 3 = xyzuvw
x	Target x absolute position
y	Target y absolute position
z	Target z absolute position
withdraw	Incremental withdraw distance from the start position on z axis
overlapControlA	0..0.9 Specify rounding on corner A (optional)
overlapControlB	0..0.9 Specify rounding on corner B (optional)
u	Target u absolute position (optional)
v	Target v absolute position (optional)
w	Target w absolute position (optional)
withdrawControl	0..1 Adjust the speed for withdraw section (optional)
xyControl	0..1 Adjust the speed for xy section (optional)
approachControl	0..1 Adjust the speed for approach section (optional)

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.444 MoveRel()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveRel (
            double pos,
            int32_t axis = -1 )  [pure virtual]
```

Incremental move of single axis.

Parameters

pos	Distance to move axis in user units
axis	Index of the axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.445 MoveRelMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveRelMulti (
            const double dist[],
            uint32_t distCount,
            int32_t axis = -1 )  [pure virtual]
```

Incremental move of multiple axes.

Parameters

dist	Array of distances in user units to move axes defined by SetMultiAxisBase
distCount	Number of positions contained in array.
axis	Index of the first axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.446 MoveRelSeq()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveRelSeq (
            int32_t axis,
            uint32_t tablePointer,
```

```
            uint32_t nAxes,
            uint32_t nPoints,
            int32_t options,
            double radius,
            int32_t tableOutput = 0,
            double transitionAngle = 0 )  [pure virtual]
```

Function allows a sequence of relative movements to be loaded via TABLE values. The moves can be automatically merged together using a circular or spherical arc. Movement on 2 to 6 axes may be controlled in this way.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used
tablePointer	Start of the move sequence in the TABLE data.
nAxes	Number of axes 2 to 6
nPoints	The number of points, each point requires 2 or 3 table values
options	Modifiers for the movements. See the TrioBASIC MOVEABSSEQ documentation for more details.
radius	The merging/filleting radius to be applied. 0 for no filleting
tableOutput	Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.
transitionAngle	Transition angle (in Radians) for any filleted arcs that are generated. Set to 0 if no transition curve is to be applied.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.447 MoveRelSp()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveRelSp (
            double pos,
            int32_t axis = -1 )  [pure virtual]
```

Incremental move of single axis with speed changed between moves.

Parameters

pos	Distance to move axis in user units
axis	Index of the axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.448 MoveRelSpMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveRelSpMulti (
            const double dist[],
```

```
            uint32_t distCount,
            int32_t axis = -1 )  [pure virtual]
```

Incremental move of multiple axes with speed changed between moves.

Parameters

dist	Array of distances in user units to move axes defined by SetMultiAxisBase
distCount	Number of positions contained in array.
axis	Index of the first axis for which to be subject of move operation or -1 to use default base axis

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.449 MoveSpherical()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveSpherical (
            int32_t axis,
            double endMidX,
            double endMidY,
            double endMidZ,
            double centreX,
            double centreY,
            double centreZ,
            int32_t mode = 0,
            int32_t gtpi = 0,
            double rotau = 0,
            double rotav = 0,
            double rotaw = 0,
            double rotap = 0 )  [pure virtual]
```

Moves the three axis group defined in BASE along a spherical path with a vector speed determined by the SPEED set in the first axis of the BASE array. There are 4 modes of operation with the option of finishing the move at an endpoint different to the start, or returning to the start point to complete a circle. The path of the movement in 3D space can be defined either by specifying a point somewhere along the path, or by specifying the centre of the sphere.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used.
endMidX	End position of the first axis
endMidY	End position of the second axis
endMidZ	End position of the third axis
centreX	Centre position of the first axis
centreY	Centre position of the second axis
centreZ	Centre position of the third axis
mode	Mode of operation. See the MSPHERICAL command documentation for more details.
gtpi	If this optional parameter is non zero, modes 0 and 1 will perform a move taking the opposite way around a 360 degree circle to the same endpoint
rotau	If this optional parameter is non zero, a fourth axis will perform linear interpolation at the same time as the spherical move. The axis is the next in the BASE sequence

Parameters

rotav	If this optional parameter is non zero, a fifth axis will perform linear interpolation at the same time as the spherical move
rotaw	If this optional parameter is non zero, a sixth axis will perform linear interpolation at the same time as the spherical move
rotap	If this optional parameter is non zero, a seventh axis will perform linear interpolation at the same time as the spherical move

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.450 MoveSphericalSp()

```
virtual void Trio::UnifiedApi::ITrioConnection::MoveSphericalSp (
            int32_t axis,
            double endMidX,
            double endMidY,
            double endMidZ,
            double centreX,
            double centreY,
            double centreZ,
            int32_t mode = 0,
            int32_t gtpi = 0,
            double rotau = 0,
            double rotav = 0,
            double rotaw = 0,
            double rotap = 0 )  [pure virtual]
```

Performs a spherical move the same as MSPHERICAL and additionally allows vector speed to be changed when using multiple moves in the look ahead buffer when MERGE = ON, using additional parameters FORCE_SPEED, ENDMOVE_SPEED and STARTMOVE_SPEED.

Parameters

axis	Base axis for this command. If value is -1 then the default base axis is used.
endMidX	End position of the first axis
endMidY	End position of the second axis
endMidZ	End position of the third axis
centreX	Centre position of the first axis
centreY	Centre position of the second axis
centreZ	Centre position of the third axis
mode	Mode of operation. See the MSPHERICAL command documentation for more details.
gtpi	If this optional parameter is non zero, modes 0 and 1 will perform a move taking the opposite way around a 360 degree circle to the same endpoint
rotau	If this optional parameter is non zero, a fourth axis will perform linear interpolation at the same time as the spherical move. The axis is the next in the BASE sequence
rotav	If this optional parameter is non zero, a fifth axis will perform linear interpolation at the same time as the spherical move
rotaw	If this optional parameter is non zero, a sixth axis will perform linear interpolation at the same time as the spherical move
rotap	If this optional parameter is non zero, a seventh axis will perform linear interpolation at the same time as the spherical move

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.451 New()

```
virtual void Trio::UnifiedApi::ITrioConnection::New (
            const std::string_view & target )  [pure virtual]
```

Deletes a file, all files or the TABLE data the Motion Coordinator.

Parameters

target	The target to be cleared (deleted) "file_name" - deletes the file with given name from the Motion Coordinator "ALL" - deletes all user files from the Motion Coordinator. System files are not deleted. "TABLE" - sets all TABLE values to 0 .

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.452 OpenConnection()

```
virtual void Trio::UnifiedApi::ITrioConnection::OpenConnection ( )  [pure virtual]
```

Open a connection to the Motion Coordinator.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

In case of TCP connection SetTcpConnectTimeout function can be used to adjust the maximum connection delay

6.3.3.453 ParamBlockAppendAxis() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendAxis (
            const std::string_view axisParams[],
            uint32_t count,
            int32_t blk )  [pure virtual]
```

Append a parameter block of Axis parameter values.

Parameters

axisParams	Axis parameter name array
count	The array size
blk	Previously create block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.454 ParamBlockAppendAxis() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendAxis (
            const std::vector< std::string_view > & axisParams,
            int32_t blk )  [pure virtual]
```

Append a parameter block of Axis parameter values.

Parameters

axisParams	Axis parameter name array
blk	Previously create block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.455 ParamBlockAppendIO() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendIO (
            const Range ranges[],
            uint32_t count,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of Digital IO.

Parameters

ranges	Digital IO range array
count	The array size
blk	Previously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.456 ParamBlockAppendIO() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendIO (
            const std::vector< Range > & ranges,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of Digital IO.

Parameters

ranges	Digital IO range array
blk	Previously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.457 ParamBlockAppendProgVar() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendProgVar (
            const std::string_view & program,
            int32_t process,
            const std::string_view variables[],
            uint32_t count,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of Program variables.

Parameters

program	Program name
process	Optional process ID or -1 to select first running process
variables	Program variable name array
count	The array size
blk	Previously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.458 ParamBlockAppendProgVar() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendProgVar (
            const std::string_view & program,
            int32_t process,
            const std::vector< std::string_view > & variables,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of Program variables.

Parameters

program	Program name
process	Optional process ID or -1 to select first running process
variables	Program variable name array
blk	Previously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.459 ParamBlockAppendSystem() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendSystem (
            const std::string_view sysParams[],
            uint32_t count,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of System parameter values.

Parameters

sysParams	System parameter name array
count	The array size
blk	Previously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.460 ParamBlockAppendSystem() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendSystem (
            const std::vector< std::string_view > & sysParams,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of System parameter values.

Parameters

sysParams	System parameter name array
blk	Previously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.461 ParamBlockAppendTable() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendTable (
            const std::vector< uint32_t > & indices,
            int32_t blk )  [pure virtual]
```

Append a parameter block of TABLE values.

Parameters

indices	TABLE indices array
blk	Prevously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.462 ParamBlockAppendTable() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendTable (
            const uint32_t indices[],
            uint32_t count,
            int32_t blk )  [pure virtual]
```

Append a parameter block of TABLE values.

Parameters

indices	TABLE indices array
count	The array size
blk	Prevously created block index

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.463 ParamBlockAppendVr() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendVr (
            const std::vector< uint32_t > & indices,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of VR values.

Parameters

indices	VR indices array
blk	Previously created block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.464 ParamBlockAppendVr() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockAppendVr (
            const uint32_t indices[],
            uint32_t count,
            int32_t blk )  [pure virtual]
```

Append to a parameter block of VR values.

Parameters

indices	VR indices array
count	The array size
blk	Previously created block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.465 ParamBlockDefineAxis() [1/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineAxis (
            const std::string_view axisParams[],
            uint32_t count )  [pure virtual]
```

Define or re-define a parameter block of Axis parameter values.

Parameters

axisParams	Axis parameter name array
count	The array size

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.466 ParamBlockDefineAxis() [2/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineAxis (
            const std::vector< std::string_view > & axisParams )  [pure virtual]
```

Define or re-define a parameter block of Axis parameter values.

Parameters

axisParams	Axis parameter name array

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.467 ParamBlockDefineIO() [1/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineIO (
            const Range ranges[],
            uint32_t count )  [pure virtual]
```

Define or re-define a parameter block of Digital IO.

Parameters

ranges	Digital IO range array
count	The array size

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.468 ParamBlockDefineIO() [2/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineIO (
            const std::vector< Range > & ranges )  [pure virtual]
```

Define or re-define a parameter block of Digital IO.

Parameters

ranges	Digital IO range array

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.469 ParamBlockDefineProgVar() [1/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineProgVar (
            const std::string_view & program,
            int32_t process,
            const std::string_view variables[],
            uint32_t count )  [pure virtual]
```

Define or re-define a parameter block of Program variables.

Parameters

program	Program name
process	Optional process ID or -1 to select any process running the program
variables	Program variable name array
count	The array size

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.470 ParamBlockDefineProgVar() [2/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineProgVar (
            const std::string_view & program,
            int32_t process,
            const std::vector< std::string_view > & variables )  [pure virtual]
```

Define or re-define a parameter block of Program variables.

Parameters

program	Program name
process	Optional process ID or -1 to select any process running the program
variables	Program variable name array

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.471 ParamBlockDefineSystem() [1/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineSystem (
            const std::string_view sysParams[],
            uint32_t count )  [pure virtual]
```

Define or re-define a parameter block of System parameter values.

Parameters

sysParams	System parameter name array
count	The array size

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.472 ParamBlockDefineSystem() [2/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineSystem (
            const std::vector< std::string_view > & sysParams )  [pure virtual]
```

Define or re-define a parameter block of System parameter values.

Parameters

sysParams	System parameter name array

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.473 ParamBlockDefineTable() [1/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineTable (
            const std::vector< uint32_t > & indices )  [pure virtual]
```

Define or re-define a parameter block of TABLE values.

Parameters

indices	TABLE indices array

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.474 ParamBlockDefineTable() [2/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineTable (
            const uint32_t indices[],
            uint32_t count )  [pure virtual]
```

Define or re-define a parameter block of TABLE values.

Parameters

indices	TABLE indices array
count	The array size

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.475 ParamBlockDefineVr() [1/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineVr (
            const std::vector< uint32_t > & indices )  [pure virtual]
```

Define or re-define a parameter block of VR values.

Parameters

indices	VR indices array

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.476 ParamBlockDefineVr() [2/2]

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockDefineVr (
            const uint32_t indices[],
            uint32_t count )  [pure virtual]
```

Define or re-define a parameter block of VR values.

Parameters

indices	VR indices array
count	The array size

Returns

A placeholder for the newly defined block

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.477 ParamBlockDelete()

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockDelete (
            int32_t blk )  [pure virtual]
```

Deletes a defined parameter block.

Parameters

blk	The block to delete. It should have been defined previously by some of "ParamBlockDefine" functions

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.478 ParamBlockDeleteAll()

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockDeleteAll ( )  [pure virtual]
```

Deletes all defined parameter blocks.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.479 ParamBlockFetchAllAxisParam() [1/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            const int32_t axes[],
            uint32_t axis_count,
            double values[],
            uint32_t value_count )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
axes	Array of axis indices
axis_count	The size of axis indices array
values	Result placeholder array
value_count	The size of values array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range. Values array size should be multiple of the number of axes by the number of parameters in the block.

6.3.3.480 ParamBlockFetchAllAxisParam() [2/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            const Range & axes,
```

```
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
axes	Axis range
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Values array size should be multiple of the number of axes by the number of parameters in the block

6.3.3.481 ParamBlockFetchAllAxisParam() [3/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            const Range & axes,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
axes	Axis range
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Values array size should be multiple of the number of axes by the number of parameters in the block

6.3.3.482 ParamBlockFetchAllAxisParam() [4/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            const Range ranges[],
            uint32_t range_count,
            double values[],
            uint32_t value_count )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
ranges	Array of axis ranges
range_count	Axis range array size
values	Result placeholder array
value_count	The size of values array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range. Values array size should be multiple of the number of axes by the number of parameters in the block.

6.3.3.483 ParamBlockFetchAllAxisParam() [5/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            const std::vector< int32_t > & axes,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
axes	Array of axis indices
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range. Values array size should be multiple of the number of axes by the number of parameters in the block.

6.3.3.484 ParamBlockFetchAllAxisParam() [6/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            const std::vector< Range > & ranges,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
ranges	Array of axis ranges
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range. Values array size should be multiple of the number of axes by the number of parameters in the block.

6.3.3.485 ParamBlockFetchAllAxisParam() [7/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            int32_t axis,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
axis	Axis index
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.486 ParamBlockFetchAllAxisParam() [8/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllAxisParam (
            int32_t blk,
            int32_t axis,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in an Axis parameter block.

Parameters

blk	A valid parameter block
axis	Axis index
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.487 ParamBlockFetchAllIO() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllIO (
            int32_t blk,
            bool outputs,
            std::vector< uint64_t > & values )  [pure virtual]
```

Fetches all values in a Digital IO parameter block.

Parameters

blk	A valid parameter block
outputs	If true Digital Outputs are fetched. If false Digital Inputs are fetched.
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.488 ParamBlockFetchAllIO() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllIO (
            int32_t blk,
            bool outputs,
            uint64_t values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in a Digital IO parameter block.

Parameters

blk	A valid parameter block
outputs	If true Digital Outputs are fetched. If false Digital Inputs are fetched.
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.489 ParamBlockFetchAllProgVar() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllProgVar (
            int32_t blk,
            const char * values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in a block of program variables.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.490 ParamBlockFetchAllProgVar() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllProgVar (
            int32_t blk,
            std::vector< std::string_view > & values )  [pure virtual]
```

Fetches all values in a block of program variables.

Parameters

blk	A valid parameter block
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.491 ParamBlockFetchAllSystem() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllSystem (
            int32_t blk,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in a System parameter block.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.492 ParamBlockFetchAllSystem() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllSystem (
            int32_t blk,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in a System parameter block.

Parameters

blk	A valid parameter block
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.493 ParamBlockFetchAllTable() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllTable (
            int32_t blk,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in a TABLE parameter block.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.494 ParamBlockFetchAllTable() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllTable (
            int32_t blk,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in a TABLE parameter block.

Parameters

blk	A valid parameter block
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.495 ParamBlockFetchAllVr() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllVr (
            int32_t blk,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches all values in a VR parameter block.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.496 ParamBlockFetchAllVr() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchAllVr (
            int32_t blk,
            std::vector< double > & values )  [pure virtual]
```

Fetches all values in a VR parameter block.

Parameters

blk	A valid parameter block
values	Result placeholder array

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.497 ParamBlockFetchChangedAxisParam() [1/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            const int32_t axes[],
            uint32_t axis_count,
            double values[],
            uint32_t value_count,
            uint64_t changed_bitmasks[] )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
axes	Array of axis indices
axis_count	The size of axis indices array
values	Result placeholder array
value_count	The size of values array
changed_bitmasks	An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range. Values array size should be multiple of the number of axes by the number of parameters in the block. Changed bitmasks array size should be equal to the number of axes.

6.3.3.498 ParamBlockFetchChangedAxisParam() [2/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            const Range & axes,
            double values[],
            uint32_t count,
            uint64_t changed_bitmasks[] )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
axes	Axis range
values	Result placeholder array
count	The array size
changed_bitmasks	An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Values array size should be multiple of the number of axes by the number of parameters in the block. Changed bitmasks array size should be equal to the number of axes.

6.3.3.499 ParamBlockFetchChangedAxisParam() [3/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            const Range & axes,
            std::vector< double > & values,
            std::vector< uint64_t > & changed_bitmasks )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
axes	Axis range
values	Result placeholder array
changed_bitmasks	An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Values array size should be multiple of the number of axes by the number of parameters in the block. Changed bitmasks array size should be equal to the number of axes.

6.3.3.500 ParamBlockFetchChangedAxisParam() [4/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            const Range ranges[],
            uint32_t range_count,
            double values[],
            uint32_t value_count,
            uint64_t changed_bitmasks[] )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block

Parameters

ranges	Array of axis ranges
range_count	Axis range array size
values	Result placeholder array
value_count	The array size
changed_bitmasks	An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Values array size should be multiple of the number of axes by the number of parameters in the block. Changed bitmasks array size should be equal to the number of axes.

6.3.3.501 ParamBlockFetchChangedAxisParam() [5/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            const std::vector< int32_t > & axes,
            std::vector< double > & values,
            std::vector< uint64_t > & changed_bitmasks )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
axes	Array of axis indices
values	Result placeholder array
changed_bitmasks	An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range. Values array size should be multiple of the number of axes by the number of parameters in the block. Changed bitmasks array size should be equal to the number of axes.

6.3.3.502 ParamBlockFetchChangedAxisParam() [6/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
```

```
const std::vector< Range > & ranges,
std::vector< double > & values,
std::vector< uint64_t > & changed_bitmasks )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
ranges	Array of axis ranges
values	Result placeholder array
changed_bitmasks	An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

Values array size should be multiple of the number of axes by the number of parameters in the block. Changed bitmasks array size should be equal to the number of axes.

6.3.3.503 ParamBlockFetchChangedAxisParam() [7/8]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            int32_t axis,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
axis	Axis index
values	Result placeholder array
count	The array size

Returns

The "changed" state bit-mask. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.504 ParamBlockFetchChangedAxisParam() [8/8]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedAxisParam (
            int32_t blk,
            int32_t axis,
            std::vector< double > & values )  [pure virtual]
```

Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
axis	Axis index
values	Result placeholder array

Returns

The "changed" state bit-mask. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.505 ParamBlockFetchChangedIO() [1/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedIO (
            int32_t blk,
            bool outputs,
            std::vector< uint64_t > & values )  [pure virtual]
```

Fetches only changed values in a Digital IO parameter block.

Parameters

blk	A valid parameter block
outputs	If true Digital Outputs are fetched. If false Digital Inputs are fetched.
values	Result placeholder array

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.506 ParamBlockFetchChangedIO() [2/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedIO (
            int32_t blk,
            bool outputs,
            uint64_t values[],
            uint32_t count )  [pure virtual]
```

Fetches only changed values in a Digital IO parameter block.

Parameters

blk	A valid parameter block
outputs	If true Digital Outputs are fetched. If false Digital Inputs are fetched.
values	Result placeholder array
count	The array size

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.507 ParamBlockFetchChangedProgVar() [1/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedProgVar (
            int32_t blk,
            const char * values[],
            uint32_t count )  [pure virtual]
```

Fetches only changed values of a block of program variables. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.508 ParamBlockFetchChangedProgVar() [2/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedProgVar (
            int32_t blk,
            std::vector< std::string_view > & values )  [pure virtual]
```

Fetches only changed values of a block of program variables. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.509 ParamBlockFetchChangedSystem() [1/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedSystem (
            int32_t blk,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches only changed values of a System parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.510 ParamBlockFetchChangedSystem() [2/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedSystem (
            int32_t blk,
            std::vector< double > & values )  [pure virtual]
```

Fetches only changed values of a System parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.511 ParamBlockFetchChangedTable() [1/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedTable (
            int32_t blk,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches only changed values of a TABLE parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.512 ParamBlockFetchChangedTable() [2/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedTable (
            int32_t blk,
            std::vector< double > & values )  [pure virtual]
```

Fetches only changed values of a TABLE parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.513 ParamBlockFetchChangedVr() [1/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedVr (
            int32_t blk,
            double values[],
            uint32_t count )  [pure virtual]
```

Fetches only changed values of a VR parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array
count	The array size

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.514 ParamBlockFetchChangedVr() [2/2]

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockFetchChangedVr (
            int32_t blk,
            std::vector< double > & values )  [pure virtual]
```

Fetches only changed values of a VR parameter block. A value is considered changed if it differs from the previous fetch.

Parameters

blk	A valid parameter block
values	Result placeholder array

Returns

The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.515 ParamBlockGetProgBoolValue()

```
virtual bool Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgBoolValue (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Returns

The result

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.516 ParamBlockGetProgDoubleValue()

```
virtual double Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgDoubleValue (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Returns

The result

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.517 ParamBlockGetProgInt32Value()

```
virtual int32_t Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgInt32Value (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Returns

The result

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.518 ParamBlockGetProgInt64Value()

```
virtual int64_t Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgInt64Value (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Returns

The result

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.519 ParamBlockGetProgStrValue() [1/2]

```
virtual std::string_view Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgStrValue (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.520 ParamBlockGetProgStrValue() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgStrValue (
            int32_t blk,
            uint32_t index,
            char * buff,
            uint32_t & buff_size )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.
buff	Buffer to receive the response string value
buff_size	Size of the buffer on input and on exit required buffer size including null terminator

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.521 ParamBlockGetProgUInt32Value()

```
virtual uint32_t Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgUInt32Value (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Returns

The result

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.522 ParamBlockGetProgUInt64Value()

```
virtual uint64_t Trio::UnifiedApi::ITrioConnection::ParamBlockGetProgUInt64Value (
            int32_t blk,
            uint32_t index )  [pure virtual]
```

Retrieves a fetched value from a parameter block.

Parameters

blk	A valid parameter block
index	The index of the value in the parameter block. Please refer to the block definition.

Returns

The result

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.523 ProjectKey()

```
virtual void Trio::UnifiedApi::ITrioConnection::ProjectKey (
            const std::string_view & key,
            SecurityCodeType security_code = SecurityCodeType::Controller )  [pure virtual]
```

Sets a project key on the controller.

Parameters

key	The project key string.
security_code	The project key security code (key type).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.524 ProjectKeyRegistryClear()

```
virtual void Trio::UnifiedApi::ITrioConnection::ProjectKeyRegistryClear ( ) [pure virtual]
```

Remove all non-default encryption authorities from database. The default (pre-installed) authorities are not removed.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.525 ProjectKeyRegistryList()

```
virtual void Trio::UnifiedApi::ITrioConnection::ProjectKeyRegistryList (
            StringCallback && handler ) [pure virtual]
```

List the authorities that have project keys stored on the Motion Coordinator.

Parameters

handler	The handler of each returned name.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.526 ProjectKeyRegistryStore()

```
virtual void Trio::UnifiedApi::ITrioConnection::ProjectKeyRegistryStore (
            const std::string_view & authority,
            const std::string_view & key,
            SecurityCodeType security_code = SecurityCodeType::Controller ) [pure virtual]
```

Stores a project key on the Motion Coordinator under the specified authority name.

Parameters

authority	The authority name.
key	The project key string.
security_code	The project key security code (key type).

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.527 ProjectKeyRegistryUse()

```
virtual void Trio::UnifiedApi::ITrioConnection::ProjectKeyRegistryUse (
            const std::string_view & authority )  [pure virtual]
```

Loads a project key for use (project/program load) on the Motion Coordinator.

Parameters

authority	The authority name of the project key to load.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.528 PSwitch()

```
virtual void Trio::UnifiedApi::ITrioConnection::PSwitch (
            int32_t switchNumber,
            PSwitchMode mode,
            int32_t axis,
            uint32_t outputNumber,
            bool outputState,
            double setPosition,
            double resetPosition )  [pure virtual]
```

The PSWITCH command allows an output to be set when a predefined position is reached, and to be reset when a second position is reached.

Parameters

switchNumber	The switch number in the range 0..63
mode	Mode of operation. See PSwitchMode enumeration for more information
axis	Axis to link the PSWITCH to, may be any real or virtual axis
outputNumber	Selects the output to set, can be any real or virtual output
outputState	False - turn the output OFF at setpos, or true - turn the output ON at setpos
setPosition	The position at which output is set or reset, in user units
resetPosition	The position at which output is set or reset, in user units

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.529 PSwitchMulti()

```
virtual void Trio::UnifiedApi::ITrioConnection::PSwitchMulti (
            int32_t switchNumber,
            PSwitchMultiMode mode,
            int32_t axis,
            uint32_t outputNumber,
            bool outputState,
            uint32_t tableStart,
            uint32_t tableEnd,
            uint32_t pulseTime = 0 )  [pure virtual]
```

The PSWITCHMULTI command allows an output to be set that turns the output on and off at selected positions loaded from the TABLE memory.

Parameters

switchNumber	The switch number in the range 0..63
mode	Mode of operation. See PSwitchMultiMode enumeration for more information
axis	Axis to link the PSWITCH to, may be any real or virtual axis
outputNumber	Selects the output to set, can be any real or virtual output
outputState	False - turn the output OFF at setpos, or true - turn the output ON at setpos
tableStart	TABLE index for start of PSWITCH data
tableEnd	TABLE index for end of PSWITCH data
pulseTime	Time in servo-cycles when pulse output mode bit 4 is set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.530 PSwitchOff()

```
virtual void Trio::UnifiedApi::ITrioConnection::PSwitchOff (
            int32_t switchNumber,
            bool hold = false )  [pure virtual]
```

The PSWITCHOFF command is used to disable.

Parameters

switchNumber	The switch number in the range 0..63
hold	false = The PSWITCH output will hold in the same state it was when the PSWITCH is set to OFF . (Default) true = The PSWITCH output is forced OFF even if it was ON when the PSWTICH is set to OFF .

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.531 RapidStop()

```
virtual void Trio::UnifiedApi::ITrioConnection::RapidStop (
            int32_t mode = 0 )  [pure virtual]
```

The RAPIDSTOP command operates like the CANCEL command but does so for ALL axes using a subset of cancel modes.

Parameters

mode	0 Cancels axis commands from the MTYPE buffers(default) 1 Cancels all buffered moves on all axis(excluding the PMOVE) 2 Cancels all active and buffered moves including the PMOVE

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.532 Regist() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::Regist (
            RegistMode mode,
            int32_t axis = -1 )  [pure virtual]
```

The REGIST command initiates a capture of an axis position when it sees a registration input or the Z mark on the encoder. Once a registration event is captured MARK is set and the position and speed at the event can be read back.

Parameters

mode	Registration mode. See TrioBASIC REGIST documentation for more details
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

This variant is used with all other modes except 20, 21 and 22 and is not recommended for new applications.

6.3.3.533 Regist() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::Regist (
            RegistMode mode,
```

```
            uint32_t channel,
            uint32_t source,
            RegistEdge edge,
            RegistWindow window,
            uint32_t repeatCount = 0,
            uint32_t tableStart = 0,
            int32_t axis = -1 )  [pure virtual]
```

The REGIST command initiates a capture of an axis position when it sees a registration input or the Z mark on the encoder. Once a registration event is captured MARK is set and the position and speed at the event can be read back.

Parameters

mode	Registration mode. See TrioBASIC REGIST documentation for more details.
channel	Registration channel. See TrioBASIC REGIST documentation for more details.
source	Registration source. See TrioBASIC REGIST documentation for more details.
edge	0 - raising edge, 1 - falling edge.
window	Window of operation. See TrioBASIC REGIST documentation for more details.
repeatCount	Quantity of registration captures to store in the TABLE.
tableStart	Start position in the TABLE for the registration positions.
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

This variant is used with modes 20, 21 and 22 and is recommended for new applications.

6.3.3.534 Reverse()

```
virtual void Trio::UnifiedApi::ITrioConnection::Reverse (
            int32_t axis = -1 )  [pure virtual]
```

Sets continuous reverse movement.

Parameters

axis	Axis index to set reverse movement to or -1 to apply to default base axis.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.535 RunProgram() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::RunProgram (
            const std::string_view & name )  [pure virtual]
```

Runs a program.

Parameters

name	Name of the program to run.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.536 RunProgram() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::RunProgram (
            const std::string_view & name,
            int32_t process_number )  [pure virtual]
```

Runs a program.

Parameters

name	Name of the program to run.
process_number	Optional process number to run the procces on.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.537 ScopeOff()

```
virtual void Trio::UnifiedApi::ITrioConnection::ScopeOff ( )  [pure virtual]
```

Disable SCOPE.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.538 ScopeOn() [1/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::ScopeOn ( )  [pure virtual]
```

Enable SCOPE with previous settings.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.539 ScopeOn() [2/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::ScopeOn (
            uint32_t period_cycles )  [pure virtual]
```

Enable SCOPE with previous settings and new capture period.

Parameters

period_cycles	The number of servo periods between data samples.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.540 ScopeOn() [3/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::ScopeOn (
            uint32_t period_cycles,
            const std::string_view & fifo_name,
            uint32_t fifo_size,
            uint32_t sample_count,
            const std::string_view prms[],
            uint32_t prm_count )  [pure virtual]
```

Enable SCOPE and configure for capture to FIFO.

Parameters

period_cycles	The number of servo periods between data samples.
fifo_name	Name of the FIFO file to be written. If this file already exists on the Motion Coordinator it must be a FIFO file.
fifo_size	Size of the FIFO file to be created. If the FIFO file already exists with a different size then it will be deleted and recreated with the correct size.
sample_count	Number of samples to store to the FIFO file.
prms	Array of parameters to be captured.
prm_count	Number of parameters to be captured.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.541 ScopeOn() [4/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::ScopeOn (
            uint32_t period_cycles,
            uint32_t table_start,
            uint32_t table_end,
```

```
            const std::string_view prms[],
            uint32_t prm_count )  [pure virtual]
```

Enable SCOPE and configure for capture to TABLE range.

Parameters

period_cycles	The number of servo periods between data samples.
table_start	Start of table range to use store captured data.
table_end	End of table range to use (inclusive).
prms	Array of parameters to be captured.
prm_count	Number of parameters to be captured.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.542 Select()

```
virtual void Trio::UnifiedApi::ITrioConnection::Select (
            const std::string_view & remote_file,
            FileType file_type )  [pure virtual]
```

Makes the named program the currently selected program on the Motion Coordinator, if the named program does not exist then it makes a program of that name. The selected program cannot be changed when programs are running. When a program is selected any previously selected program is compiled. When program_type is specified is to make a program of that name and type, it cannot be possible to change the type of an existing program.

Parameters

remote_file	The file (program) name
file_type	The type of the file (program) to be created

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.543 SetAOut()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAOut (
            uint32_t channel,
            int32_t value )  [pure virtual]
```

Set the state of the Analogue Output. Note that the actual change will happen on the next servo cycle. Therefore in immediate read may give a wrong result on some (fast) systems.

Parameters

channel	Analogue output channel number
value	New value for the analogue output channel

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.544 SetAxisBase()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisBase (
            int32_t axis )  [pure virtual]
```

The BASE command is used to direct all subsequent motion commands and axis parameter read/writes to a particular axis, or group of axes. The default setting is a sequence: 0, 1, 2, 3...

Parameters

axis	The axes to send the motion commands to

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.545 SetAxisParameter_ACCEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ACCEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.546 SetAxisParameter_AFF_GAIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AFF_GAIN (
```

```
            int32_t axis,
            double value )  [pure virtual]
```

Setter for AFF_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.547 SetAxisParameter_ATYPE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ATYPE (
            int32_t axis,
            AxisType value )  [pure virtual]
```

Setter for ATYPE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.548 SetAxisParameter_AXIS_ACCEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_ACCEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for AXIS_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.549 SetAxisParameter_AXIS_ADDRESS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_ADDRESS (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for AXIS_ADDRESS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.550 SetAxisParameter_AXIS_DECEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_DECEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for AXIS_DECEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.551 SetAxisParameter_AXIS_DISPLAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_DISPLAY (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for AXIS_DISPLAY axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.552 SetAxisParameter_AXIS_ENABLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_ENABLE (
            int32_t axis,
            bool value )  [pure virtual]
```

Setter for AXIS_ENABLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.553 SetAxisParameter_AXIS_ENABLE_OVERRIDE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_ENABLE_OVERRIDE (
            int32_t axis,
            bool value )  [pure virtual]
```

Setter for AXIS_ENABLE_OVERRIDE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.554 SetAxisParameter_AXIS_ERROR_COUNT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_ERROR_COUNT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for AXIS_ERROR_COUNT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.555 SetAxisParameter_AXIS_FS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_FS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for AXIS_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.556 SetAxisParameter_AXIS_MAX_JOG_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_MAX_JOG_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for AXIS_MAX_JOG_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.557 SetAxisParameter_AXIS_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_MODE (
            int32_t axis,
            AxisMode value ) [pure virtual]
```

Setter for AXIS_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.558 SetAxisParameter_AXIS_RS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_RS_LIMIT (
            int32_t axis,
            double value ) [pure virtual]
```

Setter for AXIS_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.559 SetAxisParameter_AXIS_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_SPEED (
            int32_t axis,
            double value ) [pure virtual]
```

Setter for AXIS_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.560 SetAxisParameter_AXIS_UNITS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_UNITS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for AXIS_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.561 SetAxisParameter_AXIS_Z_OUTPUT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_AXIS_Z_OUTPUT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for AXIS_Z_OUTPUT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.562 SetAxisParameter_BACKLASH_DIST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_BACKLASH_DIST (
            int32_t axis,
            int64_t value )  [pure virtual]
```

Setter for BACKLASH_DIST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.563 SetAxisParameter_CHANGE_DIR_LAST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_CHANGE_DIR_LAST (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for CHANGE_DIR_LAST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.564 SetAxisParameter_CLOSE_WIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_CLOSE_WIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for CLOSE_WIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.565 SetAxisParameter_CLUTCH_RATE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_CLUTCH_RATE (
            int32_t axis,
            double value ) [pure virtual]
```

Setter for CLUTCH_RATE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.566 SetAxisParameter_CORNER_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_CORNER_MODE (
            int32_t axis,
            CornerMode value ) [pure virtual]
```

Setter for CORNER_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.567 SetAxisParameter_CORNER_STATE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_CORNER_STATE (
            int32_t axis,
            CornerState value ) [pure virtual]
```

Setter for CORNER_STATE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.568 SetAxisParameter_CREEP()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_CREEP (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for CREEP axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.569 SetAxisParameter_D_GAIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_D_GAIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for D_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.570 SetAxisParameter_DAC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DAC (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DAC axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.571 SetAxisParameter_DAC_SCALE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DAC_SCALE (
            int32_t axis,
            int64_t value )  [pure virtual]
```

Setter for DAC_SCALE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.572 SetAxisParameter_DATUM_IN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DATUM_IN (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for DATUM_IN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.573 SetAxisParameter_DECEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DECEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for DECEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.574 SetAxisParameter_DECEL_ANGLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DECEL_ANGLE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for DECEL_ANGLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.575 SetAxisParameter_DRIVE_CONTROL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_CONTROL (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.576 SetAxisParameter_DRIVE_CONTROLWORD()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_CONTROLWORD (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_CONTROLWORD axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.577 SetAxisParameter_DRIVE_CW_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_CW_MODE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_CW_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.578 SetAxisParameter_DRIVE_ENABLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_ENABLE (
            int32_t axis,
            int64_t value )  [pure virtual]
```

Setter for DRIVE_ENABLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.579 SetAxisParameter_DRIVE_FE_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_FE_LIMIT (
            int32_t axis,
            int64_t value )  [pure virtual]
```

Setter for DRIVE_FE_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.580 SetAxisParameter_DRIVE_INDEX()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_INDEX (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_INDEX axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.581 SetAxisParameter_DRIVE_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_MODE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.582 SetAxisParameter_DRIVE_NEG_TORQUE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_NEG_TORQUE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_NEG_TORQUE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.583 SetAxisParameter_DRIVE_PARAMETER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_PARAMETER (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_PARAMETER axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.584 SetAxisParameter_DRIVE_POS_TORQUE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_POS_TORQUE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_POS_TORQUE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.585 SetAxisParameter_DRIVE_PROFILE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_PROFILE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_PROFILE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.586 SetAxisParameter_DRIVE_SET_VAL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_SET_VAL (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_SET_VAL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.587 SetAxisParameter_DRIVE_TYPE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_TYPE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_TYPE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.588 SetAxisParameter_DRIVE_VALUE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_DRIVE_VALUE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for DRIVE_VALUE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.589 SetAxisParameter_ENCODER_BITS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENCODER_BITS (
            int32_t axis,
            uint16_t value )  [pure virtual]
```

Setter for ENCODER_BITS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.590 SetAxisParameter_ENCODER_CONTROL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENCODER_CONTROL (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for ENCODER_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.591 SetAxisParameter_ENCODER_FILTER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENCODER_FILTER (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ENCODER_FILTER axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.592 SetAxisParameter_ENCODER_ID()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENCODER_ID (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for ENCODER_ID axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.593 SetAxisParameter_ENCODER_STATUS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENCODER_STATUS (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for ENCODER_STATUS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.594 SetAxisParameter_ENCODER_TURNS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENCODER_TURNS (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for ENCODER_TURNS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.595 SetAxisParameter_END_DIR_LAST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_END_DIR_LAST (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for END_DIR_LAST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.596 SetAxisParameter_ENDMOVE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENDMOVE (
            int32_t axis,
            int64_t value )  [pure virtual]
```

Setter for ENDMOVE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.597 SetAxisParameter_ENDMOVE_BUFFER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENDMOVE_BUFFER (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ENDMOVE_BUFFER axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.598 SetAxisParameter_ENDMOVE_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ENDMOVE_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ENDMOVE_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.599 SetAxisParameter_ERRORMASK()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ERRORMASK (
            int32_t axis,
            AxisStatus value )  [pure virtual]
```

Setter for ERRORMASK axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.600 SetAxisParameter_FAST_JOG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FAST_JOG (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for FAST_JOG axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.601 SetAxisParameter_FASTDEC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FASTDEC (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FASTDEC axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.602 SetAxisParameter_FE_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FE_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FE_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.603 SetAxisParameter_FE_LIMIT_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FE_LIMIT_MODE (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for FE_LIMIT_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.604 SetAxisParameter_FE_RANGE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FE_RANGE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FE_RANGE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.605 SetAxisParameter_FEED_OVERRIDE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FEED_OVERRIDE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FEED_OVERRIDE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.606 SetAxisParameter_FHOLD_IN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FHOLD_IN (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for FHOLD_IN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.607 SetAxisParameter_FHSPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FHSPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FHSPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.608 SetAxisParameter_FORCE_ACCEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FORCE_ACCEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FORCE_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.609 SetAxisParameter_FORCE_DECEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FORCE_DECEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FORCE_DECEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.610 SetAxisParameter_FORCE_DWELL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FORCE_DWELL (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for FORCE_DWELL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.611 SetAxisParameter_FORCE_JERK()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FORCE_JERK (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FORCE_JERK axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.612 SetAxisParameter_FORCE_RAMP()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FORCE_RAMP (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FORCE_RAMP axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.613 SetAxisParameter_FORCE_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FORCE_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FORCE_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.614 SetAxisParameter_FRAME()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FRAME (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for FRAME axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.615 SetAxisParameter_FRAME_REP_DIST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FRAME_REP_DIST (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FRAME_REP_DIST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.616 SetAxisParameter_FRAME_SCALE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FRAME_SCALE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FRAME_SCALE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.617 SetAxisParameter_FS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.618 SetAxisParameter_FULL_SP_RADIUS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FULL_SP_RADIUS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for FULL_SP_RADIUS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.619 SetAxisParameter_FWD_IN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FWD_IN (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for FWD_IN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.620 SetAxisParameter_FWD_JOG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FWD_JOG (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for FWD_JOG axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.621 SetAxisParameter_FWD_START()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_FWD_START (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for FWD_START axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.622 SetAxisParameter_I_GAIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_I_GAIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for I_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.623 SetAxisParameter_IN_POS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_IN_POS (
            int32_t axis,
            uint32_t value )  [pure virtual]
```

Setter for IN_POS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.624 SetAxisParameter_IN_POS_DIST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_IN_POS_DIST (
            int32_t axis,
            int64_t value )  [pure virtual]
```

Setter for IN_POS_DIST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.625 SetAxisParameter_IN_POS_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_IN_POS_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for IN_POS_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.626 SetAxisParameter_INTERP_FACTOR()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_INTERP_FACTOR (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for INTERP_FACTOR axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.627 SetAxisParameter_INVERT_STEP()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_INVERT_STEP (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for INVERT_STEP axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.628 SetAxisParameter_JERK()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_JERK (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for JERK axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.629 SetAxisParameter_JOGSPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_JOGSPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for JOGSPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.630 SetAxisParameter_LOOKAHEAD_FACTOR()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_LOOKAHEAD_FACTOR (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for LOOKAHEAD_FACTOR axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.631 SetAxisParameter_MERGE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MERGE (
            int32_t axis,
            bool value )  [pure virtual]
```

Setter for MERGE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.632 SetAxisParameter_MOVE_COUNT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_COUNT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for MOVE_COUNT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.633 SetAxisParameter_MOVE_PA()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_PA (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for MOVE_PA axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.634 SetAxisParameter_MOVE_PA_CONT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_PA_CONT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for MOVE_PA_CONT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.635 SetAxisParameter_MOVE_PA_IDLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_PA_IDLE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for MOVE_PA_IDLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.636 SetAxisParameter_MOVE_PB()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_PB (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for MOVE_PB axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.637 SetAxisParameter_MOVE_PB_CONT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_PB_CONT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for MOVE_PB_CONT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.638 SetAxisParameter_MOVE_PB_IDLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVE_PB_IDLE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for MOVE_PB_IDLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.639 SetAxisParameter_MOVELINK_MODIFY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MOVELINK_MODIFY (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for MOVELINK_MODIFY axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.640 SetAxisParameter_MSPEED_FILTER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_MSPEED_FILTER (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for MSPEED_FILTER axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.641 SetAxisParameter_OFFPOS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_OFFPOS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for OFFPOS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.642 SetAxisParameter_OPEN_WIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_OPEN_WIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for OPEN_WIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.643 SetAxisParameter_OUTLIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_OUTLIMIT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for OUTLIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.644 SetAxisParameter_OV_GAIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_OV_GAIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for OV_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.645 SetAxisParameter_P_GAIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_P_GAIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for P_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.646 SetAxisParameter_POS_DELAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_POS_DELAY (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for POS_DELAY axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.647 SetAxisParameter_POSI_SEQ_DELAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_POSI_SEQ_DELAY (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for POSI_SEQ_DELAY axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.648 SetAxisParameter_POSI_SEQ_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_POSI_SEQ_MODE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for POSI_SEQ_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.649 SetAxisParameter_PP_STEP()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_PP_STEP (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for PP_STEP axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.650 SetAxisParameter_PWM_CYCLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_PWM_CYCLE (
            int32_t axis,
            uint32_t value )  [pure virtual]
```

Setter for PWM_CYCLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.651 SetAxisParameter_PWM_MARK()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_PWM_MARK (
            int32_t axis,
            uint32_t value )  [pure virtual]
```

Setter for PWM_MARK axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.652 SetAxisParameter_RAISE_ANGLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_RAISE_ANGLE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for RAISE_ANGLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.653 SetAxisParameter_REG_INPUTS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REG_INPUTS (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for REG_INPUTS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.654 SetAxisParameter_REGIST_CONTROL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REGIST_CONTROL (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for REGIST_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.655 SetAxisParameter_REGIST_DELAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REGIST_DELAY (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for REGIST_DELAY axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.656 SetAxisParameter_REP_DIST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REP_DIST (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for REP_DIST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.657 SetAxisParameter_REP_OPTION()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REP_OPTION (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for REP_OPTION axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.658 SetAxisParameter_REV_IN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REV_IN (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for REV_IN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.659 SetAxisParameter_REV_JOG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REV_JOG (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for REV_JOG axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.660 SetAxisParameter_REV_START()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_REV_START (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for REV_START axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.661 SetAxisParameter_ROBOT_FE_RANGE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ROBOT_FE_RANGE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ROBOT_FE_RANGE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.662 SetAxisParameter_ROBOT_FS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ROBOT_FS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ROBOT_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.663 SetAxisParameter_ROBOT_RS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ROBOT_RS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ROBOT_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.664 SetAxisParameter_ROBOT_SP_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ROBOT_SP_MODE (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for ROBOT_SP_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.665 SetAxisParameter_ROBOT_UNITS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_ROBOT_UNITS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for ROBOT_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.666 SetAxisParameter_RS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_RS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.667 SetAxisParameter_S_REF()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_S_REF (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for S_REF axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.668 SetAxisParameter_S_REF_OUT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_S_REF_OUT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for S_REF_OUT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.669 SetAxisParameter_SERVO()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SERVO (
            int32_t axis,
            bool value )  [pure virtual]
```

Setter for SERVO axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.670 SetAxisParameter_SLOT_NUMBER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SLOT_NUMBER (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for SLOT_NUMBER axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.671 SetAxisParameter_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.672 SetAxisParameter_SRAMP()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SRAMP (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for SRAMP axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.673 SetAxisParameter_START_DIR_LAST()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_START_DIR_LAST (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for START_DIR_LAST axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.674 SetAxisParameter_STARTMOVE_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_STARTMOVE_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for STARTMOVE_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.675 SetAxisParameter_STOP_ANGLE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_STOP_ANGLE (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for STOP_ANGLE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.676 SetAxisParameter_SYNC_AXIS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_AXIS (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for SYNC_AXIS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.677 SetAxisParameter_SYNC_CONTROL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_CONTROL (
            int32_t axis,
            SyncControl value ) [pure virtual]
```

Setter for SYNC_CONTROL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.678 SetAxisParameter_SYNC_DPOS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_DPOS (
            int32_t axis,
            double value ) [pure virtual]
```

Setter for SYNC_DPOS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.679 SetAxisParameter_SYNC_DWELL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_DWELL (
            int32_t axis,
            int32_t value ) [pure virtual]
```

Setter for SYNC_DWELL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.680 SetAxisParameter_SYNC_FLIGHT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_FLIGHT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for SYNC_FLIGHT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.681 SetAxisParameter_SYNC_PAUSE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_PAUSE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for SYNC_PAUSE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.682 SetAxisParameter_SYNC_TIME()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_TIME (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for SYNC_TIME axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.683 SetAxisParameter_SYNC_WITHDRAW()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_SYNC_WITHDRAW (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for SYNC_WITHDRAW axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.684 SetAxisParameter_T_REF()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_T_REF (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for T_REF axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.685 SetAxisParameter_T_REF_OUT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_T_REF_OUT (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for T_REF_OUT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.686 SetAxisParameter_TABLE_POINTER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_TABLE_POINTER (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for TABLE_POINTER axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.687 SetAxisParameter_TANG_DIRECTION()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_TANG_DIRECTION (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for TANG_DIRECTION axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.688 SetAxisParameter_TCP_FS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_TCP_FS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for TCP_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.689 SetAxisParameter_TCP_RS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_TCP_RS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for TCP_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.690 SetAxisParameter_TCP_UNITS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_TCP_UNITS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for TCP_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.691 SetAxisParameter_UNITS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_UNITS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for UNITS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.692 SetAxisParameter_VECTOR_BUFFERED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_VECTOR_BUFFERED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for VECTOR_BUFFERED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.693 SetAxisParameter_VERIFY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_VERIFY (
            int32_t axis,
            int16_t value )  [pure virtual]
```

Setter for VERIFY axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.694 SetAxisParameter_VFF_GAIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_VFF_GAIN (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for VFF_GAIN axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.695 SetAxisParameter_VP_MODE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_VP_MODE (
            int32_t axis,
            int32_t value )  [pure virtual]
```

Setter for VP_MODE axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.696 SetAxisParameter_WORLD_ACCEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_ACCEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_ACCEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.697 SetAxisParameter_WORLD_DECEL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_DECEL (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_DECEL axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.698 SetAxisParameter_WORLD_FS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_FS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_FS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.699 SetAxisParameter_WORLD_JOGSPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_JOGSPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_JOGSPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.700 SetAxisParameter_WORLD_RS_LIMIT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_RS_LIMIT (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_RS_LIMIT axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.701 SetAxisParameter_WORLD_SPEED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_SPEED (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_SPEED axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.702 SetAxisParameter_WORLD_UNITS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameter_WORLD_UNITS (
            int32_t axis,
            double value )  [pure virtual]
```

Setter for WORLD_UNITS axis parameter.

Parameters

axis	Target axis whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.703 SetAxisParameterBoolValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameterBoolValue (
            const std::string_view & param_name,
            int32_t axis,
            bool value )  [pure virtual]
```

Set boolean value to axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.704 SetAxisParameterDoubleValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameterDoubleValue (
            const std::string_view & param_name,
            int32_t axis,
            double value )  [pure virtual]
```

Set double value to axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.705 SetAxisParameterInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameterInt32Value (
            const std::string_view & param_name,
            int32_t axis,
            int32_t value )  [pure virtual]
```

Set 32-bit signed integer value to axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.706 SetAxisParameterInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameterInt64Value (
            const std::string_view & param_name,
```

```
            int32_t axis,
            int64_t value )  [pure virtual]
```

Set 64 bit signed integer value to axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.707 SetAxisParameterUInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameterUInt32Value (
            const std::string_view & param_name,
            int32_t axis,
            uint32_t value )  [pure virtual]
```

Set 32 bit unsigned integer value to axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.708 SetAxisParameterUInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetAxisParameterUInt64Value (
            const std::string_view & param_name,
            int32_t axis,
            uint64_t value )  [pure virtual]
```

Set 64 bit unsigned integer value to axis parameter by name.

Parameters

param_name	Parameter name
axis	Axis index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.709 SetDOut() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetDOut (
            uint32_t channel,
            bool state )  [pure virtual]
```

Set the state of the Digital Output. Note that the actual change will happen on the next servo cycle. Therefore in immediate read may give a wrong result on some (fast) systems.

Parameters

channel	Digital output channel number
state	New state for the digital output channel

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.710 SetDOut() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetDOut (
            uint32_t firstChannel,
            uint32_t lastChannel,
            uint32_t states )  [pure virtual]
```

Set the state of the Digital Outputs.

Parameters

firstChannel	First digital output channel number
lastChannel	Last digital output channel number
states	Binary sum of digital output states

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.711 SetDOut32()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetDOut32 (
            uint32_t states32 )  [pure virtual]
```

Set the state of the first 32 Digital Outputs. Note that the actual change will happen on the next servo cycle. Therefore in immediate read may give a wrong result on some (fast) systems.

Parameters

states32	Bit sum of new the states for the first 32 digital output channels

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.712 SetFrameGroup() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetFrameGroup (
            uint32_t group,
            uint32_t tableOffset,
            const int32_t axes[],
            uint32_t axisCount )  [pure virtual]
```

Define new frame group.

Parameters

group	Frame group number to define
tableOffset	Start offset of TABLE range used by the frame group
axes	Array of axes to be added to the frame group
axisCount	Number of axes to be added to the frame group

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.713 SetFrameGroup() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetFrameGroup (
            uint32_t group,
            uint32_t tableOffset,
            const std::vector< int32_t > & axes )  [pure virtual]
```

Define new frame group.

Parameters

group	Frame group number to define
tableOffset	Start offset of TABLE range used by the frame group
axes	Array of axes to be added to the frame group

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.714 SetInvertIN()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetInvertIN (
            uint32_t input,
            bool invert )  [pure virtual]
```

The INVERT_IN command allows the input channels to be individually inverted in software.

Parameters

input	The input to invert
invert	True to invert, false not to invert

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.715 SetMotion() [1/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotion (
            int32_t axis,
            double speed )  [pure virtual]
```

Function used to pre-set the parameter list for a subsequent move using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
speed	The value for SPEED to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.716 SetMotion() [2/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotion (
            int32_t axis,
            double speed,
            double accel )  [pure virtual]
```

Function used to pre-set the parameter list for a subsequent move using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
speed	The value for SPEED to be set
accel	The value for ACCEL to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.717 SetMotion() [3/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotion (
            int32_t axis,
            double speed,
            double accel,
            double decel )  [pure virtual]
```

Function used to pre-set the parameter list for a subsequent move using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
speed	The value for SPEED to be set
accel	The value for ACCEL to be set
decel	The value for DECEL to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.718 SetMotion() [4/4]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotion (
            int32_t axis,
            double speed,
            double accel,
            double decel,
            double jerk )  [pure virtual]
```

Function used to pre-set the parameter list for a subsequent move using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
speed	The value for SPEED to be set
accel	The value for ACCEL to be set
decel	The value for DECEL to be set
jerk	The value for JERK to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.719 SetMotionSp() [1/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.720 SetMotionSp() [2/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed,
            double startMoveSpeed )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.721 SetMotionSp() [3/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
```

```
            double forceSpeed,
            double startMoveSpeed,
            double endMoveSpeed )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set
endMoveSpeed	The value for ENDMOVE_SPEED to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.722 SetMotionSp() [4/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed,
            double startMoveSpeed,
            double endMoveSpeed,
            double forceAccel )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set
endMoveSpeed	The value for ENDMOVE_SPEED to be set
forceAccel	The value for FORCE_ACCEL to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.723 SetMotionSp() [5/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed,
            double startMoveSpeed,
```

```
            double endMoveSpeed,
            double forceAccel,
            double forceDecel )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set
endMoveSpeed	The value for ENDMOVE_SPEED to be set
forceAccel	The value for FORCE_ACCEL to be set
forceDecel	The value for FORCE_DECEL to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.724 SetMotionSp() [6/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed,
            double startMoveSpeed,
            double endMoveSpeed,
            double forceAccel,
            double forceDecel,
            double forceJerk )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set
endMoveSpeed	The value for ENDMOVE_SPEED to be set
forceAccel	The value for FORCE_ACCEL to be set
forceDecel	The value for FORCE_DECEL to be set
forceJerk	The value for FORCE_JERK to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.725 SetMotionSp() [7/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed,
            double startMoveSpeed,
            double endMoveSpeed,
            double forceAccel,
            double forceDecel,
            double forceJerk,
            double forceRamp )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set
endMoveSpeed	The value for ENDMOVE_SPEED to be set
forceAccel	The value for FORCE_ACCEL to be set
forceDecel	The value for FORCE_DECEL to be set
forceJerk	The value for FORCE_JERK to be set
forceRamp	The value for FORCE_RAMP to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.726 SetMotionSp() [8/8]

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMotionSp (
            int32_t axis,
            double forceSpeed,
            double startMoveSpeed,
            double endMoveSpeed,
            double forceAccel,
            double forceDecel,
            double forceJerk,
            double forceRamp,
            double forceDwell )  [pure virtual]
```

SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.

Parameters

axis	Target axis for whose parameters are to be modified
forceSpeed	The value for FORCE_SPEED to be set
startMoveSpeed	The value for STARTMOVE_SPEED to be set

Parameters

endMoveSpeed	The value for ENDMOVE_SPEED to be set
forceAccel	The value for FORCE_ACCEL to be set
forceDecel	The value for FORCE_DECEL to be set
forceJerk	The value for FORCE_JERK to be set
forceRamp	The value for FORCE_RAMP to be set
forceDwell	The value for FORCE_DWELL to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.727 SetMultiAxisBase()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMultiAxisBase (
            const int32_t axes[],
            uint32_t axisCount )  [pure virtual]
```

Set multiple base axes.

Parameters

axes	Array of axis indexes to set as base
axisCount	Number of axis indexes to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.728 SetMultiTableValues()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMultiTableValues (
            uint32_t start_index,
            uint32_t count,
            const double values[] )  [pure virtual]
```

Set multiple table entries.

Parameters

start_index	Start index of the table range to be written
count	Number of table entries to be written
values	Array of values to be written

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.729 SetMultiVrValues()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetMultiVrValues (
            uint32_t start_index,
            uint32_t count,
            const double values[] )  [pure virtual]
```

Set multiple VR entries.

Parameters

start_index	Start index of the VR range to be written
count	Number of VR entries to be written
values	Array of VR values to be written

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.730 SetPortParameterBoolValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetPortParameterBoolValue (
            const std::string_view & param_name,
            int32_t port,
            bool value )  [pure virtual]
```

Set boolean value to port parameter by name.

Parameters

param_name	Parameter name
port	Port index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.731 SetPortParameterDoubleValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetPortParameterDoubleValue (
            const std::string_view & param_name,
```

```
            int32_t port,
            double value )  [pure virtual]
```

Set double value to port parameter by name.

Parameters

param_name	Parameter name
port	Port index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.732 SetPortParameterInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetPortParameterInt32Value (
            const std::string_view & param_name,
            int32_t port,
            int32_t value )  [pure virtual]
```

Set 32-bit signed integer value to port parameter by name.

Parameters

param_name	Parameter name
port	Port index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.733 SetPortParameterInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetPortParameterInt64Value (
            const std::string_view & param_name,
            int32_t port,
            int64_t value )  [pure virtual]
```

Set 64 bit signed integer value to port parameter by name.

Parameters

param_name	Parameter name
port	Port index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.734 SetPortParameterUInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetPortParameterUInt32Value (
            const std::string_view & param_name,
            int32_t port,
            uint32_t value )  [pure virtual]
```

Set 32 bit unsigned integer value to port parameter by name.

Parameters

param_name	Parameter name
port	Port index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.735 SetPortParameterUInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetPortParameterUInt64Value (
            const std::string_view & param_name,
            int32_t port,
            uint64_t value )  [pure virtual]
```

Set 64 bit unsigned integer value to port parameter by name.

Parameters

param_name	Parameter name
port	Port index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.736 SetProcessParameter_INDEVICE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameter_INDEVICE (
```

```
            int32_t process,
            int32_t value )  [pure virtual]
```

Setter for INDEVICE process parameter.

Parameters

process	Target process whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.737 SetProcessParameter_OUTDEVICE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameter_OUTDEVICE (
            int32_t process,
            int32_t value )  [pure virtual]
```

Setter for OUTDEVICE process parameter.

Parameters

process	Target process whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.738 SetProcessParameter_TICKS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameter_TICKS (
            int32_t process,
            int64_t value )  [pure virtual]
```

Setter for TICKS process parameter.

Parameters

process	Target process whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.739 SetProcessParameterBoolValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameterBoolValue (
            const std::string_view & param_name,
            int32_t process,
            bool value )  [pure virtual]
```

Set boolean value to process parameter by name.

Parameters

param_name	Parameter name
process	Process index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.740 SetProcessParameterDoubleValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameterDoubleValue (
            const std::string_view & param_name,
            int32_t process,
            double value )  [pure virtual]
```

Set double value to process parameter by name.

Parameters

param_name	Parameter name
process	Process index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.741 SetProcessParameterInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameterInt32Value (
            const std::string_view & param_name,
            int32_t process,
            int32_t value )  [pure virtual]
```

Set 32-bit signed integer value to process parameter by name.

Parameters

param_name	Parameter name
process	Process index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.742 SetProcessParameterInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameterInt64Value (
            const std::string_view & param_name,
            int32_t process,
            int64_t value )  [pure virtual]
```

Set 64 bit signed integer value to process parameter by name.

Parameters

param_name	Parameter name
process	Process index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.743 SetProcessParameterUInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameterUInt32Value (
            const std::string_view & param_name,
            int32_t process,
            uint32_t value )  [pure virtual]
```

Set 32 bit unsigned integer value to process parameter by name.

Parameters

param_name	Parameter name
process	Process index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.744 SetProcessParameterUInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetProcessParameterUInt64Value (
            const std::string_view & param_name,
            int32_t process,
            uint64_t value )  [pure virtual]
```

Set 64 bit unsigned integer value to process parameter by name.

Parameters

param_name	Parameter name
process	Process index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.745 SetSlotParameter_SERIAL_NUMBER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameter_SERIAL_NUMBER (
            int32_t slot,
            int32_t value )  [pure virtual]
```

Setter for SERIAL_NUMBER slot parameter.

Parameters

slot	Target slot whose parameter is modified
value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.746 SetSlotParameterBoolValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameterBoolValue (
            const std::string_view & param_name,
            int32_t slot,
            bool value )  [pure virtual]
```

Set boolean value to slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.747 SetSlotParameterDoubleValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameterDoubleValue (
            const std::string_view & param_name,
            int32_t slot,
            double value )  [pure virtual]
```

Set double value to slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.748 SetSlotParameterInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameterInt32Value (
            const std::string_view & param_name,
            int32_t slot,
            int32_t value )  [pure virtual]
```

Set 32-bit signed integer value to slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.749 SetSlotParameterInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameterInt64Value (
            const std::string_view & param_name,
```

```
            int32_t slot,
            int64_t value )  [pure virtual]
```

Set 64 bit signed integer value to slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.750 SetSlotParameterUInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameterUInt32Value (
            const std::string_view & param_name,
            int32_t slot,
            uint32_t value )  [pure virtual]
```

Set 32 bit unsigned integer value to slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.751 SetSlotParameterUInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSlotParameterUInt64Value (
            const std::string_view & param_name,
            int32_t slot,
            uint64_t value )  [pure virtual]
```

Set 64 bit unsigned integer value to slot parameter by name.

Parameters

param_name	Parameter name
slot	Slot index
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.752 SetSystemParameter_ADDRESS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_ADDRESS (
            uint32_t value ) [pure virtual]
```

Setter for ADDRESS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.753 SetSystemParameter_AIN0()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AIN0 (
            uint32_t value ) [pure virtual]
```

Setter for AIN0 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.754 SetSystemParameter_AIN1()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AIN1 (
            uint32_t value ) [pure virtual]
```

Setter for AIN1 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.755 SetSystemParameter_AIN2()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AIN2 (
            uint32_t value ) [pure virtual]
```

Setter for AIN2 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.756 SetSystemParameter_AIN3()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AIN3 (
            uint32_t value ) [pure virtual]
```

Setter for AIN3 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.757 SetSystemParameter_AINBI0()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AINBI0 (
            uint32_t value ) [pure virtual]
```

Setter for AINBI0 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.758 SetSystemParameter_AINBI1()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AINBI1 (
            uint32_t value ) [pure virtual]
```

Setter for AINBI1 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.759 SetSystemParameter_AINBI2()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AINBI2 (
            uint32_t value ) [pure virtual]
```

Setter for AINBI2 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.760 SetSystemParameter_AINBI3()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AINBI3 (
            uint32_t value ) [pure virtual]
```

Setter for AINBI3 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.761 SetSystemParameter_AOUT0()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AOUT0 (
            uint32_t value ) [pure virtual]
```

Setter for AOUT0 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.762 SetSystemParameter_AOUT1()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AOUT1 (
            uint32_t value ) [pure virtual]
```

Setter for AOUT1 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.763 SetSystemParameter_AOUT2()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AOUT2 (
            uint32_t value ) [pure virtual]
```

Setter for AOUT2 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.764 SetSystemParameter_AOUT3()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AOUT3 (
            uint32_t value ) [pure virtual]
```

Setter for AOUT3 system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.765 SetSystemParameter_AUTO_ETHERCAT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AUTO_ETHERCAT (
            int32_t value ) [pure virtual]
```

Setter for AUTO_ETHERCAT system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.766 SetSystemParameter_AVERAGE_JITTER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_AVERAGE_JITTER (
            int64_t value ) [pure virtual]
```

Setter for AVERAGE_JITTER system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.767 SetSystemParameter_CPU_EXCEPTIONS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_CPU_EXCEPTIONS (
            int64_t value ) [pure virtual]
```

Setter for CPU_EXCEPTIONS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.768 SetSystemParameter_CURRENT_JITTER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_CURRENT_JITTER (
            int64_t value ) [pure virtual]
```

Setter for CURRENT_JITTER system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.769 SetSystemParameter_DISPLAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_DISPLAY (
            int32_t value ) [pure virtual]
```

Setter for DISPLAY system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.770 SetSystemParameter_FPU_EXCEPTIONS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_FPU_EXCEPTIONS (
            int64_t value )  [pure virtual]
```

Setter for FPU_EXCEPTIONS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.771 SetSystemParameter_HMI_PROC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_HMI_PROC (
            int32_t value )  [pure virtual]
```

Setter for HMI_PROC system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.772 SetSystemParameter_IP_ADDRESS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_ADDRESS (
            uint32_t value )  [pure virtual]
```

Setter for IP_ADDRESS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.773 SetSystemParameter_IP_GATEWAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_GATEWAY (
            uint32_t value ) [pure virtual]
```

Setter for IP_GATEWAY system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.774 SetSystemParameter_IP_MAC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_MAC (
            uint64_t value ) [pure virtual]
```

Setter for IP_MAC system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.775 SetSystemParameter_IP_MEMORY_CONFIG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_MEMORY_CONFIG (
            uint32_t value ) [pure virtual]
```

Setter for IP_MEMORY_CONFIG system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.776 SetSystemParameter_IP_NETMASK()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_NETMASK (
            uint32_t value )  [pure virtual]
```

Setter for IP_NETMASK system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.777 SetSystemParameter_IP_PROTOCOL_CONFIG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_PROTOCOL_CONFIG (
            uint32_t value )  [pure virtual]
```

Setter for IP_PROTOCOL_CONFIG system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.778 SetSystemParameter_IP_PROTOCOL_CTRL()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_PROTOCOL_CTRL (
            uint32_t value )  [pure virtual]
```

Setter for IP_PROTOCOL_CTRL system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.779 SetSystemParameter_IP_TCP_TIMEOUT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_TCP_TIMEOUT (
            uint64_t value ) [pure virtual]
```

Setter for IP_TCP_TIMEOUT system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.780 SetSystemParameter_IP_TCP_TX_THRESHOLD()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_TCP_TX_THRESHOLD (
            uint32_t value ) [pure virtual]
```

Setter for IP_TCP_TX_THRESHOLD system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.781 SetSystemParameter_IP_TCP_TX_TIMEOUT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_IP_TCP_TX_TIMEOUT (
            uint32_t value ) [pure virtual]
```

Setter for IP_TCP_TX_TIMEOUT system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.782 SetSystemParameter_LAST_AXIS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_LAST_AXIS (
            int32_t value ) [pure virtual]
```

Setter for LAST_AXIS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.783 SetSystemParameter_LIMIT_BUFFERED()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_LIMIT_BUFFERED (
            int16_t value ) [pure virtual]
```

Setter for LIMIT_BUFFERED system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.784 SetSystemParameter_MAX_JITTER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_MAX_JITTER (
            int64_t value ) [pure virtual]
```

Setter for MAX_JITTER system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.785 SetSystemParameter_MISSED_INTERRUPTS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_MISSED_INTERRUPTS (
            int64_t value ) [pure virtual]
```

Setter for MISSED_INTERRUPTS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.786 SetSystemParameter_MISSED_RECV_FRAMES()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_MISSED_RECV_FRAMES (
            int64_t value ) [pure virtual]
```

Setter for MISSED_RECV_FRAMES system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.787 SetSystemParameter_MISSED_SEND_FRAMES()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_MISSED_SEND_FRAMES (
            int64_t value ) [pure virtual]
```

Setter for MISSED_SEND_FRAMES system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.788 SetSystemParameter_MODBUS_TX_DELAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_MODBUS_TX_DELAY (
            uint16_t value )  [pure virtual]
```

Setter for MODBUS_TX_DELAY system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.789 SetSystemParameter_PCMCAT_PROC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_PCMCAT_PROC (
            int32_t value )  [pure virtual]
```

Setter for PCMCAT_PROC system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.790 SetSystemParameter_PLC_CONFIG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_PLC_CONFIG (
            int32_t value )  [pure virtual]
```

Setter for PLC_CONFIG system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.791 SetSystemParameter_REMOTE_PROC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_REMOTE_PROC (
            int32_t value ) [pure virtual]
```

Setter for REMOTE_PROC system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.792 SetSystemParameter_SCHEDULE_TYPE()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SCHEDULE_TYPE (
            uint32_t value ) [pure virtual]
```

Setter for SCHEDULE_TYPE system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.793 SetSystemParameter_SCOPE_CYCLE_COUNT()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SCOPE_CYCLE_COUNT (
            int32_t value ) [pure virtual]
```

Setter for SCOPE_CYCLE_COUNT system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.794 SetSystemParameter_SCOPE_DELAY()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SCOPE_DELAY (
            int64_t value )  [pure virtual]
```

Setter for SCOPE_DELAY system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.795 SetSystemParameter_SCOPE_POS()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SCOPE_POS (
            int32_t value )  [pure virtual]
```

Setter for SCOPE_POS system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.796 SetSystemParameter_SERIAL_NUMBER()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SERIAL_NUMBER (
            int32_t value )  [pure virtual]
```

Setter for SERIAL_NUMBER system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.797 SetSystemParameter_SERVO_OFFSET()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SERVO_OFFSET (
            int32_t value )  [pure virtual]
```

Setter for SERVO_OFFSET system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.798 SetSystemParameter_SERVO_PERIOD()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SERVO_PERIOD (
            int32_t value )  [pure virtual]
```

Setter for SERVO_PERIOD system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.799 SetSystemParameter_SLOT0_TIME()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SLOT0_TIME (
            uint32_t value )  [pure virtual]
```

Setter for SLOT0_TIME system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.800 SetSystemParameter_SLOT1_TIME()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SLOT1_TIME (
            uint32_t value )  [pure virtual]
```

Setter for SLOT1_TIME system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.801 SetSystemParameter_SLOT2_TIME()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SLOT2_TIME (
            uint32_t value )  [pure virtual]
```

Setter for SLOT2_TIME system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.802 SetSystemParameter_SLOT3_TIME()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SLOT3_TIME (
            uint32_t value )  [pure virtual]
```

Setter for SLOT3_TIME system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.803 SetSystemParameter_SYSTEM_ERROR_MASK()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SYSTEM_ERROR_MASK (
            uint32_t value ) [pure virtual]
```

Setter for SYSTEM_ERROR_MASK system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.804 SetSystemParameter_SYSTEM_LOAD_MAX()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_SYSTEM_LOAD_MAX (
            uint32_t value ) [pure virtual]
```

Setter for SYSTEM_LOAD_MAX system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.805 SetSystemParameter_TEXT_FILE_LOADER_PROC()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_TEXT_FILE_LOADER_PROC (
            int32_t value ) [pure virtual]
```

Setter for TEXT_FILE_LOADER_PROC system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.806 SetSystemParameter_WDOG()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameter_WDOG (
            bool value )  [pure virtual]
```

Setter for WDOG system parameter.

Parameters

value	New value to set to parameter

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.807 SetSystemParameterBoolValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterBoolValue (
            const std::string_view & param_name,
            bool value )  [pure virtual]
```

Set boolean value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.808 SetSystemParameterDoubleValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterDoubleValue (
            const std::string_view & param_name,
            double value )  [pure virtual]
```

Set double value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.809 SetSystemParameterInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterInt32Value (
            const std::string_view & param_name,
            int32_t value ) [pure virtual]
```

Set 32-bit signed integer value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.810 SetSystemParameterInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterInt64Value (
            const std::string_view & param_name,
            int64_t value ) [pure virtual]
```

Set 64 bit signed integer value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.811 SetSystemParameterStringValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterStringValue (
```

```
            const std::string_view & param_name,
            const std::string_view & value )  [pure virtual]
```

Set string value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.812 SetSystemParameterUInt32Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterUInt32Value (
            const std::string_view & param_name,
            uint32_t value )  [pure virtual]
```

Set 32 bit unsigned integer value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.813 SetSystemParameterUInt64Value()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetSystemParameterUInt64Value (
            const std::string_view & param_name,
            uint64_t value )  [pure virtual]
```

Set 64 bit unsigned integer value to system parameter by name.

Parameters

param_name	Parameter name
value	New value

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.814 SetTableValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetTableValue (
            uint32_t index,
            double value )  [pure virtual]
```

Set single table value.

Parameters

index	Index of the table entry to be written
value	New value to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.815 SetTcpCommandTimeout()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetTcpCommandTimeout (
            uint32_t timeout_ms )  [pure virtual]
```

Set new timeout for command execution. Valid only for TCP connection.

Parameters

timeout_ms	New value for the command execution timeout in milliseconds

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.816 SetTcpConnectTimeout()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetTcpConnectTimeout (
            uint32_t timeout_ms )  [pure virtual]
```

Set new timeout for opening TCP connection. Valid only for TCP connection.

Parameters

timeout_ms	New value for the open connection timeout in milliseconds

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.817 SetTcpHostName()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetTcpHostName (
            const std::string_view & host )  [pure virtual]
```

Set host name or IP to connection object. Valid only for TCP connection.

Parameters

host	New value for the host name

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.818 SetTcpPort()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetTcpPort (
            uint16_t port )  [pure virtual]
```

Set TCP port to connection object. Valid only for TCP connection.

Parameters

port	New value for the TCP port

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.819 SetVrBit()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetVrBit (
            uint16_t bit,
            uint32_t vr_variable )  [pure virtual]
```

The function can be used to set the value of a single bit within a VR variable. All other bits are unchanged.

Parameters

bit	The bit number to set, valid range is 0 to 52.
vr_variable	The VR which to operate on.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.820 SetVrString()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetVrString (
            uint32_t vr,
            const std::string_view & str )  [pure virtual]
```

Set a null-terminated string into consecutive VRs.

Parameters

vr	Starting VR index
str	String to be set

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.821 SetVrValue()

```
virtual void Trio::UnifiedApi::ITrioConnection::SetVrValue (
            uint32_t index,
            double value )  [pure virtual]
```

Set single VR value.

Parameters

index	Index of the VR entry to be written
value	New value to be written

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.822 StepRatio()

```
virtual void Trio::UnifiedApi::ITrioConnection::StepRatio (
            int64_t output_count,
            int64_t dpos_count,
            int32_t axis = -1 )  [pure virtual]
```

This command sets up an integer ratio for the axis' demanded position. Usually this is used on a stepper (pulse+direction) output. It can also be applied to a networked axis in positon mode, such as EtherCAT and RTEX position servo. Every servo-period the change in position is passed through the STEP_RATIO function before it goes to the step pulse or network drive output. The STEP_RATIO function operates before the divide by 16 factor(divide by 256 in Flexslice P375) in the stepper axis.This maintains the good timing resolution of the stepper output circuit. STEP_RATIO does not replace UNITS.

Parameters

output_count	Number of counts to output for the given dpos_count value. Range: 0 to 16777215
dpos_count	Change in DPOS value for corresponding output count. Range: 0 to 16777215.
axis	Base axis for this command. If value is -1 then the default base axis is used

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.823 StopProgram() [1/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::StopProgram (
            const std::string_view & name )  [pure virtual]
```

Stops a program at its current line.

Parameters

name	Name of the program to be stopped.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.824 StopProgram() [2/2]

```
virtual void Trio::UnifiedApi::ITrioConnection::StopProgram (
            const std::string_view & name,
            int32_t process_number )  [pure virtual]
```

Stops a program at its current line.

Parameters

name	Name of the program to be stopped.
process_number	Optional process number to be used when multiple instances of the program are running and only one is to be stopped.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.825 Trigger()

```
virtual void Trio::UnifiedApi::ITrioConnection::Trigger (
            bool rearm = false )  [pure virtual]
```

Starts a previously set up SCOPE command. This allows you to manually start the scope capture.

Parameters

rearm	If true the SCOPE will automatically retrigger at the end of each scan.

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

6.3.3.826 UploadFile()

```
virtual void Trio::UnifiedApi::ITrioConnection::UploadFile (
            const std::string_view & local_file,
            const std::string_view & remote_file,
            FileType remote_file_type,
            FileTransferOption opt,
            ProgressHandlerCallback && handler )  [pure virtual]
```

Uploads a file to the Motion Coordinator.

Parameters

local_file	Local file name
remote_file	Remote file name. Can be prefixed by: "RAM:" - the location of the file is in the controller RAM. "SD:" - the location of the file is on the SD card. "FIFO:" - The target file is a FIFO file on the controller. It is expected that the controller (a program) will consume the data. If no prefix "RAM:" is assumed.
remote_file_type	The type of the file. Ignored when uploading to the SD card or with for FIFO transfers.
opt	File transfer option
handler	Callback handler with the progress advance

Exceptions

TrioConnectionException	Thrown if error occurs in communication module.

The documentation for this class was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.4 Trio::UnifiedApi::ProcessInfo Struct Reference

Data used with a callback for processing the response of Process command.

```
#include <Trio_UnifiedApi_CPP.h>
```

Public Member Functions

- ProcessInfo ()

  Constructor.

Data Fields

- int32_t process

  The process number.
- ProcessStatus status

  Status of the process on the controller.
- int32_t line

  The current line - when the process is in step mode.
- uint64_t time_millisecond

  The process running time.
- double cpu_usage

  CPU usage in percent (%)

6.4.1 Detailed Description

Data used with a callback for processing the response of Process command.

Definition at line 2520 of file Trio_UnifiedApi_CPP.h.

6.4.2 Constructor & Destructor Documentation

6.4.2.1 ProcessInfo()

```
Trio::UnifiedApi::ProcessInfo::ProcessInfo ( ) [inline]
```

Constructor.

Definition at line 2546 of file Trio_UnifiedApi_CPP.h.

6.4.3 Field Documentation

6.4.3.1 cpu_usage

```
double Trio::UnifiedApi::ProcessInfo::cpu_usage
```

CPU usage in percent (%)

Definition at line 2541 of file Trio_UnifiedApi_CPP.h.

6.4.3.2 line

```
int32_t Trio::UnifiedApi::ProcessInfo::line
```

The current line - when the process is in step mode.

Definition at line 2533 of file Trio_UnifiedApi_CPP.h.

6.4.3.3 process

```
int32_t Trio::UnifiedApi::ProcessInfo::process
```

The process number.

Definition at line 2525 of file Trio_UnifiedApi_CPP.h.

6.4.3.4 status

```
ProcessStatus Trio::UnifiedApi::ProcessInfo::status
```

Status of the process on the controller.

Definition at line 2529 of file Trio_UnifiedApi_CPP.h.

6.4.3.5 time_millisecond

```
uint64_t Trio::UnifiedApi::ProcessInfo::time_millisecond
```

The process running time.

Definition at line 2537 of file Trio_UnifiedApi_CPP.h.

The documentation for this struct was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.5 Trio::UnifiedApi::ProgressInfo Struct Reference

Progress info helper.

```
#include <Trio_UnifiedApi_CPP.h>
```

Public Member Functions

- ProgressInfo ()

Data Fields

- uint32_t total_length

  The length of the process. If 0 the length is unknown.
- uint32_t current_pos

  The current progress position.
- ProgressEvent evt

  Progress event.

6.5.1 Detailed Description

Progress info helper.

Definition at line 2462 of file Trio_UnifiedApi_CPP.h.

6.5.2 Constructor & Destructor Documentation

6.5.2.1 ProgressInfo()

```
Trio::UnifiedApi::ProgressInfo::ProgressInfo ( ) [inline]
```

Definition at line 2478 of file Trio_UnifiedApi_CPP.h.

6.5.3 Field Documentation

6.5.3.1 current_pos

```
uint32_t Trio::UnifiedApi::ProgressInfo::current_pos
```

The current progress position.

Definition at line 2471 of file Trio_UnifiedApi_CPP.h.

6.5.3.2 evt

```
ProgressEvent Trio::UnifiedApi::ProgressInfo::evt
```

Progress event.

Definition at line 2475 of file Trio_UnifiedApi_CPP.h.

6.5.3.3 total_length

```
uint32_t Trio::UnifiedApi::ProgressInfo::total_length
```

The length of the process. If 0 the length is unknown.

Definition at line 2467 of file Trio_UnifiedApi_CPP.h.

The documentation for this struct was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.6 Trio::UnifiedApi::Range Struct Reference

A structure representing a range of consecutive values.

```
#include <Trio_UnifiedApi_CPP.h>
```

Public Member Functions

- Range ()

  Constructor.
- Range (uint32_t s, uint32_t c)

  Constructor.

Data Fields

- uint32_t start

  Start value.
- uint32_t count

  Value count.

6.6.1 Detailed Description

A structure representing a range of consecutive values.

Definition at line 2493 of file Trio_UnifiedApi_CPP.h.

6.6.2 Constructor & Destructor Documentation

6.6.2.1 Range() [1/2]

```
Trio::UnifiedApi::Range::Range ( ) [inline]
```

Constructor.

Definition at line 2507 of file Trio_UnifiedApi_CPP.h.

6.6.2.2 Range() [2/2]

```
Trio::UnifiedApi::Range::Range (
            uint32_t s,
            uint32_t c ) [inline]
```

Constructor.

Parameters

s	Start index
c	Count

Definition at line 2514 of file Trio_UnifiedApi_CPP.h.

6.6.3 Field Documentation

6.6.3.1 count

```
uint32_t Trio::UnifiedApi::Range::count
```

Value count.

Definition at line 2502 of file Trio_UnifiedApi_CPP.h.

6.6.3.2 start

```
uint32_t Trio::UnifiedApi::Range::start
```

Start value.

Definition at line 2498 of file Trio_UnifiedApi_CPP.h.

The documentation for this struct was generated from the following file:

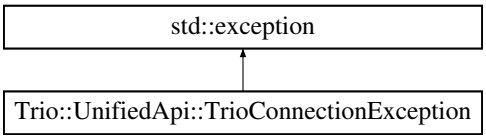
- Trio_UnifiedApi_CPP.h

6.7 Trio::UnifiedApi::TrioConnectionException Class Reference

Exception raised when error occurs in communication library.

```
#include <Trio_UnifiedApi_CPP.h>
```

Inheritance diagram for Trio::UnifiedApi::TrioConnectionException:



Public Member Functions

- TRIO_DECLSPEC TrioConnectionException (ErrorCode errorCode)
- ErrorCode errorCode () const
- virtual TRIO_DECLSPEC char const ∗ what () const noexcept override

6.7.1 Detailed Description

Exception raised when error occurs in communication library.

Definition at line 2563 of file Trio_UnifiedApi_CPP.h.

6.7.2 Constructor & Destructor Documentation

6.7.2.1 TrioConnectionException()

```
TRIO_DECLSPEC Trio::UnifiedApi::TrioConnectionException::TrioConnectionException (
            ErrorCode errorCode )
```

6.7.3 Member Function Documentation

6.7.3.1 errorCode()

```
ErrorCode Trio::UnifiedApi::TrioConnectionException::errorCode ( ) const  [inline]
```

Definition at line 2568 of file Trio_UnifiedApi_CPP.h.

6.7.3.2 what()

```
virtual TRIO_DECLSPEC char const * Trio::UnifiedApi::TrioConnectionException::what ( ) const
[override], [virtual], [noexcept]
```

The documentation for this class was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.8 Trio::UnifiedApi::TrioConnectionFactory Class Reference

Factory for creating communication instances.

```
#include <Trio_UnifiedApi_CPP.h>
```

Static Public Member Functions

- static TRIO_DECLSPEC ITrioConnection *TRIO_CALLCONV CreateTCP (EventCallback &&event_callback, const std::string_view &host_name=std::string_view(), uint16_t port=0)

  Create TCP based connection to Motion Coordinator.
- static TRIO_DECLSPEC ITrioConnection *TRIO_CALLCONV CreatePCMCAT (EventCallback &&event_callback)

  Create PC-MCAT based connection to Motion Coordinator.
- static TRIO_DECLSPEC ITrioConnection *TRIO_CALLCONV CreateFlex7 (EventCallback &&event_callback)

  Create Flex7 based connection to Motion Coordinator.

6.8.1 Detailed Description

Factory for creating communication instances.

Definition at line 2578 of file Trio_UnifiedApi_CPP.h.

6.8.2 Member Function Documentation

6.8.2.1 CreateFlex7()

```
static TRIO_DECLSPEC ITrioConnection *TRIO_CALLCONV Trio::UnifiedApi::TrioConnectionFactory::↩
CreateFlex7 (
            EventCallback && event_callback )  [static]
```

Create Flex7 based connection to Motion Coordinator.

Parameters

event_callback	Callback function to receive various events from the API

Returns

On success an instance of connection interface is returned

6.8.2.2 CreatePCMCAT()

```
static TRIO_DECLSPEC ITrioConnection *TRIO_CALLCONV Trio::UnifiedApi::TrioConnectionFactory::↩
CreatePCMCAT (
            EventCallback && event_callback )  [static]
```

Create PC-MCAT based connection to Motion Coordinator.

Parameters

event_callback	Callback function to receive various events from the API

Returns

On success an instance of connection interface is returned

6.8.2.3 CreateTCP()

```
static TRIO_DECLSPEC ITrioConnection *TRIO_CALLCONV Trio::UnifiedApi::TrioConnectionFactory::↩
CreateTCP (
            EventCallback && event_callback,
            const std::string_view & host_name = std::string_view(),
            uint16_t port = 0 )  [static]
```

Create TCP based connection to Motion Coordinator.

Parameters

event_callback	Callback function to receive various events from the API
host_name	Host name or IP of the Motion Coordinator to connect via TCP protocol
port	Port on Motion Coordinator open for TCP connections. It is normally 3240 unless Motion Coorinator has custom configuration

Returns

On success an instance of connection interface is returned

The documentation for this class was generated from the following file:

- Trio_UnifiedApi_CPP.h

6.9 Trio::UnifiedApi::VersionInfo Struct Reference

Version info.

```
#include <Trio_UnifiedApi_CPP.h>
```

Data Fields

- uint16_t major
- uint16_t minor
- uint16_t patch
- uint16_t build

6.9.1 Detailed Description

Version info.

Definition at line 2374 of file Trio_UnifiedApi_CPP.h.

6.9.2 Field Documentation

6.9.2.1 build

```
uint16_t Trio::UnifiedApi::VersionInfo::build
```

Definition at line 2379 of file Trio_UnifiedApi_CPP.h.

6.9.2.2 major

```
uint16_t Trio::UnifiedApi::VersionInfo::major
```

Definition at line 2376 of file Trio_UnifiedApi_CPP.h.

6.9.2.3 minor

```
uint16_t Trio::UnifiedApi::VersionInfo::minor
```

Definition at line 2377 of file Trio_UnifiedApi_CPP.h.

6.9.2.4 patch

```
uint16_t Trio::UnifiedApi::VersionInfo::patch
```

Definition at line 2378 of file Trio_UnifiedApi_CPP.h.

The documentation for this struct was generated from the following file:

- Trio_UnifiedApi_CPP.h

Chapter 7

File Documentation

7.1 Trio_UnifiedApi_CPP.h File Reference

```
#include <string>
#include <functional>
#include <cstdint>
```

Data Structures

- struct Trio::UnifiedApi::VersionInfo

  Version info.
- struct Trio::UnifiedApi::FileInfo

  Data used with a callback for processing the response of Dir command.
- struct Trio::UnifiedApi::DataStreamPacket

  Data used with a callback for streaming back the response of commands that return string data.
- struct Trio::UnifiedApi::ProgressInfo

  Progress info helper.
- struct Trio::UnifiedApi::Range

  A structure representing a range of consecutive values.
- struct Trio::UnifiedApi::ProcessInfo

  Data used with a callback for processing the response of Process command.
- class Trio::UnifiedApi::TrioConnectionException

  Exception raised when error occurs in communication library.
- class Trio::UnifiedApi::TrioConnectionFactory

  Factory for creating communication instances.
- class Trio::UnifiedApi::ITrioConnection

  Communication object intarface.

Namespaces

- namespace Trio
- namespace Trio::UnifiedApi

Macros

- #define TRIO_DECLSPEC
- #define TRIO_CALLCONV

Typedefs

- typedef std::function< void TRIO_CALLCONV(EventType event_id, uint64_t integer_data, const std::string↩ _view &string_data) Trio::UnifiedApi::EventCallback)

  Prototype for the callback function that will be called when events happen on the underlying connection.
- typedef std::function< void TRIO_CALLCONV(const FileInfo &file, const std::string_view &file_name) Trio::UnifiedApi::FileCallback)

  Prototype of a callback for processing the response of Dir command.
- typedef std::function< void TRIO_CALLCONV(const DataStreamPacket &data_info, const char *data) Trio::UnifiedApi::StreamHandlerCallback)

  Prototype of a callback for streaming back the response of commands that return string data.
- typedef std::function< void TRIO_CALLCONV(const ProgressInfo &progress) Trio::UnifiedApi::ProgressHandlerCallback)

  Prototype of a callback for progress report.
- typedef std::function< void TRIO_CALLCONV(const ProcessInfo &process, const std::string_view &program_name) Trio::UnifiedApi::ProcessInfoCallback)
- typedef std::function< bool TRIO_CALLCONV(const std::string_view &string) Trio::UnifiedApi::StringCallback)

Enumerations

- enum class Trio::UnifiedApi::ErrorCode : uint64_t {
  Trio::UnifiedApi::NoError = 0 , Trio::UnifiedApi::_Unknown = 0xFFFFFFFF , Trio::UnifiedApi::_Fatal = 0x8000000000 , Trio::UnifiedApi::_ApiError = 0x100000000 ,
  Trio::UnifiedApi::Api_Unknown = 0 | _ApiError | _Fatal , Trio::UnifiedApi::Fail = 1 | _ApiError | _Fatal ,
  Trio::UnifiedApi::NotSupported = 2 | _ApiError , Trio::UnifiedApi::ConnectionModule = 3 | _ApiError | _Fatal ,
  Trio::UnifiedApi::ConnectionContext = 4 | _ApiError | _Fatal , Trio::UnifiedApi::TokenTable = 5 | _ApiError | _Fatal , Trio::UnifiedApi::Timeout = 6 | _ApiError | _Fatal , Trio::UnifiedApi::UnknownFloatingPointFormat = 7 | _ApiError | _Fatal ,
  Trio::UnifiedApi::Remote_TokenBuffer = 8 | _ApiError | _Fatal , Trio::UnifiedApi::UnsupportedController = 10 | _ApiError | _Fatal , Trio::UnifiedApi::BadResponse = 11 | _ApiError | _Fatal , Trio::UnifiedApi::BadArgument = 12 | _ApiError ,
  Trio::UnifiedApi::Reopen = 13 | _ApiError , Trio::UnifiedApi::BufferTooSmall = 14 | _ApiError ,
  Trio::UnifiedApi::SocketRead = 15 | _ApiError | _Fatal , Trio::UnifiedApi::SocketWrite = 16 | _ApiError | _Fatal ,
  Trio::UnifiedApi::LowLevelError = 17 | _ApiError | _Fatal , Trio::UnifiedApi::VersionMismatch = 18 | _ApiError | _Fatal , Trio::UnifiedApi::Compression = 19 | _ApiError | _Fatal , Trio::UnifiedApi::Abort = 20 | _ApiError | _Fatal ,
  Trio::UnifiedApi::Disconnected = 21 | _ApiError , Trio::UnifiedApi::InvalidParamBlock = 22 | _ApiError ,
  Trio::UnifiedApi::OutOfRange = 23 | _ApiError , Trio::UnifiedApi::CapacityExceeded = 24 | _ApiError ,
  Trio::UnifiedApi::Project = 25 | _ApiError , Trio::UnifiedApi::_ApiLast = 25 | _ApiError | _Fatal ,
  Trio::UnifiedApi::_SystemError = 0x200000000 , Trio::UnifiedApi::Sys_Unknown = (_SystemError | _↩ Unknown) ,
  Trio::UnifiedApi::_ControllerError = 0x400000000 , Trio::UnifiedApi::ControllerError_COMMAND_NOT_RECOGNISED = 1 | _ControllerError , Trio::UnifiedApi::ControllerError_INVALID_TRANSFER_TYPE = 2 | _ControllerError ,
  Trio::UnifiedApi::ControllerError_FLASH_PROGRAMMING_ERROR = 3 | _ControllerError ,
  Trio::UnifiedApi::ControllerError_OPERAND_EXPECTED = 4 | _ControllerError , Trio::UnifiedApi::ControllerError_ASSIGNMEN
  = 5 | _ControllerError , Trio::UnifiedApi::ControllerError_QUOTES_EXPECTED = 6 | _ControllerError ,
  Trio::UnifiedApi::ControllerError_STACK_OVERFLOW = 7 | _ControllerError ,
  Trio::UnifiedApi::ControllerError_TOO_MANY_VARIABLES = 8 | _ControllerError , Trio::UnifiedApi::ControllerError_DIVISION_E
  = 9 | _ControllerError , Trio::UnifiedApi::ControllerError_EXTRA_CHARS_AT_EOLN = 10 | _ControllerError ,

Trio::UnifiedApi::ControllerError_CLOSE_SQUARE_BRACKET_EXPECTED_IN_PRINT = 11 | _Controller↩
Error ,
Trio::UnifiedApi::ControllerError_CANNOT_MODIOFY_SPECIAL_PROGRAM = 12 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_THEN_EXPECTED_IN_IF_ELSEIF = 13 | _ControllerError ,
Trio::UnifiedApi::ControllerError_FLASH_ERASE_ERROR = 14 | _ControllerError , Trio::UnifiedApi::ControllerError_START_OF
= 15 | _ControllerError ,
Trio::UnifiedApi::ControllerError_CLOSE_ROUND_BRACKET_EXPECTED = 16 | _ControllerError ,
Trio::UnifiedApi::ControllerError_COMMA_EXPECTED = 17 | _ControllerError , Trio::UnifiedApi::ControllerError_COMMAND_L
= 18 | _ControllerError , Trio::UnifiedApi::ControllerError_PARAMETER_OUT_OF_RANGE = 19 | _↩
ControllerError ,
Trio::UnifiedApi::ControllerError_NO_PROCESS_AVAILABLE = 20 | _ControllerError , Trio::UnifiedApi::ControllerError_VALUE_
= 21 | _ControllerError , Trio::UnifiedApi::ControllerError_MODIFIER_NOT_ALLOWED = 22 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_REMOTE_AXIS_IS_IN_USE = 23 | _ControllerError ,
Trio::UnifiedApi::ControllerError_COMMAND_IS_COMMAND_LINE_ONLY = 24 | _ControllerError ,
Trio::UnifiedApi::ControllerError_COMMAND_IS_RUNTIME_ONLY = 25 | _ControllerError , Trio::UnifiedApi::ControllerError_LA
= 26 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_NOT_FOUND = 27 | _ControllerError ,
Trio::UnifiedApi::ControllerError_DUPLICATE_IDENTIFIER = 28 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM
= 29 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAMS_RUNNING = 30 | _ControllerError ,
Trio::UnifiedApi::ControllerError_PROGRAM_IS_STOPPED = 31 | _ControllerError ,
Trio::UnifiedApi::ControllerError_CANNOT_SELECT_PROGRAM = 32 | _ControllerError , Trio::UnifiedApi::ControllerError_NO_
= 33 | _ControllerError , Trio::UnifiedApi::ControllerError_NO_MORE_PROGRAMS_AVAILABLE = 34 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_OUT_OF_MEMORY = 35 | _ControllerError ,
Trio::UnifiedApi::ControllerError_NO_CODE_AVAILABLE_FOR_PROGRAM = 36 | _ControllerError ,
Trio::UnifiedApi::ControllerError_COMMAND_OUT_OF_CONTEXT = 37 | _ControllerError , Trio::UnifiedApi::ControllerError_TO
= 38 | _ControllerError , Trio::UnifiedApi::ControllerError_STRUCTURE_NESTING_ERROR = 39 | _↩
ControllerError ,
Trio::UnifiedApi::ControllerError_ELSE_ELSEIF_ENDIF_WITHOUT_PREVIOUS_IF = 40 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_WEND_WITHOUT_PREVIOUS_WHILE = 41 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_UNTIL_WITHOUT_PREVIOUS_REPEAT = 42 | _ControllerError
, Trio::UnifiedApi::ControllerError_IDENTIFIER_EXPECTED = 43 | _ControllerError ,
Trio::UnifiedApi::ControllerError_TO_EXPECTED_AFTER_FOR = 44 | _ControllerError , Trio::UnifiedApi::ControllerError_TOO_
= 45 | _ControllerError , Trio::UnifiedApi::ControllerError_NEXT_WITHOUT_FOR = 46 | _ControllerError ,
Trio::UnifiedApi::ControllerError_UNTIL_IDLE_LOADED_EXPECTED_AFTER_WAIT = 47 | _ControllerError
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Trio::UnifiedApi::ControllerError_GOTO_GOSUB_EXEC_EXPECTED = 48 | _ControllerError , Trio::UnifiedApi::ControllerError_
= 49 | _ControllerError , Trio::UnifiedApi::ControllerError_RETURN_WITHOUT_GOSUB = 50 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_LABEL_MUST_BE_AT_START_OF_LINE = 51 | _ControllerError ,
Trio::UnifiedApi::ControllerError_CANNOT_NEST_ONE_LINE_IF = 52 | _ControllerError , Trio::UnifiedApi::ControllerError_LAB
= 53 | _ControllerError , Trio::UnifiedApi::ControllerError_LINE_NUMBER_CANNOT_HAVE_DECIMAL_POINT
= 54 | _ControllerError , Trio::UnifiedApi::ControllerError_MAXIMUM_NUMBER_OF_PROGRAM_INSTANCES_ACTIVE
= 55 | _ControllerError ,
Trio::UnifiedApi::ControllerError_INVALID_USE_OF_DOLAR_SIGN = 56 | _ControllerError , Trio::UnifiedApi::ControllerError_V
= 57 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_ALREADY_EXISTS = 58 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_PROCESS_ALREADY_SELECTED = 59 | _ControllerError
,
Trio::UnifiedApi::ControllerError_DUPLICATE_AXES_NOT_PERMITTED = 60 | _ControllerError , Trio::UnifiedApi::ControllerErr
= 61 | _ControllerError , Trio::UnifiedApi::ControllerError_EVALUATION_ERROR = 62 | _ControllerError ,
Trio::UnifiedApi::ControllerError_RESERVED_KEYWORD_NOT_AVAILABLE_ON_THIS_CONTROLLER =
63 | _ControllerError ,
Trio::UnifiedApi::ControllerError_VARIABLE_NOT_FOUND = 64 | _ControllerError , Trio::UnifiedApi::ControllerError_TABLE_IN
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Trio::UnifiedApi::ControllerError_STRING_EXCEEDS_PERMITTED_LENGTH = 68 | _ControllerError ,
Trio::UnifiedApi::ControllerError_SCOPE_PERIOD_SHOULD_EXCEED_NUMBER_OF_AIN_PARAMS =
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Trio::UnifiedApi::ControllerError_FLASH_VALUE_CANNOT_BE_SET = 72 | _ControllerError , Trio::UnifiedApi::ControllerError_
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ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_NOT_RUNNING_ON_THIS_PROCESS = 75 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_NOT_RUNNING = 76 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGR = 77 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_NOT_PAUSED = 78 | _ControllerError , Trio::UnifiedApi::ControllerError_COMMAND_NOT_ALLOWED_WHEN_RUNNING_MOTION_PERFECT = 79 | _ControllerError , Trio::UnifiedApi::ControllerError_DIRECTORY_STRUCTURE_INVALID = 80 | _ControllerError , Trio::UnifiedApi::ControllerError = 81 | _ControllerError , Trio::UnifiedApi::ControllerError_CANNOT_EDIT_PROGRAM = 82 | _ControllerError , Trio::UnifiedApi::ControllerError_TOO_MANY_NESTED_OPERANDS = 83 | _ControllerError , Trio::UnifiedApi::ControllerError_CANNOT_RESET_WHEN_DRIVE_SERVO_ON = 84 | _ControllerError , Trio::UnifiedApi::ControllerError_FLASH_STICK_BLANK = 85 | _ControllerError , Trio::UnifiedApi::ControllerError_FLASH_STIC = 86 | _ControllerError 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Trio::UnifiedApi::ControllerError_DEVICE_ERROR__ERROR_WRITING_TO_DEVICE = 122 | _↩ ControllerError , Trio::UnifiedApi::ControllerError_DEVICE_ERROR__INVALID_RESPONSE_FROM_DEVICE = 123 | _ControllerError , Trio::UnifiedApi::ControllerError_FIRMWARE_ERROR__CANNOT_REFERENCE_DATA_OUTSIDE_CURRENT_BLOCK = 124 | _ControllerError , Trio::UnifiedApi::ControllerError_DISK_ERROR__INVALID_MBR = 125 | _↩ ControllerError , Trio::UnifiedApi::ControllerError_DISK_ERROR__INVALID_BOOT_SECTOR = 126 | _↩ ControllerError , Trio::UnifiedApi::ControllerError_DISK_ERROR__INVALID_SECTOR_CLUSTER_REFERENCE = 127 | _ControllerError , Trio::UnifiedApi::ControllerError_FILE_ERROR__DISK_FULL = 128 | _ControllerError , Trio::UnifiedApi::ControllerError_FILE_E = 129 | _ControllerError , Trio::UnifiedApi::ControllerError_FILE_ERROR__FILENAME_ALREADY_EXISTS = 130 | _ControllerError , Trio::UnifiedApi::ControllerError_FILE_ERROR__INVALID_FILENAME = 131 | _ControllerError , Trio::UnifiedApi::ControllerError_FILE_ERROR__DIRECTORY_FULL = 132 | _ControllerError , Trio::UnifiedApi::ControllerError = 133 | _ControllerError , Trio::UnifiedApi::ControllerError_HASH_EXPECTED = 134 | _ControllerError ,

Trio::UnifiedApi::ControllerError_FOR_EXPECTED = 135 | _ControllerError ,
Trio::UnifiedApi::ControllerError_INPUT__OUTPUT__APPEND__FIFO_READ__FIFO_WRITE_EXPECTED
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Trio::UnifiedApi::ControllerError_END_OF_FILE = 138 | _ControllerError , Trio::UnifiedApi::ControllerError_FILE_ALREADY_OP
= 139 | _ControllerError ,
Trio::UnifiedApi::ControllerError_INVALID_STORAGE_AREA = 140 | _ControllerError , Trio::UnifiedApi::ControllerError_INVALI
= 141 | _ControllerError , Trio::UnifiedApi::ControllerError_INVALID_SYSTEM_CODE_WRONG_CONTROLLER
= 142 | _ControllerError , Trio::UnifiedApi::ControllerError_IEC_ERROR__INVALID_VARIABLE_ACCESS =
143 | _ControllerError ,
Trio::UnifiedApi::ControllerError_NUMERICAL_ERROR__NAN_USED = 144 | _ControllerError , Trio::UnifiedApi::ControllerErro
= 145 | _ControllerError , Trio::UnifiedApi::ControllerError_NUMERICAL_ERROR__SUBNORMAL_VALUE_USED
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ControllerError ,
Trio::UnifiedApi::ControllerError_INVALID_MIX_OF_DATA_TYPES = 148 | _ControllerError , Trio::UnifiedApi::ControllerError_IN
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ControllerError , Trio::UnifiedApi::ControllerError_KINEMATICS_RUNTIME_LIMITED_TO_1_HOUR_FEC_22
= 151 | _ControllerError ,
Trio::UnifiedApi::ControllerError_IEC_RUNTIME_LIMITED_TO_1_HOUR_FEC_21 = 152 | _ControllerError
, Trio::UnifiedApi::ControllerError_COMMAND_NOT_ALLOWED_WITH_CURRENT_ATYPE = 153 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_WILDCARD_LENGTH_MUST_BE_1 = 154 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_INCOMPATIBLE_ARRAY_DIMENSIONS = 155 | _ControllerError ,
Trio::UnifiedApi::ControllerError_MATRIX_IS_SINGULAR = 156 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM
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158 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_IS_STOPPED_HALT_FORCED = 159
| _ControllerError ,
Trio::UnifiedApi::ControllerError_VARIABLE_NOT_DEFINED_USING_DIM_OR_GLOBAL_CONSTANT =
160 | _ControllerError , Trio::UnifiedApi::ControllerError_VARIABLE_NOT_DEFINED_USING_DIM_WITHIN_INCLUDE_FILE
= 161 | _ControllerError , Trio::UnifiedApi::ControllerError_PARAMETER_CAN_ONLY_BE_CHANGED_USING_MC_CONFIG
= 162 | _ControllerError , Trio::UnifiedApi::ControllerError_IEC_TASK_MEMORY_ALLOCATION_ERROR =
163 | _ControllerError ,
Trio::UnifiedApi::ControllerError_IEC_IO_BINDING_INDEX_OUT_OF_RANGE = 164 | _ControllerError ,
Trio::UnifiedApi::ControllerError_IEC_VR_BINDING_INDEX_OUT_OF_RANGE = 165 | _ControllerError ,
Trio::UnifiedApi::ControllerError_IEC_TABLE_BINDING_INDEX_OUT_OF_RANGE = 166 | _ControllerError
, Trio::UnifiedApi::ControllerError_FAILED_OPEN_CONNECTION = 167 | _ControllerError ,
Trio::UnifiedApi::ControllerError_MCCONFIG_CHANGED_EX_REQUIRED = 168 | _ControllerError ,
Trio::UnifiedApi::ControllerError_NO_RESPONSE_FROM_CAN_MODULE = 169 | _ControllerError ,
Trio::UnifiedApi::ControllerError_INVALID_FPGA_VERSION = 170 | _ControllerError , Trio::UnifiedApi::ControllerError_VARIAB
= 171 | _ControllerError ,
Trio::UnifiedApi::ControllerError_PARAMETER_CAN_ONLY_BE_CHANGED_USING_MC_CONFIG_OR_COMMAND_LINE
= 172 | _ControllerError , Trio::UnifiedApi::ControllerError_ETHERCAT_AXES_CANNOT_BE_SET_USING_ATYPE_PLEASE_U
= 173 | _ControllerError , Trio::UnifiedApi::ControllerError_BIT_NUMBER_OUT_OF_RANGE = 174 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_FUNCTION_NAME_EXPECTED = 175 | _ControllerError ,
Trio::UnifiedApi::ControllerError_FUNCTION_MUST_BE_DEFINED_WITHIN_A_BASIC_LIBRARY = 176
| _ControllerError , Trio::UnifiedApi::ControllerError_TOO_MANY_FUNCTIONS_DEFINED = 177 | ↩
_ControllerError , Trio::UnifiedApi::ControllerError_MISSING_ENDFUNC = 178 | _ControllerError ,
Trio::UnifiedApi::ControllerError_ENDFUNC_WITHOUT_PREVIOUS_FUNCTION = 179 | _ControllerError ,
Trio::UnifiedApi::ControllerError_PARAMETER_NAME_EXPECTED = 180 | _ControllerError , Trio::UnifiedApi::ControllerError_I
= 181 | _ControllerError , Trio::UnifiedApi::ControllerError_DEVICE_ERROR_SDCARD_WRITE_PROTECTED
= 182 | _ControllerError , Trio::UnifiedApi::ControllerError_FAILED_ATTEMPT_1 = 183 | _ControllerError ,
Trio::UnifiedApi::ControllerError_FAILED_ATTEMPT_2 = 184 | _ControllerError , Trio::UnifiedApi::ControllerError_FAILED_TOO
= 185 | _ControllerError , Trio::UnifiedApi::ControllerError_LINE_MUST_BE_INSIDE_FUNCTION = 186 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_DUPLICATE_NAME_USED_IN_LABEL_AND_VARIABLE
= 187 | _ControllerError ,
Trio::UnifiedApi::ControllerError_STRUCTURE_FIELD_NAME_NOT_RECOGNISED = 188 | _ControllerError
, Trio::UnifiedApi::ControllerError_ID_NOT_DEFINED = 189 | _ControllerError , Trio::UnifiedApi::ControllerError_ROBOT_CONF
= 190 | _ControllerError , Trio::UnifiedApi::ControllerError_TOOL_OFFSET_ID_NOT_DEFINED = 191 | _↩
ControllerError ,
Trio::UnifiedApi::ControllerError_OBJECT_FRAME_ID_NOT_DEFINED = 192 | _ControllerError , Trio::UnifiedApi::ControllerErr

= 193 | _ControllerError , Trio::UnifiedApi::ControllerError_COMMAND_NOT_PERMITTED_IN_FUNCTION
= 194 | _ControllerError , Trio::UnifiedApi::ControllerError_TOO_MANY_FUNCTION_PARAMETERS = 195 |
_ControllerError ,
Trio::UnifiedApi::ControllerError_OPERATION_CANNOT_BE_PERFORMED_UNTIL_IDLE = 196 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_SINGULARITY = 197 | _ControllerError , Trio::UnifiedApi::ControllerError_TOO
= 198 | _ControllerError , Trio::UnifiedApi::ControllerError_OBJECT_FRAME_CANNOT_BE_MODIFIED_OR_DELETED_WHIL
= 199 | _ControllerError ,
Trio::UnifiedApi::ControllerError_ROBOT_FRAME_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED
= 200 | _ControllerError , Trio::UnifiedApi::ControllerError_ROBOT_MOVES_CANNOT_BE_PERFORMED_UNTIL_JOG_OPER
= 201 | _ControllerError , Trio::UnifiedApi::ControllerError_CYCLIC_INCLUDE_DEPENDENCY = 202 | ↩
_ControllerError , Trio::UnifiedApi::ControllerError_OBJECT_FRAME_ALREADY_ATTACHED = 203 | _↩
ControllerError ,
Trio::UnifiedApi::ControllerError_OBJECT_FRAME_HAS_NOT_BEEN_ATTACHED = 204 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_OBJECT_FRAME_ALREADY_SYNCED = 205 | _ControllerError
, Trio::UnifiedApi::ControllerError_BASE_AXIS_DOES_NOT_BELONG_TO_ANY_KINEMATIC_GROUP =
206 | _ControllerError , Trio::UnifiedApi::ControllerError_SYNC_IN_PROCESS = 207 | _ControllerError ,
Trio::UnifiedApi::ControllerError_ROBOT_PROGRAM_STOPPED = 208 | _ControllerError , Trio::UnifiedApi::ControllerError_TA
= 209 | _ControllerError , Trio::UnifiedApi::ControllerError_PROGRAM_FAILED_TO_RUN = 210 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_RETURN_STATEMENT_EXPECTED_BEFORE_ENDFUNC
= 211 | _ControllerError ,
Trio::UnifiedApi::ControllerError_CONTINUE_WITHOUT_PREVIOUS_FOR_WHILE_OR_REPEAT = 212 |
_ControllerError , Trio::UnifiedApi::ControllerError_EXIT_WITHOUT_PREVIOUS_FOR_WHILE_OR_REPEAT
= 213 | _ControllerError , Trio::UnifiedApi::ControllerError_TOO_MANY_EXIT_STATEMENTS_WITHIN_SINGLE_BLOCK
= 214 | _ControllerError , Trio::UnifiedApi::ControllerError_CASE_OR_END_CASE_WITHOUT_PREVIOUS_SELECT_CASE
= 215 | _ControllerError ,
Trio::UnifiedApi::ControllerError_FIXED_INTEGER_OR_ELSE_EXPECTED = 216 | _ControllerError ,
Trio::UnifiedApi::ControllerError_TOO_MANY_CASE_OPTIONS = 217 | _ControllerError , Trio::UnifiedApi::ControllerError_ROB
= 218 | _ControllerError , Trio::UnifiedApi::ControllerError_COLLISION_OBJECTS_CANNOT_BE_MODIFIED_OR_DELETED_
= 219 | _ControllerError ,
Trio::UnifiedApi::ControllerError_TOO_MANY_COLLISION_OBJECTS_ACTIVE = 220 | _ControllerError ,
Trio::UnifiedApi::ControllerError_ATYPE_MUST_BE_CONFIGURED_USING_MC_CONFIG_PARAMETER_DRIVE_MODE
= 221 | _ControllerError , Trio::UnifiedApi::ControllerError_COMMAND_NOT_ALLOWED_IN_FRAME = 222
| _ControllerError , Trio::UnifiedApi::ControllerError_COMMAND_LINE_TOO_LONG = 223 | _ControllerError
,
Trio::UnifiedApi::ControllerError_TABLE_LOCATION_WRITE_PROTECTED = 224 | _ControllerError ,
Trio::UnifiedApi::ControllerError_ENCRYPTION_AUTHORITY_NOT_FOUND = 225 | _ControllerError
, Trio::UnifiedApi::ControllerError_NO_FREE_ENCRYPTION_AUTHORITY = 226 | _ControllerError ,
Trio::UnifiedApi::ControllerError_LOCAL_TARGET_VALUE_HAS_NOT_BEEN_DEFINED = 227 | _↩
ControllerError ,
Trio::UnifiedApi::ControllerError_COMMAND_NOT_PERMITTED_IN_BASIC_LIBRARY = 228 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_INVALID_MOVEC_INPUT = 229 | _ControllerError ,
Trio::UnifiedApi::ControllerError_FUNCTION_EXPECTED = 230 | _ControllerError , Trio::UnifiedApi::ControllerError_CAM_TAB
= 231 | _ControllerError ,
Trio::UnifiedApi::ControllerError_RUNNING_OF_RPS_PROGRAMS_INHIBITED_USE_DISABLE_ROBOT =
232 | _ControllerError , Trio::UnifiedApi::ControllerError_U_AND_V_ORIENTATION_INHIBITED_WHILE_WRIST_SINGULARIT
= 233 | _ControllerError , Trio::UnifiedApi::ControllerError_TARGET_POINT_OUT_OF_ROBOT_WORKING_AREA
= 234 | _ControllerError , Trio::UnifiedApi::ControllerError_AXIS_LOCKED_TO_A_ROBOT = 235 | _↩
ControllerError ,
Trio::UnifiedApi::ControllerError_ROBOT_CONTROLLED_BY_ANOTHER_PROCESS = 236 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_ROBOT_NOT_DEFINED_USING_KINEMATIC_GROUP = 237 | _↩
ControllerError , Trio::UnifiedApi::ControllerError_COMMAND_INHIBITED_ROBOT_MUST_BE_USED_TO_ACTIVATE_A_KIN
= 238 | _ControllerError , Trio::UnifiedApi::ControllerError_UNDEFINED_VARIABLE_IN_DIM_INITIALISER
= 239 | _ControllerError ,
Trio::UnifiedApi::ControllerError_THE_SPECIFIED_ROBOT_CONFIGURATION_IS_NOT_ALLOWED = 240
| _ControllerError , Trio::UnifiedApi::ControllerError_TOO_MANY_AXES_ACTIVATED = 241 | _Controller↩
Error , Trio::UnifiedApi::ControllerError_MULTIPLE_MOVE_TRIGGERS_SPECIFIED = 242 | _ControllerError
, Trio::UnifiedApi::ControllerError_MOVE_TRIGGER_RESPONSE_NOT_SPECIFIED = 243 | _Controller↩
Error ,

Trio::UnifiedApi::ControllerError_MOVE_TRIGGER_NOT_SPECIFIED = 244 | _ControllerError , Trio::UnifiedApi::ControllerError = 245 | _ControllerError , Trio::UnifiedApi::ControllerError_DUPLICATE_VR_INDEX = 246 | _ControllerError , Trio::UnifiedApi::ControllerError_CONSTANT_VALUE_EXPECTED = 247 | _ControllerError , Trio::UnifiedApi::ControllerError_RPS_KERNEL_V2_NOT_ENABLED = 248 | _ControllerError , Trio::UnifiedApi::ControllerError = 249 | _ControllerError , Trio::UnifiedApi::ControllerError_TARGET_TAUGHT_WITH_ANOTHER_ROBOT = 250 | _ControllerError , Trio::UnifiedApi::ControllerError_TARGET_ROBOT_CONFIG_MISMATCH = 251 | _ControllerError , Trio::UnifiedApi::ControllerError_TARGET_TOOL_OFFSET_MISMATCH = 252 | _ControllerError , Trio::UnifiedApi::ControllerError_TARGET_OBJECT_FRAME_MISMATCH = 253 | _ControllerError , Trio::UnifiedApi::ControllerError_TARGET_ROBOT_FRAME_MISMATCH = 254 | _ControllerError , Trio::UnifiedApi::ControllerError_INVALID_FRAME_NUMBER = 255 | _ControllerError , Trio::UnifiedApi::ControllerError_FRAME_NOT_SUPPORTED_BY_RPS = 256 | _ControllerError , Trio::UnifiedApi::ControllerError_FORWARD_TRANSFORM_FUNCTION_NOT_DEFINED = 257 | _ ↩ ControllerError , Trio::UnifiedApi::ControllerError_INVERSE_TRANSFORM_FUNCTION_NOT_DEFINED = 258 | _ControllerError , Trio::UnifiedApi::ControllerError_INVALID_MOVEOPTION = 259 | _ControllerError , Trio::UnifiedApi::ControllerError_MAX_OPEN_KEYWORDS_REACHED = 260 | _ControllerError , Trio::UnifiedApi::ControllerError_KEYWORD_INACTIVE = 261 | _ControllerError , Trio::UnifiedApi::ControllerError_KEYWORD = 262 | _ControllerError , Trio::UnifiedApi::ControllerError_KEYWORD_SAME_AS_SYSTEM_WORD = 263 | _ControllerError , Trio::UnifiedApi::ControllerError_UNEXPECTED_CHARACTERS_AFTER_LINE_CONTINUATION = 264 | _ControllerError , Trio::UnifiedApi::ControllerError_THIS_COMMAND_CAN_BE_EXECUTED_ONLY_ONCE = 265 | _ControllerError , Trio::UnifiedApi::ControllerError_TABLE_MEMORY_CONFLICT = 266 | _ ↩ ControllerError , Trio::UnifiedApi::ControllerError_MAX_OPEN_MOVES_REACHED = 267 | _ControllerError , Trio::UnifiedApi::ControllerError_USER_DEFINED_ERROR = 268 | _ControllerError , Trio::UnifiedApi::ControllerError_INVALID = 269 | _ControllerError , Trio::UnifiedApi::ControllerError_OPEN_MOVE_FUNCTIONS_NOT_DEFINED = 270 | _ControllerError , Trio::UnifiedApi::ControllerError_TOO_MANY_CHARS_IN_STR = 271 | _Controller ↩ Error , Trio::UnifiedApi::ControllerError_INVALID_OPEN_ARCH_FUNC_DEFINITION = 272 | _ControllerError , Trio::UnifiedApi::ControllerError_PROCESS_IN_USE = 273 | _ControllerError , Trio::UnifiedApi::ControllerError_VR_LOCATION = 274 | _ControllerError }

Enumeration of error codes used in Unified API.

- enum class Trio::UnifiedApi::EventType { Trio::UnifiedApi::Error , Trio::UnifiedApi::Warning , Trio::UnifiedApi::Message }

  Events that may occur on a connection.
- enum class Trio::UnifiedApi::ConnectionType : uint32_t { Trio::UnifiedApi::Unknown = 0x00 , Trio::UnifiedApi::TCP = 0x01 , Trio::UnifiedApi::PCMCAT = 0x02 , Trio::UnifiedApi::FLEX7 = 0x04 }

  Connection types.
- enum class Trio::UnifiedApi::ProgressEvent { Trio::UnifiedApi::Start , Trio::UnifiedApi::Progress , Trio::UnifiedApi::Finish }

  Progress event.
- enum class Trio::UnifiedApi::Co_ObjectType { Trio::UnifiedApi::Boolean = 1 , Trio::UnifiedApi::Integer8 = 2 , Trio::UnifiedApi::Integer16 = 3 , Trio::UnifiedApi::Integer32 = 4 , Trio::UnifiedApi::Unsigned8 = 5 , Trio::UnifiedApi::Unsigned16 = 6 , Trio::UnifiedApi::Unsigned32 = 7 , Trio::UnifiedApi::Real32 = 8 , Trio::UnifiedApi::Real64 = 17 , Trio::UnifiedApi::Integer64 = 21 , Trio::UnifiedApi::Unsigned64 = 27 }

  Co Read/Write operation object types.
- enum class Trio::UnifiedApi::EthercatState { Trio::UnifiedApi::Initial = 0 , Trio::UnifiedApi::PreOperational = 1 , Trio::UnifiedApi::SafeOperational = 2 , Trio::UnifiedApi::Operational = 3 }

  EtherCAT state.
- enum class Trio::UnifiedApi::EthercatStartOption : uint32_t { Trio::UnifiedApi::None = 0x0000 , Trio::UnifiedApi::PrintOutProgress = 0x0001 , Trio::UnifiedApi::ForceRescan = 0x0002 , Trio::UnifiedApi::ExpressStartup = 0x0004 }

  EtherCAT start option.
- enum class Trio::UnifiedApi::FileTransferOption : uint32_t { Trio::UnifiedApi::None = 0x0000 , Trio::UnifiedApi::CompressOnly = 0x0001 , Trio::UnifiedApi::CompressDecompress = 0x0002 }

File transfer option.

- enum class Trio::UnifiedApi::FileType : int32_t {
  Trio::UnifiedApi::Undefined = -1 , Trio::UnifiedApi::BASIC = 0 , Trio::UnifiedApi::EncryptedBASIC = 1 , Trio::UnifiedApi::GCode = 2 ,
  Trio::UnifiedApi::TextFile = 3 , Trio::UnifiedApi::IecProgram = 4 , Trio::UnifiedApi::FifoFile = 5 , Trio::UnifiedApi::TempFile = 6 ,
  Trio::UnifiedApi::Project = 7 , Trio::UnifiedApi::Binary = 8 , Trio::UnifiedApi::MCConfig = 9 , Trio::UnifiedApi::IecLibrary = 10 ,
  Trio::UnifiedApi::HmiDesign = 11 , Trio::UnifiedApi::BASICLibrary = 12 , Trio::UnifiedApi::EncryptedBASICibrary = 13 , Trio::UnifiedApi::HmiLibrary = 14 ,
  Trio::UnifiedApi::TempVolatileFile = 15 , Trio::UnifiedApi::Robot = 16 , Trio::UnifiedApi::RobotBasic = 17 , Trio::UnifiedApi::TempTextFile = 18 ,
  Trio::UnifiedApi::RobotLibrary = 19 , Trio::UnifiedApi::RobotBasicLibrary = 20 , Trio::UnifiedApi::Recipe = 21 , Trio::UnifiedApi::Alarms = 22 ,
  Trio::UnifiedApi::RealtimeBASICLibrary = 23 , Trio::UnifiedApi::EncryptedRealtimeBasicLibrary = 24 , Trio::UnifiedApi::EncryptedIecProgram = 25 , Trio::UnifiedApi::EncryptedIecLibrary = 26 ,
  Trio::UnifiedApi::__Size }

  File types on the Motion Coordinator.

- enum class Trio::UnifiedApi::FileFlags : uint32_t { Trio::UnifiedApi::None = 0x00 , Trio::UnifiedApi::Editable = 0x01 , Trio::UnifiedApi::Autorun = 0x02 }

  Flags related to files and programs in internal storage on the Motion Coordinator.

- enum class Trio::UnifiedApi::ConnectMode : uint32_t { Trio::UnifiedApi::MPOS = 0 , Trio::UnifiedApi::DPOS = 1 }

  Mode for CONNECT command.

- enum class Trio::UnifiedApi::PSwitchMode : uint32_t { Trio::UnifiedApi::None = 0x00 , Trio::UnifiedApi::Invert = 0x02 , Trio::UnifiedApi::DPOS = 0x04 , Trio::UnifiedApi::Mask = Invert | DPOS }

  Flags PSWITCH command in single/cyclic mode.

- enum class Trio::UnifiedApi::PSwitchMultiMode : uint32_t {
  Trio::UnifiedApi::None = 0x00 , Trio::UnifiedApi::DPOS = 0x04 , Trio::UnifiedApi::TimerMode = 0x10 , Trio::UnifiedApi::NegativeDirection = 0x20 ,
  Trio::UnifiedApi::ZOut = 0x40 , Trio::UnifiedApi::Mask = DPOS | TimerMode | NegativeDirection | ZOut }

  Flags PSWITCH command in multi/sequence mode.

- enum class Trio::UnifiedApi::CamBoxLinkOptions : uint32_t {
  Trio::UnifiedApi::None = 0 , Trio::UnifiedApi::LinkMark = 1 , Trio::UnifiedApi::LinkAbsolute = 2 , Trio::UnifiedApi::LinkRepeat = 4 ,
  Trio::UnifiedApi::LinkPattern = 8 , Trio::UnifiedApi::LinkPositiveMove = 32 , Trio::UnifiedApi::LinkForceStart = 128 , Trio::UnifiedApi::LinkMarkB = 256 ,
  Trio::UnifiedApi::LinkR_Mark = 512 , Trio::UnifiedApi::LinkThreshold = 16384 }

  Link options in CamBox command.

- enum class Trio::UnifiedApi::AxisType : uint16_t {
  Trio::UnifiedApi::None = 0 , Trio::UnifiedApi::Stepper = 1 , Trio::UnifiedApi::Servo = 2 , Trio::UnifiedApi::Encoder = 3 ,
  Trio::UnifiedApi::StepperEncoderInt = 4 , Trio::UnifiedApi::Resolver = 5 , Trio::UnifiedApi::VOut = 6 , Trio::UnifiedApi::SSI = 7 ,
  Trio::UnifiedApi::CAN = 8 , Trio::UnifiedApi::CAN_Remote = 9 , Trio::UnifiedApi::PSwitch = 10 , Trio::UnifiedApi::DLink_Remote = 11 ,
  Trio::UnifiedApi::Servo2 = 12 , Trio::UnifiedApi::Yaskawa = 13 , Trio::UnifiedApi::EncoderOut = 14 , Trio::UnifiedApi::HighestMotionType = EncoderOut ,
  Trio::UnifiedApi::Dummy15 = 15 , Trio::UnifiedApi::Dummy16 = 16 , Trio::UnifiedApi::Dummy17 = 17 , Trio::UnifiedApi::DS402PosType1 = 18 ,
  Trio::UnifiedApi::DS402VelType1 = 19 , Trio::UnifiedApi::First_Comms = 20 , Trio::UnifiedApi::CAN_Comms = First_Comms , Trio::UnifiedApi::USB_Comms = 21 ,
  Trio::UnifiedApi::DLink_Comms = 22 , Trio::UnifiedApi::Profibus_Comms = 23 , Trio::UnifiedApi::Dummy24 = 24 , Trio::UnifiedApi::Dummy25 = 25 ,
  Trio::UnifiedApi::DS402PosType2 = 26 , Trio::UnifiedApi::DS402VelType2 = 27 , Trio::UnifiedApi::Dummy28 = 28 , Trio::UnifiedApi::Dummy29 = 29 ,
  Trio::UnifiedApi::DACout_ANALin = 30 , Trio::UnifiedApi::ANALout_ANALin = 31 , Trio::UnifiedApi::Dummy32

= 32 , Trio::UnifiedApi::Dummy33 = 33 ,
Trio::UnifiedApi::Dummy34 = 34 , Trio::UnifiedApi::SDout_PWMout = 35 , Trio::UnifiedApi::SDout_PWMout2
= 36 , Trio::UnifiedApi::PWMout_SDFREQin = 37 ,
Trio::UnifiedApi::PWMout_SDin = 37 , Trio::UnifiedApi::Dummy38 = 38 , Trio::UnifiedApi::Dummy39 = 39 ,
Trio::UnifiedApi::MechatroPos = 40 ,
Trio::UnifiedApi::MechatroSpeed = 41 , Trio::UnifiedApi::MechatroTorque = 42 , Trio::UnifiedApi::SDout_Zout
= 43 , Trio::UnifiedApi::DACout_QUADin = 44 ,
Trio::UnifiedApi::QUADout_Zout = 45 , Trio::UnifiedApi::DACout_TAMAin = 46 , Trio::UnifiedApi::DACout_ENDATin
= 47 , Trio::UnifiedApi::DACout_SSIin = 48 ,
Trio::UnifiedApi::MechatroInv = 49 , Trio::UnifiedApi::PanasonicPos = 50 , Trio::UnifiedApi::PanasonicSpeed
= 51 , Trio::UnifiedApi::PanasonicTorque = 52 ,
Trio::UnifiedApi::SercosSpeed = 53 , Trio::UnifiedApi::SercosPos = 54 , Trio::UnifiedApi::SercosTorque = 55 ,
Trio::UnifiedApi::SercosOpen = 56 ,
Trio::UnifiedApi::SercosSpeedReg = 57 , Trio::UnifiedApi::SercosPosReg = 58 , Trio::UnifiedApi::SercosEncoder
= 59 , Trio::UnifiedApi::DACout_SDin = 60 ,
Trio::UnifiedApi::SLM32 = 61 , Trio::UnifiedApi::PLM32 = 62 , Trio::UnifiedApi::SDout_Zin = 63 ,
Trio::UnifiedApi::QUADout_Zin = 64 ,
Trio::UnifiedApi::EthercatPos = 65 , Trio::UnifiedApi::EthercatSpeed = 66 , Trio::UnifiedApi::EthercatTorque =
67 , Trio::UnifiedApi::EthercatOpenSpeed = 68 ,
Trio::UnifiedApi::EthercatEncoderIn = 69 , Trio::UnifiedApi::EthercatInAnalOut = 70 , Trio::UnifiedApi::EthercatPosTrio
= 71 , Trio::UnifiedApi::EthercatSpeedTrio = 72 ,
Trio::UnifiedApi::EthercatDummy5 = 73 , Trio::UnifiedApi::EthercatDummy6 = 74 , Trio::UnifiedApi::DACout_SSISLAVEin
= 75 , Trio::UnifiedApi::QUADin = 76 ,
Trio::UnifiedApi::DACout_QUADin_Zout = 77 , Trio::UnifiedApi::SDout_VFFDACout = 78 , Trio::UnifiedApi::SDin
= 79 , Trio::UnifiedApi::G5Pos = 80 ,
Trio::UnifiedApi::G5Speed = 81 , Trio::UnifiedApi::G5Torque = 82 , Trio::UnifiedApi::Jacquard = 83 ,
Trio::UnifiedApi::QUADout_VFFDACout = 84 ,
Trio::UnifiedApi::AxesDiff = 85 , Trio::UnifiedApi::TAMAin = 86 , Trio::UnifiedApi::ENDATin = 87 ,
Trio::UnifiedApi::SSIin = 88 ,
Trio::UnifiedApi::SDFREQout_ANALin = 89 , Trio::UnifiedApi::DACout_SDFREQin = 90 , Trio::UnifiedApi::AFFDACout
= 91 , Trio::UnifiedApi::DACout_BISSin = 92 ,
Trio::UnifiedApi::BISSin = 93 , Trio::UnifiedApi::DACPWMout_QUADin = 94 , Trio::UnifiedApi::SDout_TAMAin
= 95 , Trio::UnifiedApi::Dummy96 = 96 ,
Trio::UnifiedApi::QUADin_Zout = 97 , Trio::UnifiedApi::Dummy98 = 98 , Trio::UnifiedApi::Dummy99 = 99 ,
Trio::UnifiedApi::SDout_HD = 100 ,
Trio::UnifiedApi::QUADout_HD = 101 , Trio::UnifiedApi::DACout_QUADin_HD = 102 , Trio::UnifiedApi::DACout_SDin_HD
= 103 , Trio::UnifiedApi::QUADin_HD = 104 ,
Trio::UnifiedApi::SSIin_HD = 105 , Trio::UnifiedApi::TAMAin_HD = 106 , Trio::UnifiedApi::Dummy107 = 107 ,
Trio::UnifiedApi::Dummy108 = 108 ,
Trio::UnifiedApi::Dummy109 = 109 , Trio::UnifiedApi::SDout_SDin_HD = 110 , Trio::UnifiedApi::SDout_QUADin_HD
= 111 , Trio::UnifiedApi::QUADout_SDin_HD = 112 ,
Trio::UnifiedApi::QUADout_QUADin_HD = 113 , Trio::UnifiedApi::SDout_TAMAin_HD = 114 , Trio::UnifiedApi::SDout_ENDATin_
= 115 , Trio::UnifiedApi::SDout_SSIin_HD = 116 ,
Trio::UnifiedApi::SDout_BISSin_HD = 117 , Trio::UnifiedApi::QUADout_TAMAin_HD = 118 , Trio::UnifiedApi::QUADout_ENDATin
= 119 , Trio::UnifiedApi::QUADout_SSIin_HD = 120 ,
Trio::UnifiedApi::QUADout_BISSin_HD = 121 , Trio::UnifiedApi::Dummy122 = 122 , Trio::UnifiedApi::Dummy123
= 123 , Trio::UnifiedApi::Dummy124 = 124 ,
Trio::UnifiedApi::Dummy125 = 125 , Trio::UnifiedApi::Dummy126 = 126 , Trio::UnifiedApi::Dummy127 = 127 ,
Trio::UnifiedApi::Dummy128 = 128 ,
Trio::UnifiedApi::Dummy129 = 129 , Trio::UnifiedApi::SLMvirtual = 200 , Trio::UnifiedApi::PLMvirtual = 201 }

Enumeration of known axis types used with ATYPE axis parameter.

- enum class Trio::UnifiedApi::AxisMode : uint32_t {
Trio::UnifiedApi::NONE = 0 , Trio::UnifiedApi::NOT_USED_1 = ((uint32_t)1 << 0) , Trio::UnifiedApi::ALLOW_CONNECT_MOVE
= ((uint32_t)1 << 1) , Trio::UnifiedApi::DIRECTION_MODE_3D = ((uint32_t)1 << 2) ,
Trio::UnifiedApi::CANCEL_ADDAX_ON_LIMIT = ((uint32_t)1 << 3) , Trio::UnifiedApi::DISABLE_LOOKAHEAD
= ((uint32_t)1 << 4) , Trio::UnifiedApi::NOT_USED_2 = ((uint32_t)1 << 5) , Trio::UnifiedApi::USE_NON_SIGNEXTENDED_A
= ((uint32_t)1 << 6) ,
Trio::UnifiedApi::NOT_USED_3 = ((uint32_t)1 << 7) , Trio::UnifiedApi::SOFTWARE_LIMIT_USES_DPOS =

((uint32_t)1 << 8) , Trio::UnifiedApi::DISABLE_FE_ACCUMULATION_DURING_MOVE = ((uint32_t)1 << 9) , Trio::UnifiedApi::USE_FASTDEC_WHEN_CANCELLING_RPS_JOG = ((uint32_t)1 << 10) , Trio::UnifiedApi::DELAY_POSITION_EVALUATION_TO_LOAD_TIME = ((uint32_t)1 << 11) , Trio::UnifiedApi::DIRECTION_SH = ((uint32_t)1 << 12) , Trio::UnifiedApi::MONITOR_FE = ((uint32_t)1 << 13) , Trio::UnifiedApi::OVERRIDE_FE_LIMITS = ((uint32_t)1 << 14) , Trio::UnifiedApi::DUAL_AXIS_AVERAGE_POS = ((uint32_t)1 << 15) , Trio::UnifiedApi::INVERT_AXIS_DIRECTION = ((uint32_t)1 << 16) , Trio::UnifiedApi::CANCEL_MTYPE_ON_MASTER_IF_ADDAX_HITS_LIM = ((uint32_t)1 << 17) }

Enumeration of axis mode flags used with AXIS_MODE axis parameter.

- enum class Trio::UnifiedApi::CornerMode : uint32_t { Trio::UnifiedApi::NONE = 0 , Trio::UnifiedApi::RESERVED = ((uint32_t)1 << 0) , Trio::UnifiedApi::AUTO_SPEED_CONTROL = ((uint32_t)1 << 1) , Trio::UnifiedApi::CORNER_STATE_MACHINE = ((uint32_t)1 << 2) , Trio::UnifiedApi::AUTO_RADIUS_SPEED_CONTROL = ((uint32_t)1 << 3) , Trio::UnifiedApi::AUTO_RADIUS_SPEED_CONTF = ((uint32_t)1 << 4) , Trio::UnifiedApi::AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE_USE_ACCEL = ((uint32_t)1 << 5) , Trio::UnifiedApi::AUTO_BLEND_MODE_0_USING_CORNER_RADIUS = ((uint32_t)1 << 6) , Trio::UnifiedApi::AUTO_BLEND_MODE_1_USING_CORNER_RADIUS = ((uint32_t)1 << 7) }

  Enumeration of corner mode flags used with CORNER_MODE axis parameter.

- enum class Trio::UnifiedApi::CornerState : uint32_t { Trio::UnifiedApi::NONE = 0 , Trio::UnifiedApi::HELD = ((uint32_t)1 << 0) , Trio::UnifiedApi::LOAD = ((uint32_t)1 << 1) , Trio::UnifiedApi::LOAD_ZERO_SPEED = ((uint32_t)1 << 2) }

  Enumeration of corner state flags used with CORNER_STATE axis parameter.

- enum class Trio::UnifiedApi::MotionType : uint32_t { Trio::UnifiedApi::IDLE = 0 , Trio::UnifiedApi::MOVE = 1 , Trio::UnifiedApi::MOVEABS = 2 , Trio::UnifiedApi::MHELICAL = 3 , Trio::UnifiedApi::MOVECIRC = 4 , Trio::UnifiedApi::MOVEMODIFY = 5 , Trio::UnifiedApi::MOVESP = 6 , Trio::UnifiedApi::MOVEABSSP = 7 , Trio::UnifiedApi::MOVECIRCSP = 8 , Trio::UnifiedApi::MHELICALSP = 9 , Trio::UnifiedApi::FORWARD = 10 , Trio::UnifiedApi::REVERSE = 11 , Trio::UnifiedApi::DATUM = 12 , Trio::UnifiedApi::CAM = 13 , Trio::UnifiedApi::FORWARD_JOG = 14 , Trio::UnifiedApi::REVERSE_JOG = 15 , Trio::UnifiedApi::WETCAM = 16 , Trio::UnifiedApi::CAMBOX = 20 , Trio::UnifiedApi::CONNECT = 21 , Trio::UnifiedApi::MOVELINK = 22 , Trio::UnifiedApi::CONNPATH = 23 , Trio::UnifiedApi::FLEXLINK = 24 , Trio::UnifiedApi::CAMLINK = 25 , Trio::UnifiedApi::COMPENSATE_XY = 26 , Trio::UnifiedApi::PICKLINK = 27 , Trio::UnifiedApi::CAMIN = 28 , Trio::UnifiedApi::MOVETANG = 30 , Trio::UnifiedApi::MSPHERICAL = 31 , Trio::UnifiedApi::MTRANSCIRC = 32 , Trio::UnifiedApi::MOVESEQ = 33 , Trio::UnifiedApi::MOVEABSSEQ = 34 , Trio::UnifiedApi::MOVETURN = 35 , Trio::UnifiedApi::MOVEPICK = 36 , Trio::UnifiedApi::MOVECONTOUR = 37 , Trio::UnifiedApi::MOVEDISPENSE = 38 , Trio::UnifiedApi::MSPHERICALSP = 40 , Trio::UnifiedApi::MTRANSCIRCSP = 41 , Trio::UnifiedApi::MOVESEQSP = 42 , Trio::UnifiedApi::MOVEABSSEQSP = 43 , Trio::UnifiedApi::MOVEJ = 44 , Trio::UnifiedApi::MOVEL = 45 , Trio::UnifiedApi::MOVEC = 46 , Trio::UnifiedApi::MOVEBLEND = 47 , Trio::UnifiedApi::MOVEJREL = 48 , Trio::UnifiedApi::MOVELREL = 49 , Trio::UnifiedApi::MOVETEST = 50 , Trio::UnifiedApi::MOVEP = 51 , Trio::UnifiedApi::MOVEU = 52 , Trio::UnifiedApi::MOVEB = 53 , Trio::UnifiedApi::MOVESPCONT = 60 , Trio::UnifiedApi::MOVEABSSPCONT = 61 , Trio::UnifiedApi::GEARIN = 62 , Trio::UnifiedApi::ACCPROFILE = 70 , Trio::UnifiedApi::VELPROFILE = 71 , Trio::UnifiedApi::POSPROFILE = 72 , Trio::UnifiedApi::COMBINEAXES = 73 , Trio::UnifiedApi::OPENMOVE_BASE = 100 , Trio::UnifiedApi::OPENMOVE_1 = OPENMOVE_BASE , Trio::UnifiedApi::OPENMOVE_2 = OPENMOVE_BASE + 1 , Trio::UnifiedApi::OPENMOVE_3 = OPENMOVE_BASE + 2 , Trio::UnifiedApi::OPENMOVE_4 = OPENMOVE_BASE + 3 , Trio::UnifiedApi::OPENMOVE_5 = OPENMOVE↩_BASE + 4 }

  Motion types returned by MTYPE and NTYPE axis parameters.

- enum class Trio::UnifiedApi::AxisStatus : uint64_t {
  Trio::UnifiedApi::NONE = 0 , Trio::UnifiedApi::LOOKAHEAD_OVERRIDE = ((uint64_t)1 << 0) ,
  Trio::UnifiedApi::FE_WARNING = ((uint64_t)1 << 1) , Trio::UnifiedApi::COMMS_ERROR = ((uint64_t)1
  << 2) ,
  Trio::UnifiedApi::REMOTE_DRIVE_ERROR = ((uint64_t)1 << 3) , Trio::UnifiedApi::IN_FORWARD_LIMIT =
  ((uint64_t)1 << 4) , Trio::UnifiedApi::IN_REVERSE_LIMIT = ((uint64_t)1 << 5) , Trio::UnifiedApi::DATUMING
  = ((uint64_t)1 << 6) ,
  Trio::UnifiedApi::FEEDHOLD = ((uint64_t)1 << 7) , Trio::UnifiedApi::FE_EXCEEDS_LIMIT = ((uint64_t)1
  << 8) , Trio::UnifiedApi::FORWARD_SOFTWARE_LIMIT = ((uint64_t)1 << 9) , Trio::UnifiedApi::REVERSE_SOFTWARE_LIM
  = ((uint64_t)1 << 10) ,
  Trio::UnifiedApi::CANCEL_CURRENT = ((uint64_t)1 << 11) , Trio::UnifiedApi::PULSE_OVER_SPEED =
  ((uint64_t)1 << 12) , Trio::UnifiedApi::OVERRIDE_TO_ZERO = ((uint64_t)1 << 13) , Trio::UnifiedApi::MOVE_CANCEL_DONE
  = ((uint64_t)1 << 14) ,
  Trio::UnifiedApi::VOLUME_LIMIT = ((uint64_t)1 << 15) , Trio::UnifiedApi::AXIS_FS_LIMIT = ((uint64_t)1
  << 16) , Trio::UnifiedApi::AXIS_RS_LIMIT = ((uint64_t)1 << 17) , Trio::UnifiedApi::ENCODER_OVER_I =
  ((uint64_t)1 << 18) ,
  Trio::UnifiedApi::PSWITCH_FIFO_NOT_EMPTY = ((uint64_t)1 << 19) , Trio::UnifiedApi::PSWITCH_FIFO_FULL
  = ((uint64_t)1 << 20) , Trio::UnifiedApi::KINEMATICS_1HOUR = ((uint64_t)1 << 21) , Trio::UnifiedApi::BISS_CLEAN_WARNIN
  = ((uint64_t)1 << 22) ,
  Trio::UnifiedApi::WORLD_FS_LIMIT = ((uint64_t)1 << 23) , Trio::UnifiedApi::WORLD_RS_LIMIT =
  ((uint64_t)1 << 24) , Trio::UnifiedApi::CANCEL_CURRENT_AT_SPEED = ((uint64_t)1 << 25) ,
  Trio::UnifiedApi::REMOTE_NODE_STATUS_ERROR = ((uint64_t)1 << 26) ,
  Trio::UnifiedApi::PSWITCH_FIFO_HALF_FULL = ((uint64_t)1 << 27) , Trio::UnifiedApi::MOVEADD =
  ((uint64_t)1 << 28) , Trio::UnifiedApi::CANCEL_MOVEADD = ((uint64_t)1 << 29) , Trio::UnifiedApi::REMOTE_NODE_ERROR
  = ((uint64_t)1 << 30) ,
  Trio::UnifiedApi::SINGULARITY = ((uint64_t)1 << 31) , Trio::UnifiedApi::OPEN_FRAME_ERROR =
  ((uint64_t)1 << 32) , Trio::UnifiedApi::VP_ERROR = ((uint64_t)1 << 33) , Trio::UnifiedApi::CANCEL_BUFFERED
  = ((uint64_t)1 << 34) ,
  Trio::UnifiedApi::CANCEL_BUFFERED_DECEL_INTO_BUFFERED_MOVES = ((uint64_t)1 << 35) ,
  Trio::UnifiedApi::CANCEL_BUFFERED_DECEL_WITHIN_CURRENT_MOVE = ((uint64_t)1 << 36) ,
  Trio::UnifiedApi::CANCEL_LAST_BUFFERED = ((uint64_t)1 << 37) }
- enum class Trio::UnifiedApi::RobotStatus : uint32_t {
  Trio::UnifiedApi::NONE = 0 , Trio::UnifiedApi::WORLD_FS_LIMIT = ((uint32_t)1 << 0) , Trio::UnifiedApi::WORLD_RS_LIMIT
  = ((uint32_t)1 << 1) , Trio::UnifiedApi::ROBOT_FS_LIMIT = ((uint32_t)1 << 2) ,
  Trio::UnifiedApi::ROBOT_RS_LIMIT = ((uint32_t)1 << 3) , Trio::UnifiedApi::TCP_FS_LIMIT = ((uint32_t)1
  << 4) , Trio::UnifiedApi::TCP_RS_LIMIT = ((uint32_t)1 << 5) , Trio::UnifiedApi::FE_WARNING = ((uint32↩
  _t)1 << 6) ,
  Trio::UnifiedApi::FE_EXCEEDS_LIMIT = ((uint32_t)1 << 7) , Trio::UnifiedApi::WRIST_SINGULARITY =
  ((uint32_t)1 << 8) , Trio::UnifiedApi::ALIGNMENT_SINGULARITY = ((uint32_t)1 << 9) , Trio::UnifiedApi::ELBOW_SINGULAR
  = ((uint32_t)1 << 10) ,
  Trio::UnifiedApi::MAX_SPEED_LIMIT = ((uint32_t)1 << 11) , Trio::UnifiedApi::COLLIDED = ((uint32_t)1 <<
  12) , Trio::UnifiedApi::NOT_IN_POS = ((uint32_t)1 << 13) , Trio::UnifiedApi::ELBOW_CLOSED_SINGULARITY
  = ((uint32_t)1 << 14) ,
  Trio::UnifiedApi::DISALLOW_JOG = ((uint32_t)1 << 15) }
- enum class Trio::UnifiedApi::SyncControl : int32_t {
  Trio::UnifiedApi::WAITING_TO_START = -1 , Trio::UnifiedApi::NO_SYNC = 0 , Trio::UnifiedApi::STARTING_SYNC
  = 1 , Trio::UnifiedApi::PERFORMING_SYNC = 2 ,
  Trio::UnifiedApi::SYNCHRONISED = 3 , Trio::UnifiedApi::STOPPING_SYNC = 4 , Trio::UnifiedApi::STARTING_INTERP_MOVE
  = 5 , Trio::UnifiedApi::PERFORMING_INTERP_MOVE = 6 ,
  Trio::UnifiedApi::STARTING_RESYNC = 10 , Trio::UnifiedApi::PERFORMING_RESYNC = 11 , Trio::UnifiedApi::STARTING_RES
  = 20 , Trio::UnifiedApi::PERFORMING_RESYNC_TO_USERFRAME = 21 }
- enum class Trio::UnifiedApi::RegistMode : int32_t {
  Trio::UnifiedApi::Cancel = 0 , Trio::UnifiedApi::ZAMarkRising = 1 , Trio::UnifiedApi::ZAMarkFalling = 2 ,
  Trio::UnifiedApi::RAInputRising = 3 ,
  Trio::UnifiedApi::RAInputFalling = 4 , Trio::UnifiedApi::RAInputRising_ZBMarkRising = 6 , Trio::UnifiedApi::RAInputRising_ZBMa
  = 7 , Trio::UnifiedApi::RAInputFalling_ZBMarkRising = 8 ,
  Trio::UnifiedApi::RAInputFalling_ZBMarkFalling = 9 , Trio::UnifiedApi::RAInputRising_RBInputRising = 10 ,
  Trio::UnifiedApi::RAInputRising_RBInputFalling = 11 , Trio::UnifiedApi::RAInputFalling_RBInputRising = 12 ,

Trio::UnifiedApi::RAInputFalling_RBInputFalling = 13 , Trio::UnifiedApi::ZBMarkRising = 14 , Trio::UnifiedApi::ZBMarkFalling = 15 , Trio::UnifiedApi::RBInputRising = 16 , Trio::UnifiedApi::RBInputFalling = 17 , Trio::UnifiedApi::HardwareRegistration = 20 , Trio::UnifiedApi::TimeInterpolatedRegistratio = 21 , Trio::UnifiedApi::MultiChannelRegistration = 22 , Trio::UnifiedApi::EnableFilter = 23 , Trio::UnifiedApi::DisableFilter = 24 , Trio::UnifiedApi::InsideOpenCloseWindow = 256 , Trio::UnifiedApi::OutsideOpenCloseWindow = 768 }

- enum class Trio::UnifiedApi::RegistEdge : int32_t { Trio::UnifiedApi::Rising = 0 , Trio::UnifiedApi::Falling = 1 }
- enum class Trio::UnifiedApi::RegistWindow : int32_t { Trio::UnifiedApi::No = 0 , Trio::UnifiedApi::InsideOpenCloseWin = 1 , Trio::UnifiedApi::OutsideOpenCloseWin = 2 , Trio::UnifiedApi::VirtualInputHigh = 3 , Trio::UnifiedApi::VirtualInputLow = 4 }
- enum class Trio::UnifiedApi::ProcessStatus : uint32_t { Trio::UnifiedApi::Stopped = 0 , Trio::UnifiedApi::Ready , Trio::UnifiedApi::Stepping , Trio::UnifiedApi::Paused , Trio::UnifiedApi::Pausing , Trio::UnifiedApi::Aborted , Trio::UnifiedApi::Suspend , Trio::UnifiedApi::Suspended , Trio::UnifiedApi::Sleeping }

  The execution state of the process.
- enum class Trio::UnifiedApi::FileExistsFlags : int32_t { Trio::UnifiedApi::NoFile = 0 , Trio::UnifiedApi::HasFileNotOpen = -1 , Trio::UnifiedApi::HasFileOpenForReading = -2 , Trio::UnifiedApi::HasFileOpenForWriting = -3 , Trio::UnifiedApi::HasFileOpenForReadingAndWriting = -4 }

  Enumeration of results of IS_FILE/IS_PROG command.
- enum class Trio::UnifiedApi::SecurityCodeType : int32_t { Trio::UnifiedApi::Controller = 0 , Trio::UnifiedApi::OEM = 1 , Trio::UnifiedApi::User = 2 }

  The security code type used with project decryption key functions.

7.1.1 Macro Definition Documentation

7.1.1.1 TRIO_CALLCONV

```
#define TRIO_CALLCONV
```

Definition at line 25 of file Trio_UnifiedApi_CPP.h.

7.1.1.2 TRIO_DECLSPEC

```
#define TRIO_DECLSPEC
```

Definition at line 24 of file Trio_UnifiedApi_CPP.h.

7.2 Trio_UnifiedApi_CPP.h

Go to the documentation of this file.

```
00001 ////////////////////////////////////////////////////////////////////////////////////////////
00002 // Project   : UnifiedApi
00003 // Copyright : Trio Motion Technology 2024
00004 ////////////////////////////////////////////////////////////////////////////////////////////
00005
00006 #ifndef __TRIO_UNIFIEADAPI_CPLUSPLUS_H__
00007 #define __TRIO_UNIFIEADAPI_CPLUSPLUS_H__
00008
00009 #include <string>
00010 #include <functional>
00011 #include <cstdint>
```

```

/*-----------------------------------------------------------------------
  -- macros
  -----------------------------------------------------------------------*/
#ifdef _MSC_VER
#ifdef TRIO_DLL_EXPORT
#define TRIO_DECLSPEC __declspec(dllexport)
#else
#define TRIO_DECLSPEC __declspec(dllimport)
#endif
#define TRIO_CALLCONV __stdcall
#else
#define TRIO_DECLSPEC
#define TRIO_CALLCONV
#endif

namespace Trio::UnifiedApi
{
    ////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

    /// <summary>
    /// Enumeration of error codes used in Unified API
    /// </summary>
    enum class ErrorCode : uint64_t
    {
        /// <summary>
        /// No error
        /// </summary>
        NoError = 0,

        /// <summary>
        /// An error code that cannot be determined
        /// </summary>
        _Unknown = 0xFFFFFFFF,

        /// <summary>
        /// A fatal error flag
        /// </summary>
        _Fatal = 0x8000000000,

        /// <summary>
        /// UnifiedAPI error flag. The actual error code is in the lowest 32 bits.
        /// </summary>
        _ApiError = 0x100000000,

        /// <summary>
        /// An UnifiedAPI error that cannot be determined
        /// </summary>
        Api_Unknown = 0 | _ApiError | _Fatal,

        /// <summary>
        /// General error
        /// </summary>
        Fail = 1 | _ApiError | _Fatal,

        /// <summary>
        /// Error indicating that operation is not supported
        /// </summary>
        NotSupported = 2 | _ApiError,

        /// <summary>
        /// Requested connection type (module) cannot be load/initialized or is missing
        /// </summary>
        ConnectionModule = 3 | _ApiError | _Fatal,

        /// <summary>
        /// A bad connection context is used
        /// </summary>
        ConnectionContext = 4 | _ApiError | _Fatal,

        /// <summary>
        /// TOKENTABLE reading failed
        /// </summary>
        TokenTable = 5 | _ApiError | _Fatal,

        /// <summary>
        /// The operation timed out
        /// </summary>
        Timeout = 6 | _ApiError | _Fatal,

        /// <summary>
        /// The data is of unknown floating point format
        /// </summary>
        UnknownFloatingPointFormat = 7 | _ApiError | _Fatal,

        /// <summary>
```

```
        /// Error in the token buffer for the Remote protocol
        /// </summary>
        Remote_TokenBuffer = 8 | _ApiError | _Fatal,

        /// <summary>
        /// This Motion Coordinator model/version is not supported
        /// </summary>
        UnsupportedController = 10 | _ApiError | _Fatal,

        /// <summary>
        /// Bad/unrecognized response received
        /// </summary>
        BadResponse = 11 | _ApiError | _Fatal,

        /// <summary>
        /// A bad argument has been passed
        /// </summary>
        BadArgument = 12 | _ApiError,

        /// <summary>
        /// The connection or object is already opened
        /// </summary>
        Reopen = 13 | _ApiError,

        /// <summary>
        /// The passed buffer is too small
        /// </summary>
        BufferTooSmall = 14 | _ApiError,

        /// <summary>
        /// The read to the socket failed
        /// </summary>
        SocketRead = 15 | _ApiError | _Fatal,

        /// <summary>
        /// The write to the socket failed
        /// </summary>
        SocketWrite = 16 | _ApiError | _Fatal,

        /// <summary>
        /// Low level API call failed
        /// </summary>
        LowLevelError = 17 | _ApiError | _Fatal,

        /// <summary>
        /// Versions of modules do not match
        /// </summary>
        VersionMismatch = 18 | _ApiError | _Fatal,

        /// <summary>
        /// Compress/Decompress data error
        /// </summary>
        Compression = 19 | _ApiError | _Fatal,

        /// <summary>
        /// The operation got aborted.
        /// </summary>
        Abort = 20 | _ApiError | _Fatal,

        /// <summary>
        /// No connection
        /// </summary>
        Disconnected = 21 | _ApiError,

        /// <summary>
        /// The parameter block is invalid
        /// </summary>
        InvalidParamBlock = 22 | _ApiError,

        /// <summary>
        /// Value is out of range
        /// </summary>
        OutOfRange = 23 | _ApiError,

        /// <summary>
        /// Capacity exceeded
        /// </summary>
        CapacityExceeded = 24 | _ApiError,

        /// <summary>
        /// Project error
        /// </summary>
        Project = 25 | _ApiError,

        /// <summary>
        /// The last API error
        /// </summary>
```

```
        _ApiLast = 25 | _ApiError | _Fatal,

        /// <summary>
        /// A system error flag. The actual system error code is in the lowest 32 bits.
        /// </summary>
        _SystemError = 0x200000000,

        /// <summary>
        /// Unknown system error.
        /// </summary>
        Sys_Unknown = (_SystemError | _Unknown),

        /// <summary>
        /// A controller error flag. The actual controller error code is in the lowest 32 bits.
        /// </summary>
        _ControllerError = 0x400000000,

        /// <summary>
        /// Controller Error codes
        /// </summary>
        ControllerError_COMMAND_NOT_RECOGNISED = 1 | _ControllerError,

        ControllerError_INVALID_TRANSFER_TYPE = 2 | _ControllerError,

        ControllerError_FLASH_PROGRAMMING_ERROR = 3 | _ControllerError,

        ControllerError_OPERAND_EXPECTED = 4 | _ControllerError,

        ControllerError_ASSIGNMENT_EXPECTED = 5 | _ControllerError,

        ControllerError_QUOTES_EXPECTED = 6 | _ControllerError,

        ControllerError_STACK_OVERFLOW = 7 | _ControllerError,

        ControllerError_TOO_MANY_VARIABLES = 8 | _ControllerError,

        ControllerError_DIVISION_BY_ZERO = 9 | _ControllerError,

        ControllerError_EXTRA_CHARS_AT_EOLN = 10 | _ControllerError,

        ControllerError_CLOSE_SQUARE_BRACKET_EXPECTED_IN_PRINT = 11 | _ControllerError,

        ControllerError_CANNOT_MODIOFY_SPECIAL_PROGRAM = 12 | _ControllerError,

        ControllerError_THEN_EXPECTED_IN_IF_ELSEIF = 13 | _ControllerError,

        ControllerError_FLASH_ERASE_ERROR = 14 | _ControllerError,

        ControllerError_START_OF_EXPRESSION_EXPECTED = 15 | _ControllerError,

        ControllerError_CLOSE_ROUND_BRACKET_EXPECTED = 16 | _ControllerError,

        ControllerError_COMMA_EXPECTED = 17 | _ControllerError,

        ControllerError_COMMAND_LINE_BROKEN_BY_ESC = 18 | _ControllerError,

        ControllerError_PARAMETER_OUT_OF_RANGE = 19 | _ControllerError,

        ControllerError_NO_PROCESS_AVAILABLE = 20 | _ControllerError,

        ControllerError_VALUE_IS_READ_ONLY = 21 | _ControllerError,

        ControllerError_MODIFIER_NOT_ALLOWED = 22 | _ControllerError,

        ControllerError_REMOTE_AXIS_IS_IN_USE = 23 | _ControllerError,

        ControllerError_COMMAND_IS_COMMAND_LINE_ONLY = 24 | _ControllerError,

        ControllerError_COMMAND_IS_RUNTIME_ONLY = 25 | _ControllerError,

        ControllerError_LABEL_EXPECTED = 26 | _ControllerError,

        ControllerError_PROGRAM_NOT_FOUND = 27 | _ControllerError,

        ControllerError_DUPLICATE_IDENTIFIER = 28 | _ControllerError,

        ControllerError_PROGRAM_IS_LOCKED = 29 | _ControllerError,

        ControllerError_PROGRAMS_RUNNING = 30 | _ControllerError,

        ControllerError_PROGRAM_IS_STOPPED = 31 | _ControllerError,

        ControllerError_CANNOT_SELECT_PROGRAM = 32 | _ControllerError,

        ControllerError_NO_PROGRAM_SELECTED = 33 | _ControllerError,

        ControllerError_NO_MORE_PROGRAMS_AVAILABLE = 34 | _ControllerError,
```

```
00272
00273         ControllerError_OUT_OF_MEMORY = 35 | _ControllerError,
00274
00275         ControllerError_NO_CODE_AVAILABLE_FOR_PROGRAM = 36 | _ControllerError,
00276
00277         ControllerError_COMMAND_OUT_OF_CONTEXT = 37 | _ControllerError,
00278
00279         ControllerError_TOO_MANY_NESTED_STRUCTURES = 38 | _ControllerError,
00280
00281         ControllerError_STRUCTURE_NESTING_ERROR = 39 | _ControllerError,
00282
00283         ControllerError_ELSE_ELSEIF_ENDIF_WITHOUT_PREVIOUS_IF = 40 | _ControllerError,
00284
00285         ControllerError_WEND_WITHOUT_PREVIOUS_WHILE = 41 | _ControllerError,
00286
00287         ControllerError_UNTIL_WITHOUT_PREVIOUS_REPEAT = 42 | _ControllerError,
00288
00289         ControllerError_IDENTIFIER_EXPECTED = 43 | _ControllerError,
00290
00291         ControllerError_TO_EXPECTED_AFTER_FOR = 44 | _ControllerError,
00292
00293         ControllerError_TOO_MANY_NESTED_FOR_NEXT = 45 | _ControllerError,
00294
00295         ControllerError_NEXT_WITHOUT_FOR = 46 | _ControllerError,
00296
00297         ControllerError_UNTIL_IDLE_LOADED_EXPECTED_AFTER_WAIT = 47 | _ControllerError,
00298
00299         ControllerError_GOTO_GOSUB_EXEC_EXPECTED = 48 | _ControllerError,
00300
00301         ControllerError_TOO_MANY_NESTED_GOSUB = 49 | _ControllerError,
00302
00303         ControllerError_RETURN_WITHOUT_GOSUB = 50 | _ControllerError,
00304
00305         ControllerError_LABEL_MUST_BE_AT_START_OF_LINE = 51 | _ControllerError,
00306
00307         ControllerError_CANNOT_NEST_ONE_LINE_IF = 52 | _ControllerError,
00308
00309         ControllerError_LABEL_NOT_FOUND = 53 | _ControllerError,
00310
00311         ControllerError_LINE_NUMBER_CANNOT_HAVE_DECIMAL_POINT = 54 | _ControllerError,
00312
00313         ControllerError_MAXIMUM_NUMBER_OF_PROGRAM_INSTANCES_ACTIVE = 55 | _ControllerError,
00314
00315         ControllerError_INVALID_USE_OF_DOLAR_SIGN = 56 | _ControllerError,
00316
00317         ControllerError_VR_X_OR_STRING_VARIABLE_EXPECTED = 57 | _ControllerError,
00318
00319         ControllerError_PROGRAM_ALREADY_EXISTS = 58 | _ControllerError,
00320
00321         ControllerError_PROCESS_ALREADY_SELECTED = 59 | _ControllerError,
00322
00323         ControllerError_DUPLICATE_AXES_NOT_PERMITTED = 60 | _ControllerError,
00324
00325         ControllerError_PLC_TYPE_IS_INVALID = 61 | _ControllerError,
00326
00327         ControllerError_EVALUATION_ERROR = 62 | _ControllerError,
00328
00329         ControllerError_RESERVED_KEYWORD_NOT_AVAILABLE_ON_THIS_CONTROLLER = 63 | _ControllerError,
00330
00331         ControllerError_VARIABLE_NOT_FOUND = 64 | _ControllerError,
00332
00333         ControllerError_TABLE_INDEX_RANGE_ERROR = 65 | _ControllerError,
00334
00335         ControllerError_FEATURES_ENABLED_DO_NOT_ALLOW_ATYPE_CHANGE = 66 | _ControllerError,
00336
00337         ControllerError_INVALID_LINE_NUMBER = 67 | _ControllerError,
00338
00339         ControllerError_STRING_EXCEEDS_PERMITTED_LENGTH = 68 | _ControllerError,
00340
00341         ControllerError_SCOPE_PERIOD_SHOULD_EXCEED_NUMBER_OF_AIN_PARAMS = 69 | _ControllerError,
00342
00343         ControllerError_VALUE_IS_INCORRECT = 70 | _ControllerError,
00344
00345         ControllerError_INVALID_IO_CHANNEL = 71 | _ControllerError,
00346
00347         ControllerError_FLASH_VALUE_CANNOT_BE_SET = 72 | _ControllerError,
00348
00349         ControllerError_DIRECTORY_NOT_LOCKED = 73 | _ControllerError,
00350
00351         ControllerError_DIRECTORY_ALREADY_LOCKED = 74 | _ControllerError,
00352
00353         ControllerError_PROGRAM_NOT_RUNNING_ON_THIS_PROCESS = 75 | _ControllerError,
00354
00355         ControllerError_PROGRAM_NOT_RUNNING = 76 | _ControllerError,
00356
00357         ControllerError_PROGRAM_NOT_PAUSED_ON_THIS_PROCESS = 77 | _ControllerError,
00358
```

```
        ControllerError_PROGRAM_NOT_PAUSED = 78 | _ControllerError,

        ControllerError_COMMAND_NOT_ALLOWED_WHEN_RUNNING_MOTION_PERFECT = 79 | _ControllerError,

        ControllerError_DIRECTORY_STRUCTURE_INVALID = 80 | _ControllerError,

        ControllerError_DIRECTORY_IS_LOCKED = 81 | _ControllerError,

        ControllerError_CANNOT_EDIT_PROGRAM = 82 | _ControllerError,

        ControllerError_TOO_MANY_NESTED_OPERANDS = 83 | _ControllerError,

        ControllerError_CANNOT_RESET_WHEN_DRIVE_SERVO_ON = 84 | _ControllerError,

        ControllerError_FLASH_STICK_BLANK = 85 | _ControllerError,

        ControllerError_FLASH_STICK_NOT_AVAILABLE_ON_THIS_CONTROLLER = 86 | _ControllerError,

        ControllerError_SLAVE_ERROR = 87 | _ControllerError,

        ControllerError_MASTER_ERROR = 88 | _ControllerError,

        ControllerError_NETWORK_TIMEOUT = 89 | _ControllerError,

        ControllerError_NETWORK_PROTOCOL_ERROR = 90 | _ControllerError,

        ControllerError_GLOBAL_DEFINITION_IS_DIFFERENT = 91 | _ControllerError,

        ControllerError_INVALID_PROGRAM_NAME = 92 | _ControllerError,

        ControllerError_PROGRAM_CORRUPT = 93 | _ControllerError,

        ControllerError_MORE_THAN_ONE_PROGRAM_RUNNING_WHEN_TRYING_TO_SET_GLOBAL_OR_CONSTANT = 94 |
    _ControllerError,

        ControllerError_PROGRAM_ENCRYPTED = 95 | _ControllerError,

        ControllerError_BASIC_TOKEN_DEFINITION_INCORRECT = 96 | _ControllerError,

        ControllerError_OPEN_BRACKET_EXPECTED = 97 | _ControllerError,

        ControllerError_NUMBER_EXPECTED = 98 | _ControllerError,

        ControllerError_AS_EXPECTED = 99 | _ControllerError,

        ControllerError_DATA_TYPE_EXPECTED = 100 | _ControllerError,

        ControllerError_STRING_EXPECTED = 101 | _ControllerError,

        ControllerError_DOWNLOAD_ABORT_OR_TIMEOUT = 102 | _ControllerError,

        ControllerError_CANNOT_SPECIFY_PROGRAM_TYPE_FOR_AN_EXISTING_PROGRAM = 103 | _ControllerError,

        ControllerError_INVALID_COFF_IMAGE_FILE = 104 | _ControllerError,

        ControllerError_VARIABLE_DEFINED_OUTSIDE_INCLUDE_FILE = 105 | _ControllerError,

        ControllerError_COMMAND_NOT_ALLOWED_WITHIN_INCLUDE_FILE = 106 | _ControllerError,

        ControllerError_SERIAL_NUMBER_MUST_BE_MINUS_1 = 107 | _ControllerError,

        ControllerError_APPEND_BLOCK_INCONSISTENT = 108 | _ControllerError,

        ControllerError_INVALID_RANGE_SPECIFIED = 109 | _ControllerError,

        ControllerError_TOO_MANY_ITEMS_DEFINED_FOR_BLOCK = 110 | _ControllerError,

        ControllerError_INVALID_MSPHERICAL_INPUT = 111 | _ControllerError,

        ControllerError_TOO_MANY_LABELS = 112 | _ControllerError,

        ControllerError_SYMBOL_TABLE_LOCKED = 113 | _ControllerError,

        ControllerError_INCORRECT_SYMBOL_TYPE = 114 | _ControllerError,

        ControllerError_VARIABLES_NOT_PERMITTED_ON_COMMAND_LINE = 115 | _ControllerError,

        ControllerError_INVALID_PROGRAM_TYPE = 116 | _ControllerError,

        ControllerError_PARAMETER_EXPECTED = 117 | _ControllerError,

        ControllerError_FIRMWARE_ERROR__DEVICE_IN_USE = 118 | _ControllerError,

        ControllerError_DEVICE_ERROR__TIMEOUT_WAITING_FOR_DEVICE = 119 | _ControllerError,

        ControllerError_DEVICE_ERROR__COMMAND_NOT_SUPPORTED_BY_DEVICE = 120 | _ControllerError,

```

```
        ControllerError_DEVICE_ERROR__CRC_ERROR = 121 | _ControllerError,

        ControllerError_DEVICE_ERROR__ERROR_WRITING_TO_DEVICE = 122 | _ControllerError,

        ControllerError_DEVICE_ERROR__INVALID_RESPONSE_FROM_DEVICE = 123 | _ControllerError,

        ControllerError_FIRMWARE_ERROR__CANNOT_REFERENCE_DATA_OUTSIDE_CURRENT_BLOCK = 124 |
    _ControllerError,

        ControllerError_DISK_ERROR__INVALID_MBR = 125 | _ControllerError,

        ControllerError_DISK_ERROR__INVALID_BOOT_SECTOR = 126 | _ControllerError,

        ControllerError_DISK_ERROR__INVALID_SECTOR_CLUSTER_REFERENCE = 127 | _ControllerError,

        ControllerError_FILE_ERROR__DISK_FULL = 128 | _ControllerError,

        ControllerError_FILE_ERROR__FILE_NOT_FOUND = 129 | _ControllerError,

        ControllerError_FILE_ERROR__FILENAME_ALREADY_EXISTS = 130 | _ControllerError,

        ControllerError_FILE_ERROR__INVALID_FILENAME = 131 | _ControllerError,

        ControllerError_FILE_ERROR__DIRECTORY_FULL = 132 | _ControllerError,

        ControllerError_COMMAND_ONLY_ALLOWED_WHEN_RUNNING_MOTION_PERFECT = 133 | _ControllerError,

        ControllerError_HASH_EXPECTED = 134 | _ControllerError,

        ControllerError_FOR_EXPECTED = 135 | _ControllerError,

        ControllerError_INPUT__OUTPUT__APPEND__FIFO_READ__FIFO_WRITE_EXPECTED = 136 |
    _ControllerError,

        ControllerError_FILE_NOT_OPEN = 137 | _ControllerError,

        ControllerError_END_OF_FILE = 138 | _ControllerError,

        ControllerError_FILE_ALREADY_OPEN = 139 | _ControllerError,

        ControllerError_INVALID_STORAGE_AREA = 140 | _ControllerError,

        ControllerError_INVALID_FLOATING_POINT_OPERATION = 141 | _ControllerError,

        ControllerError_INVALID_SYSTEM_CODE_WRONG_CONTROLLER = 142 | _ControllerError,

        ControllerError_IEC_ERROR__INVALID_VARIABLE_ACCESS = 143 | _ControllerError,

        ControllerError_NUMERICAL_ERROR__NAN_USED = 144 | _ControllerError,

        ControllerError_NUMERICAL_ERROR__INF_USED = 145 | _ControllerError,

        ControllerError_NUMERICAL_ERROR__SUBNORMAL_VALUE_USED = 146 | _ControllerError,

        ControllerError_MAC_EEPROM_IS_LOCKED = 147 | _ControllerError,

        ControllerError_INVALID_MIX_OF_DATA_TYPES = 148 | _ControllerError,

        ControllerError_INVALID_STARTUP_CONFIGURATION_COMMAND = 149 | _ControllerError,

        ControllerError_SYMBOL_IS_NOT_A_VARIABLE = 150 | _ControllerError,

        ControllerError_KINEMATICS_RUNTIME_LIMITED_TO_1_HOUR_FEC_22 = 151 | _ControllerError,

        ControllerError_IEC_RUNTIME_LIMITED_TO_1_HOUR_FEC_21 = 152 | _ControllerError,

        ControllerError_COMMAND_NOT_ALLOWED_WITH_CURRENT_ATYPE = 153 | _ControllerError,

        ControllerError_WILDCARD_LENGTH_MUST_BE_1 = 154 | _ControllerError,

        ControllerError_INCOMPATIBLE_ARRAY_DIMENSIONS = 155 | _ControllerError,

        ControllerError_MATRIX_IS_SINGULAR = 156 | _ControllerError,

        ControllerError_PROGRAM_IS_NOT_EXECUTABLE_TYPE = 157 | _ControllerError,

        ControllerError_DISK_ERROR__FORMAT_MUST_BE_FAT32 = 158 | _ControllerError,

        ControllerError_PROGRAM_IS_STOPPED_HALT_FORCED = 159 | _ControllerError,

        ControllerError_VARIABLE_NOT_DEFINED_USING_DIM_OR_GLOBAL_CONSTANT = 160 | _ControllerError,

        ControllerError_VARIABLE_NOT_DEFINED_USING_DIM_WITHIN_INCLUDE_FILE = 161 | _ControllerError,

        ControllerError_PARAMETER_CAN_ONLY_BE_CHANGED_USING_MC_CONFIG = 162 | _ControllerError,

        ControllerError_IEC_TASK_MEMORY_ALLOCATION_ERROR = 163 | _ControllerError,
```

```
00530
00531         ControllerError_IEC_IO_BINDING_INDEX_OUT_OF_RANGE = 164 | _ControllerError,
00532
00533         ControllerError_IEC_VR_BINDING_INDEX_OUT_OF_RANGE = 165 | _ControllerError,
00534
00535         ControllerError_IEC_TABLE_BINDING_INDEX_OUT_OF_RANGE = 166 | _ControllerError,
00536
00537         ControllerError_FAILED_OPEN_CONNECTION = 167 | _ControllerError,
00538
00539         ControllerError_MCCONFIG_CHANGED_EX_REQUIRED = 168 | _ControllerError,
00540
00541         ControllerError_NO_RESPONSE_FROM_CAN_MODULE = 169 | _ControllerError,
00542
00543         ControllerError_INVALID_FPGA_VERSION = 170 | _ControllerError,
00544
00545         ControllerError_VARIABLE_NOT_CREATED_USING_GLOBAL_CONSTANT = 171 | _ControllerError,
00546
00547         ControllerError_PARAMETER_CAN_ONLY_BE_CHANGED_USING_MC_CONFIG_OR_COMMAND_LINE = 172 |
      _ControllerError,
00548
00549         ControllerError_ETHERCAT_AXES_CANNOT_BE_SET_USING_ATYPE_PLEASE_USE_DRIVE_MODE = 173 |
      _ControllerError,
00550
00551         ControllerError_BIT_NUMBER_OUT_OF_RANGE = 174 | _ControllerError,
00552
00553         ControllerError_FUNCTION_NAME_EXPECTED = 175 | _ControllerError,
00554
00555         ControllerError_FUNCTION_MUST_BE_DEFINED_WITHIN_A_BASIC_LIBRARY = 176 | _ControllerError,
00556
00557         ControllerError_TOO_MANY_FUNCTIONS_DEFINED = 177 | _ControllerError,
00558
00559         ControllerError_MISSING_ENDFUNC = 178 | _ControllerError,
00560
00561         ControllerError_ENDFUNC_WITHOUT_PREVIOUS_FUNCTION = 179 | _ControllerError,
00562
00563         ControllerError_PARAMETER_NAME_EXPECTED = 180 | _ControllerError,
00564
00565         ControllerError_DEVICE_ERROR_NO_SDCARD_INSERTED = 181 | _ControllerError,
00566
00567         ControllerError_DEVICE_ERROR_SDCARD_WRITE_PROTECTED = 182 | _ControllerError,
00568
00569         ControllerError_FAILED_ATTEMPT_1 = 183 | _ControllerError,
00570
00571         ControllerError_FAILED_ATTEMPT_2 = 184 | _ControllerError,
00572
00573         ControllerError_FAILED_TOO_MANY_TIMES_PLEASE_RESET = 185 | _ControllerError,
00574
00575         ControllerError_LINE_MUST_BE_INSIDE_FUNCTION = 186 | _ControllerError,
00576
00577         ControllerError_DUPLICATE_NAME_USED_IN_LABEL_AND_VARIABLE = 187 | _ControllerError,
00578
00579         ControllerError_STRUCTURE_FIELD_NAME_NOT_RECOGNISED = 188 | _ControllerError,
00580
00581         ControllerError_ID_NOT_DEFINED = 189 | _ControllerError,
00582
00583         ControllerError_ROBOT_CONFIGURATION_CANNOT_BE_CHANGED_WHILE_MOVEL = 190 | _ControllerError,
00584
00585         ControllerError_TOOL_OFFSET_ID_NOT_DEFINED = 191 | _ControllerError,
00586
00587         ControllerError_OBJECT_FRAME_ID_NOT_DEFINED = 192 | _ControllerError,
00588
00589         ControllerError_ROBOT_FRAME_ID_NOT_DEFINED = 193 | _ControllerError,
00590
00591         ControllerError_COMMAND_NOT_PERMITTED_IN_FUNCTION = 194 | _ControllerError,
00592
00593         ControllerError_TOO_MANY_FUNCTION_PARAMETERS = 195 | _ControllerError,
00594
00595         ControllerError_OPERATION_CANNOT_BE_PERFORMED_UNTIL_IDLE = 196 | _ControllerError,
00596
00597         ControllerError_SINGULARITY = 197 | _ControllerError,
00598
00599         ControllerError_TOOL_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED = 198 | _ControllerError,
00600
00601         ControllerError_OBJECT_FRAME_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED = 199 |
      _ControllerError,
00602
00603         ControllerError_ROBOT_FRAME_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED = 200 |
      _ControllerError,
00604
00605         ControllerError_ROBOT_MOVES_CANNOT_BE_PERFORMED_UNTIL_JOG_OPERATION_IS_DISABLED = 201 |
      _ControllerError,
00606
00607         ControllerError_CYCLIC_INCLUDE_DEPENDENCY = 202 | _ControllerError,
00608
00609         ControllerError_OBJECT_FRAME_ALREADY_ATTACHED = 203 | _ControllerError,
00610
00611         ControllerError_OBJECT_FRAME_HAS_NOT_BEEN_ATTACHED = 204 | _ControllerError,
```

```
        ControllerError_OBJECT_FRAME_ALREADY_SYNCED = 205 | _ControllerError,

        ControllerError_BASE_AXIS_DOES_NOT_BELONG_TO_ANY_KINEMATIC_GROUP = 206 | _ControllerError,

        ControllerError_SYNC_IN_PROCESS = 207 | _ControllerError,

        ControllerError_ROBOT_PROGRAM_STOPPED = 208 | _ControllerError,

        ControllerError_TARGET_POINT_NOT_ACTIVATED = 209 | _ControllerError,

        ControllerError_PROGRAM_FAILED_TO_RUN = 210 | _ControllerError,

        ControllerError_RETURN_STATEMENT_EXPECTED_BEFORE_ENDFUNC = 211 | _ControllerError,

        ControllerError_CONTINUE_WITHOUT_PREVIOUS_FOR_WHILE_OR_REPEAT = 212 | _ControllerError,

        ControllerError_EXIT_WITHOUT_PREVIOUS_FOR_WHILE_OR_REPEAT = 213 | _ControllerError,

        ControllerError_TOO_MANY_EXIT_STATEMENTS_WITHIN_SINGLE_BLOCK = 214 | _ControllerError,

        ControllerError_CASE_OR_END_CASE_WITHOUT_PREVIOUS_SELECT_CASE = 215 | _ControllerError,

        ControllerError_FIXED_INTEGER_OR_ELSE_EXPECTED = 216 | _ControllerError,

        ControllerError_TOO_MANY_CASE_OPTIONS = 217 | _ControllerError,

        ControllerError_ROBOTICS_RUNTIME_LIMITED_TO_1_HOUR = 218 | _ControllerError,

        ControllerError_COLLISION_OBJECTS_CANNOT_BE_MODIFIED_OR_DELETED_WHILE_SELECTED = 219 |
    _ControllerError,

        ControllerError_TOO_MANY_COLLISION_OBJECTS_ACTIVE = 220 | _ControllerError,

        ControllerError_ATYPE_MUST_BE_CONFIGURED_USING_MC_CONFIG_PARAMETER_DRIVE_MODE = 221 |
    _ControllerError,

        ControllerError_COMMAND_NOT_ALLOWED_IN_FRAME = 222 | _ControllerError,

        ControllerError_COMMAND_LINE_TOO_LONG = 223 | _ControllerError,

        ControllerError_TABLE_LOCATION_WRITE_PROTECTED = 224 | _ControllerError,

        ControllerError_ENCRYPTION_AUTHORITY_NOT_FOUND = 225 | _ControllerError,

        ControllerError_NO_FREE_ENCRYPTION_AUTHORITY = 226 | _ControllerError,

        ControllerError_LOCAL_TARGET_VALUE_HAS_NOT_BEEN_DEFINED = 227 | _ControllerError,

        ControllerError_COMMAND_NOT_PERMITTED_IN_BASIC_LIBRARY = 228 | _ControllerError,

        ControllerError_INVALID_MOVEC_INPUT = 229 | _ControllerError,

        ControllerError_FUNCTION_EXPECTED = 230 | _ControllerError,

        ControllerError_CAM_TABLE_ID_NOT_DEFINED = 231 | _ControllerError,

        ControllerError_RUNNING_OF_RPS_PROGRAMS_INHIBITED_USE_DISABLE_ROBOT = 232 | _ControllerError,

        ControllerError_U_AND_V_ORIENTATION_INHIBITED_WHILE_WRIST_SINGULARITY = 233 |
    _ControllerError,

        ControllerError_TARGET_POINT_OUT_OF_ROBOT_WORKING_AREA = 234 | _ControllerError,

        ControllerError_AXIS_LOCKED_TO_A_ROBOT = 235 | _ControllerError,

        ControllerError_ROBOT_CONTROLLED_BY_ANOTHER_PROCESS = 236 | _ControllerError,

        ControllerError_ROBOT_NOT_DEFINED_USING_KINEMATIC_GROUP = 237 | _ControllerError,

        ControllerError_COMMAND_INHIBITED_ROBOT_MUST_BE_USED_TO_ACTIVATE_A_KINEMATIC_GROUP = 238 |
    _ControllerError,

        ControllerError_UNDEFINED_VARIABLE_IN_DIM_INITIALISER = 239 | _ControllerError,

        ControllerError_THE_SPECIFIED_ROBOT_CONFIGURATION_IS_NOT_ALLOWED = 240 | _ControllerError,

        ControllerError_TOO_MANY_AXES_ACTIVATED = 241 | _ControllerError,

        ControllerError_MULTIPLE_MOVE_TRIGGERS_SPECIFIED = 242 | _ControllerError,

        ControllerError_MOVE_TRIGGER_RESPONSE_NOT_SPECIFIED = 243 | _ControllerError,

        ControllerError_MOVE_TRIGGER_NOT_SPECIFIED = 244 | _ControllerError,

        ControllerError_INVALID_ARRAY_DIMENSION = 245 | _ControllerError,
```

```
        ControllerError_DUPLICATE_VR_INDEX = 246 | _ControllerError,

        ControllerError_CONSTANT_VALUE_EXPECTED = 247 | _ControllerError,

        ControllerError_RPS_KERNEL_V2_NOT_ENABLED = 248 | _ControllerError,

        ControllerError_INCLUDE_FILE_MUST_CONTAIN_DEFCONST_DIM_STATEMENTS_ONLY = 249 | _ControllerError,

        ControllerError_TARGET_TAUGHT_WITH_ANOTHER_ROBOT = 250 | _ControllerError,

        ControllerError_TARGET_ROBOT_CONFIG_MISMATCH = 251 | _ControllerError,

        ControllerError_TARGET_TOOL_OFFSET_MISMATCH = 252 | _ControllerError,

        ControllerError_TARGET_OBJECT_FRAME_MISMATCH = 253 | _ControllerError,

        ControllerError_TARGET_ROBOT_FRAME_MISMATCH = 254 | _ControllerError,

        ControllerError_INVALID_FRAME_NUMBER = 255 | _ControllerError,

        ControllerError_FRAME_NOT_SUPPORTED_BY_RPS = 256 | _ControllerError,

        ControllerError_FORWARD_TRANSFORM_FUNCTION_NOT_DEFINED = 257 | _ControllerError,

        ControllerError_INVERSE_TRANSFORM_FUNCTION_NOT_DEFINED = 258 | _ControllerError,

        ControllerError_INVALID_MOVEOPTION = 259 | _ControllerError,

        ControllerError_MAX_OPEN_KEYWORDS_REACHED = 260 | _ControllerError,

        ControllerError_KEYWORD_INACTIVE = 261 | _ControllerError,

        ControllerError_KEYWORD_FUNCTION_NOT_DEFINED = 262 | _ControllerError,

        ControllerError_KEYWORD_SAME_AS_SYSTEM_WORD = 263 | _ControllerError,

        ControllerError_UNEXPECTED_CHARACTERS_AFTER_LINE_CONTINUATION = 264 | _ControllerError,

        ControllerError_THIS_COMMAND_CAN_BE_EXECUTED_ONLY_ONCE = 265 | _ControllerError,

        ControllerError_TABLE_MEMORY_CONFLICT = 266 | _ControllerError,

        ControllerError_MAX_OPEN_MOVES_REACHED = 267 | _ControllerError,

        ControllerError_USER_DEFINED_ERROR = 268 | _ControllerError,

        ControllerError_INVALID_FORMAT_SPECIFIER = 269 | _ControllerError,

        ControllerError_OPEN_MOVE_FUNCTIONS_NOT_DEFINED = 270 | _ControllerError,

        ControllerError_TOO_MANY_CHARS_IN_STR = 271 | _ControllerError,

        ControllerError_INVALID_OPEN_ARCH_FUNC_DEFINITION = 272 | _ControllerError,

        ControllerError_PROCESS_IN_USE = 273 | _ControllerError,

        ControllerError_VR_LOCATION_WRITE_PROTECTED = 274 | _ControllerError,
    };

    /// <summary>
    /// Events that may occur on a connection.
    /// </summary>
    enum class EventType
    {
        /// <summary>
        /// Error processing a command.
        /// </summary>
        Error,

        /// <summary>
        /// Warning for non critical problem
        /// </summary>
        Warning,

        /// <summary>
        /// Information about the command being processed.
        /// </summary>
        Message,
    };

    /// <summary>
    /// Connection types.
    /// </summary>
    enum class ConnectionType : uint32_t
    {
        /// <summary>
```

```
        /// Unknown connection type. Returned when there is no connection context
        /// </summary>
        Unknown = 0x00,

        /// <summary>
        /// TCP connection via Remote protocol
        /// </summary>
        TCP = 0x01,

        /// <summary>
        /// Shared memory connection for PC-MCAT devices
        /// </summary>
        PCMCAT = 0x02,

        /// <summary>
        /// Shared memory connection for Flex7 devices
        /// </summary>
        FLEX7 = 0x04,
    };

    /// <summary>
    /// Progress event
    /// </summary>
    enum class ProgressEvent
    {
        /// <summary>
        /// Progress started.
        /// </summary>
        Start,

        /// <summary>
        /// Progress.
        /// </summary>
        Progress,

        /// <summary>
        /// Progress finished.
        /// </summary>
        Finish,
    };

    /// <summary>
    /// Co Read/Write operation object types
    /// </summary>
    enum class Co_ObjectType
    {
        /// <summary>
        /// Boolean type
        /// </summary>
        Boolean = 1,

        /// <summary>
        /// 8 bit integer
        /// </summary>
        Integer8 = 2,

        /// <summary>
        /// 16 bit integer
        /// </summary>
        Integer16 = 3,

        /// <summary>
        /// 32 bit integer
        /// </summary>
        Integer32 = 4,

        /// <summary>
        /// 8 bit unsigned integer
        /// </summary>
        Unsigned8 = 5,

        /// <summary>
        /// 16 bit unsigned integer
        /// </summary>
        Unsigned16 = 6,

        /// <summary>
        /// 32 bit unsigned integer
        /// </summary>
        Unsigned32 = 7,

        /// <summary>
        /// 32 bit floating point
        /// </summary>
        Real32 = 8,

        /// <summary>
```

```
        /// 64 bit floating point
        /// </summary>
        Real64 = 17,

        /// <summary>
        /// 64 bit integer
        /// </summary>
        Integer64 = 21,

        /// <summary>
        /// 64 bit unsigned integer
        /// </summary>
        Unsigned64 = 27,
    };

    /// <summary>
    /// EtherCAT state
    /// </summary>
    enum class EthercatState
    {
        /// <summary>
        /// Initial
        /// </summary>
        Initial = 0,

        /// <summary>
        /// Pre operational
        /// </summary>
        PreOperational = 1,

        /// <summary>
        /// Safe operational
        /// </summary>
        SafeOperational = 2,

        /// <summary>
        /// Operational
        /// </summary>
        Operational = 3,
    };

    /// <summary>
    /// EtherCAT start option
    /// </summary>
    enum class EthercatStartOption : uint32_t
    {
        /// <summary>
        /// No option is selected
        /// </summary>
        None = 0x0000,

        /// <summary>
        /// Print out progress information
        ///  on - Print out progress information (default)
        ///  off - Do not print progress information
        /// </summary>
        PrintOutProgress = 0x0001,

        /// <summary>
        /// Network re-scan
        ///  off - Normal state change (default)
        ///  on - Force a re-scan of the network
        /// </summary>
        ForceRescan = 0x0002,

        /// <summary>
        /// Start-up type
        ///  on - Express start-up; do not wait for PLL to lock between state changes
        ///  off - Standard start-up
        /// </summary>
        ExpressStartup = 0x0004,
    };

    /// <summary>
    /// File transfer option
    /// </summary>
    enum class FileTransferOption : uint32_t
    {
        /// <summary>
        /// No option is selected
        /// </summary>
        None = 0x0000,

        /// <summary>
        /// The data gets compressed and transferred. The other side receives and stores the compressed data.
        /// </summary>
```

```
        CompressOnly = 0x0001,

        /// <summary>
        /// The data gets compressed, transferred and decompressed. This saves bandwidth.
        /// </summary>
        CompressDecompress = 0x0002,
    };

    /// <summary>
    /// File types on the Motion Coordinator
    /// </summary>
    enum class FileType : int32_t
    {
        /// <summary>
        /// The file type is not recognized
        /// </summary>
        Undefined = -1,

        /// <summary>
        /// TrioBASIC program
        /// </summary>
        BASIC = 0,

        /// <summary>
        /// Encrypted TrioBASIC program.
        /// </summary>
        EncryptedBASIC = 1,

        /// <summary>
        /// GCode program.
        /// </summary>
        GCode = 2,

        /// <summary>
        /// Unknown contents.
        /// </summary>
        TextFile = 3,

        /// <summary>
        /// IEC program
        /// </summary>
        IecProgram = 4,

        /// <summary>
        /// FIFO (not valid in project)
        /// </summary>
        FifoFile = 5,

        /// <summary>
        /// Temporary File (not valid in project)
        /// </summary>
        TempFile = 6,

        /// <summary>
        /// Project file.
        /// </summary>
        Project = 7,

        /// <summary>
        /// Container for anything that the controller does not understand
        /// </summary>
        Binary = 8,

        /// <summary>
        /// Special for the MC_CONFIG file
        /// </summary>
        MCConfig = 9,

        /// <summary>
        /// IEC library
        /// </summary>
        IecLibrary = 10,

        /// <summary>
        /// HMI
        /// </summary>
        HmiDesign = 11,

        /// <summary>
        /// Trio BASIC library
        /// </summary>
        BASICLibrary = 12,

        /// <summary>
        /// Encrypted Trio BASIC library
        /// </summary>
        EncryptedBASICibrary = 13,
```

```
        /// <summary>
        /// HMI Library
        /// </summary>
        HmiLibrary = 14,

        /// <summary>
        /// Volatile temporary controller file stored in RAM (not valid in project)
        /// </summary>
        TempVolatileFile = 15,

        /// <summary>
        /// Robot program
        /// </summary>
        Robot = 16,

        /// <summary>
        /// Robot basic program
        /// </summary>
        RobotBasic = 17,

        /// <summary>
        /// Temporary Text File (not valid in project)
        /// </summary>
        TempTextFile = 18,

        /// <summary>
        /// Robot library
        /// </summary>
        RobotLibrary = 19,

        /// <summary>
        /// Robot basic library
        /// </summary>
        RobotBasicLibrary = 20,

        /// <summary>
        /// Recipe data
        /// </summary>
        Recipe = 21,

        /// <summary>
        /// Alarm data
        /// </summary>
        Alarms = 22,

        /// <summary>
        /// Real-time basic library
        /// </summary>
        RealtimeBASICLibrary = 23,

        /// <summary>
        /// Encrypted real-time basic library
        /// </summary>
        EncryptedRealtimeBasicLibrary = 24,

        /// <summary>
        /// Encrypted IEC program
        /// </summary>
        EncryptedIecProgram = 25,

        /// <summary>
        /// Encrypted IEC library
        /// </summary>
        EncryptedIecLibrary = 26,

        /// <summary>
        /// Not a file type. Used to size the FileType enumeration.
        /// </summary>
        __Size,
    };

    /// <summary>
    /// Flags related to files and programs in internal storage on the Motion Coordinator
    /// </summary>
    enum class FileFlags : uint32_t
    {
        /// <summary>
        /// No flags are set
        /// </summary>
        None = 0x00,

        /// <summary>
        /// If set file is editable on the controller
        /// </summary>
        Editable = 0x01,

```

```
        /// <summary>
        /// If set program can be auto-run
        /// </summary>
        Autorun = 0x02,
    };

    /// <summary>
    /// Mode for CONNECT command
    /// </summary>
    enum class ConnectMode : uint32_t
    {
        /// <summary>
        /// Measured position (MPOS) is followed by connected axis
        /// </summary>
        MPOS = 0,

        /// <summary>
        /// Demand position (DPOS) is followed by connected axis
        /// </summary>
        DPOS = 1,
    };

    /// <summary>
    /// Flags PSWITCH command in single/cyclic mode
    /// </summary>
    enum class PSwitchMode : uint32_t
    {
        /// <summary>
        /// No flags are set
        /// </summary>
        None = 0x00,

        /// <summary>
        /// If set - set position should be bigger than reset position
        ///  If not set - set position should be less than reset position
        /// </summary>
        Invert = 0x02,

        /// <summary>
        /// If set DPOS is used
        ///  If not set MPOS is used
        /// </summary>
        DPOS = 0x04,

        /// <summary>
        /// Mask of allowed bits
        /// </summary>
        Mask = Invert | DPOS,
    };

    /// <summary>
    /// Flags PSWITCH command in multi/sequence mode
    /// </summary>
    enum class PSwitchMultiMode : uint32_t
    {
        /// <summary>
        /// No flags are set
        /// </summary>
        None = 0x00,

        /// <summary>
        /// If set DPOS is used
        ///  If not set MPOS is used
        /// </summary>
        DPOS = 0x04,

        /// <summary>
        /// If set timer mode
        ///  If not set set and reset positions mode
        /// </summary>
        TimerMode = 0x10,

        /// <summary>
        /// If set negative direction
        ///  If not set positive direction
        /// </summary>
        NegativeDirection = 0x20,

        /// <summary>
        /// If set Z output mode
        ///  If not set digital output mode
        /// </summary>
        ZOut = 0x40,

        /// <summary>
        /// Mask of allowed bits
        /// </summary>
```

```
        Mask = DPOS | TimerMode | NegativeDirection | ZOut,
    };

    /// <summary>
    /// Link options in CamBox command
    /// </summary>
    enum class CamBoxLinkOptions : uint32_t
    {
        /// <summary>
        /// No flags are set
        /// </summary>
        None = 0,

        /// <summary>
        /// link commences exactly when registration event MARK occurs on link axis
        /// </summary>
        LinkMark = 1,

        /// <summary>
        /// link commences at an absolute position on link axis
        /// </summary>
        LinkAbsolute = 2,

        /// <summary>
        /// CAMBOX repeats automatically and bi-directionally when this bit is set.
        /// </summary>
        LinkRepeat = 4,

        /// <summary>
        /// PATTERN mode. Advanced use of CAMBOX: allows multiple scale values to be used
        /// </summary>
        LinkPattern = 8,

        /// <summary>
        /// Link is only active during a positive move on the link axis
        /// </summary>
        LinkPositiveMove = 32,

        /// <summary>
        /// Forces the profile to start at a defined point in the link_dist
        /// </summary>
        LinkForceStart = 128,

        /// <summary>
        /// link commences exactly when registration event MARKB occurs on link axis
        /// </summary>
        LinkMarkB = 256,

        /// <summary>
        /// link commences exactly when registration event R_MARK occurs on link axis.
        /// </summary>
        LinkR_Mark = 512,

        /// <summary>
        /// Link is only active during a negative move on the link axisLink is active during positive movement from a threshold.
        /// </summary>
        LinkThreshold = 16384,
    };

    /// <summary>
    /// Enumeration of known axis types used with ATYPE axis parameter
    /// </summary>
    enum class AxisType : uint16_t
    {
        None = 0,

        Stepper = 1,

        Servo = 2,

        Encoder = 3,

        StepperEncoderInt = 4,

        Resolver = 5,

        VOut = 6,

        SSI = 7,

        CAN = 8,

        CAN_Remote = 9,

        PSwitch = 10,

```

```
        DLink_Remote = 11,

        Servo2 = 12,

        Yaskawa = 13,

        EncoderOut = 14,

        HighestMotionType = EncoderOut,

        Dummy15 = 15,

        Dummy16 = 16,

        Dummy17 = 17,

        DS402PosType1 = 18,

        DS402VelType1 = 19,

        First_Comms = 20,

        CAN_Comms = First_Comms,

        USB_Comms = 21,

        DLink_Comms = 22,

        Profibus_Comms = 23,

        Dummy24 = 24,

        Dummy25 = 25,

        DS402PosType2 = 26,

        DS402VelType2 = 27,

        Dummy28 = 28,

        Dummy29 = 29,

        DACout_ANALin = 30,

        ANALout_ANALin = 31,

        Dummy32 = 32,

        Dummy33 = 33,

        Dummy34 = 34,

        SDout_PWMout = 35,

        /// <summary>
        /// PWM variant that uses path distance to generate output
        /// </summary>
        SDout_PWMout2 = 36,

        PWMout_SDFREQin = 37,

        PWMout_SDin = 37,

        Dummy38 = 38,

        Dummy39 = 39,

        /// <summary>
        /// This is the first MC464 usable type
        /// </summary>
        MechatroPos = 40,

        MechatroSpeed = 41,

        MechatroTorque = 42,

        /// <summary>
        /// Standard stepper axis
        /// </summary>
        SDout_Zout = 43,

        /// <summary>
        /// Standard analogue servo axis with encoder feedback
        /// </summary>
        DACout_QUADin = 44,

        QUADout_Zout = 45,
```

```
01388
01389         DACout_TAMAin = 46,
01390
01391         DACout_ENDATin = 47,
01392
01393         DACout_SSIin = 48,
01394
01395         MechatroInv = 49,
01396
01397         PanasonicPos = 50,
01398
01399         PanasonicSpeed = 51,
01400
01401         PanasonicTorque = 52,
01402
01403         SercosSpeed = 53,
01404
01405         SercosPos = 54,
01406
01407         SercosTorque = 55,
01408
01409         SercosOpen = 56,
01410
01411         SercosSpeedReg = 57,
01412
01413         SercosPosReg = 58,
01414
01415         SercosEncoder = 59,
01416
01417         DACout_SDin = 60,
01418
01419         SLM32 = 61,
01420
01421         PLM32 = 62,
01422
01423         SDout_Zin = 63,
01424
01425         QUADout_Zin = 64,
01426
01427         EthercatPos = 65,
01428
01429         EthercatSpeed = 66,
01430
01431         EthercatTorque = 67,
01432
01433         EthercatOpenSpeed = 68,
01434
01435         EthercatEncoderIn = 69,
01436
01437         EthercatInAnalOut = 70,
01438
01439         EthercatPosTrio = 71,
01440
01441         EthercatSpeedTrio = 72,
01442
01443         EthercatDummy5 = 73,
01444
01445         EthercatDummy6 = 74,
01446
01447         DACout_SSISLAVEin = 75,
01448
01449         /// <summary>
01450         /// Standard encoder only axis
01451         /// </summary>
01452         QUADin = 76,
01453
01454         DACout_QUADin_Zout = 77,
01455
01456         SDout_VFFDACout = 78,
01457
01458         SDin = 79,
01459
01460         G5Pos = 80,
01461
01462         G5Speed = 81,
01463
01464         G5Torque = 82,
01465
01466         Jacquard = 83,
01467
01468         QUADout_VFFDACout = 84,
01469
01470         AxesDiff = 85,
01471
01472         TAMAin = 86,
01473
01474         ENDATin = 87,
```

```
01475
01476         SSIin = 88,
01477
01478         SDFREQout_ANALin = 89,
01479
01480         DACout_SDFREQin = 90,
01481
01482         AFFDACout = 91,
01483
01484         DACout_BISSin = 92,
01485
01486         BISSin = 93,
01487
01488         DACPWMout_QUADin = 94,
01489
01490         SDout_TAMAin = 95,
01491
01492         Dummy96 = 96,
01493
01494         QUADin_Zout = 97,
01495
01496         Dummy98 = 98,
01497
01498         Dummy99 = 99,
01499
01500         SDout_HD = 100,
01501
01502         QUADout_HD = 101,
01503
01504         DACout_QUADin_HD = 102,
01505
01506         DACout_SDin_HD = 103,
01507
01508         QUADin_HD = 104,
01509
01510         SSIin_HD = 105,
01511
01512         TAMAin_HD = 106,
01513
01514         Dummy107 = 107,
01515
01516         Dummy108 = 108,
01517
01518         Dummy109 = 109,
01519
01520         SDout_SDin_HD = 110,
01521
01522         SDout_QUADin_HD = 111,
01523
01524         QUADout_SDin_HD = 112,
01525
01526         QUADout_QUADin_HD = 113,
01527
01528         SDout_TAMAin_HD = 114,
01529
01530         SDout_ENDATin_HD = 115,
01531
01532         SDout_SSIin_HD = 116,
01533
01534         SDout_BISSin_HD = 117,
01535
01536         QUADout_TAMAin_HD = 118,
01537
01538         QUADout_ENDATin_HD = 119,
01539
01540         QUADout_SSIin_HD = 120,
01541
01542         QUADout_BISSin_HD = 121,
01543
01544         Dummy122 = 122,
01545
01546         Dummy123 = 123,
01547
01548         Dummy124 = 124,
01549
01550         Dummy125 = 125,
01551
01552         Dummy126 = 126,
01553
01554         Dummy127 = 127,
01555
01556         Dummy128 = 128,
01557
01558         Dummy129 = 129,
01559
01560         SLMvirtual = 200,
01561
```

```
        PLMvirtual = 201,
    };

    /// <summary>
    /// Enumeration of axis mode flags used with AXIS_MODE axis parameter
    /// </summary>
    enum class AxisMode : uint32_t
    {
        NONE = 0,

        NOT_USED_1 = ((uint32_t)1 « 0 ),

        ALLOW_CONNECT_MOVES_IN_LIMIT = ((uint32_t)1 « 1 ),

        DIRECTION_MODE_3D = ((uint32_t)1 « 2 ),

        CANCEL_ADDAX_ON_LIMIT = ((uint32_t)1 « 3 ),

        DISABLE_LOOKAHEAD = ((uint32_t)1 « 4 ),

        NOT_USED_2 = ((uint32_t)1 « 5 ),

        USE_NON_SIGNEXTENDED_ANALOGUE_FEEDBACK = ((uint32_t)1 « 6 ),

        NOT_USED_3 = ((uint32_t)1 « 7 ),

        SOFTWARE_LIMIT_USES_DPOS = ((uint32_t)1 « 8 ),

        DISABLE_FE_ACCUMULATION_DURING_MOVE = ((uint32_t)1 « 9 ),

        USE_FASTDEC_WHEN_CANCELLING_RPS_JOG = ((uint32_t)1 « 10),

        DELAY_POSITION_EVALUATION_TO_LOAD_TIME = ((uint32_t)1 « 11),

        DIRECTION_SHORTEST = ((uint32_t)1 « 12),

        MONITOR_FE = ((uint32_t)1 « 13),

        OVERRIDE_FE_LIMITS = ((uint32_t)1 « 14),

        DUAL_AXIS_AVERAGE_POS = ((uint32_t)1 « 15),

        INVERT_AXIS_DIRECTION = ((uint32_t)1 « 16),

        CANCEL_MTYPE_ON_MASTER_IF_ADDAX_HITS_LIM = ((uint32_t)1 « 17),
    };

    /// <summary>
    /// Enumeration of corner mode flags used with CORNER_MODE axis parameter
    /// </summary>
    enum class CornerMode : uint32_t
    {
        NONE = 0,

        RESERVED = ((uint32_t)1 « 0),

        AUTO_SPEED_CONTROL = ((uint32_t)1 « 1),

        CORNER_STATE_MACHINE = ((uint32_t)1 « 2),

        AUTO_RADIUS_SPEED_CONTROL = ((uint32_t)1 « 3),

        AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE = ((uint32_t)1 « 4),

        AUTO_RADIUS_SPEED_CONTROL_CENTRIPETAL_MODE_USE_ACCEL = ((uint32_t)1 « 5),

        AUTO_BLEND_MODE_0_USING_CORNER_RADIUS = ((uint32_t)1 « 6),

        AUTO_BLEND_MODE_1_USING_CORNER_RADIUS = ((uint32_t)1 « 7),
    };

    /// <summary>
    /// Enumeration of corner state flags used with CORNER_STATE axis parameter
    /// </summary>
    enum class CornerState : uint32_t
    {
        NONE = 0,

        HELD = ((uint32_t)1 « 0),

        LOAD = ((uint32_t)1 « 1),

        LOAD_ZERO_SPEED = ((uint32_t)1 « 2),
    };

    /// <summary>
    /// Motion types returned by MTYPE and NTYPE axis parameters
```

```
    /// </summary>
    enum class MotionType : uint32_t
    {
        IDLE = 0,

        MOVE = 1,

        MOVEABS = 2,

        MHELICAL = 3,

        MOVECIRC = 4,

        MOVEMODIFY = 5,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVESP = 6,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVEABSSP = 7,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVECIRCSP = 8,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MHELICALSP = 9,

        FORWARD = 10,

        REVERSE = 11,

        DATUM = 12,

        CAM = 13,

        FORWARD_JOG = 14,

        REVERSE_JOG = 15,

        WETCAM = 16,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        CAMBOX = 20,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        CONNECT = 21,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        MOVELINK = 22,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        CONNPATH = 23,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        FLEXLINK = 24,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        CAMLINK = 25,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        COMPENSATE_XY = 26,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
```

```
        PICKLINK = 27,

        /// <summary>
        /// Linked move (no VPU required)
        /// </summary>
        CAMIN = 28,

        /// <summary>
        /// ??? From FORWARD to less than this number, no lookahead
        /// </summary>
        MOVETANG = 30,

        MSPHERICAL = 31,

        MTRANSCIRC = 32,

        MOVESEQ = 33,

        MOVEABSSEQ = 34,

        MOVETURN = 35,

        MOVEPICK = 36,

        MOVECONTOUR = 37,

        MOVEDISPENSE = 38,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MSPHERICALSP = 40,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MTRANSCIRCSP = 41,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVESEQSP = 42,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVEABSSEQSP = 43,

        /// <summary>
        /// MoveJoint for RPS
        /// </summary>
        MOVEJ = 44,

        /// <summary>
        /// MoveLinear for RPS
        /// </summary>
        MOVEL = 45,

        /// <summary>
        /// MoveCircular for RPS
        /// </summary>
        MOVEC = 46,

        /// <summary>
        /// MoveBlend for RPS
        /// </summary>
        MOVEBLEND = 47,

        /// <summary>
        /// MoveJointRelative for RPS
        /// </summary>
        MOVEJREL = 48,

        /// <summary>
        /// MoveLinearRelative for RPS
        /// </summary>
        MOVELREL = 49,

        MOVETEST = 50,

        /// <summary>
        /// MovePick for RPS. Note : if you don't see an implementation, I have just reserved these
    MTYPES to implement them (within 3 months) - SV
        /// </summary>
        MOVEP = 51,

        /// <summary>
```

```
        /// MoveUndulate for RPS.
        /// </summary>
        MOVEU = 52,

        /// <summary>
        /// MoveBezier for RPS. Note : if you don't see an implementation, I have just reserved these MTYPES to implement them (within 3 months) - SV
        /// </summary>
        MOVEB = 53,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVESPCONT = 60,

        /// <summary>
        /// Lookahead scannable
        /// </summary>
        MOVEABSSPCONT = 61,

        GEARIN = 62,

        ACCPROFILE = 70,

        VELPROFILE = 71,

        POSPROFILE = 72,

        COMBINEAXES = 73,

        OPENMOVE_BASE = 100,

        OPENMOVE_1 = OPENMOVE_BASE,

        OPENMOVE_2 = OPENMOVE_BASE + 1,

        OPENMOVE_3 = OPENMOVE_BASE + 2,

        OPENMOVE_4 = OPENMOVE_BASE + 3,

        OPENMOVE_5 = OPENMOVE_BASE + 4,
    };

    enum class AxisStatus : uint64_t
    {
        NONE = 0,

        /// <summary>
        /// 'L' - Speed limit active
        /// </summary>
        LOOKAHEAD_OVERRIDE = ((uint64_t)1 « 0 ),

        /// <summary>
        /// 'W' - Following error warning range
        /// </summary>
        FE_WARNING = ((uint64_t)1 « 1 ),

        /// <summary>
        /// 'A' - Communications error to remote drive
        /// </summary>
        COMMS_ERROR = ((uint64_t)1 « 2 ),

        /// <summary>
        /// 'M' - Remote drive error (EtherCAT Use: Set if status word Fault Flag raised)
        /// </summary>
        REMOTE_DRIVE_ERROR = ((uint64_t)1 « 3 ),

        /// <summary>
        /// 'F' - In forward hardware limit
        /// </summary>
        IN_FORWARD_LIMIT = ((uint64_t)1 « 4 ),

        /// <summary>
        /// 'R' - In reverse hardware limit
        /// </summary>
        IN_REVERSE_LIMIT = ((uint64_t)1 « 5 ),

        /// <summary>
        /// 'D' - Datuming in progress
        /// </summary>
        DATUMING = ((uint64_t)1 « 6 ),

        /// <summary>
        /// 'H' - Feedhold active
        /// </summary>
        FEEDHOLD = ((uint64_t)1 « 7 ),

```

```
        /// <summary>
        /// 'E' - Following error exceeds limit
        /// </summary>
        FE_EXCEEDS_LIMIT = ((uint64_t)1 « 8 ),

        /// <summary>
        /// 'X' - Forward software limit active
        /// </summary>
        FORWARD_SOFTWARE_LIMIT = ((uint64_t)1 « 9 ),

        /// <summary>
        /// 'Y' - Reverse software limit active
        /// </summary>
        REVERSE_SOFTWARE_LIMIT = ((uint64_t)1 « 10),

        /// <summary>
        /// 'C' - Cancelling the move (from CANCEL(0), CANCEL(2) and CANCEL(5), the latter two also setting AS_CANCEL_BUFFERED and AS_CANCEL_MOVEADD respectively)
        /// </summary>
        CANCEL_CURRENT = ((uint64_t)1 « 11),

        /// <summary>
        /// 'O' - Pulse output axis overspeed
        /// </summary>
        PULSE_OVER_SPEED = ((uint64_t)1 « 12),

        /// <summary>
        /// 'T' - Tangential move decelerating
        /// </summary>
        OVERRIDE_TO_ZERO = ((uint64_t)1 « 13),

        /// <summary>
        /// ' ' - TODO: Doesn't seem to be used
        /// </summary>
        MOVE_CANCEL_DONE = ((uint64_t)1 « 14),

        /// <summary>
        /// 'V' - Volume limit active (TODO: I can't see where this is used - Remove?)
        /// </summary>
        VOLUME_LIMIT = ((uint64_t)1 « 15),

        /// <summary>
        /// 'I' - Axis forward software limit active
        /// </summary>
        AXIS_FS_LIMIT = ((uint64_t)1 « 16),

        /// <summary>
        /// 'J' - Axis reverse software limit active
        /// </summary>
        AXIS_RS_LIMIT = ((uint64_t)1 « 17),

        /// <summary>
        /// 'P' - Encoder poer supply overload
        /// </summary>
        ENCODER_OVER_I = ((uint64_t)1 « 18),

        /// <summary>
        /// 'N' - Hardware P-switch FIFO not empty
        /// </summary>
        PSWITCH_FIFO_NOT_EMPTY = ((uint64_t)1 « 19),

        /// <summary>
        /// 'B' - Hardware P-switch FIFO full
        /// </summary>
        PSWITCH_FIFO_FULL = ((uint64_t)1 « 20),

        /// <summary>
        /// 'K' - Kinematic frame stopped (due to 1 hour timeout)
        /// </summary>
        KINEMATICS_1HOUR = ((uint64_t)1 « 21),

        /// <summary>
        /// ' ' - BISS encoder warning flag
        /// </summary>
        BISS_CLEAN_WARNING = ((uint64_t)1 « 22),

        /// <summary>
        /// ' ' - World forward software limit active (TODO: This is a duplicate of RS_WORLD_FS_LIMIT)
        /// </summary>
        WORLD_FS_LIMIT = ((uint64_t)1 « 23),

        /// <summary>
        /// ' ' - World reverse software limit active (TODO: This is a duplicate of RS_WORLD_RS_LIMIT)
        /// </summary>
        WORLD_RS_LIMIT = ((uint64_t)1 « 24),

        /// <summary>
```

```
        /// ' ' - Mode 3 cancel status (from CANCEL(3))
        /// </summary>
        CANCEL_CURRENT_AT_SPEED = ((uint64_t)1 « 25),

        /// <summary>
        /// ' ' - Remote node status error (EtherCAT Use: Set based on masked flags in status word)
        /// </summary>
        REMOTE_NODE_STATUS_ERROR = ((uint64_t)1 « 26),

        /// <summary>
        /// ' ' - Hardware P-switch FIFO half full
        /// </summary>
        PSWITCH_FIFO_HALF_FULL = ((uint64_t)1 « 27),

        /// <summary>
        /// ' ' - MOVEADD additional offset active (axis has MOVEADD in progress)
        /// </summary>
        MOVEADD = ((uint64_t)1 « 28),

        /// <summary>
        /// ' ' - MOVEADD is being cancelled (from CANCEL(5))
        /// </summary>
        CANCEL_MOVEADD = ((uint64_t)1 « 29),

        /// <summary>
        /// ' ' - EtherCAT emergency message received from remote drive (EtherCAT Use: Set if emergency message received)
        /// </summary>
        REMOTE_NODE_ERROR2 = ((uint64_t)1 « 30),

        /// <summary>
        /// ' ' - TODO: Doesn't seem to be used (all the code is inside a #if 0)
        /// </summary>
        SINGULARITY = ((uint64_t)1 « 31),

        /// <summary>
        /// ' ' - A user frame has thrown an error
        /// </summary>
        OPEN_FRAME_ERROR = ((uint64_t)1 « 32),

        /// <summary>
        /// ' ' - Indicates that the VPU is in an error state
        /// </summary>
        VP_ERROR = ((uint64_t)1 « 33),

        /// <summary>
        /// ' ' - Cancel all buffered moves (from CANCEL(1) or CANCEL(2), the latter also setting AS_CANCEL_CURRENT)
        /// </summary>
        CANCEL_BUFFERED = ((uint64_t)1 « 34),

        /// <summary>
        /// ' ' - Signifies that deceleration will stray into the buffered moves (stopping distance bigger than distance to go)
        /// </summary>
        CANCEL_BUFFERED_DECEL_INTO_BUFFERED_MOVES = ((uint64_t)1 « 35),

        /// <summary>
        /// ' ' - Signifies that deceleration will finish at the end of the current move (stopping distance less than distance to go)
        /// </summary>
        CANCEL_BUFFERED_DECEL_WITHIN_CURRENT_MOVE = ((uint64_t)1 « 36),

        /// <summary>
        /// ' ' - Cancel last buffered move (from CANCEL(4))
        /// </summary>
        CANCEL_LAST_BUFFERED = ((uint64_t)1 « 37),
    };

    enum class RobotStatus : uint32_t
    {
        NONE = 0,

        WORLD_FS_LIMIT = ((uint32_t)1 « 0),

        WORLD_RS_LIMIT = ((uint32_t)1 « 1),

        ROBOT_FS_LIMIT = ((uint32_t)1 « 2),

        ROBOT_RS_LIMIT = ((uint32_t)1 « 3),

        TCP_FS_LIMIT = ((uint32_t)1 « 4),

        TCP_RS_LIMIT = ((uint32_t)1 « 5),

        FE_WARNING = ((uint32_t)1 « 6),

```

```
        FE_EXCEEDS_LIMIT = ((uint32_t)1 « 7),

        WRIST_SINGULARITY = ((uint32_t)1 « 8),

        ALIGNMENT_SINGULARITY = ((uint32_t)1 « 9),

        ELBOW_SINGULARITY = ((uint32_t)1 « 10),

        MAX_SPEED_LIMIT = ((uint32_t)1 « 11),

        COLLIDED = ((uint32_t)1 « 12),

        NOT_IN_POS = ((uint32_t)1 « 13),

        ELBOW_CLOSED_SINGULARITY = ((uint32_t)1 « 14),

        DISALLOW_JOG = ((uint32_t)1 « 15),
    };

    enum class SyncControl : int32_t
    {
        WAITING_TO_START = -1,

        NO_SYNC = 0,

        STARTING_SYNC = 1,

        PERFORMING_SYNC = 2,

        SYNCHRONISED = 3,

        STOPPING_SYNC = 4,

        STARTING_INTERP_MOVE = 5,

        PERFORMING_INTERP_MOVE = 6,

        STARTING_RESYNC = 10,

        PERFORMING_RESYNC = 11,

        STARTING_RESYNC_TO_USERFRAME = 20,

        PERFORMING_RESYNC_TO_USERFRAME = 21,
    };

    enum class RegistMode : int32_t
    {
        /// <summary>
        /// Cancel any previous registration commands
        /// </summary>
        Cancel = 0,

        /// <summary>
        /// Z Mark rising into REG_POS
        /// </summary>
        ZAMarkRising = 1,

        /// <summary>
        /// Z Mark falling into REG_POS
        /// </summary>
        ZAMarkFalling = 2,

        /// <summary>
        /// RA input rising into REG_POS
        /// </summary>
        RAInputRising = 3,

        /// <summary>
        /// RA input falling into REG_POS
        /// </summary>
        RAInputFalling = 4,

        /// <summary>
        /// RA input rising into REG_POS and Z Mark rising into REG_POSB
        /// </summary>
        RAInputRising_ZBMarkRising = 6,

        /// <summary>
        /// RA input rising into REG_POS and Z Mark falling into REG_POSB
        /// </summary>
        RAInputRising_ZBMarkFalling = 7,

        /// <summary>
        /// RA input falling into REG_POS and Z Mark rising into REG_POSB
        /// </summary>
        RAInputFalling_ZBMarkRising = 8,
```

```
        /// <summary>
        /// RA input falling into REG_POS and Z Mark falling into REG_POSB
        /// </summary>
        RAInputFalling_ZBMarkFalling = 9,

        /// <summary>
        /// RA input rising into REG_POS and RB input rising into REG_POSB
        /// </summary>
        RAInputRising_RBInputRising = 10,

        /// <summary>
        /// RA input rising into REG_POS and RB input rising into REG_POSB
        /// </summary>
        RAInputRising_RBInputFalling = 11,

        /// <summary>
        /// RA input falling into REG_POS and RB input falling into REG_POSB
        /// </summary>
        RAInputFalling_RBInputRising = 12,

        /// <summary>
        /// RA input falling into REG_POS and RB input falling into REG_POSB
        /// </summary>
        RAInputFalling_RBInputFalling = 13,

        /// <summary>
        /// Z Mark rising into REG_POSB
        /// </summary>
        ZBMarkRising = 14,

        /// <summary>
        /// Z Mark falling into REG_POSB
        /// </summary>
        ZBMarkFalling = 15,

        /// <summary>
        /// RB input rising into REG_POSB
        /// </summary>
        RBInputRising = 16,

        /// <summary>
        /// RB input falling into REG_POSB
        /// </summary>
        RBInputFalling = 17,

        /// <summary>
        /// Replacement mode for 1..16 with additional configuration options
        /// </summary>
        HardwareRegistration = 20,

        /// <summary>
        /// Time interpolated registration
        /// </summary>
        TimeInterpolatedRegistration = 21,

        /// <summary>
        /// Replacement for modes 32..39 and 64..71
        /// </summary>
        MultiChannelRegistration = 22,

        /// <summary>
        /// This mode assigns a 2.4 microseconds minimum pulse width to the axis. This affects any REGIST mode that is used.
        /// </summary>
        EnableFilter = 23,

        /// <summary>
        /// This mode assigns a 0.15 microseconds minimum pulse width to the axis (default). This affects any REGIST mode that is used.
        /// </summary>
        DisableFilter = 24,

        /// <summary>
        /// Position must be inside OPEN_WIN .. CLOSE_WIN. To be combined with modes 1..17
        /// </summary>
        InsideOpenCloseWindow = 256,

        /// <summary>
        /// Position must be outside OPEN_WIN .. CLOSE_WIN. To be combined with modes 1..17
        /// </summary>
        OutsideOpenCloseWindow = 768,
    };

    enum class RegistEdge : int32_t
    {
        Rising = 0,
```

```
        Falling = 1,
    };

    enum class RegistWindow : int32_t
    {
        No = 0,

        InsideOpenCloseWin = 1,

        OutsideOpenCloseWin = 2,

        VirtualInputHigh = 3,

        VirtualInputLow = 4,
    };

    /// <summary>
    /// The execution state of the process
    /// </summary>
    enum class ProcessStatus : uint32_t
    {
        /// <summary>
        /// No program assigned to the process. not called by scheduler
        /// </summary>
        Stopped = 0,

        /// <summary>
        /// Program assigned and being processed by the scheduler
        /// </summary>
        Ready,

        /// <summary>
        /// Step through the program (one command) and return to Paused
        /// </summary>
        Stepping,

        /// <summary>
        /// A program executing halted
        /// </summary>
        Paused,

        /// <summary>
        /// Program about to be paused
        /// </summary>
        Pausing,

        /// <summary>
        /// A request has been made for the program to stop
        /// </summary>
        Aborted,

        /// <summary>
        /// Request to suspend program
        /// </summary>
        Suspend,

        /// <summary>
        /// Program is suspended
        /// </summary>
        Suspended,

        /// <summary>
        /// Program is sleeping
        /// </summary>
        Sleeping,
    };

    /// <summary>
    /// Enumeration of results of IS_FILE/IS_PROG command
    /// </summary>
    enum class FileExistsFlags : int32_t
    {
        /// <summary>
        /// File or program does not exist on Motion Coordinator
        /// </summary>
        NoFile = 0,

        /// <summary>
        /// File or program exists and is not open
        /// </summary>
        HasFileNotOpen = -1,

        /// <summary>
        /// File or program exists and is open for reading
        /// </summary>
        HasFileOpenForReading = -2,
```

```
        /// <summary>
        /// File or program exists and is open for writing
        /// </summary>
        HasFileOpenForWriting = -3,

        /// <summary>
        /// File or program exists and is open for reading and writing
        /// </summary>
        HasFileOpenForReadingAndWriting = -4,
    };

    /// <summary>
    /// The security code type used with project decryption key functions
    /// </summary>
    enum class SecurityCodeType : int32_t
    {
        /// <summary>
        /// Controller security code
        /// </summary>
        Controller = 0,

        /// <summary>
        /// OEM security code
        /// </summary>
        OEM = 1,

        /// <summary>
        /// User security code
        /// </summary>
        User = 2,
    };


////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

    /// <summary>
    /// Version info
    /// </summary>
    struct VersionInfo
    {
        uint16_t major;
        uint16_t minor;
        uint16_t patch;
        uint16_t build;
    };

    /// <summary>
    /// Prototype for the callback function that will be called when events happen on the underlying connection.
    /// </summary>
    typedef std::function<void TRIO_CALLCONV(EventType event_id, uint64_t integer_data, const std::string_view& string_data)> EventCallback;

    /// <summary>
    /// Data used with a callback for processing the response of Dir command
    /// </summary>
    /// <remarks>
    /// NOTE: (BD) There are mirror structurs in Trio_UnifiedApi_C.h, Trio_UnifiedApi_CPP.h and ITrioConnection.cs
    ///            make sure they are in sync when doing changes here
    /// </remarks>
    struct FileInfo
    {
        /// <summary>
        /// Type of the file on the controller
        /// </summary>
        FileType type;
        /// <summary>
        /// Text code size
        /// </summary>
        uint32_t size;
        /// <summary>
        /// Compiled binary size
        /// </summary>
        uint32_t compiled_size;
        /// <summary>
        /// Autorun process number
        /// </summary>
        int32_t autorun_process;
        /// <summary>
        /// Flags related to files and programs on the controller
        /// </summary>
        FileFlags flags;

        FileInfo()
        {
```

```
            type = FileType::BASIC;
            size = compiled_size = 0;
            autorun_process = -1;
            flags = FileFlags::None;
        }

    };
    /// <summary>
    /// Prototype of a callback for processing the response of Dir command
    /// </summary>
    typedef std::function<void TRIO_CALLCONV(const FileInfo& file, const std::string_view& file_name)> FileCallback;

    /// <summary>
    /// Data used with a callback for streaming back the response of commands that return string data
    /// </summary>
    struct DataStreamPacket
    {
        /// <summary>
        /// The length of the stream, if available. 0 otherwise
        /// </summary>
        uint32_t total_length;
        /// <summary>
        /// The packet number
        /// </summary>
        uint32_t packet_number;
        /// <summary>
        /// The total count of the packets of the stream, if available. 0 otherwise.
        /// </summary>
        uint32_t packet_count;
        /// <summary>
        /// The number of bytes
        /// </summary>
        uint32_t data_length;
    };

    /// <summary>
    /// Prototype of a callback for streaming back the response of commands that return string data
    /// </summary>
    typedef std::function<void TRIO_CALLCONV(const DataStreamPacket& data_info, const char* data)> StreamHandlerCallback;

    /// <summary>
    /// Progress info helper
    /// </summary>
    struct ProgressInfo
    {
        /// <summary>
        /// The length of the process. If 0 the length is unknown.
        /// </summary>
        uint32_t total_length;
        /// <summary>
        /// The current progress position.
        /// </summary>
        uint32_t current_pos;
        /// <summary>
        /// Progress event
        /// </summary>
        ProgressEvent evt;

        // Constructor
        ProgressInfo()
        {
            total_length = current_pos = 0;
            evt = ProgressEvent::Finish;
        }
    };

    /// <summary>
    /// Prototype of a callback for progress report
    /// </summary>
    typedef std::function<void TRIO_CALLCONV(const ProgressInfo& progress)> ProgressHandlerCallback;

    /// <summary>
    /// A structure representing a range of consecutive values
    /// </summary>
    struct Range
    {
        /// <summary>
        /// Start value
        /// </summary>
        uint32_t start;
        /// <summary>
        /// Value count
        /// </summary>
        uint32_t count;

```

```
        /// <summary>
        /// Constructor
        /// </summary>
        Range() : start(0), count(0) {}

        /// <summary>
        /// Constructor
        /// </summary>
        /// <param name="s">Start index</param>
        /// <param name="c">Count</param>
        Range(uint32_t s, uint32_t c) : start(s), count(c) {}
    };

    /// <summary>
    /// Data used with a callback for processing the response of Process command
    /// </summary>
    struct ProcessInfo
    {
        /// <summary>
        /// The process number
        /// </summary>
        int32_t process;
        /// <summary>
        /// Status of the process on the controller
        /// </summary>
        ProcessStatus status;
        /// <summary>
        /// The current line - when the process is in step mode
        /// </summary>
        int32_t line;
        /// <summary>
        /// The process running time
        /// </summary>
        uint64_t time_millisecond;
        /// <summary>
        /// CPU usage in percent (%)
        /// </summary>
        double cpu_usage;

        /// <summary>
        /// Constructor
        /// </summary>
        ProcessInfo()
        {
            process = 0;
            status = (ProcessStatus)0;
            line = 0;
            time_millisecond = 0;
            cpu_usage = 0;
        }
    };
    typedef std::function<void TRIO_CALLCONV(const ProcessInfo& process, const std::string_view&
program_name)> ProcessInfoCallback;

    typedef std::function<bool TRIO_CALLCONV(const std::string_view& string)> StringCallback;


    /// <summary>
    /// Exception raised when error occurs in communication library
    /// </summary>
    class TrioConnectionException : public std::exception
    {
        ErrorCode _errorCode;
    public:
        TRIO_DECLSPEC TrioConnectionException(ErrorCode errorCode);
        ErrorCode errorCode() const { return _errorCode; }

        TRIO_DECLSPEC virtual char const* what() const noexcept override;
    };

    class ITrioConnection;

    /// <summary>
    /// Factory for creating communication instances
    /// </summary>
    class TrioConnectionFactory
    {
    public:
        /// <summary>
        /// Create TCP based connection to Motion Coordinator
        /// </summary>
        /// <param name="event_callback">Callback function to receive various events from the
API</param>
        /// <param name="host_name">Host name or IP of the Motion Coordinator to connect via TCP
protocol</param>
        /// <param name="port">Port on Motion Coordinator open for TCP connections. It is normally
3240 unless Motion Coorinator has custom configuration</param>
```

```
        /// <returns>On success an instance of connection interface is returned</returns>
        static TRIO_DECLSPEC ITrioConnection* TRIO_CALLCONV CreateTCP(EventCallback&& event_callback, const std::string_view& host_name = std::string_view(), uint16_t port = 0);
        /// <summary>
        /// Create PC-MCAT based connection to Motion Coordinator
        /// </summary>
        /// <param name="event_callback">Callback function to receive various events from the API</param>
        /// <returns>On success an instance of connection interface is returned</returns>
        static TRIO_DECLSPEC ITrioConnection* TRIO_CALLCONV CreatePCMCAT(EventCallback&& event_callback);
        /// <summary>
        /// Create Flex7 based connection to Motion Coordinator
        /// </summary>
        /// <param name="event_callback">Callback function to receive various events from the API</param>
        /// <returns>On success an instance of connection interface is returned</returns>
        static TRIO_DECLSPEC ITrioConnection* TRIO_CALLCONV CreateFlex7(EventCallback&& event_callback);
    };

    /// <summary>
    /// Communication object intarface
    /// </summary>
    class ITrioConnection
    {
    protected:

        /// <summary>
        /// Constructor - protected to disallow direct construction
        /// </summary>
        ITrioConnection() = default;
        /// <summary>
        /// Destructor - protected to disallow direct destruction as it must happen from inside creating module
        /// </summary>
        virtual ~ITrioConnection() = default;

    public:

        /// <summary>
        /// Publicly exposed method for communication object destruction
        /// Calling it is the proper way of destoring ITrioConnection instance
        /// After calling this method other methods should not be called
        /// </summary>
        virtual void DeleteMe() = 0;

        /// <summary>
        /// Error description
        /// </summary>
        /// <param name="errorCode">Error code</param>
        /// <returns>Description from error code number</returns>
        static TRIO_DECLSPEC const std::string_view& TRIO_CALLCONV ErrorString(ErrorCode errorCode);

        /// <summary>
        /// Get C++ language module version info
        /// </summary>
        /// <returns>On success version info of C++ language module</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        static TRIO_DECLSPEC VersionInfo TRIO_CALLCONV GetModuleVersion();

        /// <summary>
        /// Get version string of C++ language module
        /// </summary>
        /// <returns>On success string representation of C++ language module version</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        static TRIO_DECLSPEC std::string_view TRIO_CALLCONV GetModuleVersionStr();


////////////
```

```
        virtual std::string_view GetTcpHostName() = 0;

        /// <summary>
        /// Set TCP port to connection object. Valid only for TCP connection.
        /// </summary>
        /// <param name="port">New value for the TCP port</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetTcpPort(uint16_t port) = 0;

        /// <summary>
        /// Get current TCP port from connection object. Valid only for TCP connection.
        /// </summary>
        /// <returns>Current value of the TCP port</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint16_t GetTcpPort() = 0;

        /// <summary>
        /// Set new timeout for command execution. Valid only for TCP connection.
        /// </summary>
        /// <param name="timeout_ms">New value for the command execution timeout in milliseconds</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetTcpCommandTimeout(uint32_t timeout_ms) = 0;

        /// <summary>
        /// Get current timeout for command execution. Valid only for TCP connection.
        /// </summary>
        /// <returns>Current value of the command execution timeout in milliseconds</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetTcpCommandTimeout() = 0;

        /// <summary>
        /// Set new timeout for opening TCP connection. Valid only for TCP connection.
        /// </summary>
        /// <param name="timeout_ms">New value for the open connection timeout in milliseconds</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetTcpConnectTimeout(uint32_t timeout_ms) = 0;

        /// <summary>
        /// Get current timeout for opening TCP connection. Valid only for TCP connection.
        /// </summary>
        /// <returns>Current value of the open connection timeout in milliseconds</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetTcpConnectTimeout() = 0;

        /// <summary>
        /// Check if a connection to the Motion Coordinator was opened.
        /// </summary>
        /// <returns>If true if there is active connection to the Motion Coordinator or false otherwise.</returns>
        virtual bool IsConnected() const = 0;

        /// <summary>
        /// Get connection type
        /// </summary>
        /// <returns>Connection type or current connection instance or Unknown if connection is disposed</returns>
        virtual ConnectionType GetConnectionType() const = 0;

        /// <summary>
        /// Get Unified API version info
        /// </summary>
        /// <returns>On success version info of the Unified API</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual VersionInfo GetApiVersion() = 0;

        /// <summary>
        /// Get Unified API version string
        /// </summary>
        /// <returns>On success string representation of the Unified API version</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual std::string_view GetApiVersionStr() = 0;

        /// <summary>
        /// Get the firmware version of the Motion Coordinator.
        /// </summary>
        /// <returns>On success Motion Coordinator version as a floating point number</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual double GetFirmwareVersionNumber() = 0;

        /// <summary>
        /// Open a connection to the Motion Coordinator.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// In case of TCP connection <see cref="SetTcpConnectTimeout"/> function can be used to adjust the maximum connection delay
        /// </remarks>
        virtual void OpenConnection() = 0;

        /// <summary>
        /// Closes previously open connection to the Motion Coordinator.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void CloseConnection() = 0;

        /// <summary>
        /// Perform software reset on Motion Coordinator.
        /// </summary>
        /// <param name="mode">Reset mode. See the TrioBASIC EX documentation for more details.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ex(int32_t mode = 0) = 0;

        /// <summary>
        /// Used to cancel current or buffered axis commands on an axis or an interpolating axis group.
        /// </summary>
        /// <param name="mode">Cancel mode</param>
        /// <param name="axis">Index of axis subject to cancel operation or -1 to cancel default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Cancel(int32_t mode, int32_t axis = -1) = 0;

        /// <summary>
        /// Links the demand position of the base axis to the measured movements of the driving axes to produce an electronic gearbox.
        /// </summary>
        /// <param name="ratio">This parameter holds the number of edges the base axis is required to move per increment of the driving axis. The ratio value can be either positive or negative. The ratio is always specified as an encoder edge ratio</param>
        /// <param name="linkAxis">This parameter specifies the axis to link to.</param>
        /// <param name="mode">
        ///  0   The axis follows the measured position(MPOS) of the driving axis. (default)<br/>
        ///  1   The axis follows the demand position (DPOS) of the driving axis.
        /// </param>
        /// <param name="axis">Index of axis subject to cancel operation or -1 to cancel default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Connect(double ratio, int32_t linkAxis, ConnectMode mode = ConnectMode::MPOS, int32_t axis = -1) = 0;

        /// <summary>
        /// Enables you to link to the path of an interpolated movement by linking the demand position of the base axis, to the interpolated path distance of the driving axis.
        /// </summary>
        /// <param name="ratio">This is the ratio between the interpolated distance moved on the driving axis to the distance moved on the base axis. </param>
        /// <param name="linkAxis">This parameter specifies the axis to link to.</param>
        /// <param name="axis">Index of axis subject to cancel operation or -1 to cancel default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ConnPath(double ratio, int32_t linkAxis, int32_t axis = -1) = 0;

        /// <summary>
        /// The PSWITCH command allows an output to be set when a predefined position is reached, and to be reset when a second position is reached
        /// </summary>
        /// <param name="switchNumber">The switch number in the range 0..63 </param>
        /// <param name="mode">Mode of operation. See PSwitchMode enumeration for more information</param>
        /// <param name="axis">Axis to link the PSWITCH to, may be any real or virtual axis </param>
        /// <param name="outputNumber">Selects the output to set, can be any real or virtual output </param>
        /// <param name="outputState">False - turn the output OFF at setpos, or true - turn the output ON at setpos</param>
        /// <param name="setPosition">The position at which output is set or reset, in user units </param>
        /// <param name="resetPosition">The position at which output is set or reset, in user units
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</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void PSwitch(int32_t switchNumber, PSwitchMode mode, int32_t axis, uint32_t outputNumber, bool outputState, double setPosition, double resetPosition) = 0;

        /// <summary>
        /// The PSWITCHOFF command is used to disable
        /// </summary>
        /// <param name="switchNumber">The switch number in the range 0..63 </param>
        /// <param name="hold">
        ///  false = The PSWITCH output will hold in the same state it was when the PSWITCH is set to OFF . (Default)<br/>
        ///  true = The PSWITCH output is forced OFF even if it was ON when the PSWTICH is set to OFF .
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void PSwitchOff(int32_t switchNumber, bool hold = false) = 0;

        /// <summary>
        /// The PSWITCHMULTI command allows an output to be set that turns the output on and off at selected positions loaded from the TABLE memory.
        /// </summary>
        /// <param name="switchNumber">The switch number in the range 0..63 </param>
        /// <param name="mode">Mode of operation. See PSwitchMultiMode enumeration for more information</param>
        /// <param name="axis">Axis to link the PSWITCH to, may be any real or virtual axis </param>
        /// <param name="outputNumber">Selects the output to set, can be any real or virtual output </param>
        /// <param name="outputState">False - turn the output OFF at setpos, or true - turn the output ON at setpos</param>
        /// <param name="tableStart">TABLE index for start of PSWITCH data </param>
        /// <param name="tableEnd">TABLE index for end of PSWITCH data </param>
        /// <param name="pulseTime">Time in servo-cycles when pulse output mode bit 4 is set </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void PSwitchMulti(int32_t switchNumber, PSwitchMultiMode mode, int32_t axis, uint32_t outputNumber, bool outputState, uint32_t tableStart, uint32_t tableEnd, uint32_t pulseTime = 0) = 0;

        /// <summary>
        /// Execute arbitrary command.
        /// </summary>
        /// <param name="command">Command to be executed</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Execute(const std::string_view& command) = 0;

        /// <summary>
        /// Execute arbitrary command with response.
        /// </summary>
        /// <param name="command">Command to be executed</param>
        /// <param name="buff">Buffer to receive the response string characters</param>
        /// <param name="buff_size">Size of the buffer on input and on exit required buffer size including null terminator</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ExecuteWithResponse(const std::string_view& command, char* buff, uint32_t& buff_size) = 0;

        /// <summary>
        /// Execute arbitrary command with response.
        /// </summary>
        /// <param name="command">Command to be executed</param>
        /// <param name="handler">Callback function to receive response string chunks</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ExecuteWithResponse(const std::string_view& command, StreamHandlerCallback&& handler) = 0;

        /// <summary>
        /// Execute arbitrary command with response.
        /// </summary>
        /// <param name="command">Command to be executed</param>
        /// <returns>Command response</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual std::string ExecuteWithResponse(const std::string_view& command) = 0;

        /// <summary>
        /// The BASE command is used to direct all subsequent motion commands and axis parameter read/writes to a particular axis, or group of axes. The default setting is a sequence: 0, 1, 2, 3...
        /// </summary>
        /// <param name="axis">The axes to send the motion commands to</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisBase(int32_t axis) = 0;
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```
        /// <summary>
        /// Set multiple base axes.
        /// </summary>
        /// <param name="axes">Array of axis indexes to set as base</param>
        /// <param name="axisCount">Number of axis indexes to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMultiAxisBase(const int32_t axes[], uint32_t axisCount) = 0;

        /// <summary>
        /// Performs one of 6 homing sequences to locate an axis to an absolute position.
        /// </summary>
        /// <param name="sequence">Bit-mask for homing sequence to be executed</param>
        /// <param name="axis">Index of the axis to be subject of datum sequence or -1 to datum default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Datum(int32_t sequence, int32_t axis = -1) = 0;

        /// <summary>
        /// Defines the current axis position as a new absolute value
        /// </summary>
        /// <param name="pos">Absolute position to set on axis in user units</param>
        /// <param name="axis">Index of the axis for which the default position to be set or -1 to set position to default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void DefPos(double pos, int32_t axis = -1) = 0;

        /// <summary>
        /// Defines the current axis position as a new absolute value to multiple axes
        /// </summary>
        /// <param name="pos">Array of absolute positions in user units to set on axes defined by SetMultiAxisBase</param>
        /// <param name="posCount">Number of positions contained in array.</param>
        /// <param name="axis">Index of the first axis for which the default position to be set or -1 to set position to default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void DefPosMulti(const double pos[], uint32_t posCount, int32_t axis = -1) = 0;

        /// <summary>
        /// Sets continuous forward movement.
        /// </summary>
        /// <param name="axis">Axis index to set forward movement to or -1 to apply to default base axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Forward(int32_t axis = -1) = 0;

        /// <summary>
        /// Sets continuous reverse movement.
        /// </summary>
        /// <param name="axis">Axis index to set reverse movement to or -1 to apply to default base axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Reverse(int32_t axis = -1) = 0;

        /// <summary>
        /// Incremental move of single axis
        /// </summary>
        /// <param name="pos">Distance to move axis in user units</param>
        /// <param name="axis">Index of the axis for which to be subject of move operation or -1 to use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveRel(double pos, int32_t axis = -1) = 0;

        /// <summary>
        /// Incremental move of multiple axes
        /// </summary>
        /// <param name="dist">Array of distances in user units to move axes defined by SetMultiAxisBase</param>
        /// <param name="distCount">Number of positions contained in array.</param>
        /// <param name="axis">Index of the first axis for which to be subject of move operation or -1 to use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveRelMulti(const double dist[], uint32_t distCount, int32_t axis = -1) = 0;

        /// <summary>
        /// Incremental move of single axis with speed changed between moves
        /// </summary>
        /// <param name="pos">Distance to move axis in user units</param>
        /// <param name="axis">Index of the axis for which to be subject of move operation or -1 to
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use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveRelSp(double pos, int32_t axis = -1) = 0;

        /// <summary>
        /// Incremental move of multiple axes with speed changed between moves
        /// </summary>
        /// <param name="dist">Array of distances in user units to move axes defined by SetMultiAxisBase</param>
        /// <param name="distCount">Number of positions contained in array.</param>
        /// <param name="axis">Index of the first axis for which to be subject of move operation or -1 to use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveRelSpMulti(const double dist[], uint32_t distCount, int32_t axis = -1) = 0;

        /// <summary>
        /// Absolute positional move of single axis
        /// </summary>
        /// <param name="pos">Absolute position to move axis to in user units</param>
        /// <param name="axis">Index of the axis for which to be subject of move operation or -1 to use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveAbs(double pos, int32_t axis = -1) = 0;

        /// <summary>
        /// Absolute positional move of multiple axes
        /// </summary>
        /// <param name="pos">Array of absolute positions in user units to move axes defined by SetMultiAxisBase</param>
        /// <param name="posCount">Number of positions contained in array.</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveAbsMulti(const double pos[], uint32_t posCount, int32_t axis = -1) = 0;

        /// <summary>
        /// Absolute positional move of single axis with speed changed between moves
        /// </summary>
        /// <param name="pos">Absolute position to move axis to in user units</param>
        /// <param name="axis">Index of the axis for which to be subject of move operation or -1 to use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveAbsSp(double pos, int32_t axis = -1) = 0;

        /// <summary>
        /// Absolute positional move of multiple axes with speed changed between moves
        /// </summary>
        /// <param name="pos">Array of absolute positions in user units to move axes defined by SetMultiAxisBase</param>
        /// <param name="posCount">Number of positions contained in array.</param>
        /// <param name="axis">Index of the first axis for which to be subject of move operation or -1 to use default base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveAbsSpMulti(const double pos[], uint32_t posCount, int32_t axis = -1) = 0;

        /// <summary>
        /// Moves 2 orthogonal axes in such a way as to produce a circular arc at the tool point.
        ///  The length and radius of the arc are defined by the five parameters in the command line.
        ///  The move parameters are always relative to the end of the last specified move.
        ///  This is the start position on the circle circumference. Axis 1 is the current BASE axis.
        ///  Axis 2 is the next axis in the BASE array.
        ///  The first 4 distance parameters are scaled according to the current unit conversion factor for the BASE axis.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <param name="finishBase">Position on BASE axis to finish at.</param>
        /// <param name="finishNext">Position on next axis in BASE array to finish at.</param>
        /// <param name="centreBase">Position on BASE about which to move.</param>
        /// <param name="centreNext">Position on next axis in BASE array about which to move.</param>
        /// <param name="direction">
        ///  0 Arc is interpolated in an anti - clockwise direction<br/>
        ///  1 Arc is interpolated in a clockwise direction<br/>
        ///  2 Arc is interpolated using the shortest path to endpoint<br/>
        ///  3 Arc is interpolated using the longest path to endpoint
        /// </param>
        /// <param name="transitionAngle">Transition angle (in Radians) for any filleted arcs that are generated.  Set to 0 if no transition curve is to be applied.</param>
        /// <param name="tableOutput">Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
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module.</exception>
        virtual void MoveCirc(int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double transitionAngle = 0, int32_t tableOutput = -1) = 0;

        /// <summary>
        /// Works as MoveCirc and additionally allows vector speed to be changed when using multiple moves in the look ahead buffer when MERGE = ON , using additional parameters FORCE_SPEED and ENDMOVE_SPEED.
        ///  Optional parameters ta and output allow the use of transition curves.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <param name="finishBase">Position on BASE axis to finish at.</param>
        /// <param name="finishNext">Position on next axis in BASE array to finish at.</param>
        /// <param name="centreBase">Position on BASE about which to move.</param>
        /// <param name="centreNext">Position on next axis in BASE array about which to move.</param>
        /// <param name="direction">
        ///  0 Arc is interpolated in an anti - clockwise direction<br/>
        ///  1 Arc is interpolated in a clockwise direction<br/>
        ///  2 Arc is interpolated using the shortest path to endpoint<br/>
        ///  3 Arc is interpolated using the longest path to endpoint
        /// </param>
        /// <param name="transitionAngle">Transition angle (in Radians) for any filleted arcs that are generated.  Set to 0 if no transition curve is to be applied.</param>
        /// <param name="tableOutput">Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveCircSp(int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double transitionAngle = 0, int32_t tableOutput = 0) = 0;

        /// <summary>
        /// Performs a helical move.
        ///  Moves 2 orthogonal axes in such a way as to produce a circular arc at the tool point with a simultaneous linear move on a third axis.
        ///  The first 5 parameters are similar to those of an MOVECIRC command.The sixth parameter defines the simultaneous linear move.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <param name="finishBase">Position on the base axis to finish at.</param>
        /// <param name="finishNext">Position on the next axis in the base vector to finish at.</param>
        /// <param name="centreBase">Position on the base axis about which to move.</param>
        /// <param name="centreNext">Position on the next axis in the base vector about which to move.</param>
        /// <param name="direction">Direction in which to move around the arc.</param>
        /// <param name="linearDistance">Distance to move on the third axis in the base vector.</param>
        /// <param name="mode">Mode of operation. See the MHELICAL command documentation for more details.</param>
        /// <param name="transitionAngle">Transition angle (in Radians) for any filleted arcs that are generated.  Set to 0 if no transition curve is to be applied.</param>
        /// <param name="tableOutput">Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveHelical(int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double linearDistance, int32_t mode, double transitionAngle = 0, int32_t tableOutput = 0) = 0;

        /// <summary>
        /// Performs a helical move the same as MHELICAL and additionally allows vector speed to be changed when using multiple moves in the buffer.
        ///  Uses additional axis parameters FORCE_SPEED, ENDMOVE_SPEED and STARTMOVE_SPEED.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <param name="finishBase">Position on the base axis to finish at.</param>
        /// <param name="finishNext">Position on the next axis in the base vector to finish at.</param>
        /// <param name="centreBase">Position on the base axis about which to move.</param>
        /// <param name="centreNext">Position on the next axis in the base vector about which to move.</param>
        /// <param name="direction">Direction in which to move around the arc.</param>
        /// <param name="linearDistance">Distance to move on the third axis in the base vector.</param>
        /// <param name="mode">Mode of operation. See the MHELICAL command documentation for more details.</param>
        /// <param name="transitionAngle">Transition angle (in Radians) for any filleted arcs that are generated.  Set to 0 if no transition curve is to be applied.</param>
        /// <param name="tableOutput">Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveHelicalSp(int32_t axis, double finishBase, double finishNext, double centreBase, double centreNext, int32_t direction, double linearDistance, int32_t mode, double
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transitionAngle = 0, int32_t tableOutput = 0) = 0;

        /// <summary>
        /// Moves the three axis group defined in BASE along a spherical path with a vector speed determined by the SPEED set in the first axis of the BASE array.
        ///  There are 4 modes of operation with the option of finishing the move at an endpoint different to the start, or returning to the start point to complete a circle.
        ///  The path of the movement in 3D space can be defined either by specifying a point somewhere along the path, or by specifying the centre of the sphere.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <param name="endMidX">End position of the first axis</param>
        /// <param name="endMidY">End position of the second axis</param>
        /// <param name="endMidZ">End position of the third axis</param>
        /// <param name="centreX">Centre position of the first axis</param>
        /// <param name="centreY">Centre position of the second axis</param>
        /// <param name="centreZ">Centre position of the third axis</param>
        /// <param name="mode">Mode of operation. See the MSPHERICAL command documentation for more details.</param>
        /// <param name="gtpi">If this optional parameter is non zero, modes 0 and 1 will perform a move taking the opposite way around a 360 degree circle to the same endpoint</param>
        /// <param name="rotau">If this optional parameter is non zero, a fourth axis will perform linear interpolation at the same time as the spherical move. The axis is the next in the BASE sequence</param>
        /// <param name="rotav">If this optional parameter is non zero, a fifth axis will perform linear interpolation at the same time as the spherical move</param>
        /// <param name="rotaw">If this optional parameter is non zero, a sixth axis will perform linear interpolation at the same time as the spherical move</param>
        /// <param name="rotap">If this optional parameter is non zero, a seventh axis will perform linear interpolation at the same time as the spherical move</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveSpherical(int32_t axis, double endMidX, double endMidY, double endMidZ, double centreX, double centreY, double centreZ, int32_t mode = 0, int32_t gtpi = 0, double rotau = 0, double rotav = 0, double rotaw = 0, double rotap = 0) = 0;

        /// <summary>
        /// Performs a spherical move the same as MSPHERICAL and additionally allows vector speed to be changed when using multiple moves in the look ahead buffer
        ///  when MERGE = ON, using additional parameters FORCE_SPEED, ENDMOVE_SPEED and STARTMOVE_SPEED
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used.</param>
        /// <param name="endMidX">End position of the first axis</param>
        /// <param name="endMidY">End position of the second axis</param>
        /// <param name="endMidZ">End position of the third axis</param>
        /// <param name="centreX">Centre position of the first axis</param>
        /// <param name="centreY">Centre position of the second axis</param>
        /// <param name="centreZ">Centre position of the third axis</param>
        /// <param name="mode">Mode of operation. See the MSPHERICAL command documentation for more details.</param>
        /// <param name="gtpi">If this optional parameter is non zero, modes 0 and 1 will perform a move taking the opposite way around a 360 degree circle to the same endpoint</param>
        /// <param name="rotau">If this optional parameter is non zero, a fourth axis will perform linear interpolation at the same time as the spherical move. The axis is the next in the BASE sequence</param>
        /// <param name="rotav">If this optional parameter is non zero, a fifth axis will perform linear interpolation at the same time as the spherical move</param>
        /// <param name="rotaw">If this optional parameter is non zero, a sixth axis will perform linear interpolation at the same time as the spherical move</param>
        /// <param name="rotap">If this optional parameter is non zero, a seventh axis will perform linear interpolation at the same time as the spherical move</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveSphericalSp(int32_t axis, double endMidX, double endMidY, double endMidZ, double centreX, double centreY, double centreZ, int32_t mode = 0, int32_t gtpi = 0, double rotau = 0, double rotav = 0, double rotaw = 0, double rotap = 0) = 0;

        /// <summary>
        /// Function allows a sequence of relative movements to be loaded via TABLE values.
        ///  The moves can be automatically merged together using a circular or spherical arc.
        ///  Movement on 2 to 6 axes may be controlled in this way.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <param name="tablePointer">Start of the move sequence in the TABLE data.</param>
        /// <param name="nAxes">Number of axes 2 to 6</param>
        /// <param name="nPoints">The number of points, each point requires 2 or 3 table values</param>
        /// <param name="options">Modifiers for the movements. See the TrioBASIC MOVEABSSEQ documentation for more details.</param>
        /// <param name="radius">The merging/filleting radius to be applied. 0 for no filleting</param>
        /// <param name="tableOutput">Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.</param>
```

```
        /// <param name="transitionAngle">Transition angle (in Radians) for any filleted arcs that are generated.  Set to 0 if no transition curve is to be applied.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveRelSeq(int32_t axis, uint32_t tablePointer, uint32_t nAxes, uint32_t nPoints, int32_t options, double radius, int32_t tableOutput = 0, double transitionAngle = 0) = 0;

        /// <summary>
        /// Function allows a sequence of absolute movements to be loaded via TABLE values.
        ///  The moves can be automatically merged together using a circular or spherical arc.
        ///  Movement on 2 to 6 axes may be controlled in this way.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <param name="tablePointer">Start of the move sequence in the TABLE data.</param>
        /// <param name="nAxes">Number of axes 2 to 6</param>
        /// <param name="nPoints">The number of points, each point requires 2 or 3 table values</param>
        /// <param name="options">Modifiers for the movements. See the TrioBASIC MOVEABSSEQ documentation for more details.</param>
        /// <param name="radius">The merging/filleting radius to be applied. 0 for no filleting</param>
        /// <param name="tableOutput">Output data to TABLE. 0 means do not output data to TABLE, otherwise is the TABLE start index for the data.</param>
        /// <param name="transitionAngle">Transition angle (in Radians) for any filleted arcs that are generated.  Set to 0 if no transition curve is to be applied.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveAbsSeq(int32_t axis, uint32_t tablePointer, uint32_t nAxes, uint32_t nPoints, int32_t options, double radius, int32_t tableOutput = 0, double transitionAngle = 0) = 0;

        /// <summary>
        /// MOVEMODIFY will change the absolute end position of a single axis MOVE , MOVEABS , MOVESP , MOVEABSSP or MOVEMODIFY that is in the last position in the movement buffer.
        ///  If there is no motion command in the movement buffers or the last movement is not a single axis linear move, then MOVEMODIFY is loaded.
        /// </summary>
        /// <param name="position">Absolute position for the current move to complete at</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveModify(double position, int32_t axis = -1) = 0;

        /// <summary>
        /// MOVEPICK is a new move type aimed at making it much simpler to program fast pick and place operations
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <param name="mode">Specify the number of axes 0 = xyz, 1 = xyzu, 2 = xyzuv, 3 = xyzuvw</param>
        /// <param name="x">Target x absolute position</param>
        /// <param name="y">Target y absolute position</param>
        /// <param name="z">Target z absolute position</param>
        /// <param name="withdraw">Incremental withdraw distance from the start position on z axis</param>
        /// <param name="overlapControlA">0..0.9 Specify rounding on corner A (optional)</param>
        /// <param name="overlapControlB">0..0.9 Specify rounding on corner B (optional)</param>
        /// <param name="u">Target u absolute position (optional)</param>
        /// <param name="v">Target v absolute position (optional)</param>
        /// <param name="w">Target w absolute position (optional)</param>
        /// <param name="withdrawControl">0..1 Adjust the speed for withdraw section (optional)</param>
        /// <param name="xyControl">0..1 Adjust the speed for xy section (optional)</param>
        /// <param name="approachControl">0..1 Adjust the speed for approach section (optional)</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MovePick(int32_t axis, int32_t mode, double x, double y, double z, double withdraw, double overlapControlA = 0.5, double overlapControlB = 0.5, double u = 0, double v = 0, double w = 0, double withdrawControl = 1, double xyControl = 1, double approachControl = 1) = 0;

        /// <summary>
        /// The linked move command is designed for controlling movements such as:
        ///  - Synchronisation to conveyors
        ///  - Flying shears
        ///  - Thread chasing, tapping etc.
        ///  - Coil winding
        ///  The motion consists of a linear movement with separately variable acceleration and deceleration phases linked via a software gearbox to the MEASURED position(MPOS) of another axis.The command uses the BASE and AXIS, and unit conversion factors in a similar way to other move commands.
        ///  The "link" axis may move in either direction to drive the output motion.The link distances specified are always positive.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
```

```
        /// <param name="distance">Incremental distance in user units to be moved on the current base axis, as a result of the measured movement on the "input" axis which drives the move. </param>
        /// <param name="link_dist">Positive incremental distance in user units which is required to be measured on the "link" axis to result in the motion on the base axis.</param>
        /// <param name="link_acc">Positive incremental distance in user units on the input axis over which the base axis accelerates.</param>
        /// <param name="link_dec">Positive incremental distance in user units on the input axis over which the base axis decelerates.</param>
        /// <param name="link_axis">Specifies the axis to "link" to. It should be set to a value between 0 and the number of available axes.</param>
        /// <param name="link_options">
        /// Bit value options to customise how your MOVELINK operates:<br/>
        ///  Bit 0 (1) - link commences exactly when registration event MARK occurs on link axis<br/>
        ///  Bit 1 (2) - link commences at an absolute position on link axis(see link_pos for start position)<br/>
        ///  Bit 2 (4) - MOVELINK repeats automatically and bi - directionally when this bit is set. (This mode can be cleared by setting bit 1 of the REP_OPTION axis parameter)<br/>
        ///  Bit 4 (16)- If this bit is set the MOVELINK acceleration and deceleration phases are constructed using an "S" speed profile not a trapezoidal speed profile<br/>
        ///  Bit 5 (32)- Link is only active during a positive move on the link axis<br/>
        ///  Bit 8 (256) - link commences exactly when registration event MARKB occurs on link axis<br/>
        ///  Bit 9 (512) - link commences exactly when registration event R_MARK occurs on link axis. (see link_pos for channel number)<br/>
        ///  Bit 10 and 11 - S-Ramp mode selection for use when bit 4 is set.Figure in() shows peak acceleration compared to linear ramp.<br/>
        ///                 0 - SIN speed Profile(1.55)<br/>
        ///                 1 - Power 9 polynomial speed(2.42)<br/>
        ///                 2 - Power 7 polynomial speed(2.16)<br/>
        ///                 3 - Power 5 polynomial speed(1.86)<br/>
        ///  Bit 13 - Slave axis follows master axis DPOS<br/>
        ///  Bit 14 - Link is active during positive movement from a threshold.The threshold is updated to the highest positive value.
        /// </param>
        /// <param name="link_pos">Link_option bit 1 - the absolute position on the link axis in user UNITS where the MOVELINK is to be start. link_option bit 9 - the registration channel to start the movement on.</param>
        /// <param name="base_dist">
        /// Firmware V2.0253 onwards. The "base distance" is the distance that is to be travelled at the base speed and is part of the total first "distance" parameter. See Examples 4 and 5 below.<br/>
        ///  MOVELINK_MODIFY cannot be used when the base distance is in use.
        /// </param>
        /// <param name="y_dist">Same as base_dist, for 2nd base axis</param>
        /// <param name="z_dist">Same as base_dist, for 3rd base axis</param>
        /// <param name="u_dist">Same as base_dist, for 4th base axis</param>
        /// <param name="v_dist">Same as base_dist, for 5th base axis</param>
        /// <param name="w_dist">Same as base_dist, for 6th base axis</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void MoveLink(int32_t axis, double distance, double link_dist, double link_acc, double link_dec, int32_t link_axis = -1, int32_t link_options = 0, double link_pos = 0, double base_dist = 0, double y_dist = 0, double z_dist = 0, double u_dist = 0, double v_dist = 0, double w_dist = 0) = 0;

        /// <summary>
        /// Function used to pre-set the parameter list for a subsequent move using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="speed">The value for SPEED to be set</param>
        /// <param name="accel">The value for ACCEL to be set</param>
        /// <param name="decel">The value for DECEL to be set</param>
        /// <param name="jerk">The value for JERK to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotion(int32_t axis, double speed, double accel, double decel, double jerk) = 0;

        /// <summary>
        /// Function used to pre-set the parameter list for a subsequent move using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="speed">The value for SPEED to be set</param>
        /// <param name="accel">The value for ACCEL to be set</param>
        /// <param name="decel">The value for DECEL to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotion(int32_t axis, double speed, double accel, double decel) = 0;

        /// <summary>
        /// Function used to pre-set the parameter list for a subsequent move using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="speed">The value for SPEED to be set</param>
        /// <param name="accel">The value for ACCEL to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotion(int32_t axis, double speed, double accel) = 0;

```

```
        /// <summary>
        /// Function used to pre-set the parameter list for a subsequent move using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="speed">The value for SPEED to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotion(int32_t axis, double speed) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <param name="endMoveSpeed">The value for ENDMOVE_SPEED to be set</param>
        /// <param name="forceAccel">The value for FORCE_ACCEL to be set</param>
        /// <param name="forceDecel">The value for FORCE_DECEL to be set</param>
        /// <param name="forceJerk">The value for FORCE_JERK to be set</param>
        /// <param name="forceRamp">The value for FORCE_RAMP to be set</param>
        /// <param name="forceDwell">The value for FORCE_DWELL to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel, double forceJerk, double forceRamp, double forceDwell) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <param name="endMoveSpeed">The value for ENDMOVE_SPEED to be set</param>
        /// <param name="forceAccel">The value for FORCE_ACCEL to be set</param>
        /// <param name="forceDecel">The value for FORCE_DECEL to be set</param>
        /// <param name="forceJerk">The value for FORCE_JERK to be set</param>
        /// <param name="forceRamp">The value for FORCE_RAMP to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel, double forceJerk, double forceRamp) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <param name="endMoveSpeed">The value for ENDMOVE_SPEED to be set</param>
        /// <param name="forceAccel">The value for FORCE_ACCEL to be set</param>
        /// <param name="forceDecel">The value for FORCE_DECEL to be set</param>
        /// <param name="forceJerk">The value for FORCE_JERK to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel, double forceJerk) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <param name="endMoveSpeed">The value for ENDMOVE_SPEED to be set</param>
        /// <param name="forceAccel">The value for FORCE_ACCEL to be set</param>
        /// <param name="forceDecel">The value for FORCE_DECEL to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel, double forceDecel) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <param name="endMoveSpeed">The value for ENDMOVE_SPEED to be set</param>
        /// <param name="forceAccel">The value for FORCE_ACCEL to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed, double forceAccel) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <param name="endMoveSpeed">The value for ENDMOVE_SPEED to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed, double endMoveSpeed) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <param name="startMoveSpeed">The value for STARTMOVE_SPEED to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed, double startMoveSpeed) = 0;

        /// <summary>
        /// SET_MOTIONSP is used to pre-set the parameter list for a subsequent MOVESP, MOVEABSSP, MOVECIRCSP etc. using just one command.
        /// </summary>
        /// <param name="axis">Target axis for whose parameters are to be modified</param>
        /// <param name="forceSpeed">The value for FORCE_SPEED to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMotionSp(int32_t axis, double forceSpeed) = 0;

        /// <summary>
        /// Set single table value.
        /// </summary>
        /// <param name="index">Index of the table entry to be written</param>
        /// <param name="value">New value to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetTableValue(uint32_t index, double value) = 0;

        /// <summary>
        /// Get single table value.
        /// </summary>
        /// <param name="index">Index of the table entry to be read</param>
        /// <returns>On success the value of table entry returned</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetTableValue(uint32_t index) = 0;

        /// <summary>
        /// Set multiple table entries.
        /// </summary>
        /// <param name="start_index">Start index of the table range to be written</param>
        /// <param name="count">Number of table entries to be written</param>
        /// <param name="values">Array of values to be written</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMultiTableValues(uint32_t start_index, uint32_t count, const double values[]) = 0;

        /// <summary>
        /// Get multiple table entries.
        /// </summary>
        /// <param name="start_index">Start index of the table range to be read</param>
        /// <param name="count">Number of table entries to be read</param>
        /// <param name="values">Placeholder array for the values to be read</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void GetMultiTableValues(uint32_t start_index, uint32_t count, double values[]) = 0;

        /// <summary>
        /// Set single VR value.
        /// </summary>
        /// <param name="index">Index of the VR entry to be written</param>
        /// <param name="value">New value to be written</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetVrValue(uint32_t index, double value) = 0;

        /// <summary>
        /// Get single VR value.
```

```
        /// </summary>
        /// <param name="index">Index of the VR entry to be read</param>
        /// <returns>On success the value VR entry returned</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetVrValue(uint32_t index) = 0;

        /// <summary>
        /// Set multiple VR entries.
        /// </summary>
        /// <param name="start_index">Start index of the VR range to be written</param>
        /// <param name="count">Number of VR entries to be written</param>
        /// <param name="values">Array of VR values to be written</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetMultiVrValues(uint32_t start_index, uint32_t count, const double values[]) = 0;

        /// <summary>
        /// Get multiple VR entries.
        /// </summary>
        /// <param name="start_index">Start index of the VR range to be read</param>
        /// <param name="count">Number of VR entries to be read</param>
        /// <param name="values">Placeholder array for the values to be read</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void GetMultiVrValues(uint32_t start_index, uint32_t count, double values[]) = 0;

        /// <summary>
        /// The INVERT_IN command allows the input channels to be individually inverted in software.
        /// </summary>
        /// <param name="input">The input to invert</param>
        /// <param name="invert">True to invert, false not to invert</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetInvertIN(uint32_t input, bool invert) = 0;

        /// <summary>
        /// Gets the invert status set by INVERT_IN command.
        /// </summary>
        /// <param name="input">The input to invert</param>
        /// <returns>True if inverted, false if not inverted</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetInvertIN(uint32_t input) = 0;

        /// <summary>
        /// The RAPIDSTOP command operates like the CANCEL command but does so for ALL axes using a subset of cancel modes
        /// </summary>
        /// <param name="mode">
        ///  0 Cancels axis commands from the MTYPE buffers(default)<br/>
        ///  1 Cancels all buffered moves on all axis(excluding the PMOVE)<br/>
        ///  2 Cancels all active and buffered moves including the PMOVE
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void RapidStop(int32_t mode = 0) = 0;

        /// <summary>
        /// Function returns the list of programs and files in internal memory of the Motion Coordinator
        /// </summary>
        /// <param name="handler">Callback handler function executed for each item in internal memory</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Dir(FileCallback&& handler) = 0;

        /// <summary>
        /// Set a null-terminated string into consecutive VRs
        /// </summary>
        /// <param name="vr">Starting VR index</param>
        /// <param name="str">String to be set</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetVrString(uint32_t vr, const std::string_view& str) = 0;

        /// <summary>
        /// Get the length of a null-terminated string from consecutive VRs
        /// </summary>
        /// <param name="vr">Starting VR index</param>
        /// <returns>Length of the string</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetVrStringLen(uint32_t vr) = 0;

```

```
        /// <summary>
        /// Get a null-terminated string from consecutive VRs
        /// </summary>
        /// <param name="vr">Starting VR index</param>
        /// <param name="buff">Buffer to receive the string characters. If NULL function succeeds and returns expected buffers size in buff_size parameter including null terminator character</param>
        /// <param name="buff_size">Size of the buffer on input and on exit required buffer size including null terminator</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void GetVrString(uint32_t vr, char* buff, uint32_t& buff_size) = 0;

        /// <summary>
        /// Get a null-terminated string from consecutive VRs using callback function to re-construct the string from received chunks
        /// </summary>
        /// <param name="vr">Starting VR index</param>
        /// <param name="handler">Callback function to receive string chunks</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void GetVrString(uint32_t vr, StreamHandlerCallback&& handler) = 0;

        /// <summary>
        /// Get a null-terminated string from consecutive VRs
        /// </summary>
        /// <param name="vr">Starting VR index</param>
        /// <returns>On success the string stored in consecutive VR entries</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual std::string GetVrString(uint32_t vr) = 0;

        /// <summary>
        /// Get a value from an Analogue Input.
        /// </summary>
        /// <param name="channel">Analogue input channel number</param>
        /// <returns>Current value of the analog input channel</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAIn(uint32_t channel) = 0;

        /// <summary>
        /// Get the state of the Analogue Output
        /// </summary>
        /// <param name="channel">Analogue output channel number</param>
        /// <returns>Current value of the analogue output channel</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAOut(uint32_t channel) = 0;

        /// <summary>
        /// Set the state of the Analogue Output.
        ///  Note that the actual change will happen on the next servo cycle. Therefore in immediate read
        ///  may give a wrong result on some (fast) systems.
        /// </summary>
        /// <param name="channel">Analogue output channel number</param>
        /// <param name="value">New value for the analogue output channel</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAOut(uint32_t channel, int32_t value) = 0;

        /// <summary>
        /// Get the state of the Digital Inputs
        /// </summary>
        /// <param name="firstChannel">First digital input channel number</param>
        /// <param name="lastChannel">Last digital input channel number. The difference between last and first channel should be at most 32</param>
        /// <returns>Binary sum of digital input states</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetDIn(uint32_t firstChannel, uint32_t lastChannel) = 0;

        /// <summary>
        /// Get the state of the Digital Input
        /// </summary>
        /// <param name="channel">Digital input channel number</param>
        /// <returns>Current state of the digital input channel</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetDIn(uint32_t channel) = 0;

        /// <summary>
        /// Get the state of the first 32 Digital Inputs
        /// </summary>
        /// <returns>Bit sum of the states of the first 32 digital input channels</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual uint32_t GetDIn32() = 0;

        /// <summary>
        /// Get the state of the Digital Outputs
        /// </summary>
        /// <param name="firstChannel">First digital output channel number</param>
        /// <param name="lastChannel">Last digital output channel number. The difference between last and first channel should be at most 32</param>
        /// <returns>Binary sum of digital output states</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetDOut(uint32_t firstChannel, uint32_t lastChannel) = 0;

        /// <summary>
        /// Get the state of the Digital Output
        /// </summary>
        /// <param name="channel">Digital output channel number</param>
        /// <returns>Current state of the digital output channel</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetDOut(uint32_t channel) = 0;

        /// <summary>
        /// Get the state of the first 32 Digital Outputs
        /// </summary>
        /// <returns>Bit sum of the states of the first 32 digital output channels</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetDOut32() = 0;

        /// <summary>
        /// Set the state of the Digital Outputs
        /// </summary>
        /// <param name="firstChannel">First digital output channel number</param>
        /// <param name="lastChannel">Last digital output channel number</param>
        /// <param name="states">Binary sum of digital output states</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetDOut(uint32_t firstChannel, uint32_t lastChannel, uint32_t states) = 0;

        /// <summary>
        /// Set the state of the Digital Output.
        ///  Note that the actual change will happen on the next servo cycle. Therefore in immediate read
        ///  may give a wrong result on some (fast) systems.
        /// </summary>
        /// <param name="channel">Digital output channel number</param>
        /// <param name="state">New state for the digital output channel</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetDOut(uint32_t channel, bool state) = 0;

        /// <summary>
        /// Set the state of the first 32 Digital Outputs.
        ///  Note that the actual change will happen on the next servo cycle. Therefore in immediate read
        ///  may give a wrong result on some (fast) systems.
        /// </summary>
        /// <param name="states32">Bit sum of new the states for the first 32 digital output channels</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetDOut32(uint32_t states32) = 0;

        /// <summary>
        /// Stops a program at its current line.
        /// </summary>
        /// <param name="name">Name of the program to be stopped.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void StopProgram(const std::string_view& name) = 0;

        /// <summary>
        /// Stops a program at its current line.
        /// </summary>
        /// <param name="name">Name of the program to be stopped.</param>
        /// <param name="process_number">Optional process number to be used when multiple instances of the program are running and only one is to be stopped.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void StopProgram(const std::string_view& name, int32_t process_number) = 0;

        /// <summary>
        /// Runs a program.
        /// </summary>
        /// <param name="name">Name of the program to run.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual void RunProgram(const std::string_view& name) = 0;

        /// <summary>
        /// Runs a program
        /// </summary>
        /// <param name="name">Name of the program to run.</param>
        /// <param name="process_number">Optional process number to run the procces on.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void RunProgram(const std::string_view& name, int32_t process_number) = 0;

        /// <summary>
        /// Get CRC16 of a file or program on Motion Coordinator.
        /// </summary>
        /// <param name="name">Name of the program to calculate the checksum for.</param>
        /// <returns>CRC-16 of the program.</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint16_t GetRemoteFileCRC(const std::string_view& name) = 0;

        /// <summary>
        /// Get CRC16 of a local file.
        /// </summary>
        /// <param name="isBinary">
        /// If true file is treated as binary and checksum is calculated directly on all bytes.<br/>
        ///  If false the file is treated as text and checksum is calculated line by line as it is done inside Motion Coordinator for programs and text files
        /// </param>
        /// <returns>CRC-16 of the file.</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint16_t GetLocalFileCRC(const std::string_view& path, bool isBinary) = 0;

        /// <summary>
        /// Check if file or program exists on Motion Coordinator
        /// </summary>
        /// <param name="name">
        /// Name of the file or program to check. Can be prefixed by:<br/>
        ///  "RAM:" - the location of the file is in the controller RAM.<br/>
        ///  "SD:" - the location of the file is on the SD card. Must be a full path and file name with extension.<br/>
        ///  If no prefix "RAM:" is assumed.
        /// </param>
        /// <returns>Result from operation.</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual FileExistsFlags FileExists(const std::string_view& name) = 0;

        /// <summary>
        /// Adds the output from the servo control block of a secondary axis to the output of the base axis.
        ///  The resulting DAC_OUT of the base axis is then the sum of the two control loop outputs.
        /// </summary>
        /// <param name="axis">Base axis for to be subject to DAC_OUT superimposition</param>
        /// <param name="secondAxis">Axis which DAC_OUT value to superimpose. -1 breaks the link with the other axis. </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void AddDac(int32_t axis, int32_t secondAxis) = 0;

        /// <summary>
        /// Adds the output from the servo control block of a secondary axis to the output of the base axis.
        ///  The resulting DAC_OUT of the base axis is then the sum of the two control loop outputs.
        ///  Axis previously set by SetAxisBase() is subject to superimposition of DAC_OUT value.
        /// </summary>
        /// <param name="secondAxis">Axis which DAC_OUT value to superimpose. -1 breaks the link with the other axis. </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void AddDac(int32_t secondAxis) = 0;

        /// <summary>
        /// Function is used to superimpose 2 or more movements to build up a more complex movement profile.
        /// </summary>
        /// <param name="axis">Target axis for to be subject to superimposition</param>
        /// <param name="linkAxis">Axis to superimpose. -1 breaks the link with the other axis. </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void AddAx(int32_t axis, int32_t linkAxis) = 0;

        /// <summary>
        /// Function is used to superimpose 2 or more movements to build up a more complex movement profile.
```

```
        ///  Axis previously set by SetAxisBase() is subject to superimposition.
        /// </summary>
        /// <param name="linkAxis">Axis to superimpose. -1 breaks the link with the other axis.
</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void AddAx(int32_t linkAxis) = 0;

        /// <summary>
        /// Sets all global (numbered) variables and VR values to 0 and sets local variables on the
process on which command is run to 0.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void ClearVariables() = 0;

        /// <summary>
        /// Delete frame group. When the group is deleted FRAME is set to 0. See the TrioBASIC
FRAME_GROUP documentation for more information
        /// </summary>
        /// <param name="group">Frame group number to delete</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void DeleteFrameGroup(uint32_t group) = 0;

        /// <summary>
        /// Define new frame group.
        /// </summary>
        /// <param name="group">Frame group number to define</param>
        /// <param name="tableOffset">Start offset of TABLE range used by the frame group</param>
        /// <param name="axes">Array of axes to be added to the frame group</param>
        /// <param name="axisCount">Number of axes to be added to the frame group</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetFrameGroup(uint32_t group, uint32_t tableOffset, const int32_t axes[],
uint32_t axisCount) = 0;

        /// <summary>
        /// Define new frame group.
        /// </summary>
        /// <param name="group">Frame group number to define</param>
        /// <param name="tableOffset">Start offset of TABLE range used by the frame group</param>
        /// <param name="axes">Array of axes to be added to the frame group</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetFrameGroup(uint32_t group, uint32_t tableOffset, const std::vector<int32_t>&
axes) = 0;

        /// <summary>
        /// Get frame group info.
        /// </summary>
        /// <param name="group">Frame group number to get info</param>
        /// <param name="tableOffset">Current start offset of TABLE range used by the frame
group</param>
        /// <param name="axes">Array of axis indices currently belonging to the frame group</param>
        /// <param name="axisCount">Number of axes currently belonging in the group</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void GetFrameGroup(uint32_t group, uint32_t& tableOffset, int32_t axes[], uint32_t&
axisCount) = 0;

        /// <summary>
        /// Get frame group info.
        /// </summary>
        /// <param name="group">Frame group number to get info</param>
        /// <param name="tableOffset">Current start offset of TABLE range used by the frame
group</param>
        /// <param name="axes">Array of axis indices currently belonging to the frame group</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void GetFrameGroup(uint32_t group, uint32_t& tableOffset, std::vector<int32_t>& axes)
= 0;

        /// <summary>
        /// Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for
more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="options">
        ///  bit 0: 0-Do not print progress information; 1-Print out progress information
(default)<br/>
        ///  bit 1: 0-Normal state change (default); 1-Force a re-scan of the network<br/>
        ///  bit 2: 0-Standard startup; 1-Express startup; do not wait for PLL to lock between state
changes
        /// </param>
        /// <param name="MAC_retries">Sets the number of times the master attempts to restart the
Ethernet auto-negotiation. Default = 2</param>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_StartNetwork(int32_t slot, EthercatStartOption options, uint32_t MAC_retries) = 0;

        /// <summary>
        /// Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="options">
        ///  bit 0: 0-Do not print progress information; 1-Print out progress information (default)<br/>
        ///  bit 1: 0-Normal state change (default); 1-Force a re-scan of the network<br/>
        ///  bit 2: 0-Standard startup; 1-Express startup; do not wait for PLL to lock between state changes
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_StartNetwork(int32_t slot, EthercatStartOption options) = 0;

        /// <summary>
        /// Initialise EtherCAT network, and put it onto operational mode. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_StartNetwork(int32_t slot) = 0;

        /// <summary>
        /// Closedown the EtherCAT network.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_StopNetwork(int32_t slot) = 0;

        /// <summary>
        /// Check whether given slave is online. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Position on the network of the slave </param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CheckSlaveOnline_VR(int32_t slot, int32_t slave, uint32_t out_vr_index) = 0;

        /// <summary>
        /// Check whether given slave is online. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Position on the network of the slave </param>
        /// <returns>On success true if slave is online or false otherwise</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool Ethercat_CheckSlaveOnline(int32_t slot, int32_t slave) = 0;

        /// <summary>
        /// Determine the number of slaves on the bus. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CheckNumberOfSlaves_VR(int32_t slot, uint32_t out_vr_index) = 0;

        /// <summary>
        /// Determine the number of slaves on the bus. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <returns>On success receives the number of slaves</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t Ethercat_CheckNumberOfSlaves(int32_t slot) = 0;

        /// <summary>
        /// Determine the configured address of the slave identified by its position on the network. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Position on the network of the slave </param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetSlaveAddress_VR(int32_t slot, int32_t slave, uint32_t out_vr_index) =
```

```
0;

        /// <summary>
        /// Determine the configured address of the slave identified by its position on the network. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Position on the network of the slave </param>
        /// <returns>On success receives the configured slave address</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t Ethercat_GetSlaveAddress(int32_t slot, int32_t slave) = 0;

        /// <summary>
        /// Determine the axis of the slave identified by its position on the network. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Position on the network of the slave </param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetSlaveAxis_VR(int32_t slot, int32_t slave, uint32_t out_vr_index) = 0;

        /// <summary>
        /// Determine the axis of the slave identified by its position on the network. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Position on the network of the slave </param>
        /// <returns>On success receives the axis assigned to the slave</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t Ethercat_GetSlaveAxis(int32_t slot, int32_t slave) = 0;

        /// <summary>
        /// This function controls the EtherCAT State Machine (ESM).
        ///  It requests the master change to given EtherCAT 'state', and hence changes all slaves to the same state. When a change to a higher state is made, the EtherCAT network will progress to the new state through the in-between states to allow correct starting of the network. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="state">
        ///  -1 : Reserved<br/>
        ///  0 : Initial (EtherCAT ESC value 0x01)<br/>
        ///  1 : Pre-Operational (0x02)<br/>
        ///  2 : Safe-Operational  (0x04)<br/>
        ///  3 : Operational  (0x08)
        /// </param>
        /// <param name="display">
        ///  0 : Writes state change information to the standard output stream.<br/>
        ///  1 : Do not write out state change information.
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetState(int32_t slot, EthercatState state, uint32_t display) = 0;

        /// <summary>
        /// Gets the present state of the EtherCAT running on the defined slot. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="out_vr_index">
        /// Result is placed in the selected VR variable. The returned value means:<br/>
        ///  0 : Initial (EtherCAT ESC value 0x01)<br/>
        ///  1 : Pre-Operational (0x02)<br/>
        ///  2 : Safe-Operational  (0x04)<br/>
        ///  3 : Operational  (0x08)
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetState_VR(int32_t slot, uint32_t out_vr_index) = 0;

        /// <summary>
        /// Gets the present state of the EtherCAT running on the defined slot. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <returns>
        /// On success redeives the state. The returned value means:<br/>
        ///  0 : Initial (EtherCAT ESC value 0x01)<br/>
        ///  1 : Pre-Operational (0x02)<br/>
        ///  2 : Safe-Operational  (0x04)<br/>
        ///  3 : Operational  (0x08)
        /// </returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual EthercatState Ethercat_GetState(int32_t slot) = 0;

        /// <summary>
        /// Low-level write to the slave's ESC registers, addressing method used is auto-increment addressing. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Auto-increment address </param>
        /// <param name="esc_offset">Register address</param>
        /// <param name="data_length">Number of data bytes to write</param>
        /// <param name="data">Data values</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetSlave_ESC_Register(int32_t slot, int32_t slave, uint32_t esc_offset, uint32_t data_length, const char* data) = 0;

        /// <summary>
        /// Low-level read from the slave's ESC registers, addressing method used is auto-increment addressing. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="slave">Auto-increment address</param>
        /// <param name="esc_offset">Register address</param>
        /// <param name="data_length">Number of bytes in the register size</param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetSlave_ESC_Register(int32_t slot, int32_t slave, uint32_t esc_offset, uint32_t data_length, uint32_t out_vr_index) = 0;

        /// <summary>
        /// Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis">The axis number of the drive to be reset</param>
        /// <param name="mode">
        ///  0 : The 'Fault Reset' (bit 7) of DS402 control word is set high and then set low again after a hard coded timeout. (default)<br/>
        ///  1 : Bit 7 is set high until the 'Fault Flag' (bit 3) of the status word goes low, or a timeout occurs.
        /// </param>
        /// <param name="timeout">Timeout in ms used during mode 1 operation. Default is 100 ms. Range is 1 to 10000 ms.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SendResetSequence(int32_t axis, uint32_t mode, uint32_t timeout) = 0;

        /// <summary>
        /// Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis">The axis number of the drive to be reset</param>
        /// <param name="mode">
        ///  0 : The 'Fault Reset' (bit 7) of DS402 control word is set high and then set low again after a hard coded timeout. (default)<br/>
        ///  1 : Bit 7 is set high until the 'Fault Flag' (bit 3) of the status word goes low, or a timeout occurs.
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Default timeout is 100 ms
        /// </remarks>
        virtual void Ethercat_SendResetSequence(int32_t axis, uint32_t mode) = 0;

        /// <summary>
        /// Reset a slave error. This function runs the error reset sequence on the drive control word. DRIVE_CONTROLWORD bit 8 is toggled high then low. This will instruct the drive to reset any errors in the drive where the cause of the error has been removed.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis">The axis number of the drive to be reset</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Default timeout is 100 ms
        /// </remarks>
        virtual void Ethercat_SendResetSequence(int32_t axis) = 0;

        /// <summary>
        /// This command enables the user to define a CoE Statusword (0x6041:00) mask and trigger pattern for raising the controller AXISSTATUS 'Remote Drive Status word Error' flag (bit 26.)
```

```
        ///  The mask defines which bits within the CoE status word are relevant, and the trigger pattern defines the state of these bits(0 or 1) which will trigger the controller AXISSTATUS flag.
        ///  If it is necessary to reset the watchdog(master enable) when this AXISISTATUS flag is set then the axis ERRORMASK must be set accordingly.
        ///  The slave mask and pattern can be set and modified from preop to operational state of the network(after the configured addresses have been assigned.) It must be reset if the network is restarted.
        ///  The slave must support the CoE statusword(0x6041:00) within the PDO cyclic data.
        ///  If the Mask is zero the controller will not execute any runtime check of the CoE statusword.
        ///  If the watchdog must be cleared to OFF when the 'remote drive statusword error' flag is raised then bit 26 of the axis ERRORMASK parameter must be set.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="addr">Configured address of the slave</param>
        /// <param name="vr_index">VR index in range [0..max_vr_index-1]. The index of the VR which contains the mask and the next VR (vr_index+1) contains the trigger_pattern.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetStatusMaskAndTriggerPattern_VR(int32_t slot, int32_t addr, uint32_t vr_index) = 0;

        /// <summary>
        /// This command enables the user to define a CoE Statusword (0x6041:00) mask and trigger pattern for raising the controller AXISSTATUS 'Remote Drive Status word Error' flag (bit 26.)
        ///  The mask defines which bits within the CoE status word are relevant, and the trigger pattern defines the state of these bits(0 or 1) which will trigger the controller AXISSTATUS flag.
        ///  If it is necessary to reset the watchdog(master enable) when this AXISISTATUS flag is set then the axis ERRORMASK must be set accordingly.
        ///  The slave mask and pattern can be set and modified from preop to operational state of the network(after the configured addresses have been assigned.) It must be reset if the network is restarted.
        ///  The slave must support the CoE statusword(0x6041:00) within the PDO cyclic data.
        ///  If the Mask is zero the controller will not execute any runtime check of the CoE statusword.
        ///  If the watchdog must be cleared to OFF when the 'remote drive statusword error' flag is raised then bit 26 of the axis ERRORMASK parameter must be set.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="addr">Configured address of the slave</param>
        /// <param name="mask">CoE statusword mask.</param>
        /// <param name="trigger_pattern">Defines the state of the relevant bits which will trigger the controller AXISSTATUS 'remote drive statusword error' flag (bit 26.)</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetStatusMaskAndTriggerPattern(int32_t slot, int32_t addr, uint32_t mask, uint32_t trigger_pattern) = 0;

        /// <summary>
        /// This command enables the user to display and check the mask and trigger pattern values set (if any) for a given slave.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="addr">Configured address of the slave</param>
        /// <param name="out_vr_index">VR index in range [0..max_vr_index-2]. Index of the array of 2 VRs where the mask and the trigger_pattern will be stored.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetStatusMaskAndTriggerPattern_VR(int32_t slot, int32_t addr, uint32_t out_vr_index) = 0;

        /// <summary>
        /// This command enables the user to display and check the mask and trigger pattern values set (if any) for a given slave.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port</param>
        /// <param name="addr">Configured address of the slave</param>
        /// <param name="mask">Current statusword mask</param>
        /// <param name="trigger_pattern">Current trigger pattern</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetStatusMaskAndTriggerPattern(int32_t slot, int32_t addr, uint32_t& mask, uint32_t& trigger_pattern) = 0;

        /// <summary>
        /// Displays the network configuration to the command line terminal in Motion Perfect. See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_DisplayNetworkConfig(int32_t slot) = 0;

```

```
        /// <summary>
        /// Set the network default Sync0 start time offset. The offset value is from the given VR.
        ///  The default start time is 1 second(1000000 microseconds).Add or subtract the required offset in microseconds.For example 1000150 is an offset of 150 microseconds and 999930 is an offset of -70 microseconds.The offset must not exceed ±50 % of the servo period.
        ///  The function returns TRUE / FALSE depending upon whether the operation was successful.
        ///  This command must be run before the EtherCAT network is started.
        ///   See Trio BASIC help for more details.
        /// </summary>
        /// <param name="vr_index">VR index in range [0..max_vr_index-1]. Use the value in the VR number specified.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetSync0StartTime_VR(uint32_t vr_index) = 0;

        /// <summary>
        /// Set the network default Sync0 start time offset. If offset value is directly provided as a parameter.
        ///  The default start time is 1 second(1000000 microseconds).Add or subtract the required offset in microseconds.For example 1000150 is an offset of 150 microseconds and 999930 is an offset of -70 microseconds.The offset must not exceed ±50 % of the servo period.
        ///  The function returns TRUE / FALSE depending upon whether the operation was successful.
        ///  This command must be run before the EtherCAT network is started.
        ///   See Trio BASIC help for more details.
        /// </summary>
        /// <param name="offset">Direct value for the start time</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetSync0StartTime(int32_t offset) = 0;

        /// <summary>
        /// Set the EtherCAT master to enable the first DC enabled node to be the Distributed Clocks master. As this requires the Trio Motion Coordinator to synchronise to the external DC master, it does not function on P876, MC4N-ECAT or MC664(X) as these have FPGA based fixed timing hardware for the EtherCAT port.
        ///  The function returns TRUE / FALSE depending upon whether the operation was successful.
        ///  This command must be run before the EtherCAT network is started.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="function">Distributed Clocks function. 2 = use first DC enabled node as clocks master.</param>
        /// <param name="vr_index">VR index in range [0..max_vr_index-1]. Use the value in the VR number specified.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetDistrClockMaster_VR(int32_t slot, uint32_t function, uint32_t vr_index) = 0;

        /// <summary>
        /// Set the EtherCAT master to enable the first DC enabled node to be the Distributed Clocks master. As this requires the Trio Motion Coordinator to synchronise to the external DC master, it does not function on P876, MC4N-ECAT or MC664(X) as these have FPGA based fixed timing hardware for the EtherCAT port.
        ///  The function returns TRUE / FALSE depending upon whether the operation was successful.
        ///  This command must be run before the EtherCAT network is started.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="function">Distributed Clocks function. 2 = use first DC enabled node as clocks master.</param>
        /// <param name="value">The value. 0 = use first DC enabled node as the clocks master.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetDistrClockMaster(int32_t slot, uint32_t function, uint32_t value) = 0;

        /// <summary>
        /// Set the response of the master to an emergency telegram transmitted by a slave node. This function enables the user to determine whether emergency messages should trigger a system error alarm and cause the WDOG to be dropped.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="mode">
        ///  0 : Log error and raise a System Error and drop WDOG off. (Default)<br/>
        ///  1 : Log error only.
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_SetEmergencyMsgControl(int32_t slot, uint32_t mode) = 0;

        /// <summary>
        /// Get the current value; indicating the response of the master to an emergency telegram transmitted by a slave node.
        ///  See Trio BASIC help for more details.
        /// </summary>
```

```
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_GetEmergencyMsgControl_VR(int32_t slot, uint32_t out_vr_index) = 0;

        /// <summary>
        /// Get the current value; indicating the response of the master to an emergency telegram transmitted by a slave node.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <returns>On success the current value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t Ethercat_GetEmergencyMsgControl(int32_t slot) = 0;

        /// <summary>
        /// This function gets a CANopen-over-EtherCAT (CoE) object from the remote drive or I/O device. The object's index and sub-index are used to request a value and that value is either placed in the VR or is displayed in the Motion Perfect terminal if the VR number is set to -1.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="address">Node address of the remote device on the network</param>
        /// <param name="object_index">CANopen Object index</param>
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoRead(int32_t slot, int32_t address, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, int32_t out_vr_index) = 0;

        /// <summary>
        /// This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. The value comes from a VR.
        ///  Use this function for integer values.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="address">Node address of the remote device on the network</param>
        /// <param name="object_index">CANopen Object index</param>
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="vr_index">VR number between 0 and max VR where the result will be stored. (-1 means the value will be printed to the terminal)</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoWrite_VR(int32_t slot, int32_t address, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, uint32_t vr_index) = 0;

        /// <summary>
        /// This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object.
        ///  Use this function for integer values.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="address">Node address of the remote device on the network</param>
        /// <param name="object_index">CANopen Object index</param>
```

```
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="value">The value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoWrite_Value(int32_t slot, int32_t address, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, int64_t value) = 0;

        /// <summary>
        /// This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object.
        ///  Use this function for floating-point values.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="slot">Slot number of EtherCAT port.</param>
        /// <param name="address">Node address of the remote device on the network</param>
        /// <param name="object_index">CANopen Object index</param>
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="value">The value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoWrite_Value(int32_t slot, int32_t address, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, double value) = 0;

        /// <summary>
        /// This function gets a CANopen-over-EtherCAT (CoE) object from the remote drive or I/O device. The object's index and sub-index are used to request a value and that value is either placed in the VR or is displayed in the Motion Perfect terminal if the VR number is set to -1.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis_number">Axis number of the EtherCAT drive.</param>
        /// <param name="object_index">CANopen Object index</param>
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="out_vr_index">Result is placed in the selected VR variable</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoReadAxis(int32_t axis_number, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, int32_t out_vr_index) = 0;

        /// <summary>
        /// This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object. The value comes from a VR.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis_number">Axis number of the EtherCAT drive.</param>
        /// <param name="object_index">CANopen Object index</param>
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        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="vr_index">VR number between 0 and max VR where the value is stored.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoWriteAxis_VR(int32_t axis_number, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, uint32_t vr_index) = 0;

        /// <summary>
        /// This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object.
        ///  Use this function for integer values.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis_number">Axis number of the EtherCAT drive.</param>
        /// <param name="object_index">CANopen Object index</param>
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="value">The value.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoWriteAxis_Value(int32_t axis_number, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, int64_t value) = 0;

        /// <summary>
        /// This function sets a CANopen-over-EtherCAT (CoE) object in the remote drive or I/O device. The object's index and sub-index are used to write a value to that object.
        ///  Use this function for floating-point values.
        ///  Refer to the remote device's manual for a list of available objects.
        ///  See Trio BASIC help for more details.
        /// </summary>
        /// <param name="axis_number">Axis number of the EtherCAT drive.</param>
        /// <param name="object_index">CANopen Object index</param>
        /// <param name="object_subindex">CANopen Object sub-index</param>
        /// <param name="type">
        ///  1 : Boolean<br/>
        ///  2 : Integer 8<br/>
        ///  3 : Integer 16<br/>
        ///  4 : Integer 32<br/>
        ///  5 : Unsigned 8<br/>
        ///  6 : Unsigned 16<br/>
        ///  7 : Unsigned 32<br/>
        ///  8 : Real 32<br/>
        ///  17 : Real 64<br/>
        ///  21 : Integer 64<br/>
        ///  27 : Unsigned 64
        /// </param>
        /// <param name="value">The value.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Ethercat_CoWriteAxis_Value(int32_t axis_number, uint16_t object_index, uint8_t object_subindex, Co_ObjectType type, double value) = 0;

        /// <summary>
        /// Uploads a file to the Motion Coordinator.
        /// </summary>
        /// <param name="local_file">Local file name</param>
        /// <param name="remote_file">
        /// Remote file name. Can be prefixed by:<br/>
        ///  "RAM:" - the location of the file is in the controller RAM.<br/>
        ///  "SD:" - the location of the file is on the SD card.<br/>
        ///  "FIFO:" - The target file is a FIFO file on the controller. It is expected that the
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controller (a program) will consume the data.<br/>
        ///  If no prefix "RAM:" is assumed.
        /// </param>
        /// <param name="remote_file_type">The type of the file. Ignored when uploading to the SD card or with for FIFO transfers.</param>
        /// <param name="opt">File transfer option</param>
        /// <param name="handler">Callback handler with the progress advance</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void UploadFile(const std::string_view& local_file, const std::string_view& remote_file, FileType remote_file_type, FileTransferOption opt, ProgressHandlerCallback&& handler) = 0;

        /// <summary>
        /// Downloads a file from the Motion Coordinator.
        /// </summary>
        /// <param name="local_file">Local source file</param>
        /// <param name="remote_file">
        /// Remote file name. Can be prefixed by:<br/>
        ///  "RAM:" - the location of the file is in the controller RAM.<br/>
        ///  "SD:" - the location of the file is on the SD card.<br/>
        ///  "FIFO:" - The target file is a FIFO file on the controller.<br/>
        ///  If no prefix "RAM:" is assumed.
        /// </param>
        /// <param name="handler">Callback handler with the progress advance</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void DownloadFile(const std::string_view& local_file, const std::string_view& remote_file, ProgressHandlerCallback&& handler) = 0;

        /// <summary>
        /// Deletes a file on the Motion Coordinator.
        /// </summary>
        /// <param name="remote_file">The name of the file</param>
        /// <param name="compiled_code_only">In case of a program: when true only the compiled code of the program will be deleted, the source code will remain. When false all data related to the program will be deleted</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Delete(const std::string_view& remote_file, bool compiled_code_only = false) = 0;

        /// <summary>
        /// Deletes a file, all files or the TABLE data the Motion Coordinator.
        /// </summary>
        /// <param name="target">
        /// The target to be cleared (deleted)<br/>
        ///  "file_name" - deletes the file with given name from the Motion Coordinator<br/>
        ///  "ALL" - deletes all user files from the Motion Coordinator. System files are not deleted.<br/>
        ///  "TABLE" - sets all TABLE values to 0 .
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void New(const std::string_view& target) = 0;

        /// <summary>
        /// Makes the named program the currently selected program on the Motion Coordinator, if the named program does not exist then it makes a program of that name.
        ///  The selected program cannot be changed when programs are running. When a program is selected any previously selected program is compiled.
        ///  When program_type is specified is to make a program of that name and type, it cannot be possible to change the type of an existing program.
        /// </summary>
        /// <param name="remote_file">The file (program) name</param>
        /// <param name="file_type">The type of the file (program) to be created</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Select(const std::string_view& remote_file, FileType file_type) = 0;

        /// <summary>
        /// This axis function disables backlash compensation.
        /// </summary>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void BacklashDisable(int32_t axis = -1) = 0;

        /// <summary>
        /// This axis function allows backlash compensation to be loaded. This is achieved by applying an offset move when the motor demand is in one direction, then reversing the offset move when the motor demand is in the opposite direction.
        ///  These moves are superimposed on the commanded axis movements.
        ///  The backlash compensation is applied after a reversal of the direction of change of the DPOS parameter.
        ///  The backlash compensation can be seen in the AXIS_DPOS axis parameter.This is effectively DPOS + backlash compensation.
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        /// </summary>
        /// <param name="distance">The distance to be offset in user units</param>
        /// <param name="speed">The ACCEL / DECEL rate at which is compensation move is applied in user units</param>
        /// <param name="acceleration">Enables/disables the backlash compensation</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Backlash(double distance, double speed, double acceleration, int32_t axis = -1) = 0;

        /// <summary>
        /// This command allows the incoming encoder count to be scaled by a non integer ratio :
        ///  MPOS = (mpos_count / input_count) x encoder_edges_input
        ///  When using the servo loop you will need to adjust the gains to maintain performance and stability.
        ///  Unlike the UNITS parameter, which only affects the scaling seen by the user programs, ENCODER_RATIO affects all motion commands.
        ///  ENCODER_RATIO does not replace UNITS.Only use ENCODER_RATIO where absolutely necessary.PP_STEP and ENCODER_RATIO cannot be used at the same time on the same axis.
        /// </summary>
        /// <param name="mpos_count">An integer number which defines the numerator</param>
        /// <param name="input_count">An integer number which defines the denominator</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void EncoderRatio(int64_t mpos_count, int64_t input_count, int32_t axis = -1) = 0;

        /// <summary>
        /// This command sets up an integer ratio for the axis' demanded position.  Usually this is used on a stepper (pulse+direction) output.  It can also be applied to a networked axis in positon mode, such as EtherCAT and RTEX position servo.
        ///  Every servo-period the change in position is passed through the STEP_RATIO function before it goes to the step pulse or network drive output.
        ///  The STEP_RATIO function operates before the divide by 16 factor(divide by 256 in Flexslice P375) in the stepper axis.This maintains the good timing resolution of the stepper output circuit.
        ///  STEP_RATIO does not replace UNITS.
        /// </summary>
        /// <param name="output_count">Number of counts to output for the given dpos_count value. Range: 0 to 16777215</param>
        /// <param name="dpos_count">Change in DPOS value for corresponding output count.  Range: 0 to 16777215.</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void StepRatio(int64_t output_count, int64_t dpos_count, int32_t axis = -1) = 0;

        /// <summary>
        /// This command is used to generate movement of an axis according to a table of positions which define a movement profile.
        /// </summary>
        /// <param name="tableStart">The start position of the cam profile in the TABLE.</param>
        /// <param name="tableStop">The end position of the cam profile in the TABLE.</param>
        /// <param name="tableMultiplier">The table values are multiplied by this value to generate the positions.</param>
        /// <param name="linkDistance">The distance parameter relates the speed of the axis to the time taken to complete the cam profile.</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Cam(uint32_t tableStart, uint32_t tableStop, double tableMultiplier, double linkDistance, int32_t axis = -1) = 0;

        /// <summary>
        /// This command is used to generate movement of an axis according to a table of POSITIONS which define the movement profile
        /// </summary>
        /// <param name="tableStart">The start position of the cam profile in the TABLE.</param>
        /// <param name="tableStop">The end position of the cam profile in the TABLE.</param>
        /// <param name="tableMultiplier">The table values are multiplied by this value to generate the positions.</param>
        /// <param name="linkDistance">The distance the link axis must move to complete CAMBOX profile.</param>
        /// <param name="linkAxis">The axis to link to.</param>
        /// <param name="linkOptions">Bit value options to customise how your CAMBOX operates.</param>
        /// <param name="linkPos">
        ///  linkOptions bit 1 - the absolute position on the link axis in user UNITS where the CAMBOX is to be start.<br/>
        ///  linkOptions bit 9 - the registration channel to start the movement on
        /// </param>
        /// <param name="offsetStart">The position defined on the link_dist where the profile will start. </param>
```

```
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void CamBox(uint32_t tableStart, uint32_t tableStop, double tableMultiplier, double linkDistance, int32_t linkAxis, CamBoxLinkOptions linkOptions, double linkPos = 0, double offsetStart = 0, int32_t axis = -1) = 0;

        /// <summary>
        /// This command is used to generate movement of multiple axas according to a table of POSITIONS which define the movement profile
        /// </summary>
        /// <param name="tableStart">The start index in TABLE.</param>
        /// <param name="axisCount">Number of axes to generate movement for.</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// That variant of CAMBOX command allow movement to be defined for with one call.
        ///  Parameters are read from a TABLE range starting at <paramref name="tableStart"/> offset.
        ///  For each axis 8 parameters should be provided in the following order:
        ///   Offset 0 - TABLE start point for CAMBOX profile position data
        ///   Offset 1 - TABLE end point for CAMBOX profile position data
        ///   Offset 2 - TABLE data multiplier
        ///   Offset 3 - Link distance
        ///   Offset 4 - Link axis
        ///   Offset 5 - Link options(as per CAMBOX single - axis command)
        ///   Offset 6 - Link position / Registration channel for link options
        ///   Offset 7 - Forces the profile to start at a defined point in the link distance(options bit 7)
        ///  The same pattern is repeated for the number of axes to be controlled.
        ///  The data will be validated and if an error is detected then no CAMBOX requests will be loaded and a run-time error will be raised.
        /// </remarks>
        virtual void CamBoxMulti(uint32_t tableStart, uint32_t axisCount, int32_t axis = -1) = 0;

        /// <summary>
        /// The REGIST command initiates a capture of an axis position when it sees a registration input or the Z mark on the encoder. Once a registration event is captured MARK is set and the position and speed at the event can be read back.
        /// </summary>
        /// <param name="mode">Registration mode. See TrioBASIC REGIST documentation for more details</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// This variant is used with all other modes except 20, 21 and 22 and is not recommended for new applications.
        /// </remarks>
        virtual void Regist(RegistMode mode, int32_t axis = -1) = 0;

        /// <summary>
        /// The REGIST command initiates a capture of an axis position when it sees a registration input or the Z mark on the encoder. Once a registration event is captured MARK is set and the position and speed at the event can be read back.
        /// </summary>
        /// <param name="mode">Registration mode. See TrioBASIC REGIST documentation for more details.</param>
        /// <param name="channel">Registration channel. See TrioBASIC REGIST documentation for more details.</param>
        /// <param name="source">Registration source. See TrioBASIC REGIST documentation for more details.</param>
        /// <param name="edge">0 - raising edge, 1 - falling edge.</param>
        /// <param name="window">Window of operation. See TrioBASIC REGIST documentation for more details.</param>
        /// <param name="repeatCount">Quantity of registration captures to store in the TABLE.</param>
        /// <param name="tableStart">Start position in the TABLE for the registration positions.</param>
        /// <param name="axis">Base axis for this command. If value is -1 then the default base axis is used</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// This variant is used with modes 20, 21 and 22 and is recommended for new applications.
        /// </remarks>
        virtual void Regist(RegistMode mode, uint32_t channel, uint32_t source, RegistEdge edge, RegistWindow window, uint32_t repeatCount = 0, uint32_t tableStart = 0, int32_t axis = -1) = 0;

        /// <summary>
        /// Define or re-define a parameter block of VR values
        /// </summary>
        /// <param name="indices">VR indices array</param>
        /// <param name="count">The array size</param>
        /// <returns>A placeholder for the newly defined block</returns>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineVr(const uint32_t indices[], uint32_t count) = 0;

        /// <summary>
        /// Define or re-define a parameter block of VR values
        /// </summary>
        /// <param name="indices">VR indices array</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineVr(const std::vector<uint32_t>& indices) = 0;

        /// <summary>
        /// Append to a parameter block of VR values
        /// </summary>
        /// <param name="indices">VR indices array</param>
        /// <param name="count">The array size</param>
        /// <param name="blk">Previously created block</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendVr(const uint32_t indices[], uint32_t count, int32_t blk) = 0;

        /// <summary>
        /// Append to a parameter block of VR values
        /// </summary>
        /// <param name="indices">VR indices array</param>
        /// <param name="blk">Previously created block</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendVr(const std::vector<uint32_t>& indices, int32_t blk) = 0;

        /// <summary>
        /// Define or re-define a parameter block of TABLE values
        /// </summary>
        /// <param name="indices">TABLE indices array</param>
        /// <param name="count">The array size</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineTable(const uint32_t indices[], uint32_t count) = 0;

        /// <summary>
        /// Define or re-define a parameter block of TABLE values
        /// </summary>
        /// <param name="indices">TABLE indices array</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineTable(const std::vector<uint32_t>& indices) = 0;

        /// <summary>
        /// Append a parameter block of TABLE values
        /// </summary>
        /// <param name="indices">TABLE indices array</param>
        /// <param name="count">The array size</param>
        /// <param name="blk">Prevously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendTable(const uint32_t indices[], uint32_t count, int32_t blk) = 0;

        /// <summary>
        /// Append a parameter block of TABLE values
        /// </summary>
        /// <param name="indices">TABLE indices array</param>
        /// <param name="blk">Prevously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendTable(const std::vector<uint32_t>& indices, int32_t blk) = 0;

        /// <summary>
        /// Define or re-define a parameter block of Axis parameter values
        /// </summary>
        /// <param name="axisParams">Axis parameter name array</param>
        /// <param name="count">The array size</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineAxis(const std::string_view axisParams[], uint32_t count) = 0;

        /// <summary>
        /// Define or re-define a parameter block of Axis parameter values
        /// </summary>
        /// <param name="axisParams">Axis parameter name array</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual int32_t ParamBlockDefineAxis(const std::vector<std::string_view>& axisParams) = 0;

        /// <summary>
        /// Append a parameter block of Axis parameter values
        /// </summary>
        /// <param name="axisParams">Axis parameter name array</param>
        /// <param name="count">The array size</param>
        /// <param name="blk">Previously create block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendAxis(const std::string_view axisParams[], uint32_t count, int32_t blk) = 0;

        /// <summary>
        /// Append a parameter block of Axis parameter values
        /// </summary>
        /// <param name="axisParams">Axis parameter name array</param>
        /// <param name="blk">Previously create block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendAxis(const std::vector<std::string_view>& axisParams, int32_t blk) = 0;

        /// <summary>
        /// Define or re-define a parameter block of System parameter values
        /// </summary>
        /// <param name="sysParams">System parameter name array</param>
        /// <param name="count">The array size</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineSystem(const std::string_view sysParams[], uint32_t count) = 0;

        /// <summary>
        /// Define or re-define a parameter block of System parameter values
        /// </summary>
        /// <param name="sysParams">System parameter name array</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineSystem(const std::vector<std::string_view>& sysParams) = 0;

        /// <summary>
        /// Append to a parameter block of System parameter values
        /// </summary>
        /// <param name="sysParams">System parameter name array</param>
        /// <param name="count">The array size</param>
        /// <param name="blk">Previously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendSystem(const std::string_view sysParams[], uint32_t count, int32_t blk) = 0;

        /// <summary>
        /// Append to a parameter block of System parameter values
        /// </summary>
        /// <param name="sysParams">System parameter name array</param>
        /// <param name="blk">Previously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendSystem(const std::vector<std::string_view>& sysParams, int32_t blk) = 0;

        /// <summary>
        /// Define or re-define a parameter block of Program variables.
        /// </summary>
        /// <param name="program">Program name</param>
        /// <param name="process">Optional process ID or -1 to select any process running the program</param>
        /// <param name="variables">Program variable name array</param>
        /// <param name="count">The array size</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineProgVar(const std::string_view& program, int32_t process, const std::string_view variables[], uint32_t count) = 0;

        /// <summary>
        /// Define or re-define a parameter block of Program variables.
        /// </summary>
        /// <param name="program">Program name</param>
        /// <param name="process">Optional process ID or -1 to select any process running the program</param>
        /// <param name="variables">Program variable name array</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual int32_t ParamBlockDefineProgVar(const std::string_view& program, int32_t process, const std::vector<std::string_view>& variables) = 0;

        /// <summary>
        /// Append to a parameter block of Program variables.
        /// </summary>
        /// <param name="program">Program name</param>
        /// <param name="process">Optional process ID or -1 to select first running process</param>
        /// <param name="variables">Program variable name array</param>
        /// <param name="count">The array size</param>
        /// <param name="blk">Previously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendProgVar(const std::string_view& program, int32_t process, const std::string_view variables[], uint32_t count, int32_t blk) = 0;

        /// <summary>
        /// Append to a parameter block of Program variables.
        /// </summary>
        /// <param name="program">Program name</param>
        /// <param name="process">Optional process ID or -1 to select first running process</param>
        /// <param name="variables">Program variable name array</param>
        /// <param name="blk">Previously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendProgVar(const std::string_view& program, int32_t process, const std::vector<std::string_view>& variables, int32_t blk) = 0;

        /// <summary>
        /// Define or re-define a parameter block of Digital IO.
        /// </summary>
        /// <param name="ranges">Digital IO range array</param>
        /// <param name="count">The array size</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineIO(const Range ranges[], uint32_t count) = 0;

        /// <summary>
        /// Define or re-define a parameter block of Digital IO.
        /// </summary>
        /// <param name="ranges">Digital IO range array</param>
        /// <returns>A placeholder for the newly defined block</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockDefineIO(const std::vector<Range>& ranges) = 0;

        /// <summary>
        /// Append to a parameter block of Digital IO.
        /// </summary>
        /// <param name="ranges">Digital IO range array</param>
        /// <param name="count">The array size</param>
        /// <param name="blk">Previously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendIO(const Range ranges[], uint32_t count, int32_t blk) = 0;

        /// <summary>
        /// Append to a parameter block of Digital IO.
        /// </summary>
        /// <param name="ranges">Digital IO range array</param>
        /// <param name="blk">Previously created block index</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockAppendIO(const std::vector<Range>& ranges, int32_t blk) = 0;

        /// <summary>
        /// Deletes a defined parameter block.
        /// </summary>
        /// <param name="blk">The block to delete. It should have been defined previously by some of "ParamBlockDefine" functions</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockDelete(int32_t blk) = 0;

        /// <summary>
        /// Deletes all defined parameter blocks.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockDeleteAll() = 0;

        /// <summary>
        /// Fetches only changed values of a VR parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
```

```
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedVr(int32_t blk, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches only changed values of a VR parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedVr(int32_t blk, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches only changed values of a TABLE parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedTable(int32_t blk, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches only changed values of a TABLE parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedTable(int32_t blk, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axis">Axis index</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedAxisParam(int32_t blk, int32_t axis, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axis">Axis index</param>
        /// <param name="values">Result placeholder array</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedAxisParam(int32_t blk, int32_t axis, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Axis range</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <param name="changed_bitmasks">An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.</param>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Values array size should be multiple of the number of axes by the number of parameters in the block.
        ///  Changed bitmasks array size should be equal to the number of axes.
        /// </remarks>
        virtual void ParamBlockFetchChangedAxisParam(int32_t blk, const Range& axes, double values[], uint32_t count, uint64_t changed_bitmasks[]) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Axis range</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="changed_bitmasks">An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Values array size should be multiple of the number of axes by the number of parameters in the block.
        ///  Changed bitmasks array size should be equal to the number of axes.
        /// </remarks>
        virtual void ParamBlockFetchChangedAxisParam(int32_t blk, const Range& axes, std::vector<double>& values, std::vector<uint64_t>& changed_bitmasks) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Array of axis indices</param>
        /// <param name="axis_count">The size of axis indices array</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="value_count">The size of values array</param>
        /// <param name="changed_bitmasks">An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range.
        ///  Values array size should be multiple of the number of axes by the number of parameters in the block.
        ///  Changed bitmasks array size should be equal to the number of axes.
        /// </remarks>
        virtual void ParamBlockFetchChangedAxisParam(int32_t blk, const int32_t axes[], uint32_t axis_count, double values[], uint32_t value_count, uint64_t changed_bitmasks[]) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Array of axis indices</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="changed_bitmasks">An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range.
        ///  Values array size should be multiple of the number of axes by the number of parameters in the block.
        ///  Changed bitmasks array size should be equal to the number of axes.
        /// </remarks>
        virtual void ParamBlockFetchChangedAxisParam(int32_t blk, const std::vector<int32_t>& axes, std::vector<double>& values, std::vector<uint64_t>& changed_bitmasks) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="ranges">Array of axis ranges</param>
        /// <param name="range_count">Axis range array size</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="value_count">The array size</param>
        /// <param name="changed_bitmasks">An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array
```

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corresponds to an axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Values array size should be multiple of the number of axes by the number of parameters in the block.
        ///  Changed bitmasks array size should be equal to the number of axes.
        /// </remarks>
        virtual void ParamBlockFetchChangedAxisParam(int32_t blk, const Range ranges[], uint32_t range_count, double values[], uint32_t value_count, uint64_t changed_bitmasks[]) = 0;

        /// <summary>
        /// Fetches only changed values of an Axis parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="ranges">Array of axis ranges</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="changed_bitmasks">An array of "changed" state bit-masks. A bit is 1 if the value of a parameter at the corresponding index has been fetched (changed). Each value in the array corresponds to an axis.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Values array size should be multiple of the number of axes by the number of parameters in the block.
        ///  Changed bitmasks array size should be equal to the number of axes.
        /// </remarks>
        virtual void ParamBlockFetchChangedAxisParam(int32_t blk, const std::vector<Range>& ranges, std::vector<double>& values, std::vector<uint64_t>& changed_bitmasks) = 0;

        /// <summary>
        /// Fetches only changed values of a System parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedSystem(int32_t blk, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches only changed values of a System parameter block. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedSystem(int32_t blk, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches only changed values of a block of program variables. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedProgVar(int32_t blk, const char* values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches only changed values of a block of program variables. A value is considered changed if it differs from the previous fetch.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedProgVar(int32_t blk, std::vector<std::string_view>& values) = 0;

        /// <summary>
        /// Fetches all values in a VR parameter block.
        /// </summary>
```

```
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllVr(int32_t blk, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in a VR parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllVr(int32_t blk, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches all values in a TABLE parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllTable(int32_t blk, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in a TABLE parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllTable(int32_t blk, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axis">Axis index</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, int32_t axis, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axis">Axis index</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, int32_t axis, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Axis range</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Values array size should be multiple of the number of axes by the number of parameters in the block
        /// </remarks>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, const Range& axes, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Axis range</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Values array size should be multiple of the number of axes by the number of parameters in the block
        /// </remarks>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, const Range& axes, std::vector<double>&
```

```
values) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Array of axis indices</param>
        /// <param name="axis_count">The size of axis indices array</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="value_count">The size of values array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range.
        ///  Values array size should be multiple of the number of axes by the number of parameters in the block.
        /// </remarks>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, const int32_t axes[], uint32_t axis_count, double values[], uint32_t value_count) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="axes">Array of axis indices</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range.
        ///  Values array size should be multiple of the number of axes by the number of parameters in the block.
        /// </remarks>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, const std::vector<int32_t>& axes, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="ranges">Array of axis ranges</param>
        /// <param name="range_count">Axis range array size</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="value_count">The size of values array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range.
        ///  Values array size should be multiple of the number of axes by the number of parameters in the block.
        /// </remarks>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, const Range ranges[], uint32_t range_count, double values[], uint32_t value_count) = 0;

        /// <summary>
        /// Fetches all values in an Axis parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="ranges">Array of axis ranges</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        /// <remarks>
        /// Maximum of 15 axes is supported by this method due to limitations of the protocol. For fetching more than 15 axes with single call use the method with range.
        ///  Values array size should be multiple of the number of axes by the number of parameters in the block.
        /// </remarks>
        virtual void ParamBlockFetchAllAxisParam(int32_t blk, const std::vector<Range>& ranges, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches all values in a System parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllSystem(int32_t blk, double values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in a System parameter block.
        /// </summary>
```

```
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllSystem(int32_t blk, std::vector<double>& values) = 0;

        /// <summary>
        /// Fetches all values in a block of program variables.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllProgVar(int32_t blk, const char* values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in a block of program variables.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllProgVar(int32_t blk, std::vector<std::string_view>& values) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <returns>The result</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double ParamBlockGetProgDoubleValue(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <returns>The result</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t ParamBlockGetProgInt64Value(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <returns>The result</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockGetProgUInt64Value(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <returns>The result</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t ParamBlockGetProgInt32Value(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <returns>The result</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t ParamBlockGetProgUInt32Value(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
```

```
        /// <returns>The result</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool ParamBlockGetProgBoolValue(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <param name="buff">Buffer to receive the response string value</param>
        /// <param name="buff_size">Size of the buffer on input and on exit required buffer size including null terminator</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockGetProgStrValue(int32_t blk, uint32_t index, char* buff, uint32_t& buff_size) = 0;

        /// <summary>
        /// Retrieves a fetched value from a parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="index">The index of the value in the parameter block. Please refer to the block definition.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual std::string_view ParamBlockGetProgStrValue(int32_t blk, uint32_t index) = 0;

        /// <summary>
        /// Fetches only changed values in a Digital IO parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="outputs">If true Digital Outputs are fetched. If false Digital Inputs are fetched.</param>
        /// <param name="values">Result placeholder array</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedIO(int32_t blk, bool outputs, std::vector<uint64_t>& values) = 0;

        /// <summary>
        /// Fetches only changed values in a Digital IO parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="outputs">If true Digital Outputs are fetched. If false Digital Inputs are fetched.</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <returns>The "changed" state bit-mask. A bit is 1 if the value at the corresponding index has been fetched (changed).</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t ParamBlockFetchChangedIO(int32_t blk, bool outputs, uint64_t values[], uint32_t count) = 0;

        /// <summary>
        /// Fetches all values in a Digital IO parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="outputs">If true Digital Outputs are fetched. If false Digital Inputs are fetched.</param>
        /// <param name="values">Result placeholder array</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllIO(int32_t blk, bool outputs, std::vector<uint64_t>& values) = 0;

        /// <summary>
        /// Fetches all values in a Digital IO parameter block.
        /// </summary>
        /// <param name="blk">A valid parameter block</param>
        /// <param name="outputs">If true Digital Outputs are fetched. If false Digital Inputs are fetched.</param>
        /// <param name="values">Result placeholder array</param>
        /// <param name="count">The array size</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ParamBlockFetchAllIO(int32_t blk, bool outputs, uint64_t values[], uint32_t count) = 0;

        /// <summary>
        /// The function can be used to set the value of a single bit within a VR variable. All other bits are unchanged.
        /// </summary>
```

```
        /// <param name="bit">The bit number to set, valid range is 0 to 52.</param>
        /// <param name="vr_variable">The VR which to operate on.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetVrBit(uint16_t bit, uint32_t vr_variable) = 0;

        /// <summary>
        /// The function can be used to clear the value of a single bit within a VR variable.
        /// </summary>
        /// <param name="bit">The bit number to clear, valid range is 0 to 52.</param>
        /// <param name="vr_variable">The VR which to operate on.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ClearVrBit(uint16_t bit, uint32_t vr_variable) = 0;

        /// <summary>
        /// The function can be used to test the value of a single bit within a VR variable.
        /// </summary>
        /// <param name="bit">The bit number to clear, valid range is 0 to 52.</param>
        /// <param name="vr_variable">The VR which to operate on.</param>
        /// <returns>On success the value VR entry returned</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetVrBit(uint16_t bit, uint32_t vr_variable) = 0;

        /// <summary>
        /// Halts execution of all running programs.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Halt() = 0;

        /// <summary>
        /// Function returns the list of processes on the Motion Coordinator
        /// </summary>
        /// <param name="handler">Callback handler function executed for each process item</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void GetProcessList(ProcessInfoCallback&& handler) = 0;

        /// <summary>
        /// Disable SCOPE.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ScopeOff() = 0;

        /// <summary>
        /// Enable SCOPE with previous settings.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ScopeOn() = 0;

        /// <summary>
        /// Enable SCOPE with previous settings and new capture period.
        /// </summary>
        /// <param name="period_cycles">The number of servo periods between data samples.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ScopeOn(uint32_t period_cycles) = 0;

        /// <summary>
        /// Enable SCOPE and configure for capture to TABLE range.
        /// </summary>
        /// <param name="period_cycles">The number of servo periods between data samples.</param>
        /// <param name="table_start">Start of table range to use store captured data.</param>
        /// <param name="table_end">End of table range to use (inclusive).</param>
        /// <param name="prms">Array of parameters to be captured.</param>
        /// <param name="prm_count">Number of parameters to be captured.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void ScopeOn(uint32_t period_cycles, uint32_t table_start, uint32_t table_end, const std::string_view prms[], uint32_t prm_count) = 0;

        /// <summary>
        /// Enable SCOPE and configure for capture to FIFO.
        /// </summary>
        /// <param name="period_cycles">The number of servo periods between data samples.</param>
        /// <param name="fifo_name">Name of the FIFO file to be written. If this file already exists on the Motion Coordinator it must be a FIFO file.</param>
        /// <param name="fifo_size">Size of the FIFO file to be created. If the FIFO file already exists with a different size then it will be deleted and recreated with the correct size.</param>
        /// <param name="sample_count">Number of samples to store to the FIFO file.</param>
        /// <param name="prms">Array of parameters to be captured.</param>
        /// <param name="prm_count">Number of parameters to be captured.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual void ScopeOn(uint32_t period_cycles, const std::string_view& fifo_name, uint32_t fifo_size, uint32_t sample_count, const std::string_view prms[], uint32_t prm_count) = 0;

        /// <summary>
        /// Starts a previously set up SCOPE command. This allows you to manually start the scope capture.
        /// </summary>
        /// <param name="rearm">If true the SCOPE will automatically retrigger at the end of each scan.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void Trigger(bool rearm = false) = 0;

        /// <summary>
        /// Get double value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetSystemParameterDoubleValue(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set double value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameterDoubleValue(const std::string_view& param_name, double value) = 0;

        /// <summary>
        /// Get 64-bit signed integer value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameterInt64Value(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set 64 bit signed integer value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameterInt64Value(const std::string_view& param_name, int64_t value) = 0;

        /// <summary>
        /// Get 64 bit unsigned integer value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetSystemParameterUInt64Value(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set 64 bit unsigned integer value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameterUInt64Value(const std::string_view& param_name, uint64_t value) = 0;

        /// <summary>
        /// Get 32-bit signed integer value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameterInt32Value(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set 32-bit signed integer value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual void SetSystemParameterInt32Value(const std::string_view& param_name, int32_t value) = 0;

        /// <summary>
        /// Get 32-bit unsigned integer value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameterUInt32Value(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set 32 bit unsigned integer value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameterUInt32Value(const std::string_view& param_name, uint32_t value) = 0;

        /// <summary>
        /// Get boolean value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetSystemParameterBoolValue(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set boolean value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameterBoolValue(const std::string_view& param_name, bool value) = 0;

        /// <summary>
        /// Get string value of system parameter by name, into provided buffer with size
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="buff">Buffer to receive the response string value</param>
        /// <param name="buff_size">Size of the buffer on input and on exit required buffer size including null terminator</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void GetSystemParameterStringValue(const std::string_view& param_name, char* buff, uint32_t& buff_size) = 0;

        /// <summary>
        /// Get string value of system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual std::string GetSystemParameterStringValue(const std::string_view& param_name) = 0;

        /// <summary>
        /// Set string value to system parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameterStringValue(const std::string_view& param_name, const std::string_view& value) = 0;

        /// <summary>
        /// Get double value of axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameterDoubleValue(const std::string_view& param_name, int32_t axis) = 0;

        /// <summary>
        /// Set double value to axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
```

```
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameterDoubleValue(const std::string_view& param_name, int32_t axis, double value) = 0;

        /// <summary>
        /// Get 64-bit signed integer value of axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameterInt64Value(const std::string_view& param_name, int32_t axis) = 0;

        /// <summary>
        /// Set 64 bit signed integer value to axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameterInt64Value(const std::string_view& param_name, int32_t axis, int64_t value) = 0;

        /// <summary>
        /// Get 64 bit unsigned integer value of axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetAxisParameterUInt64Value(const std::string_view& param_name, int32_t axis) = 0;

        /// <summary>
        /// Set 64 bit unsigned integer value to axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameterUInt64Value(const std::string_view& param_name, int32_t axis, uint64_t value) = 0;

        /// <summary>
        /// Get 32-bit signed integer value of axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameterInt32Value(const std::string_view& param_name, int32_t axis) = 0;

        /// <summary>
        /// Set 32-bit signed integer value to axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameterInt32Value(const std::string_view& param_name, int32_t axis, int32_t value) = 0;

        /// <summary>
        /// Get 32-bit unsigned integer value of axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetAxisParameterUInt32Value(const std::string_view& param_name, int32_t axis) = 0;

        /// <summary>
        /// Set 32 bit unsigned integer value to axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
```

```
        /// <param name="axis">Axis index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameterUInt32Value(const std::string_view& param_name, int32_t axis, uint32_t value) = 0;

        /// <summary>
        /// Get boolean value of axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetAxisParameterBoolValue(const std::string_view& param_name, int32_t axis) = 0;

        /// <summary>
        /// Set boolean value to axis parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="axis">Axis index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameterBoolValue(const std::string_view& param_name, int32_t axis, bool value) = 0;

        /// <summary>
        /// Get double value of process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetProcessParameterDoubleValue(const std::string_view& param_name, int32_t process) = 0;

        /// <summary>
        /// Set double value to process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameterDoubleValue(const std::string_view& param_name, int32_t process, double value) = 0;

        /// <summary>
        /// Get 64-bit signed integer value of process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetProcessParameterInt64Value(const std::string_view& param_name, int32_t process) = 0;

        /// <summary>
        /// Set 64 bit signed integer value to process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameterInt64Value(const std::string_view& param_name, int32_t process, int64_t value) = 0;

        /// <summary>
        /// Get 64 bit unsigned integer value of process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetProcessParameterUInt64Value(const std::string_view& param_name, int32_t process) = 0;

        /// <summary>
        /// Set 64 bit unsigned integer value to process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
```

```
        /// <param name="process">Process index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameterUInt64Value(const std::string_view& param_name, int32_t process, uint64_t value) = 0;

        /// <summary>
        /// Get 32-bit signed integer value of process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetProcessParameterInt32Value(const std::string_view& param_name, int32_t process) = 0;

        /// <summary>
        /// Set 32-bit signed integer value to process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameterInt32Value(const std::string_view& param_name, int32_t process, int32_t value) = 0;

        /// <summary>
        /// Get 32-bit unsigned integer value of process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetProcessParameterUInt32Value(const std::string_view& param_name, int32_t process) = 0;

        /// <summary>
        /// Set 32 bit unsigned integer value to process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameterUInt32Value(const std::string_view& param_name, int32_t process, uint32_t value) = 0;

        /// <summary>
        /// Get boolean value of process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetProcessParameterBoolValue(const std::string_view& param_name, int32_t process) = 0;

        /// <summary>
        /// Set boolean value to process parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="process">Process index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameterBoolValue(const std::string_view& param_name, int32_t process, bool value) = 0;

        /// <summary>
        /// Get double value of slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetSlotParameterDoubleValue(const std::string_view& param_name, int32_t slot) = 0;

        /// <summary>
        /// Set double value to slot parameter by name
        /// </summary>
```

```
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameterDoubleValue(const std::string_view& param_name, int32_t slot, double value) = 0;

        /// <summary>
        /// Get 64-bit signed integer value of slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSlotParameterInt64Value(const std::string_view& param_name, int32_t slot) = 0;

        /// <summary>
        /// Set 64 bit signed integer value to slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameterInt64Value(const std::string_view& param_name, int32_t slot, int64_t value) = 0;

        /// <summary>
        /// Get 64 bit unsigned integer value of slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetSlotParameterUInt64Value(const std::string_view& param_name, int32_t slot) = 0;

        /// <summary>
        /// Set 64 bit unsigned integer value to slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameterUInt64Value(const std::string_view& param_name, int32_t slot, uint64_t value) = 0;

        /// <summary>
        /// Get 32-bit signed integer value of slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSlotParameterInt32Value(const std::string_view& param_name, int32_t slot) = 0;

        /// <summary>
        /// Set 32-bit signed integer value to slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameterInt32Value(const std::string_view& param_name, int32_t slot, int32_t value) = 0;

        /// <summary>
        /// Get 32-bit unsigned integer value of slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSlotParameterUInt32Value(const std::string_view& param_name, int32_t slot) = 0;

        /// <summary>
        /// Set 32 bit unsigned integer value to slot parameter by name
```

```
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameterUInt32Value(const std::string_view& param_name, int32_t slot, uint32_t value) = 0;

        /// <summary>
        /// Get boolean value of slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetSlotParameterBoolValue(const std::string_view& param_name, int32_t slot) = 0;

        /// <summary>
        /// Set boolean value to slot parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="slot">Slot index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameterBoolValue(const std::string_view& param_name, int32_t slot, bool value) = 0;

        /// <summary>
        /// Get double value of port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetPortParameterDoubleValue(const std::string_view& param_name, int32_t port) = 0;

        /// <summary>
        /// Set double value to port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetPortParameterDoubleValue(const std::string_view& param_name, int32_t port, double value) = 0;

        /// <summary>
        /// Get 64-bit signed integer value of port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetPortParameterInt64Value(const std::string_view& param_name, int32_t port) = 0;

        /// <summary>
        /// Set 64 bit signed integer value to port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetPortParameterInt64Value(const std::string_view& param_name, int32_t port, int64_t value) = 0;

        /// <summary>
        /// Get 64 bit unsigned integer value of port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetPortParameterUInt64Value(const std::string_view& param_name, int32_t port) = 0;

        /// <summary>
        /// Set 64 bit unsigned integer value to port parameter by name
```

```
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetPortParameterUInt64Value(const std::string_view& param_name, int32_t port, uint64_t value) = 0;

        /// <summary>
        /// Get 32-bit signed integer value of port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetPortParameterInt32Value(const std::string_view& param_name, int32_t port) = 0;

        /// <summary>
        /// Set 32-bit signed integer value to port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetPortParameterInt32Value(const std::string_view& param_name, int32_t port, int32_t value) = 0;

        /// <summary>
        /// Get 32-bit unsigned integer value of port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetPortParameterUInt32Value(const std::string_view& param_name, int32_t port) = 0;

        /// <summary>
        /// Set 32 bit unsigned integer value to port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetPortParameterUInt32Value(const std::string_view& param_name, int32_t port, uint32_t value) = 0;

        /// <summary>
        /// Get boolean value of port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <returns>Variable to receive the value</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetPortParameterBoolValue(const std::string_view& param_name, int32_t port) = 0;

        /// <summary>
        /// Set boolean value to port parameter by name
        /// </summary>
        /// <param name="param_name">Parameter name</param>
        /// <param name="port">Port index</param>
        /// <param name="value">New value</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetPortParameterBoolValue(const std::string_view& param_name, int32_t port, bool value) = 0;

        /// <summary>
        /// Loads a Motion Perfect project on the Motion Coordinator. All the running programs and motion are stopped. All the existing content on the Motion Coordinator is deleted.
        /// </summary>
        /// <param name="project_file">The project file name. The folder should contain all the other project content - as saved by Motion Perfect.</param>
        /// <param name="uknown_project_key_handler">
        /// A handler receiving the project key authority name for which a project key needs to be set. It is expected the user to register the key for the passed authority.<br/>
        ///  For example by using ProjectKeyRegistryStore or ProjectKey functions. If the uknown_project_key_handler returns false the project load will be cancelled and will exit with error
        /// </param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual void LoadProject(const std::string_view& project_file, StringCallback&&
uknown_project_key_handler) = 0;

        /// <summary>
        /// Sets a project key on the controller.
        /// </summary>
        /// <param name="key">The project key string.</param>
        /// <param name="security_code">The project key security code (key type).</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void ProjectKey(const std::string_view& key, SecurityCodeType security_code =
SecurityCodeType::Controller) = 0;

        /// <summary>
        /// List the authorities that have project keys stored on the Motion Coordinator
        /// </summary>
        /// <param name="handler">The handler of each returned name.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void ProjectKeyRegistryList(StringCallback&& handler) = 0;

        /// <summary>
        /// Stores a project key on the Motion Coordinator under the specified authority name
        /// </summary>
        /// <param name="authority">The authority name.</param>
        /// <param name="key">The project key string.</param>
        /// <param name="security_code">The project key security code (key type).</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void ProjectKeyRegistryStore(const std::string_view& authority, const
std::string_view& key, SecurityCodeType security_code = SecurityCodeType::Controller) = 0;

        /// <summary>
        /// Remove all non-default encryption authorities from database. The default (pre-installed)
authorities are not removed.
        /// </summary>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void ProjectKeyRegistryClear() = 0;

        /// <summary>
        /// Loads a project key for use (project/program load) on the Motion Coordinator
        /// </summary>
        /// <param name="authority">The authority name of the project key to load.</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void ProjectKeyRegistryUse(const std::string_view& authority) = 0;


////////////
```

```
        /// <summary>
        /// Getter for AIN2 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AIN2() = 0;

        /// <summary>
        /// Setter for AIN2 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AIN2(uint32_t value) = 0;


        /// <summary>
        /// Getter for AIN3 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AIN3() = 0;

        /// <summary>
        /// Setter for AIN3 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AIN3(uint32_t value) = 0;


        /// <summary>
        /// Getter for AOUT0 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AOUT0() = 0;

        /// <summary>
        /// Setter for AOUT0 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AOUT0(uint32_t value) = 0;


        /// <summary>
        /// Getter for AOUT1 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AOUT1() = 0;

        /// <summary>
        /// Setter for AOUT1 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AOUT1(uint32_t value) = 0;


        /// <summary>
        /// Getter for AOUT2 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AOUT2() = 0;

        /// <summary>
        /// Setter for AOUT2 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AOUT2(uint32_t value) = 0;
```

```
        /// <summary>
        /// Getter for AOUT3 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AOUT3() = 0;

        /// <summary>
        /// Setter for AOUT3 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AOUT3(uint32_t value) = 0;


        /// <summary>
        /// Getter for AINBI0 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AINBI0() = 0;

        /// <summary>
        /// Setter for AINBI0 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AINBI0(uint32_t value) = 0;


        /// <summary>
        /// Getter for AINBI1 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AINBI1() = 0;

        /// <summary>
        /// Setter for AINBI1 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AINBI1(uint32_t value) = 0;


        /// <summary>
        /// Getter for AINBI2 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AINBI2() = 0;

        /// <summary>
        /// Setter for AINBI2 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AINBI2(uint32_t value) = 0;


        /// <summary>
        /// Getter for AINBI3 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_AINBI3() = 0;

        /// <summary>
        /// Setter for AINBI3 system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AINBI3(uint32_t value) = 0;


        /// <summary>
        /// Getter for INPUTS0 system parameter
```

```
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_INPUTS0() = 0;


        /// <summary>
        /// Getter for INPUTS1 system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_INPUTS1() = 0;


        /// <summary>
        /// Getter for SLOT0_TIME system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SLOT0_TIME() = 0;

        /// <summary>
        /// Setter for SLOT0_TIME system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SLOT0_TIME(uint32_t value) = 0;


        /// <summary>
        /// Getter for SLOT1_TIME system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SLOT1_TIME() = 0;

        /// <summary>
        /// Setter for SLOT1_TIME system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SLOT1_TIME(uint32_t value) = 0;


        /// <summary>
        /// Getter for SLOT2_TIME system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SLOT2_TIME() = 0;

        /// <summary>
        /// Setter for SLOT2_TIME system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SLOT2_TIME(uint32_t value) = 0;


        /// <summary>
        /// Getter for SLOT3_TIME system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SLOT3_TIME() = 0;

        /// <summary>
        /// Setter for SLOT3_TIME system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SLOT3_TIME(uint32_t value) = 0;


        /// <summary>
        /// Getter for WDOG system parameter
        /// </summary>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetSystemParameter_WDOG() = 0;

        /// <summary>
        /// Setter for WDOG system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_WDOG(bool value) = 0;


        /// <summary>
        /// Getter for SYSTEM_ERROR system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SYSTEM_ERROR() = 0;


        /// <summary>
        /// Getter for CURRENT_JITTER system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_CURRENT_JITTER() = 0;

        /// <summary>
        /// Setter for CURRENT_JITTER system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_CURRENT_JITTER(int64_t value) = 0;


        /// <summary>
        /// Getter for AVERAGE_JITTER system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_AVERAGE_JITTER() = 0;

        /// <summary>
        /// Setter for AVERAGE_JITTER system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AVERAGE_JITTER(int64_t value) = 0;


        /// <summary>
        /// Getter for MAX_JITTER system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_MAX_JITTER() = 0;

        /// <summary>
        /// Setter for MAX_JITTER system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_MAX_JITTER(int64_t value) = 0;


        /// <summary>
        /// Getter for MISSED_INTERRUPTS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_MISSED_INTERRUPTS() = 0;

        /// <summary>
        /// Setter for MISSED_INTERRUPTS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual void SetSystemParameter_MISSED_INTERRUPTS(int64_t value) = 0;


        /// <summary>
        /// Getter for MISSED_SEND_FRAMES system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_MISSED_SEND_FRAMES() = 0;

        /// <summary>
        /// Setter for MISSED_SEND_FRAMES system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_MISSED_SEND_FRAMES(int64_t value) = 0;


        /// <summary>
        /// Getter for MISSED_RECV_FRAMES system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_MISSED_RECV_FRAMES() = 0;

        /// <summary>
        /// Setter for MISSED_RECV_FRAMES system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_MISSED_RECV_FRAMES(int64_t value) = 0;


        /// <summary>
        /// Getter for LIMIT_BUFFERED system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetSystemParameter_LIMIT_BUFFERED() = 0;

        /// <summary>
        /// Setter for LIMIT_BUFFERED system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_LIMIT_BUFFERED(int16_t value) = 0;


        /// <summary>
        /// Getter for ADDRESS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_ADDRESS() = 0;

        /// <summary>
        /// Setter for ADDRESS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_ADDRESS(uint32_t value) = 0;


        /// <summary>
        /// Getter for AUTO_ETHERCAT system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_AUTO_ETHERCAT() = 0;

        /// <summary>
        /// Setter for AUTO_ETHERCAT system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_AUTO_ETHERCAT(int32_t value) = 0;
```

```
        /// <summary>
        /// Getter for CONTROL system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_CONTROL() = 0;


        /// <summary>
        /// Getter for CPU_EXCEPTIONS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_CPU_EXCEPTIONS() = 0;

        /// <summary>
        /// Setter for CPU_EXCEPTIONS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_CPU_EXCEPTIONS(int64_t value) = 0;


        /// <summary>
        /// Getter for DISPLAY system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_DISPLAY() = 0;

        /// <summary>
        /// Setter for DISPLAY system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_DISPLAY(int32_t value) = 0;


        /// <summary>
        /// Getter for FLASH_STATUS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_FLASH_STATUS() = 0;


        /// <summary>
        /// Getter for ERROR_AXIS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_ERROR_AXIS() = 0;


        /// <summary>
        /// Getter for FPU_EXCEPTIONS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_FPU_EXCEPTIONS() = 0;

        /// <summary>
        /// Setter for FPU_EXCEPTIONS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_FPU_EXCEPTIONS(int64_t value) = 0;


        /// <summary>
        /// Getter for HMI_PROC system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_HMI_PROC() = 0;
```

```
        /// <summary>
        /// Setter for HMI_PROC system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_HMI_PROC(int32_t value) = 0;


        /// <summary>
        /// Getter for IP_ADDRESS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_ADDRESS() = 0;

        /// <summary>
        /// Setter for IP_ADDRESS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_ADDRESS(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_GATEWAY system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_GATEWAY() = 0;

        /// <summary>
        /// Setter for IP_GATEWAY system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_GATEWAY(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_MAC system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetSystemParameter_IP_MAC() = 0;

        /// <summary>
        /// Setter for IP_MAC system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_MAC(uint64_t value) = 0;


        /// <summary>
        /// Getter for IP_MEMORY_CONFIG system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_MEMORY_CONFIG() = 0;

        /// <summary>
        /// Setter for IP_MEMORY_CONFIG system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_MEMORY_CONFIG(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_NETMASK system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_NETMASK() = 0;

        /// <summary>
```

```
        /// Setter for IP_NETMASK system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_NETMASK(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_PROTOCOL_CONFIG system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_PROTOCOL_CONFIG() = 0;

        /// <summary>
        /// Setter for IP_PROTOCOL_CONFIG system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_PROTOCOL_CONFIG(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_PROTOCOL_CTRL system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_PROTOCOL_CTRL() = 0;

        /// <summary>
        /// Setter for IP_PROTOCOL_CTRL system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_PROTOCOL_CTRL(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_TCP_TIMEOUT system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetSystemParameter_IP_TCP_TIMEOUT() = 0;

        /// <summary>
        /// Setter for IP_TCP_TIMEOUT system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_TCP_TIMEOUT(uint64_t value) = 0;


        /// <summary>
        /// Getter for IP_TCP_TX_THRESHOLD system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_TCP_TX_THRESHOLD() = 0;

        /// <summary>
        /// Setter for IP_TCP_TX_THRESHOLD system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_TCP_TX_THRESHOLD(uint32_t value) = 0;


        /// <summary>
        /// Getter for IP_TCP_TX_TIMEOUT system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_IP_TCP_TX_TIMEOUT() = 0;

        /// <summary>
        /// Setter for IP_TCP_TX_TIMEOUT system parameter
        /// </summary>
```

```
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_IP_TCP_TX_TIMEOUT(uint32_t value) = 0;


        /// <summary>
        /// Getter for LAST_AXIS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_LAST_AXIS() = 0;

        /// <summary>
        /// Setter for LAST_AXIS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_LAST_AXIS(int32_t value) = 0;


        /// <summary>
        /// Getter for MODBUS_TX_DELAY system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint16_t GetSystemParameter_MODBUS_TX_DELAY() = 0;

        /// <summary>
        /// Setter for MODBUS_TX_DELAY system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_MODBUS_TX_DELAY(uint16_t value) = 0;


        /// <summary>
        /// Getter for MOTION_ERROR system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint64_t GetSystemParameter_MOTION_ERROR() = 0;


        /// <summary>
        /// Getter for MOTION_ERROR_COUNT system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_MOTION_ERROR_COUNT() = 0;


        /// <summary>
        /// Getter for MPE_MODE system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetSystemParameter_MPE_MODE() = 0;


        /// <summary>
        /// Getter for NOP system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_NOP() = 0;


        /// <summary>
        /// Getter for NIN system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_NIN() = 0;


        /// <summary>
        /// Getter for NIO system parameter
```

```
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_NIO() = 0;


        /// <summary>
        /// Getter for N_ANA_IN system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_N_ANA_IN() = 0;


        /// <summary>
        /// Getter for N_ANA_OUT system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_N_ANA_OUT() = 0;


        /// <summary>
        /// Getter for PCMCAT_PROC system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_PCMCAT_PROC() = 0;

        /// <summary>
        /// Setter for PCMCAT_PROC system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_PCMCAT_PROC(int32_t value) = 0;


        /// <summary>
        /// Getter for PLC_CONFIG system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_PLC_CONFIG() = 0;

        /// <summary>
        /// Setter for PLC_CONFIG system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_PLC_CONFIG(int32_t value) = 0;


        /// <summary>
        /// Getter for PLC_ERROR system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_PLC_ERROR() = 0;


        /// <summary>
        /// Getter for PLC_OVERFLOW system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_PLC_OVERFLOW() = 0;


        /// <summary>
        /// Getter for PLC_RUN system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_PLC_RUN() = 0;


        /// <summary>
```

```
        /// Getter for REMOTE_PROC system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_REMOTE_PROC() = 0;

        /// <summary>
        /// Setter for REMOTE_PROC system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_REMOTE_PROC(int32_t value) = 0;


        /// <summary>
        /// Getter for SCHEDULE_TYPE system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SCHEDULE_TYPE() = 0;

        /// <summary>
        /// Setter for SCHEDULE_TYPE system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SCHEDULE_TYPE(uint32_t value) = 0;


        /// <summary>
        /// Getter for SCOPE_CYCLE_COUNT system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_SCOPE_CYCLE_COUNT() = 0;

        /// <summary>
        /// Setter for SCOPE_CYCLE_COUNT system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SCOPE_CYCLE_COUNT(int32_t value) = 0;


        /// <summary>
        /// Getter for SCOPE_DELAY system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetSystemParameter_SCOPE_DELAY() = 0;

        /// <summary>
        /// Setter for SCOPE_DELAY system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SCOPE_DELAY(int64_t value) = 0;


        /// <summary>
        /// Getter for SCOPE_POS system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_SCOPE_POS() = 0;

        /// <summary>
        /// Setter for SCOPE_POS system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SCOPE_POS(int32_t value) = 0;


        /// <summary>
        /// Getter for SCOPE_TRIGGER_POS system parameter
        /// </summary>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_SCOPE_TRIGGER_POS() = 0;


        /// <summary>
        /// Getter for SERIAL_NUMBER system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_SERIAL_NUMBER() = 0;

        /// <summary>
        /// Setter for SERIAL_NUMBER system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SERIAL_NUMBER(int32_t value) = 0;


        /// <summary>
        /// Getter for SERVO_OFFSET system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_SERVO_OFFSET() = 0;

        /// <summary>
        /// Setter for SERVO_OFFSET system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SERVO_OFFSET(int32_t value) = 0;


        /// <summary>
        /// Getter for SERVO_PERIOD system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_SERVO_PERIOD() = 0;

        /// <summary>
        /// Setter for SERVO_PERIOD system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SERVO_PERIOD(int32_t value) = 0;


        /// <summary>
        /// Getter for SYSTEM_ERROR_MASK system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SYSTEM_ERROR_MASK() = 0;

        /// <summary>
        /// Setter for SYSTEM_ERROR_MASK system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SYSTEM_ERROR_MASK(uint32_t value) = 0;


        /// <summary>
        /// Getter for SYSTEM_LOAD system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SYSTEM_LOAD() = 0;


        /// <summary>
        /// Getter for SYSTEM_LOAD_MAX system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_SYSTEM_LOAD_MAX() = 0;

        /// <summary>
        /// Setter for SYSTEM_LOAD_MAX system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_SYSTEM_LOAD_MAX(uint32_t value) = 0;


        /// <summary>
        /// Getter for TEXT_FILE_LOADER_PROC system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_TEXT_FILE_LOADER_PROC() = 0;

        /// <summary>
        /// Setter for TEXT_FILE_LOADER_PROC system parameter
        /// </summary>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSystemParameter_TEXT_FILE_LOADER_PROC(int32_t value) = 0;


        /// <summary>
        /// Getter for TSIZE system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_TSIZE() = 0;


        /// <summary>
        /// Getter for UNIT_ERROR system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetSystemParameter_UNIT_ERROR() = 0;


        /// <summary>
        /// Getter for VERSION system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetSystemParameter_VERSION() = 0;


        /// <summary>
        /// Getter for VRSIZE system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSystemParameter_VRSIZE() = 0;


        /// <summary>
        /// Getter for PROCNUMBER system parameter
        /// </summary>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetSystemParameter_PROCNUMBER() = 0;


    ////////////
```

```
module.</exception>
        virtual double GetAxisParameter_ACCEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ACCEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ADDAX_AXIS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_ADDAX_AXIS(int32_t axis) = 0;


        /// <summary>
        /// Getter for AFF_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AFF_GAIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for AFF_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AFF_GAIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ATYPE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual AxisType GetAxisParameter_ATYPE(int32_t axis) = 0;

        /// <summary>
        /// Setter for ATYPE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ATYPE(int32_t axis, AxisType value) = 0;


        /// <summary>
        /// Getter for AXIS_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_ACCEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_ACCEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for AXIS_ADDRESS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual int32_t GetAxisParameter_AXIS_ADDRESS(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_ADDRESS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_ADDRESS(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for AXIS_DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_DECEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_DECEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for AXIS_DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_DPOS(int32_t axis) = 0;


        /// <summary>
        /// Getter for AXIS_ENABLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetAxisParameter_AXIS_ENABLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_ENABLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_ENABLE(int32_t axis, bool value) = 0;


        /// <summary>
        /// Getter for AXIS_ERROR_COUNT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_AXIS_ERROR_COUNT(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_ERROR_COUNT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_ERROR_COUNT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for AXIS_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_FS_LIMIT(int32_t axis) = 0;
```

```
        /// <summary>
        /// Setter for AXIS_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_FS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for AXIS_MAX_JOG_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_MAX_JOG_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_MAX_JOG_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_MAX_JOG_SPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for AXIS_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual AxisMode GetAxisParameter_AXIS_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_MODE(int32_t axis, AxisMode value) = 0;


        /// <summary>
        /// Getter for AXIS_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_RS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_RS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for AXIS_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_SPEED(int32_t axis, double value) = 0;
```

```
        /// <summary>
        /// Getter for AXIS_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_AXIS_UNITS(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_UNITS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for AXISSTATUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual AxisStatus GetAxisParameter_AXISSTATUS(int32_t axis) = 0;


        /// <summary>
        /// Getter for BACKLASH_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_BACKLASH_DIST(int32_t axis) = 0;

        /// <summary>
        /// Setter for BACKLASH_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_BACKLASH_DIST(int32_t axis, int64_t value) = 0;


        /// <summary>
        /// Getter for CHANGE_DIR_LAST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_CHANGE_DIR_LAST(int32_t axis) = 0;

        /// <summary>
        /// Setter for CHANGE_DIR_LAST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_CHANGE_DIR_LAST(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for CLOSE_WIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_CLOSE_WIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for CLOSE_WIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_CLOSE_WIN(int32_t axis, double value) = 0;


        /// <summary>
```

```
        /// Getter for CLUTCH_RATE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_CLUTCH_RATE(int32_t axis) = 0;

        /// <summary>
        /// Setter for CLUTCH_RATE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_CLUTCH_RATE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for CORNER_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual CornerMode GetAxisParameter_CORNER_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for CORNER_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_CORNER_MODE(int32_t axis, CornerMode value) = 0;


        /// <summary>
        /// Getter for CORNER_STATE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual CornerState GetAxisParameter_CORNER_STATE(int32_t axis) = 0;

        /// <summary>
        /// Setter for CORNER_STATE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_CORNER_STATE(int32_t axis, CornerState value) = 0;


        /// <summary>
        /// Getter for CREEP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_CREEP(int32_t axis) = 0;

        /// <summary>
        /// Setter for CREEP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_CREEP(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for D_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_D_GAIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for D_GAIN axis parameter
        /// </summary>
```

```
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_D_GAIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for DAC axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DAC(int32_t axis) = 0;

        /// <summary>
        /// Setter for DAC axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DAC(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DAC_SCALE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_DAC_SCALE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DAC_SCALE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DAC_SCALE(int32_t axis, int64_t value) = 0;


        /// <summary>
        /// Getter for DAC_OUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DAC_OUT(int32_t axis) = 0;


        /// <summary>
        /// Getter for DATUM_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_DATUM_IN(int32_t axis) = 0;

        /// <summary>
        /// Setter for DATUM_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DATUM_IN(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_DECEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
```

```
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DECEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for DECEL_ANGLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_DECEL_ANGLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DECEL_ANGLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DECEL_ANGLE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for DEMAND_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_DEMAND_SPEED(int32_t axis) = 0;


        /// <summary>
        /// Getter for DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_DPOS(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_CONTROL(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_CONTROL(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_CONTROLWORD axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_CONTROLWORD(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_CONTROLWORD axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_CONTROLWORD(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_CW_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_CW_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_CW_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_CW_MODE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_CURRENT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_DRIVE_CURRENT(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_ENABLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_DRIVE_ENABLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_ENABLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_ENABLE(int32_t axis, int64_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_FE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_DRIVE_FE(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_FE_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_DRIVE_FE_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_FE_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_FE_LIMIT(int32_t axis, int64_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_INPUTS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetAxisParameter_DRIVE_INPUTS(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_MODE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_MONITOR axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_MONITOR(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_PROFILE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_PROFILE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_PROFILE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_PROFILE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_STATUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_STATUS(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_TORQUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_TORQUE(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_TYPE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_TYPE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_TYPE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_TYPE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_VELOCITY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_VELOCITY(int32_t axis) = 0;


        /// <summary>
        /// Getter for ENCODER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_ENCODER(int32_t axis) = 0;


        /// <summary>
        /// Getter for ENCODER_BITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint16_t GetAxisParameter_ENCODER_BITS(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENCODER_BITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENCODER_BITS(int32_t axis, uint16_t value) = 0;


        /// <summary>
        /// Getter for ENCODER_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_ENCODER_CONTROL(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENCODER_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENCODER_CONTROL(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for ENCODER_FILTER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ENCODER_FILTER(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENCODER_FILTER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENCODER_FILTER(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ENCODER_ID axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_ENCODER_ID(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENCODER_ID axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENCODER_ID(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for ENCODER_STATUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_ENCODER_STATUS(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENCODER_STATUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENCODER_STATUS(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for ENCODER_TURNS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_ENCODER_TURNS(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENCODER_TURNS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENCODER_TURNS(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for END_DIR_LAST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_END_DIR_LAST(int32_t axis) = 0;

        /// <summary>
        /// Setter for END_DIR_LAST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_END_DIR_LAST(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ENDMOVE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_ENDMOVE(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENDMOVE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENDMOVE(int32_t axis, int64_t value) = 0;


        /// <summary>
        /// Getter for ENDMOVE_BUFFER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

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module.</exception>
        virtual double GetAxisParameter_ENDMOVE_BUFFER(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENDMOVE_BUFFER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENDMOVE_BUFFER(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ENDMOVE_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ENDMOVE_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for ENDMOVE_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ENDMOVE_SPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ERRORMASK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual AxisStatus GetAxisParameter_ERRORMASK(int32_t axis) = 0;

        /// <summary>
        /// Setter for ERRORMASK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ERRORMASK(int32_t axis, AxisStatus value) = 0;


        /// <summary>
        /// Getter for FAST_JOG axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_FAST_JOG(int32_t axis) = 0;

        /// <summary>
        /// Setter for FAST_JOG axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FAST_JOG(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for FASTDEC axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FASTDEC(int32_t axis) = 0;

        /// <summary>
        /// Setter for FASTDEC axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FASTDEC(int32_t axis, double value) = 0;
```

```
        /// <summary>
        /// Getter for FE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FE(int32_t axis) = 0;


        /// <summary>
        /// Getter for FE_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FE_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for FE_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FE_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FE_LIMIT_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_FE_LIMIT_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for FE_LIMIT_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FE_LIMIT_MODE(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for FE_RANGE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FE_RANGE(int32_t axis) = 0;

        /// <summary>
        /// Setter for FE_RANGE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FE_RANGE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FE_LATCH axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FE_LATCH(int32_t axis) = 0;


        /// <summary>
        /// Getter for FHOLD_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_FHOLD_IN(int32_t axis) = 0;
```

```
        /// <summary>
        /// Setter for FHOLD_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FHOLD_IN(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for FHSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FHSPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for FHSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FHSPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FORCE_DWELL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_FORCE_DWELL(int32_t axis) = 0;

        /// <summary>
        /// Setter for FORCE_DWELL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FORCE_DWELL(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for FORCE_RAMP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FORCE_RAMP(int32_t axis) = 0;

        /// <summary>
        /// Setter for FORCE_RAMP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FORCE_RAMP(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FORCE_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FORCE_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for FORCE_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FORCE_SPEED(int32_t axis, double value) = 0;


```

```
        /// <summary>
        /// Getter for FRAME_REP_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FRAME_REP_DIST(int32_t axis) = 0;

        /// <summary>
        /// Setter for FRAME_REP_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FRAME_REP_DIST(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FRAME_SCALE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FRAME_SCALE(int32_t axis) = 0;

        /// <summary>
        /// Setter for FRAME_SCALE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FRAME_SCALE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FULL_SP_RADIUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FULL_SP_RADIUS(int32_t axis) = 0;

        /// <summary>
        /// Setter for FULL_SP_RADIUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FULL_SP_RADIUS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FWD_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_FWD_IN(int32_t axis) = 0;

        /// <summary>
        /// Setter for FWD_IN axis parameter
```

```
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FWD_IN(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for FWD_JOG axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_FWD_JOG(int32_t axis) = 0;

        /// <summary>
        /// Setter for FWD_JOG axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FWD_JOG(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for I_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_I_GAIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for I_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_I_GAIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for IDLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetAxisParameter_IDLE(int32_t axis) = 0;


        /// <summary>
        /// Getter for IN_POS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetAxisParameter_IN_POS(int32_t axis) = 0;

        /// <summary>
        /// Setter for IN_POS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_IN_POS(int32_t axis, uint32_t value) = 0;


        /// <summary>
        /// Getter for IN_POS_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_IN_POS_DIST(int32_t axis) = 0;

        /// <summary>
        /// Setter for IN_POS_DIST axis parameter
        /// </summary>
```

```
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_IN_POS_DIST(int32_t axis, int64_t value) = 0;


        /// <summary>
        /// Getter for IN_POS_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_IN_POS_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for IN_POS_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_IN_POS_SPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for INTERP_FACTOR axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_INTERP_FACTOR(int32_t axis) = 0;

        /// <summary>
        /// Setter for INTERP_FACTOR axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_INTERP_FACTOR(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for INVERT_STEP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_INVERT_STEP(int32_t axis) = 0;

        /// <summary>
        /// Setter for INVERT_STEP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_INVERT_STEP(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for JERK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_JERK(int32_t axis) = 0;

        /// <summary>
        /// Setter for JERK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_JERK(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for JOGSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_JOGSPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for JOGSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_JOGSPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for LINK_AXIS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_LINK_AXIS(int32_t axis) = 0;


        /// <summary>
        /// Getter for LOADED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetAxisParameter_LOADED(int32_t axis) = 0;


        /// <summary>
        /// Getter for LOOKAHEAD_FACTOR axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_LOOKAHEAD_FACTOR(int32_t axis) = 0;

        /// <summary>
        /// Setter for LOOKAHEAD_FACTOR axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_LOOKAHEAD_FACTOR(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MARK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_MARK(int32_t axis) = 0;


        /// <summary>
        /// Getter for MARKB axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_MARKB(int32_t axis) = 0;


        /// <summary>
        /// Getter for MERGE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetAxisParameter_MERGE(int32_t axis) = 0;

        /// <summary>
        /// Setter for MERGE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
```

```
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MERGE(int32_t axis, bool value) = 0;


        /// <summary>
        /// Getter for MOVE_COUNT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_MOVE_COUNT(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_COUNT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_COUNT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for MOVE_PA axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_MOVE_PA(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_PA axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_PA(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MOVE_PA_CONT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_MOVE_PA_CONT(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_PA_CONT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_PA_CONT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for MOVE_PA_IDLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_MOVE_PA_IDLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_PA_IDLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_PA_IDLE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MOVE_PB axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_MOVE_PB(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_PB axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_PB(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MOVE_PB_CONT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_MOVE_PB_CONT(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_PB_CONT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_PB_CONT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for MOVE_PB_IDLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_MOVE_PB_IDLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVE_PB_IDLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVE_PB_IDLE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MOVELINK_MODIFY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_MOVELINK_MODIFY(int32_t axis) = 0;

        /// <summary>
        /// Setter for MOVELINK_MODIFY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_MOVELINK_MODIFY(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MOVES_BUFFERED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_MOVES_BUFFERED(int32_t axis) = 0;


        /// <summary>
        /// Getter for MPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual double GetAxisParameter_MPOS(int32_t axis) = 0;


        /// <summary>
        /// Getter for MSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_MSPEED(int32_t axis) = 0;


        /// <summary>
        /// Getter for MSPEED_FILTER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_MSPEED_FILTER(int32_t axis) = 0;

        /// <summary>
        /// Setter for MSPEED_FILTER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_MSPEED_FILTER(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for MSPEEDF axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_MSPEEDF(int32_t axis) = 0;


        /// <summary>
        /// Getter for MTYPE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual MotionType GetAxisParameter_MTYPE(int32_t axis) = 0;


        /// <summary>
        /// Getter for NTYPE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual MotionType GetAxisParameter_NTYPE(int32_t axis) = 0;


        /// <summary>
        /// Getter for OFFPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_OFFPOS(int32_t axis) = 0;

        /// <summary>
        /// Setter for OFFPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_OFFPOS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for OPEN_WIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_OPEN_WIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for OPEN_WIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_OPEN_WIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for OUTLIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_OUTLIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for OUTLIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_OUTLIMIT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for OV_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_OV_GAIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for OV_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_OV_GAIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for P_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_P_GAIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for P_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_P_GAIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for POS_DELAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_POS_DELAY(int32_t axis) = 0;

        /// <summary>
        /// Setter for POS_DELAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual void SetAxisParameter_POS_DELAY(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for PP_STEP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_PP_STEP(int32_t axis) = 0;

        /// <summary>
        /// Setter for PP_STEP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_PP_STEP(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for PS_ENCODER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_PS_ENCODER(int32_t axis) = 0;


        /// <summary>
        /// Getter for RAISE_ANGLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_RAISE_ANGLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for RAISE_ANGLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_RAISE_ANGLE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for REG_POS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_REG_POS(int32_t axis) = 0;


        /// <summary>
        /// Getter for REG_POSB axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_REG_POSB(int32_t axis) = 0;


        /// <summary>
        /// Getter for REMAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_REMAIN(int32_t axis) = 0;


        /// <summary>
        /// Getter for REP_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
```

```
module.</exception>
        virtual double GetAxisParameter_REP_DIST(int32_t axis) = 0;

        /// <summary>
        /// Setter for REP_DIST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REP_DIST(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for REP_OPTION axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_REP_OPTION(int32_t axis) = 0;

        /// <summary>
        /// Setter for REP_OPTION axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REP_OPTION(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for REV_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_REV_IN(int32_t axis) = 0;

        /// <summary>
        /// Setter for REV_IN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REV_IN(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for REV_JOG axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_REV_JOG(int32_t axis) = 0;

        /// <summary>
        /// Setter for REV_JOG axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REV_JOG(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for ROBOT_DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ROBOT_DPOS(int32_t axis) = 0;


        /// <summary>
        /// Getter for ROBOT_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
```

```
        virtual double GetAxisParameter_ROBOT_FS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for ROBOT_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ROBOT_FS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ROBOT_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ROBOT_RS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for ROBOT_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ROBOT_RS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ROBOT_SP_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_ROBOT_SP_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for ROBOT_SP_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ROBOT_SP_MODE(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for ROBOTSTATUS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual RobotStatus GetAxisParameter_ROBOTSTATUS(int32_t axis) = 0;


        /// <summary>
        /// Getter for ROBOT_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ROBOT_UNITS(int32_t axis) = 0;

        /// <summary>
        /// Setter for ROBOT_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ROBOT_UNITS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_RS_LIMIT(int32_t axis) = 0;
```

```
        /// <summary>
        /// Setter for RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_RS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for SERVO axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetAxisParameter_SERVO(int32_t axis) = 0;

        /// <summary>
        /// Setter for SERVO axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SERVO(int32_t axis, bool value) = 0;


        /// <summary>
        /// Getter for SLOT_NUMBER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_SLOT_NUMBER(int32_t axis) = 0;

        /// <summary>
        /// Setter for SLOT_NUMBER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SLOT_NUMBER(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for SRAMP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SRAMP(int32_t axis) = 0;

        /// <summary>
        /// Setter for SRAMP axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SRAMP(int32_t axis, int32_t value) = 0;


```

```
        /// <summary>
        /// Getter for S_REF axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_S_REF(int32_t axis) = 0;

        /// <summary>
        /// Setter for S_REF axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_S_REF(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for S_REF_OUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_S_REF_OUT(int32_t axis) = 0;

        /// <summary>
        /// Setter for S_REF_OUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_S_REF_OUT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for START_DIR_LAST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_START_DIR_LAST(int32_t axis) = 0;

        /// <summary>
        /// Setter for START_DIR_LAST axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_START_DIR_LAST(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for STARTMOVE_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_STARTMOVE_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for STARTMOVE_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_STARTMOVE_SPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for STOP_ANGLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_STOP_ANGLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for STOP_ANGLE axis parameter
```

```
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_STOP_ANGLE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for SYNC_AXIS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SYNC_AXIS(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_AXIS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_AXIS(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for SYNC_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual SyncControl GetAxisParameter_SYNC_CONTROL(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_CONTROL(int32_t axis, SyncControl value) = 0;


        /// <summary>
        /// Getter for SYNC_DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_SYNC_DPOS(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_DPOS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for SYNC_DWELL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SYNC_DWELL(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_DWELL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_DWELL(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for SYNC_FLIGHT axis parameter
        /// </summary>
```

```
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SYNC_FLIGHT(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_FLIGHT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_FLIGHT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for SYNC_PAUSE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SYNC_PAUSE(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_PAUSE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_PAUSE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for SYNC_TIME axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SYNC_TIME(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_TIME axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_TIME(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for SYNC_TIMER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_SYNC_TIMER(int32_t axis) = 0;


        /// <summary>
        /// Getter for SYNC_WITHDRAW axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_SYNC_WITHDRAW(int32_t axis) = 0;

        /// <summary>
        /// Setter for SYNC_WITHDRAW axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_SYNC_WITHDRAW(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for TABLE_POINTER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
```

```
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_TABLE_POINTER(int32_t axis) = 0;

        /// <summary>
        /// Setter for TABLE_POINTER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_TABLE_POINTER(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for TANG_DIRECTION axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_TANG_DIRECTION(int32_t axis) = 0;

        /// <summary>
        /// Setter for TANG_DIRECTION axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_TANG_DIRECTION(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for T_REF axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_T_REF(int32_t axis) = 0;

        /// <summary>
        /// Setter for T_REF axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_T_REF(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for T_REF_OUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_T_REF_OUT(int32_t axis) = 0;

        /// <summary>
        /// Setter for T_REF_OUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_T_REF_OUT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_UNITS(int32_t axis) = 0;

        /// <summary>
        /// Setter for UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
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module.</exception>
        virtual void SetAxisParameter_UNITS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for VECTOR_BUFFERED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_VECTOR_BUFFERED(int32_t axis) = 0;

        /// <summary>
        /// Setter for VECTOR_BUFFERED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_VECTOR_BUFFERED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for VERIFY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_VERIFY(int32_t axis) = 0;

        /// <summary>
        /// Setter for VERIFY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_VERIFY(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for VFF_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_VFF_GAIN(int32_t axis) = 0;

        /// <summary>
        /// Setter for VFF_GAIN axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_VFF_GAIN(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for VP_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_VP_ACCEL(int32_t axis) = 0;


        /// <summary>
        /// Getter for VP_JERK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_VP_JERK(int32_t axis) = 0;


        /// <summary>
        /// Getter for VP_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
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module.</exception>
        virtual int32_t GetAxisParameter_VP_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for VP_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_VP_MODE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for VP_POSITION axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_VP_POSITION(int32_t axis) = 0;


        /// <summary>
        /// Getter for VP_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_VP_SPEED(int32_t axis) = 0;


        /// <summary>
        /// Getter for WORLD_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_WORLD_ACCEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_WORLD_ACCEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for WORLD_DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_WORLD_DECEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_WORLD_DECEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for WORLD_DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_WORLD_DPOS(int32_t axis) = 0;


        /// <summary>
        /// Getter for WORLD_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
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module.</exception>
        virtual double GetAxisParameter_WORLD_FS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_WORLD_FS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for WORLD_JOGSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_WORLD_JOGSPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_JOGSPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_WORLD_JOGSPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for WORLD_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_WORLD_RS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_WORLD_RS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for WORLD_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_WORLD_SPEED(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_WORLD_SPEED(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for WORLD_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual double GetAxisParameter_WORLD_UNITS(int32_t axis) = 0;

        /// <summary>
        /// Setter for WORLD_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication
module.</exception>
        virtual void SetAxisParameter_WORLD_UNITS(int32_t axis, double value) = 0;
```

```
        /// <summary>
        /// Getter for AXIS_ENABLE_OVERRIDE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetAxisParameter_AXIS_ENABLE_OVERRIDE(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_ENABLE_OVERRIDE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_ENABLE_OVERRIDE(int32_t axis, bool value) = 0;


        /// <summary>
        /// Getter for AXIS_DISPLAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_AXIS_DISPLAY(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_DISPLAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_DISPLAY(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for AXIS_Z_OUTPUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_AXIS_Z_OUTPUT(int32_t axis) = 0;

        /// <summary>
        /// Setter for AXIS_Z_OUTPUT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_AXIS_Z_OUTPUT(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DEMAND_EDGES axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetAxisParameter_DEMAND_EDGES(int32_t axis) = 0;


        /// <summary>
        /// Getter for DRIVE_INDEX axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_INDEX(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_INDEX axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_INDEX(int32_t axis, int32_t value) = 0;

```

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        /// <summary>
        /// Getter for DRIVE_NEG_TORQUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_NEG_TORQUE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_NEG_TORQUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_NEG_TORQUE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_PARAMETER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_PARAMETER(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_PARAMETER axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_PARAMETER(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_POS_TORQUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_POS_TORQUE(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_POS_TORQUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_POS_TORQUE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_SET_VAL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_SET_VAL(int32_t axis) = 0;

        /// <summary>
        /// Setter for DRIVE_SET_VAL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_SET_VAL(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for DRIVE_VALUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_DRIVE_VALUE(int32_t axis) = 0;

        /// <summary>
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```
        /// Setter for DRIVE_VALUE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_DRIVE_VALUE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for FEED_OVERRIDE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FEED_OVERRIDE(int32_t axis) = 0;

        /// <summary>
        /// Setter for FEED_OVERRIDE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FEED_OVERRIDE(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FORCE_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FORCE_ACCEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for FORCE_ACCEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FORCE_ACCEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FORCE_DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FORCE_DECEL(int32_t axis) = 0;

        /// <summary>
        /// Setter for FORCE_DECEL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FORCE_DECEL(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FORCE_JERK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_FORCE_JERK(int32_t axis) = 0;

        /// <summary>
        /// Setter for FORCE_JERK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FORCE_JERK(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for FRAME axis parameter
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```
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_FRAME(int32_t axis) = 0;

        /// <summary>
        /// Setter for FRAME axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FRAME(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for FWD_START axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_FWD_START(int32_t axis) = 0;

        /// <summary>
        /// Setter for FWD_START axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_FWD_START(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for POSI_SEQ_DELAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_POSI_SEQ_DELAY(int32_t axis) = 0;

        /// <summary>
        /// Setter for POSI_SEQ_DELAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_POSI_SEQ_DELAY(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for POSI_SEQ_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_POSI_SEQ_MODE(int32_t axis) = 0;

        /// <summary>
        /// Setter for POSI_SEQ_MODE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_POSI_SEQ_MODE(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for PWM_CYCLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetAxisParameter_PWM_CYCLE(int32_t axis) = 0;

        /// <summary>
        /// Setter for PWM_CYCLE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
```

```
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_PWM_CYCLE(int32_t axis, uint32_t value) = 0;


        /// <summary>
        /// Getter for PWM_MARK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual uint32_t GetAxisParameter_PWM_MARK(int32_t axis) = 0;

        /// <summary>
        /// Setter for PWM_MARK axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_PWM_MARK(int32_t axis, uint32_t value) = 0;


        /// <summary>
        /// Getter for REG_INPUTS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_REG_INPUTS(int32_t axis) = 0;

        /// <summary>
        /// Setter for REG_INPUTS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REG_INPUTS(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for REGIST_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetAxisParameter_REGIST_CONTROL(int32_t axis) = 0;

        /// <summary>
        /// Setter for REGIST_CONTROL axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REGIST_CONTROL(int32_t axis, int32_t value) = 0;


        /// <summary>
        /// Getter for REGIST_DELAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_REGIST_DELAY(int32_t axis) = 0;

        /// <summary>
        /// Setter for REGIST_DELAY axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REGIST_DELAY(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for REGIST_SPEED axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_REGIST_SPEED(int32_t axis) = 0;


        /// <summary>
        /// Getter for REGIST_SPEEDB axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_REGIST_SPEEDB(int32_t axis) = 0;


        /// <summary>
        /// Getter for REV_START axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int16_t GetAxisParameter_REV_START(int32_t axis) = 0;

        /// <summary>
        /// Setter for REV_START axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_REV_START(int32_t axis, int16_t value) = 0;


        /// <summary>
        /// Getter for TCP_DPOS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_TCP_DPOS(int32_t axis) = 0;


        /// <summary>
        /// Getter for TCP_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_TCP_FS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for TCP_FS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_TCP_FS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for TCP_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_TCP_RS_LIMIT(int32_t axis) = 0;

        /// <summary>
        /// Setter for TCP_RS_LIMIT axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_TCP_RS_LIMIT(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for TCP_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
```

```
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_TCP_UNITS(int32_t axis) = 0;

        /// <summary>
        /// Setter for TCP_UNITS axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_TCP_UNITS(int32_t axis, double value) = 0;


        /// <summary>
        /// Getter for ROBOT_FE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ROBOT_FE(int32_t axis) = 0;


        /// <summary>
        /// Getter for ROBOT_FE_LATCH axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ROBOT_FE_LATCH(int32_t axis) = 0;


        /// <summary>
        /// Getter for ROBOT_FE_RANGE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual double GetAxisParameter_ROBOT_FE_RANGE(int32_t axis) = 0;

        /// <summary>
        /// Setter for ROBOT_FE_RANGE axis parameter
        /// </summary>
        /// <param name="axis">Target axis whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetAxisParameter_ROBOT_FE_RANGE(int32_t axis, double value) = 0;



        ////////////
```

```
        /// <summary>
        /// Setter for INDEVICE process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameter_INDEVICE(int32_t process, int32_t value) = 0;


        /// <summary>
        /// Getter for OUTDEVICE process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetProcessParameter_OUTDEVICE(int32_t process) = 0;

        /// <summary>
        /// Setter for OUTDEVICE process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameter_OUTDEVICE(int32_t process, int32_t value) = 0;


        /// <summary>
        /// Getter for PMOVE process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual bool GetProcessParameter_PMOVE(int32_t process) = 0;


        /// <summary>
        /// Getter for PROC_LINE process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetProcessParameter_PROC_LINE(int32_t process) = 0;


        /// <summary>
        /// Getter for RUN_ERROR process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetProcessParameter_RUN_ERROR(int32_t process) = 0;


        /// <summary>
        /// Getter for TICKS process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int64_t GetProcessParameter_TICKS(int32_t process) = 0;

        /// <summary>
        /// Setter for TICKS process parameter
        /// </summary>
        /// <param name="process">Target process whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetProcessParameter_TICKS(int32_t process, int64_t value) = 0;



    ////////////
```

```
        /// </summary>
        /// <param name="slot">Target slot whose parameter is retrieved</param>
        /// <returns>Current value of the parameter</returns>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual int32_t GetSlotParameter_SERIAL_NUMBER(int32_t slot) = 0;

        /// <summary>
        /// Setter for SERIAL_NUMBER slot parameter
        /// </summary>
        /// <param name="slot">Target slot whose parameter is modified</param>
        /// <param name="value">New value to set to parameter</param>
        /// <exception cref="TrioConnectionException">Thrown if error occurs in communication module.</exception>
        virtual void SetSlotParameter_SERIAL_NUMBER(int32_t slot, int32_t value) = 0;


////////////
```

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